

R&S®SMBVB-K144

5G New Radio

User Manual



This document describes the following software options:

- R&S®SMBVB-K144
1423.8608.02

This manual describes firmware version FW 4.70.006.xx and later of the R&S®SMBV100B.

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1 Preface

1.1 About this Manual

This user manual provides all the information **specific to the 5G New Radio option**. All general instrument functions and settings common to all applications and operating modes are described in the main R&S SMBV100B user manual.

The main focus in this manual is on the provided settings and the tasks required to generate a signal. The following topics are included:

- **Welcome to the 5G New Radio option**
Introduction to and getting familiar with the option
- **About the 5G New Radio option**
Background information on basic terms and principles in the context of the signal generation
- **5G New Radio Configuration and Settings**
A concise description of all functions and settings available to configure signal generation with their corresponding remote control command
- **Remote Control Commands**
Remote commands required to configure and perform signal generation in a remote environment, sorted by tasks
(Commands required to set up the instrument or to perform common tasks on the instrument are provided in the main R&S SMBV100B user manual)
Programming examples demonstrate the use of many commands and can usually be executed directly for test purposes
- **List of remote commands**
Alphabetical list of all remote commands described in the manual
- **Index**
- **Abbreviations**
- **List of documents and reference information**

1.2 Documentation Overview

This section provides an overview of the R&S SMBV100B user documentation. Unless specified otherwise, you find the documents on the R&S SMBV100B product page at:

www.rohde-schwarz.com/manual/smbv100b

1.2.1 Getting Started Manual

Introduces the R&S SMBV100B and describes how to set up and start working with the product. Includes basic operations, typical measurement examples, and general information, e.g. safety instructions, etc. A printed version is delivered with the instrument.

1.2.2 User Manuals and Help

Separate manuals for the base unit and the software options are provided for download:

- Base unit manual
Contains the description of all instrument modes and functions. It also provides an introduction to remote control, a complete description of the remote control commands with programming examples, and information on maintenance, instrument interfaces and error messages. Includes the contents of the getting started manual.
- Software option manual
Contains the description of the specific functions of an option. Basic information on operating the R&S SMBV100B is not included.

All user manuals are also available for download or for immediate display on the Internet.

1.2.3 Service Manual

Describes the performance test for checking the rated specifications, module replacement and repair, firmware update, troubleshooting and fault elimination, and contains mechanical drawings and spare part lists.

The service manual is available for registered users on the global Rohde & Schwarz information system (GLORIS, <https://gloris.rohde-schwarz.com>).

1.2.4 Instrument Security Procedures

Deals with security issues when working with the R&S SMBV100B in secure areas. It is available for download on the Internet.

1.2.5 Basic Safety Instructions

Contains safety instructions, operating conditions and further important information. The printed document is delivered with the instrument.

1.2.6 Data Sheets and Brochures

The data sheet contains the technical specifications of the R&S SMBV100B. It also lists the options and their order numbers and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

See www.rohde-schwarz.com/brochure-datasheet/smbv100b

1.2.7 Release Notes and Open Source Acknowledgment (OSA)

The release notes list new features, improvements and known issues of the current firmware version, and describe the firmware installation.

The open source acknowledgment document provides verbatim license texts of the used open source software.

See www.rohde-schwarz.com/firmware/smbv100b

1.2.8 Application Notes, Application Cards, White Papers, etc.

These documents deal with special applications or background information on particular topics.

See www.rohde-schwarz.com/application/smbv100b

2 Welcome to the 5G New Radio Option

The R&S SMBV100B-K144 is a firmware application that adds functionality to generate signals in accordance with the 3GPP standard 5G New Radio.

Preamble

All supported features are in line with 3GPP Release 15; the following official 3GPP specifications are implemented:

- 3GPP TS 38.211, Version 15.4.0
- 3GPP TS 38.212, Version 15.4.0
- 3GPP TS 38.213, Version 15.4.0
- 3GPP TS 38.214, Version 15.4.0

The R&S SMBV100B-K144 key features

The R&S SMBV100B simulates signals at the physical channel level. The following list gives an overview of the functions provided for generating an 5G New Radio signal:

- Supports uplink and downlink
- Intuitive user interface with graphical display of time plan
- PSS/SSS, PBCH (incl. MIB), PDCCH (i.e. CORESET), PDSCH, CSI-RS, PDSCH PTRS, PUSCH, PUCCH, SRS supported
PDSCH, PDCCH and PUSCH incl. DMRS
- Support of downlink and uplink DCI content inside CORESET;
Supported are DCI formats 0_0, 0_1, 1_0, 1_1 and custom
- Automatic PDSCH scheduling from DCI
- PDSCH time domain scheduling
- Support of two codewords for PDSCH
- Support of SRS codebook
- PUSCH/PUCCH spatial multiplexing
- Configurable multilayer transmissions for PDSCH and PUSCH, incl. USCH channel coding and antenna ports
- Support of FRCs and TMs
- All modulation schemes supported

This user manual contains a description of the functionality that the application provides, including remote control operation.

All functions not discussed in this manual are the same as in the base unit and are described in the R&S SMBV100B user manual. The latest version is available at:

www.rohde-schwarz.com/manual/SMBV100B

Installation

You can find detailed installation instructions in the delivery of the option or in the R&S SMBV100B service manual.



To play back a signal from a waveform file created by the simulation software R&S WinIQSIM2, the corresponding R&S WinIQSIM2 digital standard option must be installed.

2.1 Accessing the 5G New Radio Dialog

To open the dialog with 5G New Radio settings

- In the block diagram of the R&S SMBV100B, select "Baseband > 5G NR".

A dialog box opens that displays the provided general settings.

The signal generation is not started immediately. To start signal generation with the default settings, select "State > On".

2.2 Scope



Tasks (in manual or remote operation) that are also performed in the base unit in the same way are not described here.

In particular, it includes:

- Managing settings and data lists, like storing and loading settings, creating and accessing data lists, or accessing files in a particular directory.
- Information on regular trigger, marker and clock signals and filter settings, if appropriate.
- General instrument configuration, such as checking the system configuration, configuring networks and remote operation
- Using the common status registers

For a description of such tasks, see the R&S SMBV100B user manual.

2.3 Notes on Screenshots

When describing the functions of the product, we use sample screenshots. These screenshots are meant to illustrate as many as possible of the provided functions and possible interdependencies between parameters. The shown values may not represent realistic usage scenarios.

The screenshots usually show a fully equipped product, that is: with all options installed. Thus, some functions shown in the screenshots may not be available in your particular product configuration.

3 About the 5G New Radio Option

The implementation of the option 5G New Radio R&S SMBVB-K144 is based on the 3GPP Rel. 15 specification group TS 38.xxx. According to these specifications, the 3GPP defines a great flexibility of the signal on the air interface.

This section is a brief summary of the main air interface parameters, see [Table 3-1](#).

Table 3-1: Main air interface parameters

Parameter	Sub-6 GHz $f \leq 6$ GHz Frequency range FR1*	mmWave $f > 6$ GHz Frequency range FR2*
Frequency bands**	3.3 GHz to 3.8 GHz 4.4 GHz to 6 GHz	24 GHz to 29 GHz 27 GHz to 43.5 GHz
Carrier aggregation	≤ 16 carriers Initially 1	≤ 16 carriers
Channel bandwidth	5, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90 MHz and 100 MHz	50, 100, 200 MHz and 400 MHz
Subcarrier spacing (SCS)	15 kHz, 30 kHz and 60 kHz	60 kHz and 120 kHz
Modulation schemes	DL: up to 256QAM UL: up to 256QAM, incl. $\pi/2$ BPSK for DFTs	256QAM
MIMO schemes	- DL: 2x2 and 4x4 MIMO (≤ 8 layers) - UL: SISO and 2x2 MIMO (≤ 4 layers)	≤ 2 layers: - DL: 2x2 MIMO - UL: SISO and 2x2 MIMO
Multiple access schemes	- DL: CP-OFDMA - UL: CP-OFDMA and optionally DFT-s-OFDM	- DL: CP-OFDMA - UL: CP-OFDMA and optionally DFT-s-OFDM
Duplexing mode	TDD FDD	TDD

*) Throughout this description, the designations are used interchangeably.

**) Frequency bands as defined by the 3GPP Rel. 15 specification group TS 38.xxx. You can however select also any other frequency within the frequency range your R&S SMBV100B supports (see [Chapter 3.1, "Required Options"](#), on page 12).

3.1 Required Options

The basic equipment layout for generating 5G New Radio signals includes the options:

- Base unit
- Baseband realtime extension (R&S SMBVB-B520)
- Option 5G New Radio (R&S SMBVB-K144)

To play back a signal from a waveform file created by the simulation software R&S WinIQSIM2, the corresponding R&S WinIQSIM2 digital standard option must be installed.

For more information, see data sheet.

3.2 5G NR Numerology

5G NR signals can be transmitted in several frequency ranges. For each of the frequency range, the 3GPP specifies the allowed subcarrier spacing (SCS) and the supported cyclic prefix (CP), see [Table 3-1](#).

However, not all combinations of frequency range, SCS and CP are allowed. Moreover, the 3GPP specification defines the allowed combinations indirectly, as the so called numerology. A numerology is thus the combination of the SCS and the CP, where the SCS is defined as follows:

$$\Delta f = 15 \text{ kHz} * 2^\mu, \text{ with } \mu = 0 \text{ to } 4.$$

Table 3-2: Supported combinations of SCS and CP per frequency range

μ	SCS, kHz (Δf)	Max. channel bandwidth, MHz	$N_{\text{symbol}}^{\text{slot}}$	$N_{\text{slot}}^{\text{frames}}$	$N_{\text{sub-frame}}^{\text{slot}}$	Symbol duration μs	Normal cyclic prefix (NCP)	Extended cyclic prefix (ECP)	Supported for data	Supported for synchronization	FR1	FR2
0	15	50	14	10	1	66.67	x	-	x	x	x	-
1	30	100	14	20	2	33.33	x	-	x	x	x	-
2	60	100 (FR1) 200 (FR2)	14 (NCP) 12 (ECP)	40	4	16.67	x	x	x	-	x	x
3	120	400	14	80	8	8.33	x	-	x	x	-	x
4	240	-	14	160	16	4.17	x	-	-	x	-	x

3.3 Multiple Access Schemes

5G NR supports two channel access methods for uplink signals, CP-OFDM and DFT-s-OFDM (see [Table 3-1](#)). Depending on the use case, one or the other has advantages over the other. For DFT-s-OFDM, an additional signal processing stage, transform precoding, is applied. Transform precoding is the term for the digital Fourier transformation (DFT) used in the 5G NR standard.

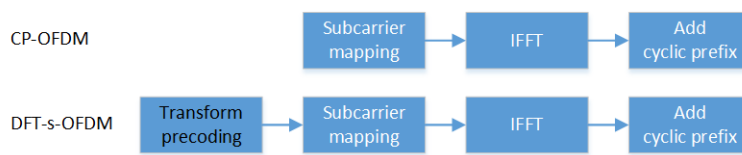


Figure 3-1: Signal processing depending on multiple access scheme

In this implementation, you can define per PUSCh which of the two access schemes is used.

3.4 Frame Structure, Slots and Slot Formats

Frame structure

As it is in LTE, the 5G NR **frame** is 10 ms long; a half-frame lasts 5 ms. Frames consist of 10 subframes, each of them with a length of 1 ms.

Subframes contain a *variable number of slots*, depending on the subcarrier spacing (SCS). Moreover, subframes can use different SCS in different bandwidth parts (BWP), see [Figure 3-4](#).

Because the frame and subframe lengths are constant but the number of slots and their lengths varies, the common time boundaries between the different numerologies are the frames or the subframes.

Slots and slots formats

The basic transmission unit in 5G NR is the **slot**. Slots can have different formats; slots comprise of 14 or 12 OFDM symbols, depending on the used cyclic prefix (CP). Hence, depending on the used CP, there are different numbers of slots per frame (and sub-frame) and the CP itself depends on the used numerology, see [Table 3-2](#).

Slots can have different formats, defined by the slot format index (SFI). The SFI is signaled by group common PDCCHs. The slot formats differ in the number of symbols that can be used for DL (D) and UL (U) transmission. Most of the symbols are usually used for transmission of user data (payload). Slots can also contain flexible symbols (x), that can be used for dynamic DL/UL transmission or as a gap for the DL to UL and vice versa switching.

The slot formats are defined in [TS 38.211](#), table 4.3.2-3.

3.5 Resource Blocks and Resource Grid

Resource grid

The basic granularity in 5G NR resource grid is the **resource element** (RE), which consists of one OFDM symbol over one subcarrier and is the same as in LTE. Another

granularity unit is a resource block (RB), which spans 12 subcarriers in frequency domain.

Resource blocks

The 5G NR specification defines different types of resource blocks:

- **Common resource blocks (CRB)**
The term CRB is typically used when referring to the whole carrier. A CRB grid is defined for each numerology.
The center of subcarrier#0 of the CRB#0 is referred as **Reference point A**; it is signaled relative to SS/PBCH or ARFCN.
See [Figure 3-2](#) and [Figure 3-3](#).
- **Usable resource blocks (URBs)**
The term URB is used to specify the subset of the CRBs of a specific numerology where transmission is allowed; that is, these RBs are not in the guard bands. The URBs define the so called transmission bandwidth (TxBW) for the respective numerology, see [Chapter 3.6, "Transmission Bandwidths TxBW"](#), on page 15.
- **Physical resource blocks (PRB)**
The term PRB describes the RBs inside the bandwidth parts (BWP). PRBs are numbered relative to the beginning of the bandwidth part.
See [Carrier Bandwidth Part \(BWP\)](#) and [Figure 5-1](#).

3.6 Transmission Bandwidths TxBW

As listed in [Table 3-2](#), there are different numerologies (that is different combinations of subcarrier spacing SCS and cyclic prefix CP) supported for each deployment and channel bandwidth combination. The different SCSs results in transmission bandwidths TxBW, which span different frequency ranges and according to [TS 38.104](#) use different number of resource blocks N_{RB} . The start of the TxBW is signaled relative to the Reference point A and is numerology-specific.

To achieve a common alignment between the different TxBW within a carrier, the 3GPP specification defines the following rules:

- Alignment is achieved, if the center of the first subcarrier (SC#0) of an RB is aligned to the center of the SC#0 of the RBs of all numerologies with a smaller μ .
- The TxBW are centered on the carrier center, that is around the center of the "Channel Bandwidth" for the corresponding carrier.
- If not applicable, the TxBW can be shifted with ± 6 subcarriers relative to the carrier center, where for each TxBW the corresponding SCS applies.

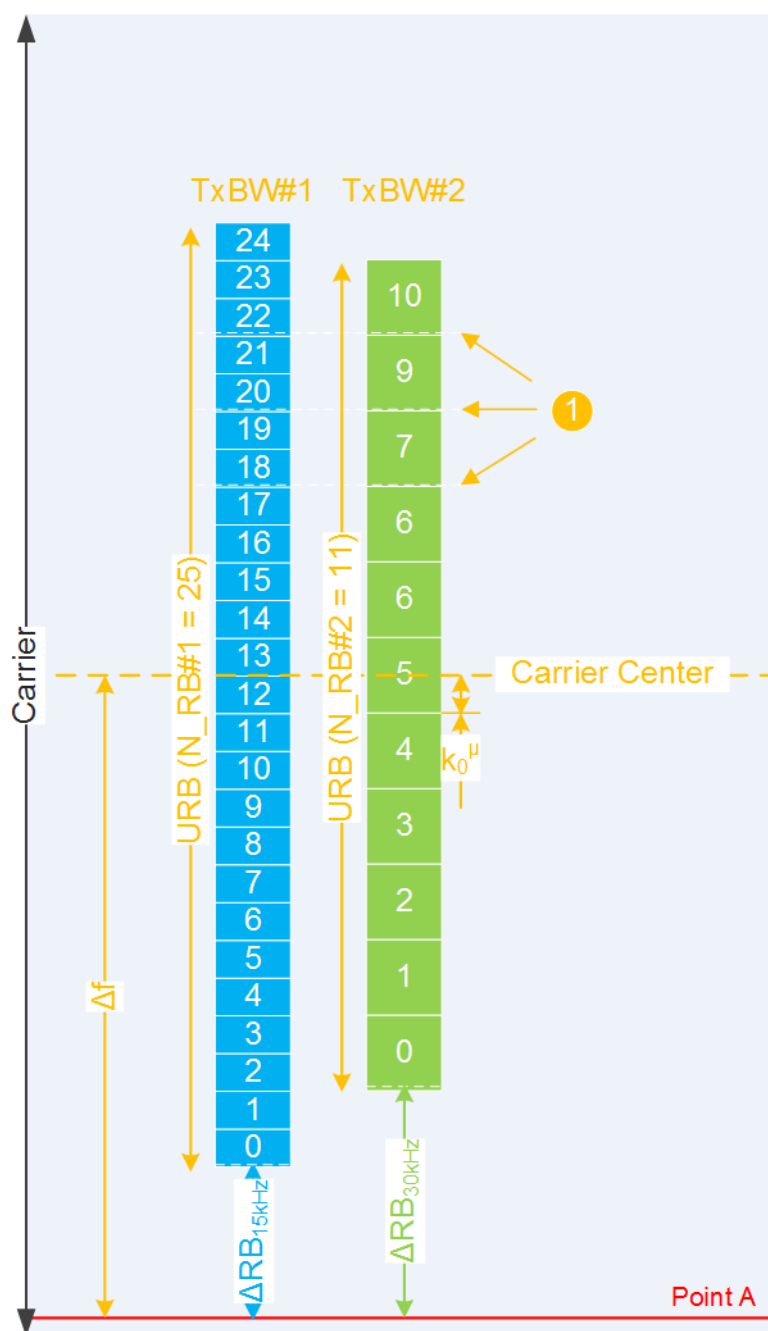


Figure 3-2: Transmission bandwidth (example)

- Carrier = Channel bandwidth = 5 MHz
 RB = Resource blocks
 SCS = Subcarrier spacing
 TxBW_s = Transmission bandwidth
 TxBW_s#1 = SCS 15 kHz, usable resource blocks USB = "N_RB = 25"; TxBW_s is not centered on the carrier center, a frequency shift with $k_0^u = -6$ subcarriers is applied
 TxBW_s#2 = SCS 30 kHz, usable resource blocks USB = "N_RB = 11"; TxBW_s is centered on the carrier center (" $k_0^u = 0$ ")
 k_0^u = Number of subcarriers the TxBW is shifted related to the carrier center, see k_0^u
 ΔRB = Usable RB to common RBs, see [TxBW Offset](#)

1	= Aligned centers of the first subcarrier (SC#0) of resource blocks
Point A	= Common reference point for all numerologies and TxBW; does not have to be within the carrier bandwidth. Point A is the center of subcarrier#0 of the CRB#0
Carrier center	= Center frequency of the carrier
Δf	= Offset between the carrier center frequency and the reference point A, see Point A to Carrier Center .

From all possible TxBW; for a specific channel bandwidth, the base station decides which TxBW; to use. The base station/network signals the position of the used TxBW; to the users as function of the following:

- The position of the Reference point A, that is common to all numerologies.
- The frequency offset ΔRB between the usable resource blocks of the TxBW; and the Reference point A, see [TxBW Offset](#).
- The used frequency shift $k_0^u = \{-6, 0, 6\}$, see [k0u](#).

3.7 Synchronization Signals and SS/PBCH Block

The 3GPP specifies two synchronization signals (SS), the primary synchronization signal (PSS) and the secondary synchronization signal (SSS). Together with the physical broadcast channel (PBCH), they are bundled in a synchronization signal block (SS/PBCH block). PSS/SSS are used for radio frame synchronization and detection of the physical layer cell ID. The location of the synchronization signal is variable.

The PBCH carries general system information. It is polar-coded and QPSK modulated. The PBCH symbols carry its own DMRS, which position depends on the physical layer cell ID.

An SS/PBCH block is transmitted on a fixed schedule. Each half-frame contains of either 4, 8 or 64 SS/PBCH blocks, depending on the subcarrier spacing (SCS). The SS/PBCH block is transmitted twice in a slot, if that slot carries the synchronization signal.

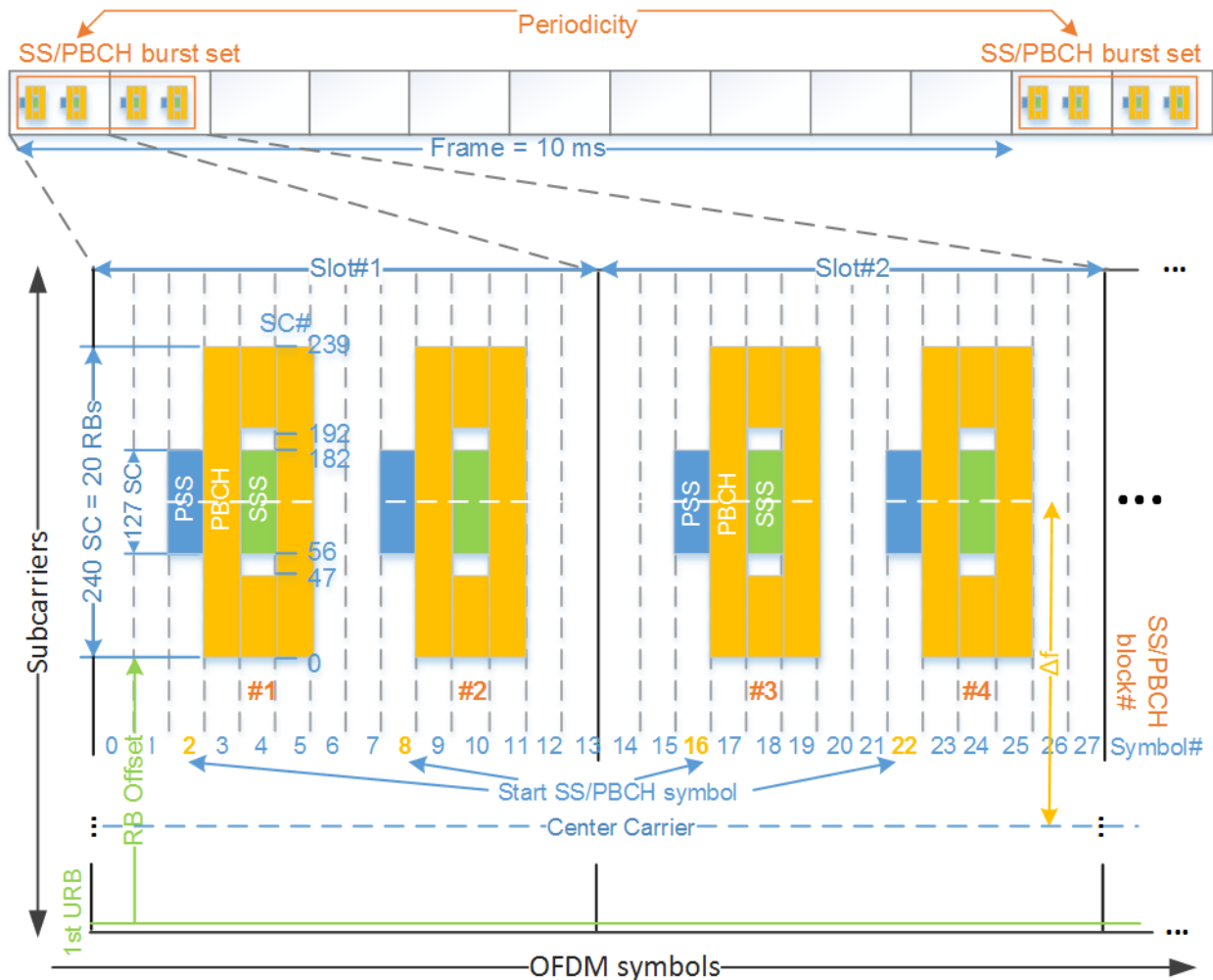


Figure 3-3: Location of synchronization signals in a succession of several slots (Case A - 15 kHz SCS, $f < 3$ GHz)

PSS/SSS	= Primary/secondary synchronization signals
PBCH	= Physical broadcast channel
SS/PBCH block	= SS and PBCH are always transmitted together in a block
SS/PBCH burst set	= A group of 4, 8 or 64 SS/PBCH blocks, depending on the SCS (in this example, burst set = 4 blocks)
Frame	= 10 ms = 10 slots
1 slot	= 1 subframe = 1 ms
SC	= Subcarrier
Start SS/PBCH symbol	= $\{2, 8\} + 14n$, where $n = 0, 1$; Start symbols = 2, 8, 16 and 22
RB Offset	= Defines the SS/PBCH allocation position in the frequency (see RB Offset/RB Offset (15 kHz SCS)/RB Offset (60 kHz SCS)); within an RB, the allocation can be further shifted (see SC Offset/SC Offset (15 kHz SCS)/SC Offset (60 kHz SCS))
Periodicity	= Defines how often the burst set is repeated (in this example, periodicity = 10 ms; the SS/PBCH burst set is repeated every frame), see Burst Set Periodicity
1st URB	= First usable resource block (URB), that is the first RB within the TxBW, see Figure 3-2
Carrier center	= Center frequency of the cell
Δf	= Offset between the center frequency of SS/PBCH block and the carrier center frequency

The SSs are allocated on predefined symbols but the allocated resource blocks and subcarriers are user-defined. The TS 38.213 specifies the number of and the start of the symbols depending on the used SCS and frequency range, see Figure 3-3. In 3GPP, these different start positions of the SS/PBCH blocks are referred as case A to case E.

3.8 Carrier Bandwidth Part (BWP)

In 5G NR, the bandwidth used for the communication with a specific user (UE) is typically smaller than the channel bandwidth. These user-specific bandwidths are managed by the different-sized, so called (carrier) bandwidth parts (BWP). Each of BWP uses its own numerology and thus can use different SCS, symbol duration and CP length (see Table 3-2).

Using BWPs with different bandwidths allows flexible resource allocation, for example, for support of UE with reduced bandwidth capabilities or supporting non-contiguous spectrum. Bandwidth parts can overlap, in which case UEs share the resource elements.

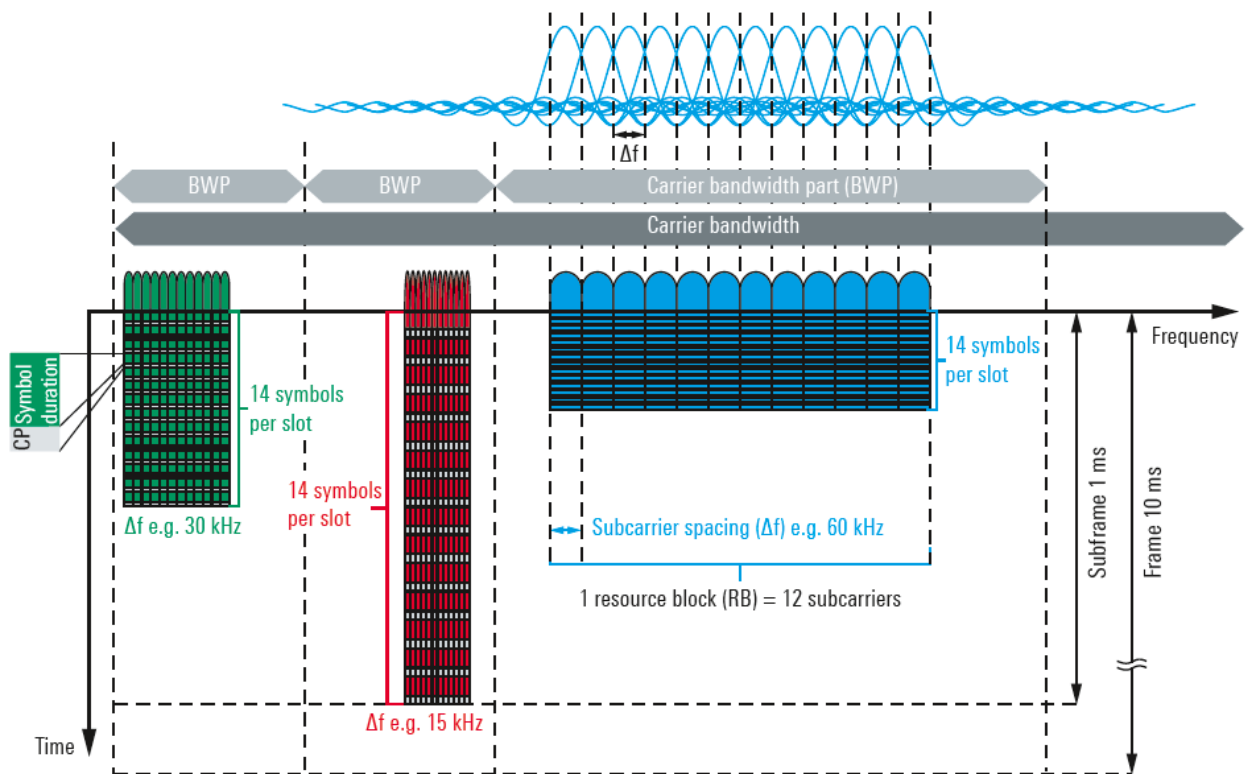


Figure 3-4: 5G NR numerology and BWP [1]

A BWP is a contiguous set of physical resource blocks that have the same subcarrier spacing and thus numerology. UEs can be configured with up to 4 BWPs in DL and UL, but only one can be active at a time. UEs are not expected to receive and transmit outside its active BWP.

3.9 Antenna Port Mapping

Antenna ports are not physical antennas, but rather are a logical concept. Each antenna port carries certain signal components (= physical channels) that should be transmitted under the same conditions. Physical channels can be transmitted on a single antenna port, or on several antenna ports. Each antenna port in turn can be mapped to one of the physical antennas. Typically, one physical antenna combines several antenna ports. However, one specific antenna port can also be transmitted on more than one physical antenna.

In this implementation, you can map the antenna ports used by the various physical channels defined by 3GPP to one or two layer configurations in the dedicated "Antenna Ports" dialogs.

These dialogs display the antenna port mapping settings in a table form, as an antenna port mapping table.

The mapping table is a matrix with number of rows equal to the number of physical Tx antennas (Basebands) and number of columns equal to the number of antenna ports (AP). The available antenna ports depend on the current configuration. The yellow matrix elements in the mapping table indicate the default antenna port to physical antenna (TX antenna/baseband) mapping.

Related settings:

- "Users/BWPs > UL BWPs > NZP CSI-RS" > [Resource-Settings > Antenna Port Mapping](#)
- "Users/BWPs > UL BWPs > SRS > SRS Resource Set Settings" > [Antenna Ports > Config](#)
- "Scheduling > PDSCH/PUSCH" > [Antenna Ports](#)

3.10 5G NR Channels and Signals Overview

Channels:

- DL: PBCH, PDCCH (CORESET), PDSCH
- UL: PRACH, PUCCH, PUSCH

Signals:

- DL: PSS/SSS, CSI-RS, DMRS (for PBCH, PDCCH, PDSCH), TRS, PTRS
- UL: DMRS (for PUCCH, PUSCH), PTRS, SRS

The following channels and signals are supported in the current firmware version:

- PSS/SSS, PBCH, PDCCH (CORESET), PDSCH, PUSCH, PUCCH; PBCH incl. MIB, PDSCH, PDCCH and PUSCH incl. DMRS, PTRS, SRS, CSI-RS

Related settings:

- [Chapter 5.2.8, "PBCH Settings"](#), on page 46
- [Chapter 5.3.4.3, "ZP CSI-RS Settings"](#), on page 67 and

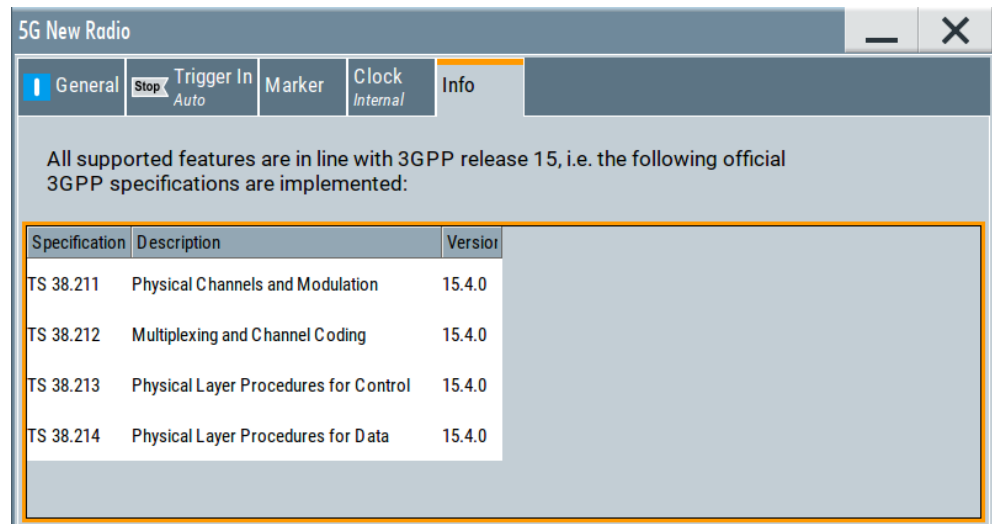
- Chapter 5.3.4.4, "NZP CSI-RS Settings", on page 70
- "PTRS Config ..." on page 64 and
Chapter 5.5.6, "PTRS Settings", on page 121
- Chapter 5.3.4.2, "PDSCH Settings", on page 59
- Chapter 5.3.5.4, "PUSCH Settings", on page 85 and
Chapter 5.3.5.5, "PUSCH UCI Settings", on page 92
- Chapter 5.3.5.6, "SRS Settings", on page 94
- Chapter 5.5, "PDSCH/PUSCH Settings", on page 111
- Chapter 5.6, "CORESET Settings", on page 128
- Chapter 5.3.5.2, "PUCCH Settings", on page 80 and
Chapter 5.7, "PUCCH Settings", on page 147
- Chapter 5.8, "PRACH Settings", on page 153

4 Find Out the Implemented 3GPP Specification

The Info dialog displays the currently supported version of the 3GPP standard.

Access:

- Select "5G New Radio > Info".



The supported features are in line with the specifications and the version displayed.

Remote command:

`[ :SOURCE ] :BB:NR5G:VERSion?` on page 199

5 5G New Radio Configuration and Settings

Access:

- Select "Baseband > 5G NR".

The remote commands required to define these settings are described in [Chapter 8, "Remote-Control Commands"](#), on page 176.

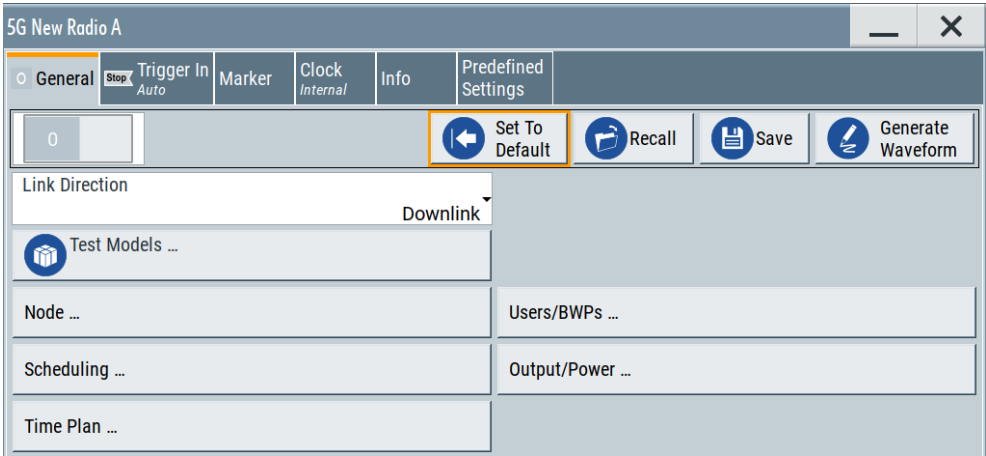
Settings:

• General Settings	23
• Node Settings	28
• Users/BWPs Settings	51
• Scheduling Settings	102
• PDSCH/PUSCH Settings	111
• CORESET Settings	128
• PUCCH Settings	147
• PRACH Settings	153

5.1 General Settings

Access:

- Select "Baseband > 5G NR".



This dialog comprises the standard general settings, the default and the "Save/ Recall" settings, as well as access to dialogs with further settings.

Settings:

State	24
Set to Default	24
Save/Recall	24

Generate Waveform File.....	25
Link Direction.....	25
Test Models.....	25
L User Files.....	25
L Filter Test Modes.....	27
Node.....	27
Users/BWPs.....	28
Scheduling.....	28
Output/Power.....	28

State

Activates the standard and deactivates all the other digital standards and digital modulation modes in the same path.

Remote command:

[:SOURce<hw>] :BB:NR5G:STATe on page 198

Set to Default

Calls the default settings. The values of the main parameters are listed in the following table.

Parameter	Values
"State"	Not affected by "Set to Default"
"Link direction"	Downlink
"Channel bandwidth"	100 MHz
"Dummy REs > State"	On
"Dummy REs > SC Spacing/CP"	30 kHz NCP

Remote command:

[:SOURce<hw>] :BB:NR5G:PRESet on page 195

Save/Recall

Accesses the "Save/Recall" dialog, that is the standard instrument function for storing and recalling the complete dialog-related (or in coupled sources entity-related) settings in a file. The provided navigation possibilities in the dialog are self-explanatory.

The filename and the directory, in which the settings are stored, are user-definable; the file extension is predefined.

On recall, the instrument checks if the saved configuration is compatible with the software and hardware options installed on the current instrument. For example, the number of block outputs (see [Output Settings](#)) and the number of internal entities (i.e. the currently selected system configuration) have to be the same.

See also, chapter "File and Data Management" in the R&S SMBV100B user manual.

Remote command:

[:SOURce<hw>] :BB:NR5G:SETTing:CATalog? on page 196

[:SOURce<hw>] :BB:NR5G:SETTing:LOAD on page 196

[:SOURce<hw>] :BB:NR5G:SETTing:STORe on page 196

[:SOURce<hw>] :BB:NR5G:SETTing:DEL on page 196

Generate Waveform File

With enabled signal generation, triggers the instrument to store the current settings as an ARB signal in a waveform file. Waveform files can be further processed by the ARB and/or as a multi-carrier or a multi-segment signal.

The filename and the directory it is stored in are user-definable; the predefined file extension for waveform files is *.wv.

Remote command:

[:SOURce<hw>] :BB:NR5G:WAVEform:CREate on page 198

Link Direction

Selects the transmission direction.

"Downlink" The transmission direction selected is base station to user equipment. The signal corresponds to that of a base station.

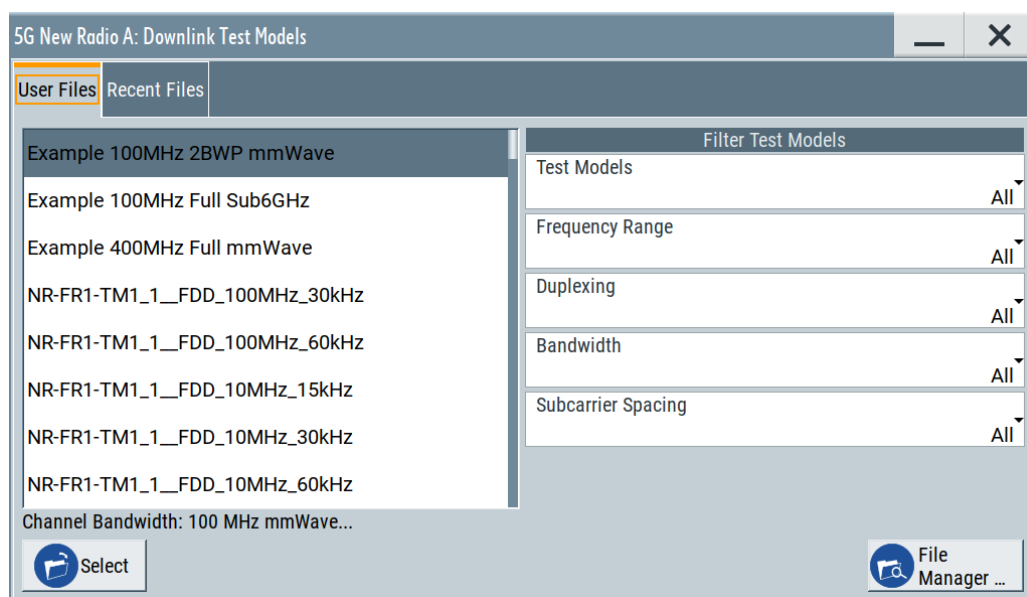
"Uplink" The transmission direction selected is user equipment to base station. The signal corresponds to that of a user equipment.

Remote command:

[:SOURce<hw>] :BB:NR5G:LINK on page 195

Test Models

Accesses a dialog for selecting of files with predefined 5G NR configurations.



User Files ← Test Models

You can select for the following:

- **One of the 5G NR Test Models (NR TM) defined in TS 38.141-1/TS 38.141-2 and TS 38.104.**

The NR FR1/FR2 test models are predefined configurations of 5G NR settings. The 3GPP specifications define three main groups of test models, the NR-RF-TM1, NR-RF-TM2 and NR-RF-TM3. Each TM is defined for a subset of different channel bandwidths and subcarrier spacings.

The test model use common physical channel parameters for PDSCH and PUCCH, as well as specific physical channels parameters per test model.

The following naming convention is used:

NR-<FR>-TM<TestModel>__<Duplexing>_<ChBW>MHz_<SCS>kHz, where:

- <FR> designates 5G NR test modules for the frequency range **FR1 ≤ 6 GHz** and **FR2 > 6 GHz**
- <TestModel> is the TM designation according to [TS 38.141-1](#)/[TS 38.141-2](#)
- <Duplexing> indicates if FDD or TDD duplexing is used
- <ChBW> is one of the defined basestation channel bandwidths: 5 MHz, 10 MHz, 15 MHz, 20 MHz, 25 MHz, 30 MHz, 40 MHz, 50 MHz, 60 MHz, 70 MHz, 70 MHz, 80, MHz, 90 MHz 100 MHz
- <SCS> indicates the subcarrier spacing of 15 kHz, 30 kHz or 60 kHz.

• **Predefined example test signal.**

"NR FR1/FR2 Test Models"

The test models are defined for the following test purpose:

5G NR Test Model	Defined for tests on
NR-FR1-TM1.1 NR-FR2-TM1.1	• BS output power • TAE • Unwanted emissions • Occupied bandwidth • ACLR • Transmitter intermodulation • Transmitter spurious emissions • Operating band unwanted emissions
NR-FR1-TM1.2	• ACLR • Unwanted emissions • Operating band unwanted emissions
NR-FR1-TM2/2a* NR-FR2-TM2	• Total power dynamic range (lower OFDM symbol power limit at min power) • EVM of single 64QAM/256QAM* PRB allocation (at min power) • Frequency error (at min power)
NR-FR1-TM3.1/3.1a* NR-FR2-TM3.1	• Output power dynamics • Transmitted signal quality (Frequency error and EVM for 64QAM/256QAM* modulation, at max power) • Transmitted signal quality • Frequency error • EVM for 64QAM/256QAM* modulation (at max power)
NR-FR1-TM3.2	• Transmitted signal quality • Frequency error • EVM for 16QAM modulation
NR-FR1-TM3.3	• Transmitted signal quality • Frequency error • EVM for QPSK modulation

Example test signals

Test signals are descriptions of specific 5G NR signals.

The following DL signals are provided:

- "Example 100MHz Full Sub6GHz"
"Channel Bandwidth = 100 MHz", "Deployment = Sub 6 GHz",
"Number of BWPs = 1", "Subcarrier Spacing = 30 kHz", "PBCH > Positions = 1100", "PDSCH > Modulation = QPSK, Type A",
"CORESET > No. Symb = 1".
- "Example 100MHz 2BWP mmWave"
"Channel Bandwidth = 100 MHz", "Deployment = > 6 GHz",
"Number of BWPs = 2", "PBCH > Positions = 1100...", "PDSCH > Modulation = QPSK, Type A", "CORESET > No. Symb = 1",
"Number of Users = 2", "Dummy RE > State > Off", where:
 - BWP#1: "Subcarrier Spacing = 120 kHz"
 - BWP#2: "Subcarrier Spacing = 60 kHz"
- "Example 400MHz Full mmWave"
"Channel Bandwidth = 400 MHz", "Deployment = > 6 GHz",
"Number of BWPs = 1", "Subcarrier Spacing = 120 kHz", "PBCH > Positions = 1100...", "PDSCH > Modulation = 256QAM, Type A",
"CORESET > No. Symb = 1"

The following UL signals are provided:

- "Example 100MHz Full Sub6GHz"
"Channel Bandwidth = 100 MHz", "Deployment = Sub 6 GHz",
"Number of BWPs = 1", "Subcarrier Spacing = 30 kHz", "PUSCH > Modulation = QPSK, Type A", "Transform Precoding = Off".
- "Example 100MHz Full TP Sub6GHz"
"Channel Bandwidth = 100 MHz", "Deployment = Sub 6 GHz",
"Number of BWPs = 1", "Subcarrier Spacing = 30 kHz", "PUSCH > Modulation = QPSK, Type A", "Transform Precoding = On".

Select a filename to observe summary information on the file content.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:FILTER:CATalog on page 197
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:DL on page 197
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:UL on page 197
```

Filter Test Modes ← Test Models

Applies the selected filter to narrow down the list of files.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:FILTER:TMODEl on page 197
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:FILTER:FREQ on page 197
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:FILTER:DUPLexing on page 198
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:FILTER:BW on page 198
[ :SOURCE<hw> ] :BB:NR5G:SETting:TMODEl:FILTER:SCS on page 198
```

Node

Accesses the "Node" dialog, see [Chapter 5.2, "Node Settings"](#), on page 28.

Remote command:

n.a.

Users/BWPs

Accesses the "Users/BWPs" dialog, see [Chapter 5.3, "Users/BWPs Settings"](#), on page 51.

Remote command:
n.a.

Scheduling

Accesses the "Scheduling" dialog, see [Chapter 5.4, "Scheduling Settings"](#), on page 102.

Remote command:
n.a.

Output/Power

Accesses the "Output/Power" dialog for setting the signal output and power-related settings.

See:

- [Chapter 6.5, "Output Settings"](#), on page 163
- [Chapter 6.6, "Time Domain Windowing Settings"](#), on page 166.

Remote command:
n.a.

5.2 Node Settings

Access:

- Select "Baseband > 5G New Radio > Node".

The settings in this dialog describe the basic physical structure of the generated signal.

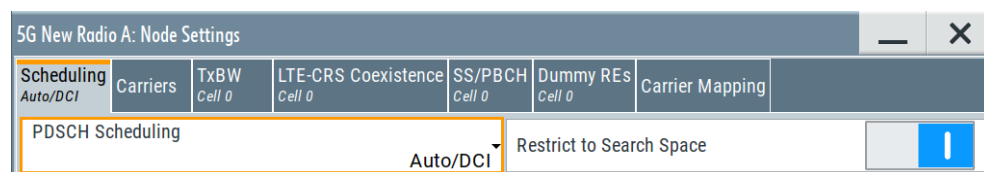
Settings:

• Scheduling	29
• Carriers Settings	30
• Transmission Bandwidth (TxBW) Settings	32
• LTE-CRS Coexistence Settings	35
• SS/PBCH Settings	37
• Dummy REs Settings	43
• Carrier-Mapping Settings	45
• PBCH Settings	46

5.2.1 Scheduling

Access:

1. Select "5G New Radio > Link > Downlink".
2. Select "5G New Radio > General > Node > Scheduling".



Settings:

PDSCH Scheduling	29
Restrict to Search Space	29

PDSCH Scheduling

Defines whether the PDSCH scheduling is performed manually or according to the configuration made for the DCIs.

- | | |
|------------|--|
| "Manual" | This mode is the default scheduling mode and the mode with full flexibility; you can configure any of the available settings.
There is no cross-reference between the settings made for the CORESET DCIs and the PDSCHs settings. You are responsible for the content of the PDSCH allocations. |
| "Auto/DCI" | This mode assures a 3GPP compliant signal; the PDSCH allocations are configured automatically according to the configuration of the CORESET DCIs.

The following settings are configured automatically: <ul style="list-style-type: none"> • "Users/BWPs > Properties" > DSCH Channel Coding = "On" • "Scheduling > CORESET > Custom DCI" > Channel Coding = "On" • For all DCI 1_1 and 1_0 in "Scheduling > CORESET > DCI Table", Create PDSCH = "On" • In the "Scheduling" dialog, you can configure only Content = "CORESET" (the PDSCH allocations are configured automatically). |

Remote command:

[\[:SOURCE<hw>\]:BB:NR5G:SCHeduling:MODE](#) on page 269

Restrict to Search Space

The term search space describes a set of CCEs that a UE monitors. The UE can decode only the control information on a PDCCH/CORESET that is transmitted over CCEs within the search space this UE monitors.

This parameter defines how the search space and the CCE indexes are defined. Together with other parameters it influences the way the search space is defined. For details, see ["Search Space"](#) on page 131.

- | | |
|-------|---|
| "Off" | Define the search space by the CCE start index, see CCE Index . |
|-------|---|

"On" The CCE start indexes are selected automatically so that they are within the search space.
The "Scheduling > CORESET > Payload" > [CCE Index](#) indicates the selected CCE index where the value is defined by the combination of [Aggregation Level](#) and candidate number ([Candidate](#)).

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHeduling:RSSpace on page 269

5.2.2 Carriers Settings

Access:

- Select "5G New Radio > General > Node > Carriers".

5G New Radio A: Node Settings											
Scheduling Auto/DCI	Carriers		TxBW Cell 0	LTE-CRS Coexistence Cell 0	SS/PBCH Cell 0	Dummy RES Cell 0	Carrier Mapping				
Number of Carriers					1	RF Phase Compensation					☒
Carrier	Cell Indicator	Cell ID	N1 ID	N2 ID	Deployment	Frequency /GHz	Channel BW	DMRS TypeA Position	SUL		
0	Cell 0	0	0	0	f <= 3GHz	1.000 000	100 MHz	2	Off		

This dialog comprises the carrier settings.

Settings:

Number of Carriers.....	30
RF Phase Compensation.....	31
Carrier table.....	31
L Carrier.....	31
L Cell Indicator.....	31
L Cell ID.....	31
L N1 ID/N2 ID.....	31
L Deployment.....	31
L Frequency in GHz.....	32
L Channel Bandwidth.....	32
L DMRS Type A Position.....	32
L SUL.....	32

Number of Carriers

Sets the number of simulated carriers. In the carrier table, there is one table row per carrier.

According to the 3GPP specifications, carrier aggregation is user-specific and thus different users served by the same base station can use different carriers. Moreover, the different users can also have different primary cell (PCell). In this implementation, however, all users use the same carrier aggregation configuration, or with other words, the primary and secondary cells (SCells) are the same for all configured users. Hence, there are *16 carriers per node, not per user*.

Set the "Number of Carriers ≥ 2 " to enable carrier aggregation.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:NODE:NCARrier` on page 199

RF Phase Compensation

If enabled, applies a frequency-related phase compensation after each symbol, as specified in [TS 38.211](#).

Use the parameter [Frequency in GHz](#) to set the carrier frequency to be compensated.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:NODE:RFPHase:STATe` on page 201

Carrier table

Comprises the carrier settings. There is one table row per carrier.

Carrier ← Carrier table

Displays the automatically assigned carrier names.

Remote command:

via suffix at `CELL<ch>`

Cell Indicator ← Carrier table

Sets the cell indicator, to identify each of up to 16 carriers.

The cell indicator is unique and unambiguous index used to address the cell, for example for cross-carrier scheduling.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:NODE:CELL<ch0>:CINDicator` on page 200

Cell ID ← Carrier table

Sets the cell identity of the selected cell.

The physical layer cell ID N_{ID}^{cell} is used for synchronization between network and user equipment. It identifies a specific radio cell in the 5G NR network. The cell ID is a value between 0 and 1007, calculated as follows:

$$N_{ID}^{cell} = 3N_{ID}^{(1)} + N_{ID}^{(2)}$$

Unlike the [Cell Indicator](#), cells can use the same "Cell ID".

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:NODE:CELL<ch0>:CELLId` on page 199

N1 ID/N2 ID ← Carrier table

Indicates the physical cell indicator group ($N_{ID}^{(1)}$) and the physical layer identity ($N_{ID}^{(2)}$), derived from the selected N_{ID}^{cell} ([Cell ID](#)).

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:NODE:CELL<ch0>:N1ID?` on page 200

`[ :SOURCE<hw> ] :BB:NR5G:NODE:CELL<ch0>:N2ID?` on page 200

Deployment ← Carrier table

Selects one of the frequency ranges, specified for 5G NR transmission, see [Table 3-1](#).

The selected frequency range (FR) determines the following parameters:

- [SC Spacing/CP](#)
- [Case](#) (i.e. the synchronization signal pattern).

Remote command:

[\[:SOURCE<hw>\]:BB:NR5G:NODE:CELL<ch0>:CARDePly](#) on page 200

Frequency in GHz ← Carrier table

If [RF Phase Compensation](#) > "On", sets the carrier frequency of the selected carrier at which the frequency phase compensation is applied.

Remote command:

[\[:SOURCE<hw>\]:BB:NR5G:NODE:CELL<ch0>:PCFReq](#) on page 201

Channel Bandwidth ← Carrier table

Selects the bandwidth of the node carrier from a list with predefined values, see [Table 3-1](#).

The available values depend on the selected [Deployment](#). Together with the selected SCS ([SCS](#) or [SC Spacing/CP](#)), the channel bandwidth affects the number of available RBs ([N_RB](#) and [No. RBs](#)).

Signals, channels (e.g. SS/PBCH) and bandwidth parts (BWPs) can only be configured within the selected channel bandwidth. Thus, the value selected here sets the upper limit for the bandwidth-related settings.

You can select channel bandwidths that are larger than the baseband bandwidth available in the R&S SMBV100B, because the used sample rate can be smaller than the channel bandwidth.

Remote command:

[\[:SOURCE<hw>\]:BB:NR5G:NODE:CELL<ch0>:CBW](#) on page 200

DMRS Type A Position ← Carrier table

Depending on the used mapping type (A or B), the demodulation reference signals (DMRS) are mapped to different symbols of the PDSCH/PUSCH allocation.

This parameter defines the position of the first DMRS symbol l_0 ([dmrs-TypeA-Position](#)) within the slot, if mapping type A is used.

Remote command:

[\[:SOURCE<hw>\]:BB:NR5G:NODE:CELL<ch0>:TAPos](#) on page 201

SUL ← Carrier table

Defines if the carrier supports supplementary uplink (SUL) or not.

Remote command:

[\[:SOURCE<hw>\]:BB:NR5G:NODE:CELL<ch0>:SUL:STATe](#) on page 202

5.2.3 Transmission Bandwidth (TxBW) Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".

2. Select "5G New Radio > General > Node > TxBW".

5G New Radio A: Node Settings						
Scheduling Auto/DCI	Carriers	TxBW Cell 0	LTE-CRS Coexistence Cell 0	SS/PBCH Cell 0	Dummy REs Cell 0	Carrier Mapping
Carrier: Cell			0			
Point A to Carrier Center			-49.140 MHz			Channel Bandwidth
			100 MHz			
Resolve Conflicts						
	Use	N_RB	TxBW Offset	k0u		
30 kHz	☒	273	0	0.0		
60 kHz	☐	135	0	-		

This dialog comprises the settings, related to the transmission bandwidths (TxBW). These include: the frequency offset between the center carrier and the Point A, the transmission bandwidth width in number of RBs and the k_0^u .

Settings:

Cell.....	33
Point A to Carrier Center.....	33
Channel Bandwidth.....	33
Resolve Conflicts.....	34
TxBW table.....	34
L SCS.....	34
L Use.....	34
L N_RB.....	34
L TxBW Offset.....	35
L k0u.....	35

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at CELL<ch>

Point A to Carrier Center

Sets the frequency offset between the reference Point A and the center carrier frequency.

The Point A is the reference point for the frequency-dependent parameters, for example TxBW Offset and RB Offset/RB Offset (15 kHz SCS)/RB Offset (60 kHz SCS).

See Chapter 3.6, "Transmission Bandwidths TxBW", on page 15.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:TXBW:POINTa on page 202

Channel Bandwidth

Selects the bandwidth of the node carrier from a list with predefined values, see Table 3-1.

The available values depend on the selected [Deployment](#). Together with the selected SCS ([SCS](#) or [SC Spacing/CP](#)), the channel bandwidth affects the number of available RBs ([N_RB](#) and [No. RBs](#)).

Signals, channels (e.g. SS/PBCH) and bandwidth parts (BWPs) can only be configured within the selected channel bandwidth. Thus, the value selected here sets the upper limit for the bandwidth-related settings.

You can select channel bandwidths that are larger than the baseband bandwidth available in the R&S SMBV100B, because the used sample rate can be smaller than the channel bandwidth.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:CBW` on page 200

Resolve Conflicts

Recalculates the frequency-dependent settings and thus redefines the frequency position of the TxBW.

Conflicts arise if:

- [TxBW Offset](#) is not a multiple integers of RBs for the corresponding numerology
- [k0u](#) ≠ {-6, 0, 6}.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:TXBW:RESolve` on page 203

TxBW table

Comprises the frequency-related settings of the subcarrier spacings (SCS) supported for the selected [Deployment](#).

Displayed are all supported SCS combinations. To enable a specific SCS, set [Use](#) > "On".

See [Chapter 3.6, "Transmission Bandwidths TxBW"](#), on page 15.

SCS ← TxBW table

Indicates the SCS to that the settings apply.

Remote command:

via block `S<SCS>K`

Use ← TxBW table

From all possible TxBW for a specific channel bandwidth, the base station decides which TxBW to use. This parameter defines whether the SCS is supported in the particular cell and if it is used by the base station.

At least one SCS must be used.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:USE` on page 203

etc.

N_RB ← TxBW table

Indicates the number of available RBs N_{RB} within the TxBW for the selected [SCS](#) and [Channel Bandwidth](#).

See [Chapter 3.6, "Transmission Bandwidths TxBW"](#), on page 15.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:NRB? on page 203
etc.

TxBW Offset ← TxBW table

Sets the usable RB to common RBs offset.

See [Chapter 3.6, "Transmission Bandwidths Tx BWs"](#), on page 15.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:OFFSet on page 204
etc.

k_{0u} ← TxBW table

Indicates the value of the parameter k_0^u , that defines the number of subcarriers the transmission bandwidth is shifted related to the carrier center.

The value is calculated automatically, based on the selected SCS and "Channel Bandwidth".

According to the specification, $k_0^u = \{-6, 0, 6\}$. If value different than these allowed values is displayed, a conflict is indicated. Use the [Resolve Conflicts](#) function to recalculate the values.

The value $k_0^u = "-"$ indicates that the SCS is not used ([Use](#) > "Off").

See [Chapter 3.6, "Transmission Bandwidths Tx BWs"](#), on page 15.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:KOMU? on page 204
etc.

5.2.4 LTE-CRS Coexistence Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".
2. Enable a carrier that **supports SCS of 15 kHz**, for example select "5G New Radio > General > Node > Carriers > Carrier x > Channel BW = 50 MHz".
3. Select "Node > TxBW > 15 kHz > Use = On".
4. Select "Resolve Conflicts".

5. Select "LTE-CRS Coexistence".

Scheduling <i>Manual</i>	Carriers	TxBW <i>Cell 1</i>	LTE-CRS Coexistence <i>Cell 1</i>	SS/PBCH <i>Cell 1</i>	Dummy REs <i>Cell 1</i>	Carrier Mapping
Carrier: Cell			1			
State			I			
Offset to Point A / 15 kHz SCs			0	LTE Bandwidth / RBs 50		
vShift			0	LTE Antenna Ports 2		

You can configure LTE CRS signals, if the carrier supports 15 kHz subcarrier spacing, because 15 kHz is the subcarrier spacing used in LTE.

In this dialog, you configure the allocation of the LTE-CRS as defined in [TS 38.214](#) in terms of:

- Frequency position as offset to Point A and bandwidth within the RS are spread.
- Number of RS and their configuration defined based on the number of used LTE antenna ports and position of the CRS patterns in the frequency domain (vShift).

The resources reserved for the transmission of the LTE CRS are omitted by 5G NR, which influences among others the PDSCH resource mapping. The LTE CRS allocation has to be known to the UE and the BS.

Observe the LTE CRS resources on the "Time Plan". In the "Scheduling" dialog, observe how enabled LTE CRS reduces the number of available physical resources for PDSCH.

Settings:

Cell.....	36
State.....	36
Offset to Point A / 15 kHz SCS.....	37
vShift.....	37
LTE Bandwidth / RBs.....	37
LTE Antenna Ports.....	37

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at `CELL<ch>`

State

Enable the LTE-CRS so that the configured resources are omitted by the 5G NR allocations.

This setting is useful if you simulate LTE and 5G NR coexistence scenarios, where the same frequency band is shared between LTE and 5G NR allocations.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:LTE:STATe on page 204

Offset to Point A / 15 kHz SCS

Sets the LTE carrier center subcarrier location (`carrierFreqDL`) as offset from (reference) point A.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:LTE:POINTa on page 205

vShift

Sets the parameter `v-Shift` required to configure the LTE-CRS allocation.

In LTE, `vShift` is defined as function of the cell ID. This parameter is used to shift the CRS patterns in LTE neighbor cells in the range of 0 to 5 subcarriers and thus prevent CRS pattern overlapping.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:LTE:VSHift on page 205

LTE Bandwidth / RBs

Sets the LTE bandwidth (`carrierBandwidthDL`) as number of RBs and thus defines the bandwidth used for the LTE-CRS. The LTE-CRS always span the whole LTE bandwidth.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:LTE:CBW on page 205

LTE Antenna Ports

Sets the parameter `nrofCRS-Ports` giving the number of antenna ports used for LTE-CRS.

The number of antenna ports defines the number and position of LTE CRS.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:LTE:NAP on page 205

5.2.5 SS/PBCH Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".

2. Select "5G New Radio > General > Node > SS/PBCH".

Scheduling	Carriers	TxBW	LTE-CRS Coexistence	SS/PBCH	Dummy REs	Carrier Mapping
Auto/DCI		Cell 0	Cell 0	Cell 0	Cell 0	
Carrier: Cell 0						
Number of SS/PBCH Patterns		Antenna Port		Offset Relative to		
1		4000		TxBW		
SC Spacing / CP	RB Offset	SC Offset	Δf to Carrier (Centers) /MHz	Case	L	Positions
30 kHz NCP	126	6	0.000 000	B	4	1100
Burst Set Periodicity	PBCH	PSS Power /dB	SSS Power /dB	PBCH Power /dB	State	
10 ms	Config...	0.00	0.00	0.00	On	

This dialog comprises the settings of the broadcast channel (PBCH) and the synchronization signals (SS). The SS and the PBCH are cell-specific and are always transmitted in a block. For more information, see [Chapter 3.7, "Synchronization Signals and SS/PBCH Block"](#), on page 17.

Settings:

Cell.....	38
Antenna Port.....	38
Number of SS/PBCH Patterns.....	39
Offset Relative to.....	39
SS/PBCH patterns table.....	39
L SC Spacing/CP.....	39
L RB Offset/RB Offset (15 kHz SCS)/RB Offset (60 kHz SCS).....	40
L SC Offset/SC Offset (15 kHz SCS)/SC Offset (60 kHz SCS).....	40
L Delta f to Carrier (Centers).....	41
L Case.....	41
L L.....	41
L Positions.....	41
L Burst Set Periodicity.....	42
L PBCH Config.....	42
L PSS/SSS Power.....	42
L PBCH Power.....	42
L State.....	42

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at CELL<ch>

Antenna Port

Indicates that antenna port 4000 is used.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CARMapping:AP4000[:ROW<ch0>]

on page 217

Number of SS/PBCH Patterns

Defines how many patterns for the synchronization signals and the PBCH are configured. There is one table row per pattern.

See also [Chapter 3.7, "Synchronization Signals and SS/PBCH Block"](#), on page 17.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:NSSPBch on page 207

Offset Relative to

Defines the reference point, relative to which the SS/PBCH is allocated in frequency domain.

"TxBW"	The frequency position of the SS/PBCH is set relative to the usable RBs that apply for the current numerology, i.e. to the start of the TxBW. If this reference point is used, you can select the SC Spacing/CP for that the following applies: • TxBW spans at least 20 RBs (i.e. N_RB > 20) • TxBW is enabled for this SCS (i.e. Use > "On").
"Point A"	The frequency position of the SS/PBCH is set relative to the position of point A.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:OFFSet on page 206

SS/PBCH patterns table

Comprises the configuration of the SS/PBCH transmissions.

You can enable different SS/PBCH patterns and configure their individual settings in the SS/PBCH pattern table. There is one table row per pattern. Each row describes a periodically repeating burst set with definable timing and frequency position, defined as follows:

- Frequency: [SC Spacing/CP](#), [RB Offset/RB Offset \(15 kHz SCS\)/RB Offset \(60 kHz SCS\)](#), [SC Offset/SC Offset \(15 kHz SCS\)/SC Offset \(60 kHz SCS\)](#) and [Delta f to Carrier \(Centers\)](#)
- Timing: [Case](#), [Positions](#) and [Burst Set Periodicity](#)

See also [Figure 3-3](#).

SC Spacing/CP ← SS/PBCH patterns table

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP), where the available values depend on the "Deployment".

See [Table 3-2](#).

The available values also depend on the following:

- SCSs that span at least 20 RBs (i.e. [N_RB](#) > 20)
- For [Offset Relative to](#) = "TxBW", only SCSs that are enabled for the TxBW (i.e. [Use](#) > "On").

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SCSPacing on page 207

RB Offset/RB Offset (15 kHz SCS)/RB Offset (60 kHz SCS) ← SS/PBCH patterns table

Sets the start resource block of the selected allocation and thus defines the allocation's position in the frequency domain.

The name of the parameter changes depending on the selected [Offset Relative to](#) as follows:

- "RB Offset": If [Offset Relative to](#) = "TxBW"
- "RB Offset (15 kHz SCS)": If [Offset Relative to](#) = "Point A" and [Deployment](#) = "f <= 3GHz" or "3GHz < f <= 6GHz"
- "RB Offset (60 kHz SCS)": If [Offset Relative to](#) = "Point A" and [Deployment](#) = "f > 6GHz"

For [Offset Relative to](#) = "TxBW", the value is set relative to the usable RBs that apply for the current numerology, i.e. to the start of the TxBW (SC Spacing/CP), see [Figure 3-3](#).

Example:

- If "RB Offset = 0", the SS/PBCH block is allocated on the first subcarrier in the channel.
- If "RB Offset = 10", the SS/PBCH block is allocated on the 10*12 = 120th subcarrier in the channel.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:RBOffset
on page 207
```

SC Offset/SC Offset (15 kHz SCS)/SC Offset (60 kHz SCS) ← SS/PBCH patterns table

Sets the start subcarrier of the selected allocation within a resource block, where the latter is defined with the parameter [RB Offset/RB Offset \(15 kHz SCS\)/RB Offset \(60 kHz SCS\)](#).

Example:

- If "RB Offset = 0" and "SC Offset = 10", the SS/PBCH block is allocated on the 10th subcarrier in the channel.
- If "RB Offset = 10" and "SC Offset = 0", the SS/PBCH block is allocated on the 10*12 = 120th subcarrier in the channel.

See also [Figure 3-3](#).

The name of the parameter changes depending on the selected [Offset Relative to](#) as follows:

- "SC Offset": If [Offset Relative to](#) = "TxBW"
- "SC Offset (15 kHz SCS)": If [Offset Relative to](#) = "Point A" and [Deployment](#) = "f <= 3GHz" or "3GHz < f <= 6GHz"
- "SC Offset (60 kHz SCS)": If [Offset Relative to](#) = "Point A" and [Deployment](#) = "f > 6GHz"

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SCOffset
on page 208
```


Delta f to Carrier (Centers) ← SS/PBCH patterns table

Indicates the frequency offset between the center of the SS/PBCH block to the center of the carrier; thus indicates the allocation's position in the frequency domain.

The value is calculated automatically, based on the selected SCS, channel bandwidth, deployment method, RB and SC offsets.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:DFreq?

on page 208

Case ← SS/PBCH patterns table

Selects one of the SS/PBCH cases, as specified in TS 38.213. The case defines which symbols in a slot carry the synchronization signals.

The SS/PBCH cases depend on the deployment and the used SCS.

SS/PBCH case	SCS	Frequency
Case A	15 KHz	< 6 GHz
Case B*	30 KHz	< 6 GHz
Case C*	30 KHz	< 6 GHz
Case D	120 KHz	> 6 GHz
Case E	240 KHz	> 6 GHz

*) The start symbol index in case B and case C is different.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CASE on page 208

L ← SS/PBCH patterns table

Sets the number of SS/PBCH blocks, transmitted per burst.

This value sets the pattern length, used to define the Positions of the transmitted SS/PBCH blocks.

The available values for "L" depend on the selected Deployment as follows:

- "Deployment" = " $f \leq 3$ GHz": L = 4 or 8, where
L = 8 is supported for the unpaired spectrum below 3 GHz if Case = "C"
- "Deployment" = " $3 \text{ GHz} < f \leq 6 \text{ GHz}$ ": L = 8
- "Deployment" = " $f > 6 \text{ GHz}$ ": L = 64

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:L on page 209

Positions ← SS/PBCH patterns table

A half-frame contains 4, 8 or 64 SS/PBCH blocks, depending on the selected subcarrier spacing and deployment and as set with the parameter L. The group of 4, 8 or 64 blocks is referred to as burst set, see Figure 3-3.

This parameter defines which of these theoretically possible maximum numbers of SS/PBCH blocks are transmitted in each burst set. The definition is a pattern, where:

- Pattern length = 4, 8 or 64 and corresponds to the maximum number of SS/PBCH blocks

- Each bit applies to one SS/PBCH block
- 1: SS/PBCH block is present
- 0: SS/PBCH transmission is suppressed.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:POSition`

on page 209

Burst Set Periodicity ← SS/PBCH patterns table

Defines how often a SS/PBCH burst set is transmitted.

The "Burst Set Periodicity = 10", for example, corresponds to a transmission in every frame.

See also [Figure 3-3](#).

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:BSPeriodicty`

on page 209

PBCH Config ← SS/PBCH patterns table

Accesses the dialog with PBCH settings, see [Chapter 5.2.8, "PBCH Settings"](#), on page 46.

PSS/SSS Power ← SS/PBCH patterns table

Sets the relative power of the allocations of the primary and the secondary synchronization signals (PSS and SSS).

The value is set relative to the power of the other resource elements.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:PSSPow`

on page 210

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SSSPow`

on page 210

PBCH Power ← SS/PBCH patterns table

Sets the relative power of the PBCH allocation.

The value is set relative to the power of the other resource elements.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:POWer` on page 210

State ← SS/PBCH patterns table

Disables the transmission of SS/PBCH.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:STATe` on page 207

5.2.6 Dummy REs Settings

Access:

- Select "5G New Radio > General > Node > Dummy REs".

Scheduling Auto/DCI	Carriers	TxBW Cell 0	LTE-CRS Coexistence Cell 0	SS/PBCH Cell 0	Dummy REs Cell 0	Carrier Mapping
Carrier: Cell						
State		0		Power dB		
SC Spacing / CP		15 kHz NCP		Modulation BPSK		
Data Source		PN9				

With the provided settings, you define if and what is transmitted in the resource elements that are not allocated.

The dummy REs are the allocations with the lowest priority. They do not include [DMRS](#) or [PTRS](#).

Settings:

Cell.....	43
State.....	43
Power.....	43
SC Spacing/CP.....	44
Modulation.....	44
Transform Precoding.....	44
Data Source.....	44

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at `CELL<ch>`

State

If enabled, the resource elements (RE) are filled in with dummy data, modulated and formatted as selected with the settings in the "Dummy REs" dialog.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:DUMRes:STATe` on page 214

Power

Sets the power of the resource elements filled in with dummy data. The value is set relative to the power of the other resource elements.

See also [Chapter 6.5, "Output Settings"](#), on page 163.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:DUMRes:POWeR` on page 214

SC Spacing/CP

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP), where the available values depend on the "Deployment".

See [Table 3-2](#).

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:DUMRes:SCSPacing on page 215

Modulation

Sets the modulation scheme for the dummy REs.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:DUMRes:MODulation on page 215

Transform Precoding

In uplink, enables using the optional DFT-S scheme.

If disabled, both DL and UL use the CP-OFDM.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:DUMRes:TPState on page 215

Data Source

Selects the data source for the dummy REs.

The following standard data sources are available:

- "All 0, All 1"
An internally generated sequence containing 0 data or 1 data.
- "PNxx"
An internally generated pseudo-random noise sequence.
- "Pattern"
An internally generated sequence according to a bit pattern.
Use the "Pattern" box to define the bit pattern.
- "Data List/Select DList"
A binary data from a data list, internally or externally generated.
Select "Select DList" to access the standard "Select List" dialog.
 - Select the "Select Data List > navigate to the list file *.dm_iqd > Select" to select an existing data list.
 - Use the "New" and "Edit" functions to create internally new data list or to edit an existing one.
 - Use the standard "File Manager" function to transfer external data lists to the instrument.

See also:

- Section "Modulation Data" in the R&S SMBV100B user manual.
- Section "File and Data Management" in the R&S SMBV100B user manual.
- Section "Data List Editor" in the R&S SMBV100B user manual

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:DUMRes:DATA on page 215

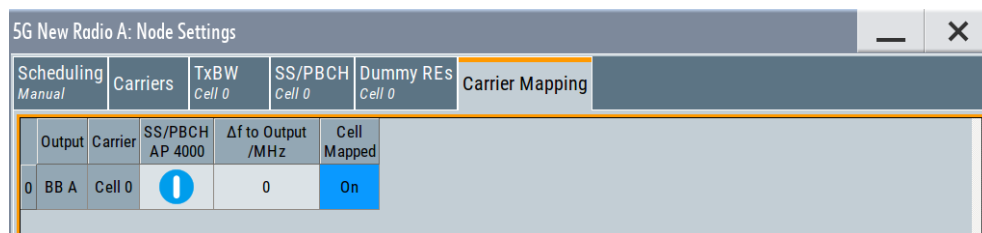
[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:DUMRes:DLISt on page 216

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:DUMRes:PATtern on page 216

5.2.7 Carrier-Mapping Settings

Access:

1. Select "5G New Radio > General > Link Direction" > **"Downlink"**.
2. Select "5G New Radio > General > Node > Carrier Mapping".



This dialog defines the carrier to basebands (baseband block outputs) mapping, where basebands are the number of output signals from the baseband block.

Settings:

Output.....	45
Carrier.....	45
SS/PBCH AP 4000.....	45
Delta f to Output.....	45
Cell Mapped.....	46

Output

Indicates the baseband (BB) output.

Remote command:

via suffix at ROW<ch0>

Carrier

Defines which carrier is output at which baseband.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CARMapping:CELL[:ROW<ch0>] on page 216

SS/PBCH AP 4000

Sets if the SS/PBCHs are output at the selected output.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CARMapping:AP4000[:ROW<ch0>]
on page 217

Delta f to Output

Shifts the signal relative to the center output frequency.

Use different frequency offsets to separate the channels of the configured and enabled cells.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CARMapping:DFReq[:ROW<ch0>] on page 217

Cell Mapped

If enabled, the signal of the selected cell is mapped to the output.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CARMapping:MAPPed[:ROW<ch0>]
on page 217

5.2.8 PBCH Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".
2. Select "5G New Radio > General > Node > SS/PBCH".
3. Select "Node > SS/PBCH > SS/PBCH Pattern Table > Pattern# > PBCH > Config".

5G New Radio A: PBCH Settings (Patt0)

Dummy Content for MIB

Dummy Content

Channel Coding

Transport Block Size 24

Data Source PN9

This dialog comprises the settings of the broadcast channel (PBCH) channel coding settings.

Settings:

Dummy Content for MIB.....	47
Dummy Content Settings.....	47
L Channel Coding.....	47
L Transport Block Size.....	47
L Data Source.....	47
MIB Content Settings.....	48
L SCS Common.....	49
L DMRS Type A Position.....	49
L Auto Subcarrier Offset.....	49
L SSPBCH Subcarrier Offset.....	50
L System Frame Number Start Offset.....	50
L CORESET Zero.....	50

L Search Space Zero.....	50
L Cell Barred.....	50
L Intra Freq Reselection.....	50
L Spare.....	51

Dummy Content for MIB

Defines how the MIB is defined:

- "On" This is a faster way to define the MIB.
You can define if channel coding is used or not and select an arbitrary data source; further settings are not required.
See "Dummy Content Settings" on page 47.
- "Off" Disable this parameter if you wish to configure the MIB content according to TS 38.331.
See "MIB Content Settings" on page 48.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:STATe
on page 211

Dummy Content Settings

Available for "Dummy Content for MIB > On".

5G New Radio A: PBCH Settings (Patt0)

Dummy Content for MIB ☒

Dummy Content

Channel Coding ☒

Transport Block Size 24

Data Source PN9

Channel Coding ← Dummy Content Settings

Enables PBCH channel coding.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:STATe
on page 210

Transport Block Size ← Dummy Content Settings

Indicates the transport block size.

Data Source ← Dummy Content Settings

Selects the PBCH data source.

The following standard data sources are available:

- "All 0, All 1"
An internally generated sequence containing 0 data or 1 data.
- "PNxx"
An internally generated pseudo-random noise sequence.
- "Pattern"
An internally generated sequence according to a bit pattern.
Use the "Pattern" box to define the bit pattern.
- "Data List/Select DList"
A binary data from a data list, internally or externally generated.
Select "Select DList" to access the standard "Select List" dialog.
 - Select the "Select Data List > navigate to the list file *.dm_iqd > Select" to select an existing data list.
 - Use the "New" and "Edit" functions to create internally new data list or to edit an existing one.
 - Use the standard "File Manager" function to transfer external data lists to the instrument.

See also:

- Section "Modulation Data" in the R&S SMBV100B user manual.
- Section "File and Data Management" in the R&S SMBV100B user manual.
- Section "Data List Editor" in the R&S SMBV100B user manual

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:DATA`
on page 210

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:DLISt`
on page 210

`[ :SOURce<hw> ] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:PATtern`
on page 211

MIB Content Settings

If "Dummy Content for MIB > Off", comprises the setting to configure the MIB as defined in the 3GPP specification.

5G New Radio A: PBCH Settings (Patt0)	
Dummy Content for MIB	0
MIB Content	
SCS Common	30 or 120 kHz
DMRS TypeA Position	2
Auto Subcarrier Offset	☒
SSPBCH Subcarrier Offset	12
System Frame Number Start Offset	0
CORESET Zero	0
Search Space Zero	0
Cell Barred	Barred
Intra Freq Reselection	Allowed
Spare	☐

SCS Common ← MIB Content Settings

Selects the common SCS (subcarrier spacing).

The selected SCS defines the default SSPBCH subcarrier offset.

Remote command:

[:SOURce<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SCSC
on page 211

DMRS Type A Position ← MIB Content Settings

Indicates the first symbol that the DMRS uses, as set with the parameter "Node > Carriers" > [DMRS Type A Position](#).

Auto Subcarrier Offset ← MIB Content Settings

Defines if the "SSPBCH Subcarrier Offset" value is derived from the selected [SCS Common](#) or set manually with [SSPBCH Subcarrier Offset](#).

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:ASOF:STATE
on page 212

SSPBCH Subcarrier Offset ← MIB Content Settings

Indicates or sets the SS/PBCH subcarrier offset:

- If **Auto Subcarrier Offset** = "On", the "SSPBCH Subcarrier Offset" is calculated automatically based on the selected "Node > SS/PBCH" > **SC Offset/SC Offset (15 kHz SCS)/SC Offset (60 kHz SCS)**.
- If **Auto Subcarrier Offset** = "Off", the SSPBCH subcarrier offset is user-definable.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SCOffset
on page 212

System Frame Number Start Offset ← MIB Content Settings

By default, the counting of the SFN (system frame number) starts with 0. Use this parameter to set a different start SFN value.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SFOffset
on page 212

CORESET Zero ← MIB Content Settings

Sets the common control resource set (CORESET) of the initial downlink BWP `controlResourceSetZero` carried by the `PDCCH-ConfigSIB1` information element, as defined in [TS 38.331](#).

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:CSZero
on page 212

Search Space Zero ← MIB Content Settings

Sets the common search space of the initial downlink BWP `searchSpaceZero` carried by the `PDCCH-ConfigSIB1` information element, as defined in [TS 38.331](#).

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SSZero
on page 213

Cell Barred ← MIB Content Settings

Cell barring (`cellBarred`) is system information that indicates if UEs can camp on the particular cell `notBarred` or not barred.

Remote command:

[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:CBARred
on page 213

Intra Freq Reselection ← MIB Content Settings

Sets the value of the system information parameter `intraFreqReselection`.

Set "Intra Freq Reselection > Allowed" to allow UEs to perform cell reselection to intra-frequency cells.

Remote command:
[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:IFRResel
on page 213

Spare ← MIB Content Settings
Sets the system information bit spare, as defined in TS 38.331.
Remote command:
[:SOURCE<hw>] :BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SPARe:STAtE
on page 213

5.3 Users/BWPs Settings

Access:
▶ Select "Baseband > 5G NR > Users/BWPs".

Settings:

- [General Settings](#)..... 51
- [Properties Settings](#).....52
- [DL/UL BWPs Settings](#).....53
- [DL BWP Configuration Settings](#).....58
- [UL BWP Configuration Settings](#).....78

5.3.1 General Settings

Access:
▶ Select "5G New Radio > General > Users/BWPs > General".

In this tab, you define the number of users to be simulated.

Settings:
[Number of Users](#)..... 51

Number of Users
Sets the number of simulated users.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:NUSer on page 218

5.3.2 Properties Settings

Access:

- Select "5G New Radio > General > Users/BWPs > Properties".

This dialog comprises the user-specific general settings.

Settings:

User.....	52
UE ID (RNTI).....	52
DSCH/USCH Channel Coding.....	52
(P)DSCH/(P)USCH Data Source.....	53

User

Selects the user to be configured.

Remote command:

via suffix at USER<ch>

UE ID (RNTI)

Sets the RNTI of the user.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:UEID on page 218

DSCH/USCH Channel Coding

Depending on the selected "Link Direction" direction, enables DSCH/USCH channel coding.

Configure the corresponding settings in "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config" > [Channel Coding Settings](#) dialog.

If [PDSCH Scheduling](#) = "Auto/DCI", the "DSCH Channel Coding" = "On".

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:DSCH:CCODing:STATe?`

on page 218

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:USCH:CCODing:STATe`

on page 218

(P)DSCH/(P)USCH Data Source

Depending on the selected "Link Direction", selects the PDSCH/PUSCH data source.

If channel coding is enabled, the parameter sets the DSCH/USCH data source.

The following standard data sources are available:

- "All 0, All 1"
An internally generated sequence containing 0 data or 1 data.
- "PNxx"
An internally generated pseudo-random noise sequence.
- "Pattern"
An internally generated sequence according to a bit pattern.
Use the "Pattern" box to define the bit pattern.
- "Data List/Select DList"
A binary data from a data list, internally or externally generated.
Select "Select DList" to access the standard "Select List" dialog.
 - Select the "Select Data List > navigate to the list file *.dm_iqd > Select" to select an existing data list.
 - Use the "New" and "Edit" functions to create internally new data list or to edit an existing one.
 - Use the standard "File Manager" function to transfer external data lists to the instrument.

See also:

- Section "Modulation Data" in the R&S SMBV100B user manual.
- Section "File and Data Management" in the R&S SMBV100B user manual.
- Section "Data List Editor" in the R&S SMBV100B user manual

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:DSCH:DATA` on page 219

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:DSCH:DList` on page 219

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:DSCH:PATtern` on page 219

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:USCH:DATA` on page 219

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:USCH:DList` on page 219

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:USCH:PATtern` on page 219

5.3.3 DL/UL BWPs Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".

2. Select "General > Users/BWPs" > "DL BWPs".

5G New Radio A: Users/BWP Settings						
General	Properties <i>User 0</i>	DL BWPs <i>User 0 / Cell 0</i>	DL BWP Config <i>User 0 / Cell 0 / BWP 0</i>	UL BWPs <i>User 0 / Cell 0</i>	UL BWP Config <i>User 0 / Cell 0 / BWP 0</i>	
User:		0		Carrier: Cell		
		0		0		
Number of DL BWPs						
1						
BWP Indicato	SC Spacing / CP	No. RBs	RB Offse in TxBW	RB Offset to PointA	Δf to Carrier (Centers) /MHz	
0	0	30 kHz NCP	273	0	0	0.000 000

The dialog comprises the settings for the DL/UL bandwidth parts (BWP) configuration. The UL BWPs dialog is displayed for DL and UL direction, whereas the DL BWP - only for DL direction.

For each user and carrier combination, you can configure one or more BWPs. You can configure BWPs in the bandwidth part configuration table. This table consists of several rows, each of which corresponds to a BWP. The size of the table therefore depends on the number of BWPs you have selected.

Each BWP is indicated by a BWP indicator. The bandwidth of the BWP depends on the number of RBs it occupies. Its allocation in the frequency, on the selected RB offset. The resulting frequency offset between the BWP center and the center of the carrier is indicated, too.

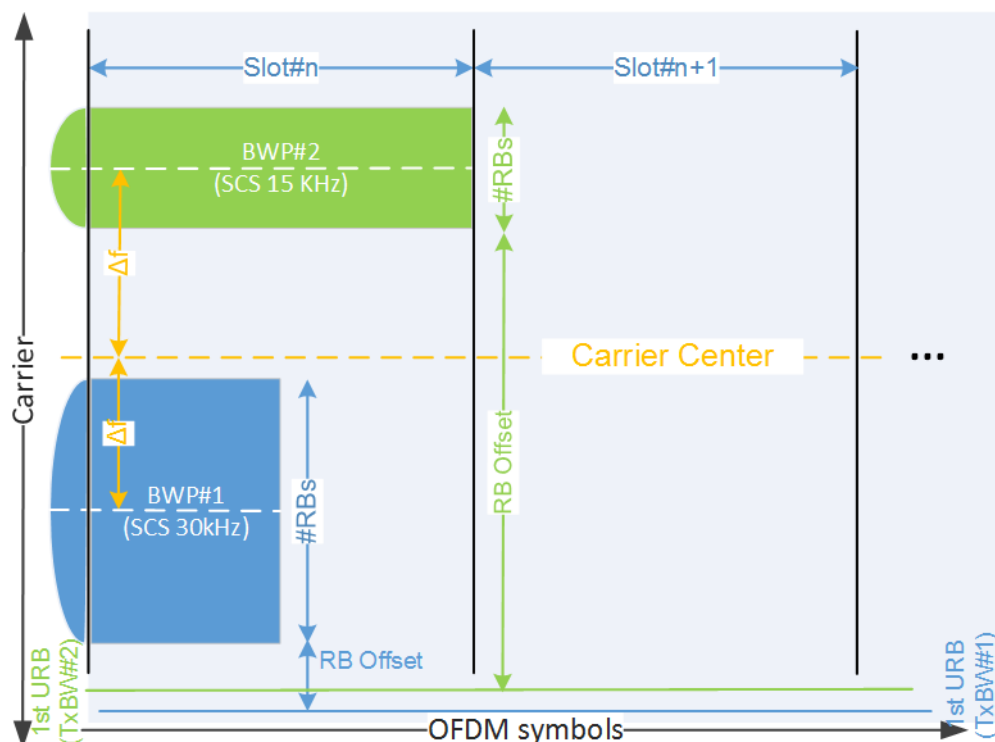


Figure 5-1: Bandwidth parts

BWP#	= Bandwidth part and indicator
SCS	= Subcarrier spacing
Center carrier	= Center frequency of the carrier (cell)
#RB	= Number of resource blocks or channel bandwidth per BWP
RB offset	= Frequency offset between the first usable resource block (URB) and the start of the BWP
1st URB	= First usable resource block (URB), that is the first RB within the transmission bandwidth TxBW, see Figure 3-2
Δf	= Frequency offset between the center frequency of the BWP and the center frequency of the carrier
Carrier	= Channel bandwidth of the carrier
OFDM symbol	= Represent the time domain

Settings:

User.....	55
Cell.....	56
Number of DL/UL BWPs.....	56
BWPs table.....	56
L BWP Indicator.....	56
L SC Spacing/CP.....	56
L No. RBs.....	56
L RB Offset in TxBW.....	57
L RB Offset to Point A.....	57
L Δf to Carrier (Centers).....	57

User

Selects the user to be configured.

Remote command:
via suffix at USER<ch>

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:
via suffix at CELL<ch>

Number of DL/UL BWPs

Sets the number of downlink bandwidth parts (BWP).

There is one table row per enabled DL/UL BWP.

See also:

- [Figure 5-1](#)
- [Chapter 7, "Observing Current Allocations On the Time Plan"](#), on page 174.

Remote command:

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:NBWParts
on page 220

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:NBWParts
on page 220

BWPs table

Comprises the BWPs settings.

There is one table row per enabled DL/UL BWP, see ["Number of DL/UL BWPs"](#) on page 56.

BWP Indicator ← BWPs table

Sets the unique indicator for the selected BWP, used to address the BWP within the DCI.

Remote command:

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
INDicator on page 220

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
INDicator on page 220

SC Spacing/CP ← BWPs table

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP), where the available values depend on the "Deployment".

See [Table 3-2](#).

Remote command:

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
SCSPacing on page 221

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
SCSPacing on page 221

No. RBs ← BWPs table

Defines the number of resource blocks (RB) the bandwidth part (BWP) occupies. This value defines the frequency width of the BWP.

The number of available RBs (N_{RB}) depends on the [SC Spacing/CP](#) and [Channel Bandwidth](#), where the latter depends on the selected deployment.

The number of available RBs is updated whenever the SCS or channel bandwidth is changed; the selected value is the maximum possible number of RB for the combination of SCS and channel bandwidth.

See also [Figure 5-1](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RBNumber on page 221
```

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
RBNumber on page 221
```

RB Offset in TxBW ← BWPs table

Defines the offset between the first resource block in the BWP relative to the first usable resource blocks (URB).

See also [Figure 5-1](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RBOffset on page 221
```

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
RBOffset on page 221
```

RB Offset to Point A ← BWPs table

Indicates the frequency starting position of the BWP relative to the point A, calculated as follows:

"RB Offset to Point A" = [RB Offset in TxBW](#) + [TxBW Offset](#)

See also [Figure 5-1](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PRBOffset? on page 222
```

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PRBOffset? on page 222
```

Δf to Carrier (Centers) ← BWPs table

Indicates the frequency offset between the BWP center and the center frequency of the carrier; it thus defines the BWP position in the frequency domain.

The value is calculated automatically, based on the selected SCS, channel bandwidth, deployment method, number of RBs and RB offset.

See also [Figure 5-1](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
DFReq? on page 222
```

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
DFReq? on page 222
```

5.3.4 DL BWP Configuration Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".
2. Select "General > Users/BWPs" > "DL BWP Config".

The dialog comprises the settings for the DL bandwidth parts (BWP) configuration.

Settings:

- [User, Carrier, BWP Settings](#).....58
- [PDSCH Settings](#).....59
- [ZP CSI-RS Settings](#).....67
- [NZP CSI-RS Settings](#).....70
- [Rate Match Settings](#).....76

5.3.4.1 User, Carrier, BWP Settings

- [User](#).....58
- [Cell](#).....59
- [BWP](#).....59

User

Selects the user to be configured.

Remote command:
via suffix at `USER<ch>`

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:
via suffix at `CELL<ch>`

BWP

Select the bandwidth parts (BWP) whose settings are displayed for configuration.

There are one or more BWPs, as set with the parameter "User/BWP Settings > DL BWPs" > [Number of DL/UL BWPs](#).

Remote command:
via suffix at `BWP<dir0>`

5.3.4.2 PDSCH Settings

Access:

- Select "User/BWP Settings > DL BWP Config > PDSCH".

The dialog comprises the DMRS and PTRS settings, divided per DMRS-mapping type A and B, as defined in [TS 38.211](#) and [TS 38.214](#).

5G New Radio A: Users/BWP Settings

General	Properties User 0	DL BWPs User 0 / Cell 0	DL BWP Config User 0 / Cell 0 / BWP 0	UL BWPs User 0 / Cell 0	UL BWP Config User 0 / Cell 0 / BWP 0
User:	0	Carrier: Cell	0	BWP	0
General Settings					
Use PDSCH Scrambling ID	☒	Data Scrambling ID	0	PDSCH	
Max. Number of Codewords Per DCI	2	VRB-to-PRB Interleaver	Non-Interleaved	ZP CSI-RS	
MCS Table	64QAM	Resource Allocation	Type 1	NZP CSI-RS	
Resource Block Group Size	Config 1			Rate Match	
DMRS for Mapping Type A					
Config Type	1	Additional Position Index	0		
Max Length	2				
Scrambling ID 0	0	Scrambling ID 1	0		
PTRS Config ...					
DMRS for Mapping Type B					
Config Type		Additional Position Index			

For description of the remote-control commands, see [Chapter 8.11.1, "PDSCH Commands"](#), on page 223.

About PDSCH and DMRS sequence generation

According to TS 38.211, the **PDSCH** sequence is generated based among others on the value of the variable n_{ID} , that can be:

- $n_{ID} = N_{ID}^{cell}$ (i.e. equal to the physical cell identifier) or
- $n_{ID} = \text{dataScramblingIdentityPDSCH}$ (higher-layer parameter).

In this implementation, you set which of the two variants is used by the parameter [Use PDSCH Scrambling ID](#) and the n_{ID} value on both case by the parameters "Node > Carriers" > [Cell ID](#) and [Data Scrambling ID](#).

According to TS 38.211, the **DMRS** sequence is generated based among others on the values of the variables N_{ID}^0, N_{ID}^1 and $N_{ID}^{n_SCID}$ or N_{ID}^0 and $N_{ID}^{n_SCID}$ with $n_SCID = \{0, 1\}$.

Which of these two combinations of variables is used depends on whether the PDSCH is scheduled by PDCCH using DCI format 1_1 or DCI format 1_0.

In this implementation, you define the values as follows:

- N_{ID}^0, N_{ID}^1 per DMRS type are set with "User/BWP > DL BWP > PDSCH" > [Scrambling ID 0/Scrambling ID 1](#).
- If N_{ID}^0 or N_{ID}^1 is used, is set with "Scheduling > PDSCH > DMRS" > [N_SCID](#).
- If $N_{ID}^{n_SCID} = N_{ID}^{cell}$ or $N_{ID}^{n_SCID} = N_{ID}^0$ or N_{ID}^1 is set with "Scheduling > PDSCH > DMRS" > [Sequence Generation](#).
- The [DCI Format 1_0](#) field "DMRS Sequence Initialization".

Settings:

Use PDSCH Scrambling ID	61
Data Scrambling ID	61
Max. Number of Codewords per DCI	61
VRB-to-PRB Interleaver	61
MCS Table	62
Resource Allocation	62
Resource Block Group Size	62
Config Type	63
Additional Position Index	63
Max Length	63
Scrambling ID 0/Scrambling ID 1	63
PTRS Config	64
L State	64
L RE-offset	65
L EPRE Ratio	65
L I_MCS_1/I_MCS_2/I_MCS_3	65
L N_RB_1/N_RB_2	65
Time Domain Allocation List	66
L Time Domain Allocations	66
L K0	66
L Mapping Type	66

L Start	66
L Length	67
L SLIV	67

Use PDSCH Scrambling ID

Defines the value of the variable n_{ID} used for the initialization of the generator of the PDSCH scrambling sequence.

"Off" $n_{ID} = N_{ID}^{cell}$ (i.e. the physical layer cell identity)

"On" n_{ID} is the values set with the parameter ["Data Scrambling ID"](#) on page 61.

See also ["About PDSCH and DMRS sequence generation"](#) on page 60.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:SCRambling:STAtE on page 225
```

Data Scrambling ID

If [Use PDSCH Scrambling ID](#) > "On", sets the identifier

`dataScramblingIdentityPDSCH` used to initialize data scrambling (`c_init`) for PDSCH.

The value of this parameter influences the PDSCH allocations.

See also ["About PDSCH and DMRS sequence generation"](#) on page 60.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DSID on page 225
```

Max. Number of Codewords per DCI

Sets if one or two codewords are transmitted.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:MCWDci on page 226
```

VRB-to-PRB Interleaver

As defined in [TS 38.211](#), the virtual resource blocks (VRB) are mapped to the physical resource blocks (PRB) interleaved or non-interleaved.

With this parameter, you set whether the interleaved or non-interleaved mapping is used in general, *for all PDSCH allocations*.

"Non-interleaved"

Sets non-interleaved VRB-to-PRB mapping. The value applies for all PDSCH allocations, also for those configured automatically from DCI format 1_0 or 1_1, if "Node > Scheduling > PDSCH Scheduling = Auto/DCI".

For all PDDCHs scheduled by DCI formats expect the 1_0 format, the VRB number n is mapped to PRB number n .

For all PDDCHs scheduled by DCI format 1_0 in common search space, the VRB number n is mapped to PRB number $n + N_{start}^{CORE-SET}$.

"2", "4"

Set one of these values to enable VRB-to-PRB interleaving in general and define the PRB bundle size, carried by the higher-level parameter `vrb-ToPRB-Interleaved`, defined in [TS 38.331](#).

You can change the *interleaving state per PDSCH* subsequently:

- If "Node > Scheduling > PDSCH Scheduling = Manual", with this parameter.
- If "Node > Scheduling > PDSCH Scheduling = Auto/DCI", for PDSCH created automatically from DCI formats 1_0 and 1_1, with the DCI field [VRB-to-PRB Mapping](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCH:VPINter on page 226
```

MCS Table

Sets the MCS table and thus defines the used modulation scheme.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCH:MCSTable on page 226
```

Resource Allocation

Sets the PDSCH resource allocation scheme and defines the allocation in the frequency domain.

"Type 0", "Type 1"

Selects the downlink resource allocation scheme, according to [TS 38.214](#).

"Dynamic Switch"

UE uses downlink resource allocation type 0 or type 1, as defined by the DCI field "Frequency domain resource assignment".

This parameter defines the PDSCH allocation in the frequency domain globally and if you configure the allocations manually.

To configure the PDSCH based on the DCI:

- Set "Node > Scheduling > PDSCH Scheduling = Auto/DCI"
- Enable a DCI Format 1_1 ("Scheduling > CORESET > Payload > DCI Format = 1_1") with "Config > Frequency Domain Resource Assignment = 262 143"
Set "Create PDSCH > On"
- Observe the allocation information for PDSCH ("Scheduling > PDSCH > Info > Content > frequency domain allocation")

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCH:RESalloc on page 227
```

Resource Block Group Size

For "Resource Allocation = Type 0/Dynamic Switch", the resource block assignment is defined in terms of resource block groups (RBG). A RBG is a set of consecutive virtual resource blocks and its size (`rbg-Size`) is defined as function of the BWP size and one of two possible configurations.

This parameter sets which of the two configurations defined in [TS 38.214](#) is used.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:RBGSize on page 227
```

Config Type

Sets the configuration type (type 1 or type 2). It thus defines the mapping of the demodulation reference signal (DMRS) to the physical resource elements in the frequency domain, as defined in TS 38.211.

The configuration type is carried by the higher-layer parameter `dmrs-Type`.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:CTYPE on page 224
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:CTYPE on page 224
```

Additional Position Index

Sets the parameter `dmrs-AdditionalPosition` that is required to select the positions of the DMRS symbols l , as defined in TS 38.211.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:APINDEX on page 224
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:APINDEX on page 224
```

Max Length

Sets the parameter `maxLength` and defines whether single- or double-symbol DMRS is used.

If "Max Length = 1", only single-symbol DMRS is allowed.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:MLENGTH on page 224
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:MLENGTH on page 224
```

Scrambling ID 0/Scrambling ID 1

Sets the higher-layer parameters `scramblingID0` and `scramblingID1` (N_{ID}^0, N_{ID}^1).

The values are used to calculate the initialization sequence of the pseudo-random sequence generator.

See also "About PDSCH and DMRS sequence generation" on page 60.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:SID0 on page 225
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:SID1 on page 225
```

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:SID0 on page 225

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:SID1 on page 225

PTRS Config ...

Comprises the settings of the phase-tracking reference signals (PTRS) for PxSCH, as defined in TS 38.214.

5G New Radio A: PTRS Settings - Mapping Type A (U0/C0/B0)	
State	0
RE-offset	00
EPRE Ratio	0
Threshold Time Density	
I_MCS_1	0
I_MCS_2	0
I_MCS_3	0
Threshold Frequency Density	
N_RB_0	1
N_RB_1	276

The phase tracking reference signal (PTRS) is a UE-specific reference signal that is used to compensate for the phase noise of the oscillator. Phase-tracking reference signals can be transmitted only in the resource blocks used for the PxSCH and if the transmission is enabled.

If the PTRS "State" is on, you can define its "Power" relative to the PxSCH and its location in the resource grid.

State ← PTRS Config ...

If [DSCH/USCH Channel Coding](#) > "On", enables the transmission of phase-tracking reference signals.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTA:PTRS:STATe on page 265
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:PTRS:STATe on page 265
```

RE-offset ← PTRS Config ...

Sets the parameter `resourceElementOffset` $k_{\text{ref}}^{\text{RE}}$ required to define the precoding and mapping to the physical resources.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTA:PTRS:REOF on page 265
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:PTRS:REOF on page 265
```

EPRE Ratio ← PTRS Config ...

Sets the higher layer parameter `epre-Ratio` `ratio`, needed to define the ratio of PTRS EPRE to PDSCH EPRE per layer per RE for PTRS port.

The abbreviation EPRE stands for energy per resource element.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTA:PTRS:EPRE on page 266
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:PTRS:EPRE on page 266
```

I_MCS_1/I_MCS_2/I_MCS_3 ← PTRS Config ...

Sets the threshold values `ptrs-MCSl` with $l = \{1, 2, 3\}$ transmitted by the higher-layer parameter `timeDensity` in `PTRS-DownlinkConfig` .

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTA:PTRS:MCS1 on page 267
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTA:PTRS:MCS2 on page 267
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTA:PTRS:MCS3 on page 267
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:PTRS:MCS1 on page 267
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:PTRS:MCS2 on page 267
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSch:DMTB:PTRS:MCS3 on page 267
```

N_RB_1/N_RB_2 ← PTRS Config ...

Sets the threshold values `NRBl` with $l = \{0, 1\}$ transmitted by the higher-layer parameter `frequencyDensity` in `PTRS-DownlinkConfig` .

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:PTRS:RB0 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:PTRS:RB1 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:PTRS:RB0 on page 268

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:PTRS:RB1 on page 268

Time Domain Allocation List

Use this settings to define the PDSCH time domain scheduling, according to TS 38.214.

Time Domain Allocation List						
Time Domain Allocations						2
	K0	Mapping Type	Start	Length	SLIV	
0	0	A	0	3	28	
1	0	B	0	2	14	

Time Domain Allocations ← Time Domain Allocation List

Sets the number of allocations.

There is one row per allocation in the time domain allocation table.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TDANum on page 227

K0 ← Time Domain Allocation List

Sets the slot offset K_0 .

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TD<grp0>:KNULL on page 227

Mapping Type ← Time Domain Allocation List

Sets the DMRS-mapping type A and B, as defined in TS 38.211

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TD<grp0>:MAPPING on page 228

Start ← Time Domain Allocation List

Sets the start OFDM symbol (S) of the allocation, where the value range depends on the selected [SC Spacing/CP](#).

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TD<grp0>:START on page 228

Length ← Time Domain Allocation List

Sets the number of consecutive OFDM symbols (L) the allocation spans, where the value depends on the selected [SC Spacing/CP](#) and the following applies:

- $S + L \leq 14$, if normal cyclic prefix is used (NCP)
- $S + L \leq 12$, if extended cyclic prefix is used (ECP)

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TD<grp0>:LENGth on page 228
```

SLIV ← Time Domain Allocation List

Indicates the resulting start and length indicator SLIV, selected automatically from the values of the parameters [K0](#), [Start](#) and [Length](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TD<grp0>:SLIV? on page 229
```

5.3.4.3 ZP CSI-RS Settings

Access:

- Select "User/BWP Settings > DL BWP Config > ZP CSI-RS".

The dialog comprises the zero-power (ZP) CSI-RS settings, as defined in [TS 38.211](#).

The screenshot shows the '5G New Radio A: Users/BWP Settings' dialog box. The 'DL BWP Config' tab is selected. Under 'DL BWP Config', the 'ZP CSI-RS' sub-tab is active. The 'User' is set to 0, 'Carrier: Cell' is 0, and 'BWP' is 0. The 'Periodic ZP CSI-RS' section shows 'State' as 'I' and 'Resources' as 2. A table displays the configuration for two resources:

	Periodicity	Slot Offset	No. RBs	Start RB	Row	Density	CDM-Type	Bitmap	Ports	I0	I1
0	4 slots	0	273	0	1	3	No CDM	1111	1	9	11
1	8 slots	1	100	50	1	3	No CDM	1111	1	9	11

On the right side of the dialog, there are buttons for 'PDSCH', 'ZP CSI-RS' (which is highlighted), 'NZP CSI-RS', and 'Rate Match'.

For description of the remote-control commands, see [Chapter 8.11.2, "ZP and NZP CSI-RS Commands"](#), on page 229.

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L No. RBs.....	68
L Start RB.....	69
L Row.....	69

L Density.....	69
L CDM-Type.....	69
L Bitmap.....	69
L Ports.....	70
L I0/I1.....	70

State

Enables the ZP CSI-RS transmission.

The CSI-RS can be used among others for time or frequency tracking or CSI calculation.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:STATe on page 230
```

Resources

Sets the number of periodicCSI-RS resources.

There is one table row per resource.

See "Periodic ZP CSI-RS resource table" on page 68.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:NRESources on page 231
```

Periodic ZP CSI-RS resource table

Each periodic ZP CSI-RS is configured individually; there is one table row per resource, where the number of rows is defined by the parameter [Resources](#).

Periodicity ← Periodic ZP CSI-RS resource table

For periodic ZP CSI-RS:

Sets the periodicity $T_{\text{CSI-RS}}$ (in slots) transmitted by the higher-layer parameter CSI-ResourcePeriodicityAndOffset.

Thus, this parameter defines how often the CSI-RS is transmitted.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:PER on page 231
```

Slot Offset ← Periodic ZP CSI-RS resource table

For periodic ZP CSI-RS:

Sets the slot offset T_{offset} transmitted by the higher-layer parameter CSI-ResourcePeriodicityAndOffset.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:OFFSet on page 231
```

No. RBs ← Periodic ZP CSI-RS resource table

Sets the periodicity $T_{\text{CSI-RS}}$ (in slots) transmitted by the higher-layer parameter CSI-ResourcePeriodicityAndOffset.

Thus, this parameter defines how often the CSI-RS is transmitted.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:RBNumber on page 231
```

Start RB ← Periodic ZP CSI-RS resource table

If the number of RB the CSI-RS spans ("No. RBs") is smaller than the available number of RB (i.e. the whole width of the BWP), sets the first RB (*startingRB*) the CSI-RS allocates.

The value is set relative to the common resource block #0 on the common RB grid.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:SRBNumber on page 232
```

Row ← Periodic ZP CSI-RS resource table

The CSI-RS location in a slot is defined according to TS 38.211 as a function of the number of ports X, the "Density" and the "CDM-Type".

The possible combinations are listed in a table and the parameter "Row" indicates one of the combinations. It is not the consequent table row number in the CSI-RS resource table. The latter is defined with the suffix *RES<user0>* in the remote control command.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:ROW on page 232
```

Density ← Periodic ZP CSI-RS resource table

Sets the value *p* that is the higher-layer parameter *density* in the CSI-RS-ResourceMapping or the CSI-RS-CellMobility messages.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:DENSITY on page 233
```

CDM-Type ← Periodic ZP CSI-RS resource table

Indicates or sets value of the higher-layer parameter *cdm-Type* in the CSI-RS-ResourceMapping.

The value is selected automatically according to TS 38.211 and depending on the selected "Row".

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:CDMType? on page 233
```

Bitmap ← Periodic ZP CSI-RS resource table

Sets the location of the CSI-RS in the frequency domain. The starting position and number of the resource blocks in which the CSI-RS is transmitted are defined by the higher-layer parameters *freqBand*.

This parameter is transmitted by the higher-layer parameters `frequencyDomainAllocation` as part of the `CSI-RS-ResourceMapping` or the `CSI-RS-CellMobility` messages.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:BITMap on page 233
```

Ports ← Periodic ZP CSI-RS resource table

Sets the number of ports x that is given by the higher-layer parameter `nrofPorts`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:PORTs? on page 232
```

I0/I1 ← Periodic ZP CSI-RS resource table

Sets the parameters I_0 and I_1 and define the CSI-RS location relative to the start of as slot. Thus, the I_0 and I_1 give the location of the CSI-RS in time.

These parameters are transmitted by the higher-layer parameters `firstOFDMSymbolInTimeDomain` and `firstOFDMSymbolInTimeDomain2` as part of the `CSI-RS-ResourceMapping` or the `CSI-RS-CellMobility` messages.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:I0 on page 234
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:ZP:SET<gr0>:RES<user0>:I1 on page 234
```

5.3.4.4 NZP CSI-RS Settings

Access:

- Select "User/BWP Settings > DL BWP Config > NZP CSI-RS".

The dialog comprises the non-zero-power (NZP) CSI-RS settings, as defined in TS 38.211.

5G New Radio A: Users/BWP Settings											
General	Properties User 0	DL BWPs User 0 / Cell 0	DL BWP Config User 0 / Cell 0 / BWP 0	UL BWPs User 0 / Cell 0	UL BWP Config User 0 / Cell 0 / BWP 0						
User:	0	Carrier:	Cell	0	BWP	0					
Periodic NZP CSI-RS											
State		I		Resources		2					
	Periodicity	Slot Offset	No. RBs	Start RB	Row	Density	CDM-Type	Bitmap	Ports	Power (/dB)	Settings
0	4 slots	0	273	0	1	3	No CDM	1111	1	0.00	Config...
1	20 slots	10	100	50	1	3	No CDM	1010	1	0.00	Config...

PDSCH
 ZP CSI-RS
NZP CSI-RS
 Rate Match

For description of the remote-control commands, see [Chapter 8.11.2, "ZP and NZP CSI-RS Commands"](#), on page 229.

State.....	71
Resources.....	71
Periodic NZP CSI-RS resource table.....	71
L Periodicity.....	71
L Offset.....	72
L No. RBs.....	72
L Start RB.....	72
L Row.....	72
L Density.....	72
L CDM-Type.....	73
L Bitmap.....	73
L Ports.....	73
L Power.....	73
L Settings.....	73
L Resource-Settings > General.....	73
L Scrambling ID.....	74
L IO/1.....	74
L Resource-Settings > Antenna Port Mapping.....	74
L Mapping Coordinates.....	75
L Mapping table.....	75

State

Enables the NZP CSI-RS transmission.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:STATe on page 230
```

Resources

Sets the number of periodic CSI-RS resources.

There is one table row per resource.

See "[Periodic ZP CSI-RS resource table](#)" on page 68.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:NRESources on page 231
```

Periodic NZP CSI-RS resource table

Each periodic NZP CSI-RS is configured individually; there is one table row per resource, where the number of rows is defined by the parameter [Resources](#).

Periodicity ← Periodic NZP CSI-RS resource table

Sets the periodicity $T_{\text{CSI-RS}}$ (in slots) transmitted by the higher-layer parameter CSI-ResourcePeriodicityAndOffset.

Thus, this parameter defines how often the CSI-RS is transmitted.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:PER on page 231
```

Offset ← Periodic NZP CSI-RS resource table

Sets the slot offset T_{offset} transmitted by the higher-layer parameter CSI-ResourcePeriodicityAndOffset.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:OFFSet on page 231
```

No. RBs ← Periodic NZP CSI-RS resource table

Sets the periodicity $T_{\text{CSI-RS}}$ (in slots) transmitted by the higher-layer parameter CSI-ResourcePeriodicityAndOffset.

Thus, this parameter defines how often the CSI-RS is transmitted.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:RBNumber on page 231
```

Start RB ← Periodic NZP CSI-RS resource table

If the number of RB the CSI-RS spans ("No. RBs") is smaller than the available number of RB (i.e. the whole width of the BWP), sets the first RB (startingRB) the CSI-RS allocates.

The value is set relative to the common resource block #0 on the common RB grid.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:SRBNumber on page 232
```

Row ← Periodic NZP CSI-RS resource table

The CSI-RS location in a slot is defined according to TS 38.211 as a function of the number of ports X , the "Density" and the "CDM-Type".

The possible combinations are listed in a table and the parameter "Row" indicates one of the combinations. It is not the consequent table row number in the CSI-RS resource table. The latter is defined with the suffix RES<user0> in the remote control command.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:ROW on page 232
```

Density ← Periodic NZP CSI-RS resource table

Sets the value p that is the higher-layer parameter density in the CSI-RS-ResourceMapping or the CSI-RS-CellMobility messages.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:DENSity on page 233
```


CDM-Type ← Periodic NZP CSI-RS resource table

Indicates or sets value of the higher-layer parameter `cdm-Type` in the `CSI-RS-ResourceMapping`.

The value is selected automatically according to [TS 38.211](#) and depending on the selected "Row".

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:CDMType? on page 233
```

Bitmap ← Periodic NZP CSI-RS resource table

Sets the location of the CSI-RS in the frequency domain. The starting position and number of the resource blocks in which the CSI-RS is transmitted are defined by the higher-layer parameters `freqBand`.

This parameter is transmitted by the higher-layer parameters `frequencyDomainAllocation` as part of the `CSI-RS-ResourceMapping` or the `CSI-RS-CellMobility` messages.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:BITMap on page 233
```

Ports ← Periodic NZP CSI-RS resource table

Sets the number of ports `x` that is given by the higher-layer parameter `nrofPorts`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:PORTs? on page 232
```

Power ← Periodic NZP CSI-RS resource table

Sets the power of the NZP resource relative to the PDSCH power.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:PWR on page 234
```

Settings ← Periodic NZP CSI-RS resource table

Opens the "Resource-Settings" dialog where you configure:

- Scrambling and I0/I1 values, see ["Resource-Settings > General"](#) on page 73.
- The antenna port mapping for the selected resource, see ["Resource-Settings > Antenna Port Mapping"](#) on page 74.

Resource-Settings > General ← Periodic NZP CSI-RS resource table

Access: "General > Users/BWPs > DL BWP Config > NZP CSI-RS" > "Resources = 1" and select "Settings > Config > General".

5G New Radio A (U0/C0/B0/RS0/R0)Resource-Settings	
General	Antenna Ports
SCrambling ID	0
I0	9
I1	11

Scrambling ID ← Resource-Settings > General ← Periodic NZP CSI-RS resource table

Sets the parameter n_{ID} that is defined by the higher-layer parameter `ScramblingID` and used for the generation of the pseudo-random reference-signal generator.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:SCID on page 234
```

I0/I1 ← Resource-Settings > General ← Periodic NZP CSI-RS resource table

Sets the parameters I_0 and I_1 and define the CSI-RS location relative to the start of a slot. Thus, the I_0 and I_1 give the location of the CSI-RS in time.

These parameters are transmitted by the higher-layer parameters

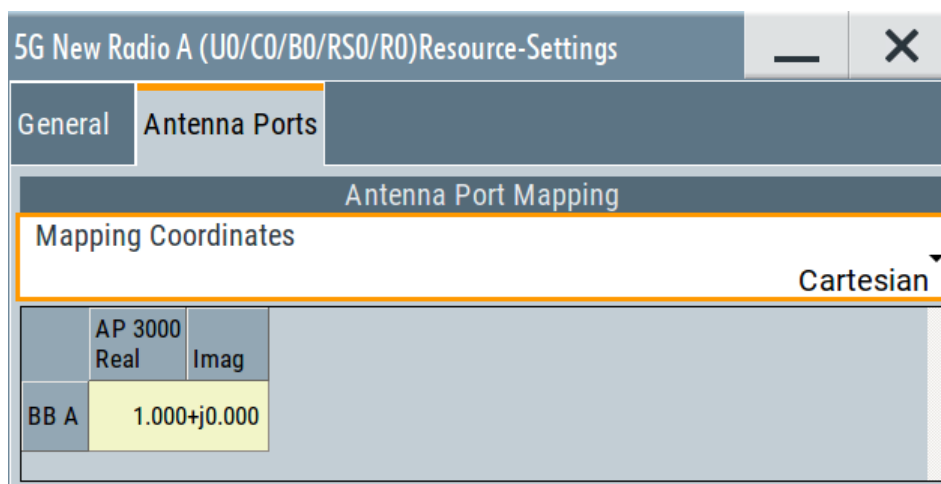
`firstOFDMSymbolInTimeDomain` and `firstOFDMSymbolInTimeDomain2` as part of the CSI-RS-ResourceMapping or the CSI-RS-CellMobility messages.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:I0 on page 234
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:I1 on page 234
```

Resource-Settings > Antenna Port Mapping ← Periodic NZP CSI-RS resource table

Access: "General > Users/BWPs > DL BWP Config > NZP CSI-RS" > "Resources = 1" and select "Settings > Config > Antenna Ports".



For details on the antenna port mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Mapping Coordinates ← Resource-Settings > Antenna Port Mapping ← Periodic NZP CSI-RS resource table

Switches representation between the "Cartesian" (Real/Imag) and "Cylindrical" (Magn./Phase) coordinates.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:APMap:MAPCoord on page 235
```

Mapping table ← Resource-Settings > Antenna Port Mapping ← Periodic NZP CSI-RS resource table

The mapping table is a matrix with number of rows equal to the number of physical Tx antennas (Basebands) and number of columns equal of the number of antenna ports (AP). The available antenna ports depend on the current configuration. The yellow matrix elements in the mapping table indicate the default antenna port to physical antenna (TX antenna/baseband) mapping.

For details on the antenna port mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Remote command:

"Mapping Coordinates = Cartesian":

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:REAL
on page 235
```

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:
IMAGinary on page 235
```

"Mapping Coordinates = Cylindrical":

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:

MAGNitude on page 236

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
CSIRs:NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:PHASe

on page 236

5.3.4.5 Rate Match Settings

Access:

- Select "User/BWP Settings > DL BWP Config > Rate Match".

The dialog comprises the PDSCH resources mapping settings, required to configure the rate match patterns, as defined in TS 38.214.

For description of the remote-control commands, see Chapter 8.11.3, "Rate Match Commands", on page 236.

State

Enables the configuration of rate match groups.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:STATe on page 237

Number of Groups

Sets the number of rate match pattern groups (rateMatchPatternGroup).

The value limits the available Group ID, the groups that can be selected for PDSCH ("Scheduling > PDSCH > Config > General" > Rate Match Pattern Group) and the bit length of the DCI format 1_1 parameter "Rate Matching Indicator".

Each rate match pattern group contains a list of RB and slot indices that build a resource set of resources. These resources are not available for the PDSCH allocations, if one of the following is true:

- The "Rate Matching Indicator" field of the DCI format 1_1 used to schedule this PDSCH is set to value different than 0.
- A "Rate Match Group" is selected for a manually configured PDSCH.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:GRPNumber on page 237
```

Resources

Sets the number of rate match resources.

There is one table row per resource.

See "Rate match resource table" on page 77.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:NREsources on page 237
```

Rate match resource table

Each rate match resource is configured individually; there is one table row per resource, where the number of rows is defined by the parameter [Resources](#).

RB Pattern ← Rate match resource table

The resource block level bitmap is defined as a binary data from a data list, internally or externally generated.

The bitmap is 275 bits long. Longer patterns are truncated; shorter are zero-padded (i.e. the remaining bits are set to 0). If no data list is loaded, a pattern with all zeros is assumed.

A bit value of 1 in the RB level bitmaps indicates that the corresponding resource is not available for PDSCH.

To load a list:

- Select "Data List"
- In the standard "Select List" dialog, select the "Select Data List > navigate to the list file *.dm_iqd > Select" to select an existing data list.
- Use the "New" and "Edit" functions to create internally new data list or to edit an existing one.
- Use the standard "File Manager" function to transfer external data lists to the instrument.

See also:

- Section "Modulation Data" in the R&S SMBV100B user manual.
- Section "File and Data Management" in the R&S SMBV100B user manual.
- Section "Data List Editor" in the R&S SMBV100B user manual

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:RBDList on page 238
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:RBPatt on page 238
```

Slot ← Rate match resource table

Sets the number of slots and thus defines the "Slot Pattern" length.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:SLOT on page 238
```

Slot Pattern ← Rate match resource table

Set the slots to be used as a pattern.

The pattern length is the 14 or 28 depending on whether "Slot = 1 or 2".

A bit value of 1 in the symbol level bitmaps indicates that the corresponding resource is not available for PDSCH.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:SLTPatt on page 239
```

Periodicity ← Rate match resource table

Sets the time-domain pattern and defines the "Periodicity Pattern" length.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:PER on page 239
```

Periodicity Pattern ← Rate match resource table

Sets the periodicity in a pattern form, where the pattern length is set with the parameter "Periodicity".

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:PERPatt on page 239
```

Group ID ← Rate match resource table

Sets the group identifier for the selected rate match pattern, where the available values depend on the selected [Number of GroupsIdPDbNr5gUserDlBwpRateMatchRsPeriodicity](#).

Up to four rate match patterns can use the same "Group ID".

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RATM:RS<gr0>:GRID on page 239
```

5.3.5 UL BWP Configuration Settings

Access:

1. Select "5G New Radio > General > Link Direction" > **"Uplink"**.
2. Select "General > Users/BWPs" > **"UL BWP Config"**.

3. Select "PUCCH".

5G New Radio A: Users/BWP Settings

General	Properties <i>User 0</i>	DL BWPs <i>User 0 / Cell 0</i>	DL BWP Config <i>User 0 / Cell 0 / BWP 0</i>	UL BWPs <i>User 0 / Cell 0</i>	UL BWP Config <i>User 0 / Cell 0 / BWP 0</i>
User:	0	Carrier: Cell	0	BWP	0
Additional DMRS	☐	$\pi/2$ -BPSK	☐	PUCCH	
Simultaneous HARQ-ACK-CSI	☐	PDSCH to HARQ Timing Map ...	1, 0, 15	FRC	
				PUSCH	
				PUSCH UCI	
				SRS	

The dialog comprises the settings for the UL bandwidth parts (BWP) configuration.

Settings:

- [User, Carrier, BWP Settings](#).....79
- [PUCCH Settings](#).....80
- [FRC Settings](#).....81
- [PUSCH Settings](#).....85
- [PUSCH UCI Settings](#).....92
- [SRS Settings](#).....94

5.3.5.1 User, Carrier, BWP Settings

[User](#).....79
[Cell](#).....79
[BWP](#).....79

User

Selects the user to be configured.

Remote command:

via suffix at `USER<ch>`

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at `CELL<ch>`

BWP

Select the bandwidth parts (BWP) whose settings are displayed for configuration.

There are one or more BWPs, as set with the parameter "User/BWP Settings > UL BWPs" > [Number of DL/UL BWPs](#).

Remote command:

via suffix at `BWP<dir0>`

5.3.5.2 PUCCH Settings

Access:

- Select "User/BWP Settings > UL BWP Config > PUCCH".

Additional DMRS.....	80
$\pi/2$ - BPSK.....	80
Simultaneous HARQ-ACK-CSI.....	80
PDSCH to HARQ Timing Map.....	81
L Number of Entries.....	81
L Value of Indicator x.....	81

Additional DMRS

Global parameter that defines if additional DMRS is used for PUCCH format 3 and 4. According to TS 38.211, the value is required to select the DMRS positions for the PUCCH format 3 and 4.

The value is ignored, if other PUCCH Format is configured.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:ADMRs:STATe on page 240
```

$\pi/2$ - BPSK

If enabled, the $\pi/2$ - BPSK modulation order is used for PUCCH, instead of the default QPSK.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:BPSK:STATe on page 240
```

Simultaneous HARQ-ACK-CSI

Reserved for future use.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:HACK:STATe on page 241
```


PDSCH to HARQ Timing Map

Select the parameter name to access a dialog where you configure the PDSCH-to-HARQ-timing-indicator field values map, as defined in TS 38.213.

The PDSCH to HARQ timing map is transmitted by the higher-layer parameter `dl-DataToUL-ACK`. It is a sequence of up to 8 entries, each four bits long, where the number of entries is set with the parameter [Number of Entries](#).

The value of the individual timing indicators is defined by the parameter [Value of Indicator x](#). Observe the resulting sequence in the overview indication together with the parameter name "PDSCH to HARQ Timing Map".

Timing Map	
Number of Entries	3
Timing Values	
Value for Indicator 0	1
Value for Indicator 1	0
Value for Indicator 2	15

Number of Entries ← PDSCH to HARQ Timing Map

Sets the number of entries in the sequence of PDSCH to HARQ timing values.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:PDSHarq:NTMentry on page 241
```

Value of Indicator x ← PDSCH to HARQ Timing Map

Sets the individual timing values.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:PDSHarq:TMIValue<gr0> on page 241
```

5.3.5.3 FRC Settings

Access:

1. Select "User/BWP Settings > Properties" > **DSCH/USCH Channel Coding** > "On".

2. Select "User/BWP Settings > UL BWP Config > FRC".

General	Properties <i>User 0</i>	DL BWPs <i>User 0 / Cell 0</i>	DL BWP Config <i>User 0 / Cell 0 / BWP 0</i>	UL BWPs <i>User 0 / Cell 0</i>	UL BWP Config <i>User 0 / Cell 0 / BWP 0</i>
User:	0	Carrier: Cell	0	BWP	0
FRC State		PUCCH			
FRC		TS 38.141: G-FR1-A1-2		Subcarrier Spacing	30 kHz
Allocated Resource Blocks		11		Modulation	QPSK
Payload Size		984		RB Offset	0
Mapping Type		A			

The dialog allows you to select and enable fixed reference measurement channels (FRC), as defined in [TS 38.141](#).

The available FRCs depend on the selected BWP SCS ([SC Spacing/CP](#)) and selected number of RBs ([No. RBs](#)).

FRC State	82
FRC	82
Subcarrier Spacing	83
Allocated Resource Blocks	83
Modulation	84
Payload Size	84
RB Offset	84
Mapping Type	84
PTRS	84

FRC State

To enable the fixed reference measurement channels (FRC), set "User/BWP Settings > Properties" > [DSCH/USCH Channel Coding](#) > "On".

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:STATE` on page 242

FRC

The available FRCs depend on the selected BWP subcarrier spacing [SC Spacing/CP](#)) and selected number of resource blocks ([No. RBs](#)).

For each particular combination, you can select from the FRCs that have the same subcarrier spacing (SCS) as the BWP and the number of RBs allocated in the FRC is less than or equal to the "No. RB" of current BWP.

The FRCs are configured as defined in [TS 38.141](#). [FRCs major parameters \[TS 38.141\]](#) list the major difference between the FRC subsets; within one subset, the FRCs further differ in the SCS and the number of allocated RBs. For details, refer to [TS 38.141](#).

Table 5-1: FRCs major parameters [TS 38.141]

Test condition	Frequency	Subset	Modulation	Reference channels
Fixed Reference Channels for receiver sensitivity and in-channel selectivity	FR1	A1	QPSK	G-FR1-A1-1 to G-FR1-A1-9
	FR2	A1	QPSK	G-FR2-A1-1 to G-FR2-A1-5
Fixed Reference Channels for dynamic range	FR1	A2	16QAM	G-FR1-A2-2 to G-FR1-A2-6
Fixed Reference Channels for performance requirements	FR1	A3	QPSK	G-FR1-A3-1 to G-FR1-A3-32
		A4	16QAM	G-FR1-A4-1 to G-FR1-A4-28
		A5	64QAM	G-FR1-A5-1 to G-FR1-A5-14
	FR2	A3	QPSK	G-FR2-A3-1 to G-FR2-A3-24
		A4	16QAM	G-FR1-A4-1 to G-FR1-A4-20
		A5	64QAM	G-FR1-A5-1 to G-FR1-A5-10

"TS 38.141: G-FR1"

Subset of FRCs defined for the FR1 (≤ 6 GHz).

"TS 38.141: G-FR2"

Subset of FRCs defined for the FR2 (> 6 GHz)

"Not Available" Indicates that there is no FRC defined for the current combination of "SC Spacing/CP" and "No. RB".

If FRC is selected, the following parameters are set automatically:

- DMRS for Mapping Type A parameters [Config Type](#), [Additional Position Index](#) and [Max Length](#).
- PUSCH allocations
- All FRC parameters in the "FRC" dialog.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:TYPE` on page 242

Subcarrier Spacing

Indicates the SCS of the selected FRC. It is the same as the SCS of the selected BWP.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:SCS?` on page 243

Allocated Resource Blocks

Indicates the number of used RBs of the selected FRC.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
ALRB? on page 244

Modulation

Indicates the modulation scheme used by the selected FRC.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
MODulation? on page 244

Payload Size

Indicates the payload size of the selected FRC.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
PASize? on page 244

RB Offset

Sets the RB offset for the selected FRC.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
RBOffset on page 244

Mapping Type

Sets the PUSCH mapping type for the selected FRC.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
MAPType on page 245

PTRS

Enables PTRS for the selected FRC.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
PTRS:STATe on page 245

5.3.5.4 PUSCH Settings

Access:

- Select "User/BWP Settings > UL BWP Config > PUSCH".

5G New Radio A: Users/BWP Settings

General Properties UL BWPs **UL BWP Config** User 0 / Cell 0 / BWP 0

User: 0 Carrier: Cell 0 BWP 0

PTRS Config ...

DMRS for Mapping Type B

Config Type 1 Additional Position Index 0

Max Length 2

Scrambling ID 0 0 Scrambling ID 1 0

PTRS Config ...

Transform Precoding ☐ Max Rank 4

TxConfig Non-codebook Codebook Subset Fully-and-partial-and-non-coherent

Use PUSCH Scrambling ID ☐ Data Scrambling ID 0

MCS Table 64QAM MCS Table Transform Precoding 64QAM

PUCCH
FRC
PUSCH
PUSCH UCI
SRS

The dialog comprises the DMRS settings, divided per DMRS-mapping type and as defined in TS 38.211.

If FRC is used ("UL BWP Config > FRC > State > On"), the settings of the "DMRS for Mapping Type A" are configured automatically and cannot be changed.

Config Type.....	86
Additional Position Index.....	86
Max Length.....	86
Scrambling ID 0/Scrambling ID 1.....	86
PTRS Config	87
L State.....	88
L RE-offset.....	88
L Power.....	88
L Max Number of Ports.....	88
L I_MCS_1/I_MCS_2/I_MCS_3.....	88
L N_RB_1/N_RB_2.....	89
PTRS Config (Transform Precoding)	89
L State.....	90
L Time Density.....	90
L N_RB_x.....	90
Transform Precoding.....	90

Max Rank.....	91
TxConfig.....	91
Codebook Subset.....	91
Use PUSCH Scrambling ID.....	91
Data Scrambling ID.....	92
MCS Table.....	92
MCS Table Transform Precoding.....	92

Config Type

Sets the configuration type (type 1 or type 2). It thus defines the mapping of the demodulation reference signal (DMRS) to the physical resource elements in the frequency domain, as defined in TS 38.211.

The configuration type is carried by the higher-layer parameter `dmrs-Type`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:CTYPE on page 247
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:CTYPE on page 247
```

Additional Position Index

Sets the parameter `dmrs-AdditionalPosition` that is required to select the positions of the DMRS symbols $\bar{l}$, as defined in TS 38.211.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:APINDEX on page 247
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:APINDEX on page 247
```

Max Length

Sets the parameter `maxLength` and defines whether single- or double-symbol DMRS is used.

If "Max Length = 1", only single-symbol DMRS is allowed.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:MLENGTH on page 247
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:MLENGTH on page 247
```

Scrambling ID 0/Scrambling ID 1

Sets the higher-layer parameters `scramblingID0` and `scramblingID1` (N_{ID}^0, N_{ID}^1).

The values are used to calculate the initialization sequence of the pseudo-random sequence generator.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:SID0 on page 247

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:SID1 on page 247

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:SID0 on page 248

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:SID1 on page 248

PTRS Config ...

Enabled if [Transform Precoding](#).

Comprises the settings of the phase-tracking reference signals (PTRS) for P_xSCH, as defined in TS 38.214.

5G New Radio A: PTRS Settings - Mapping Type A (U0/C0/B0)	
State	0
RE-offset	00
Power	00
Max Number of Ports	1
Threshold Time Density	
I_MCS_1	0
I_MCS_2	0
I_MCS_3	0
Threshold Frequency Density	
N_RB_0	1
N_RB_1	276

The phase tracking reference signal (PTRS) is a UE-specific reference signal that is used to compensate for the phase noise of the oscillator. Phase-tracking reference signals can be transmitted only in the resource blocks used for the P_xSCH and if the transmission is enabled.

If the PTRS "State" is on, you can define its "Power" relative to the PUSCH and its location in the resource grid.

State ← PTRS Config ...

If [DSCH/USCH Channel Coding](#) > "On", enables the transmission of phase-tracking reference signals.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:STATe on page 265
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:STATe on page 265
```

RE-offset ← PTRS Config ...

Sets the parameter `resourceElementOffset` $k_{\text{ref}}^{\text{RE}}$ required to define the precoding and mapping to the physical resources.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:REOF on page 265
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:REOF on page 265
```

Power ← PTRS Config ...

Sets the parameter `ptrs-Power` and thus defines the PUSCH to PTRS power ratio per layer per resource element.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:POWer on page 266
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:POWer on page 266
```

Max Number of Ports ← PTRS Config ...

Sets the maximum number of configured PTRS ports, given by the higher level parameter `maxNrofPorts`.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:PORT on page 266
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:PORT on page 266
```

I_MCS_1/I_MCS_2/I_MCS_3 ← PTRS Config ...

Sets the threshold values `ptrs-MCSl` with $l = \{1, 2, 3\}$ transmitted by the higher-layer parameter `timeDensity` in `PTRS-DownlinkConfig`.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:MCS1 on page 267
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:MCS2 on page 267
```


[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:MCS3 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:MCS1 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:MCS2 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:MCS3 on page 267

N_RB_1/N_RB_2 ← PTRS Config ...

Sets the threshold values NRB_l with $l = \{0, 1\}$ transmitted by the higher-layer parameter frequencyDensity in PTRS-DownlinkConfig .

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:RB0 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:RB1 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:RB0 on page 267

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:RB1 on page 267

PTRS Config (Transform Precoding) ...

Enabled if [Transform Precoding](#) > "On".

Comprises the settings of the phase-tracking reference signals (PTRS) for PUSCH incl. transform precoding (TP), as defined in [TS 38.214](#).

5G New Radio A: PTRS Settings - Mapping Type B (U0/C0/B0)	
State	☒
Time Density	1
Sample Density	
N_RB_0	1
N_RB_1	276
N_RB_2	276
N_RB_3	276
N_RB_4	276

State ← PTRS Config (Transform Precoding) ...

Simulates the transmission of the higher-layer parameter

`transformPrecoderEnabled`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:STAtE on page 250
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:STAtE on page 250
```

Time Density ← PTRS Config (Transform Precoding) ...

Per default, an UE is configured with time density $L_{\text{PTRS}} = 1$.

Set the value of this parameter to 2 to simulated an UE configured by the the higher-layer parameter `timeDensityTransformPrecoding`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:TMDensity on page 250
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:TMDensity on page 250
```

N_RB_x ← PTRS Config (Transform Precoding) ...

Sets the higher-layer parameter `sampleDensity` that indicates the sample density thresholds $N_{\text{RB},i}$.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB0 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB1 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB2 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB3 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB4 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB0 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB1 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB2 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB3 on page 251
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB4 on page 251
```

Transform Precoding

Sets the higher layer parameter `transformPrecoder` that defines which access method is used, the optional DFT-s-OFDM or the CP-OFDM.

See also [Chapter 3.3, "Multiple Access Schemes"](#), on page 13.

"On" Enables the precoding operation and thus enables the DFT-s-OFDM.
See ["PTRS Config \(Transform Precoding\) ..."](#) on page 89.

"Off" Disables the DFT-s-OFDM precoding operation.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:TPSTate on page 248

Max Rank

Sets the higher layer parameter `maxRank` that defines the rank of the precoding matrix, as defined in [TS 38.212](#).

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:MRANK on page 248

TxConfig

Sets the higher layer parameter `txConfig`. It defines if codebook-based or non-codebook-based transmission is used and thus defines how and which precoding matrix is used, as defined in [TS 38.212](#).

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:TXConfig on page 248

Codebook Subset

If codebook-based transmission ("TxConfig > Codebook") is used, this parameter sets the higher layer parameter `codebookSubset`, as defined in [TS 38.212](#).

"Fully-and-partial-and-non-coherent"

`codebookSubset = fullyAndPartialAndNonCoherent`

"Partial-non-coherent"

`codebookSubset = partialAndNonCoherent`

"Non-coherent" `codebookSubset= nonCoherent`

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:CBSubset on page 249

Use PUSCH Scrambling ID

Defines the value of the variable n_{ID} used for the initialization of the generator of the PUSCH scrambling sequence.

"Off" $n_{ID} = N_{ID}^{cell}$ (i.e. the physical layer cell identity)

"On" n_{ID} is the values set with the parameter [Data Scrambling ID](#).

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:SCRambling:STAtE on page 250

Data Scrambling ID

If [Use PUSCH Scrambling ID](#) > "On", sets the identifier `dataScramblingIdentityPDSCH` used to initialize data scrambling (`c_init`) for PUSCH.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DSID on page 250
```

MCS Table

Sets the MCS table and thus together with the parameter "[MCS Table Transform Precoding](#)" on page 92 defines the used modulation scheme.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:MCSTable on page 249
```

MCS Table Transform Precoding

Together with the selected "MCS Table", this parameter defines the used modulation scheme.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:MTTPrecoding on page 249
```

5.3.5.5 PUSCH UCI Settings

Access:

- Select "User/BWP Settings > UL BWP Config > PUSCH UCI".

5G New Radio A: Users/BWP Settings			
General	Properties	UL BWPs	UL BWP Config
		User 0 / Cell 0	User 0 / Cell 0 / BWP 0
User:	0	Carrier: Cell	0
BWP	0		
State	☒	Mode	UCI+UL-SCH
Scaling Alpha	1.0		
HARQ ACK			
I_HARQ_offset,0	0	I_HARQ_offset,1	0
I_HARQ_offset,2	0		
CSI			
I_CSI1_offset,0	0	I_CSI1_offset,1	0
I_CSI2_offset,0	0	I_CSI2_offset,1	0

Here you can define the PUSCH uplink control information (UCI) parameters, such as the settings of the uplink shared channel (UL-SCH), HARQ control information, and the channel state information (CSI).

State.....	93
Mode.....	93
Scaling Alpha.....	93
I_CSI1_Offset,x, I_CSI2_Offset,x.....	93
I_HARQ_Offset,x.....	94

State

Enables the transmission and the configuration of the uplink control information (UCI).

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:STATE on page 252

Mode

Defines the information transmitted on the PUSCH.

"UCI+UL-SCH" Control information and data are multiplexed into the PUSCH.

"UCI only" Only uplink control information is transmitted on PUSCH.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:MODE on page 252

Scaling Alpha

Sets the higher-layer parameter *scaling*. It defines the value of the parameter α required for the calculation of the number of coded modulation symbols per layer for HARQ-ACK and CSI transmissions.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:ALPHA on page 252

I_CSI1_Offset,x, I_CSI2_Offset,x

According to TS 38.213, the number of resources for multiplexing the CSI information in a PUSCH is defined by the CSI part 1 and CSI part 2 information offsets $\beta_{\text{offset}}^{\text{CSI-1}}$ and $\beta_{\text{offset}}^{\text{CSI-2}}$.

The information offsets are defined by the indexes $I_{\text{offset},x}^{\text{CSI-1}}$ and $I_{\text{offset},x}^{\text{CSI-2}}$, where :

- $x = 0, 1$ and
- $I_{\text{offset},0}^{\text{CSI-1}}$ and $I_{\text{offset},0}^{\text{CSI-2}}$ is used if up to 11 CSI bits are multiplexed
- $I_{\text{offset},1}^{\text{CSI-1}}$ and $I_{\text{offset},2}^{\text{CSI-2}}$ is used for more than 11 CSI bits

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF10 on page 253

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF11 on page 253

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF20 on page 253

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF21 on page 253

I_HARQ_Offset,x

According to TS 38.213, the number of resources for multiplexing the HARQ-ACK information in a PUSCH is defined by the information offsets $\beta_{\text{offset},x}^{\text{HARQ-ACK}}$.

The information offsets are defined by the indexes $I_{\text{offset},x}^{\text{HARQ-ACK}}$, where :

- $x = 0, 1, 2$ and
- $I_{\text{offset},0}^{\text{HARQ-ACK}}$ is used if up to 2 HARQ-ACK information bits are multiplexed
- $I_{\text{offset},1}^{\text{HARQ-ACK}}$ is used for multiplexing of 2 to 11 HARQ-ACK information bits
- $I_{\text{offset},2}^{\text{HARQ-ACK}}$ is used if more than 11 HARQ-ACK information bits are multiplexed

Remote command:

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:HARQ:OFF0 on page 253

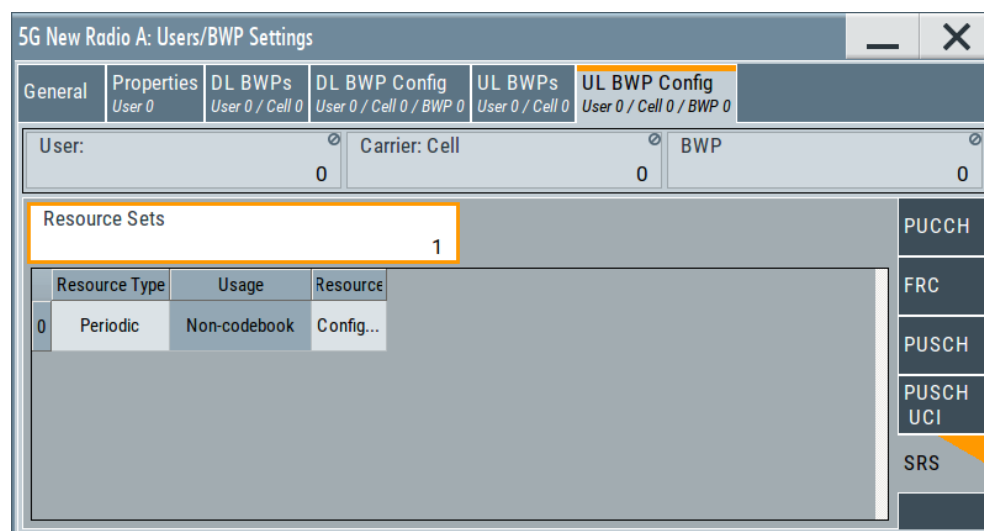
[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:HARQ:OFF1 on page 253

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:HARQ:OFF2 on page 253

5.3.5.6 SRS Settings

Access:

1. Select "User/BWP Settings > UL BWP Config > SRS".



2. Select "User/BWP Settings > UL BWP Config > SRS > Resource Set = 1".
3. Select "Resource > Config".

5G New Radio A (U0/C0/B0/R0) SRS Resource Set Settings																	
Resources																	
0	10 slots	1	4	1	0	0	0	0	0	1	2	0	Config...	1	0	0.00	Config...

4. To observe the SRS allocations on the time plan, proceed as follows:

- Select "5G NR > General > Link Direction > Uplink".
- Select "Scheduling > PUSCH > State > Off".
- Select "General > Time Plan".

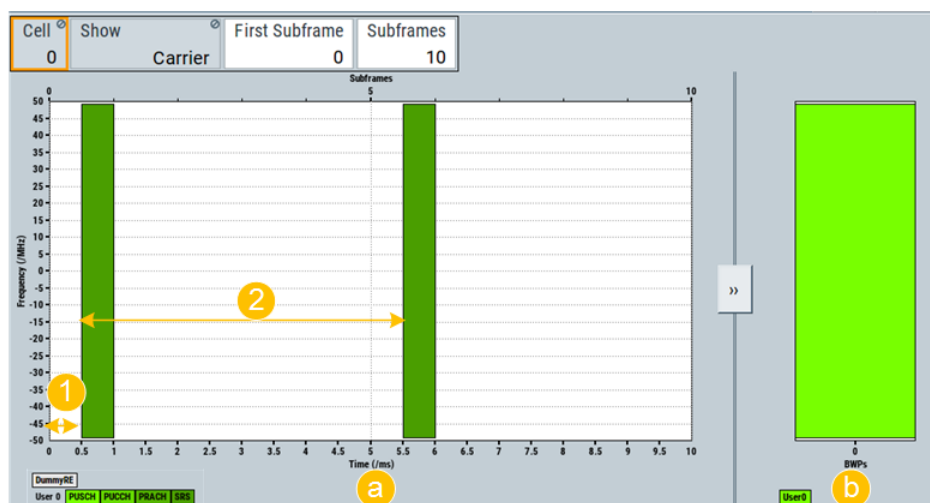


Figure 5-2: SRS representation on the Time Plan

- a = SRS (PUSCH transmission is disabled)
- b = Bandwidth part (273 RBs)
- 1 = Offset = 1 slot
- 2 = Periodicity = 10 slots

The sounding reference signal (SRS) is an UL reference signal transmitted by the UE. The BS uses the received SRS to estimate the UL channel quality for the particular user.

The SRS structure is defined in [TS 38.211](#), [TS 38.214](#) and [TS 38.331](#).

SRS structure

The SRS 5G NR structure is similar to the SRS structure used in LTE. In 5G NR, SRS is transmitted within the BWP, based on a flexible scheduling and using only portions of the overall channel bandwidth. Among other, the SRS supports the frequency hopping and transmission comb techniques known from LTE.

In the **time domain**, the SRS resource is defined as:

- 1, 2 or 4 consecutive OFDM symbols (**No. Sym.** $N_{\text{sym}}^{\text{SRS}}$)
- A starting position **Start Pos.** l_0 (i.e. the starting OFDM symbol), calculated as follows:

$$l_0 = N_{\text{symb}}^{\text{slot}} - 1 - l_{\text{offset}}$$

Where $l_{\text{offset}} = \{0 \text{ to } 5\}$ is an offset relative to the end of a slot.

In the **frequency domain**, its **starting position** is defined by the parameter k_0 and the density indication is given by the parameter **frequency shift**.

The SRS can be mapped to 1, 2 or 4 antenna ports (**No. Ports** $N_{\text{ap}}^{\text{SRS}}$) and its transmit **Power** is controlled, too.

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State

Enables the SRS transmission.

Use this function to suppress the SRS indication on the "Time Plan".

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:STATE on page 255

Resource Sets

Sets the number of SRS sets and thus defines the number of table rows in SRS resource set table.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:NRSets on page 255

SRS resource sets table

The SRS resource sets are defined in table form, one resource set per table row, where the number of rows depends on the selected number of "Resource Sets".

Resource Type ← SRS resource sets table

Indicates that periodic SRS transmission is configured.

Periodic SRS means that the SRS occurs at regular time interval. "Aperiodic" and "Semi-Persistent" SRS will be supported in a future firmware version.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RSType on page 255

Usage ← SRS resource sets table

Indicates whether a codebook based or a non-codebook based transmission is used.

The value is set automatically and resembles the transmission method selected for PUSCH, see "UL BWP Config > PUSCH" > TxConfig.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:USAGe on page 255

Resource > Config ← SRS resource sets table

Opens a dialog where you configure the SRS resources for the selected SRS resource set.

See "SRS Resource Set Settings" on page 97.

SRS Resource Set Settings

Access: "User/BWP Settings > UL BWP Config > SRS > Resource Set = 1" and "Resource > Config".

Resources ← SRS Resource Set Settings

Sets the number of SRS resources in the SRS resource set.

Each SRS resource is configured in an individual row in the SRS resource table.

The maximum number of SRS resources depends on whether TxConfig = "Codebook" or not.

Remote command:

[:SOURce<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:NRESources on page 256

Periodicity ← SRS Resource Set Settings

If **Usage** = "Periodic", sets the SRS repetition factor T_{SRS} . The periodicity is a time interval defined as number of slots that indicates how often (i.e. after how many slots) the SRS is transmitted.

As smaller the periodicity value as frequent is the SRS transmission. To enable the SRS for example in each slot, set "Periodicity = 1 slot".

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:PER on page 256
```

Offset ← SRS Resource Set Settings

If **Usage** = "Periodic" and **Periodicity** $\geq$ "2 slots", sets time position of first SRS allocation within an SRS periodicity, given as an offset in number of slots (T_{offset}) from the start of the frame.

Logically, the maximum offset value is calculated as SRS **Periodicity** - 1.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:OFFSet on page 256
```

Start Pos. ← SRS Resource Set Settings

Sets the starting position of the SRS allocation l_0 within a slot. The value gives the OFDM symbol number counted backwards from the end of a slot, see "SRS structure" on page 95.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:SPOS on page 256
```

No. Sym. ← SRS Resource Set Settings

Sets the number of consecutive OFDM symbols ($N_{\text{sym}}^{\text{SRS}}$) the SRS resource spans.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:SYMNumber on page 257
```

Freq. Pos. ← SRS Resource Set Settings

Sets the parameter k_0 that defines the starting position of the SRS allocation in the frequency domain.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:FPOS on page 257
```

Freq. Shift ← SRS Resource Set Settings

The frequency shift n_{shift} influence the SRS transmission density in the frequency domain.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:FQShift on page 257
```

B SRS ← SRS Resource Set Settings

Sets the filed b-SRS contained in the higher-layer parameter `freqHopping` and used to define the length of the SRS sequence, see TS 38.211, Table 6.4.1.4.3-1.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:BSRS on page 258
```

C SRS ← SRS Resource Set Settings

Sets the filed c-SRS contained in the higher-layer parameter `freqHopping` and used to define the length of the SRS sequence, see TS 38.211, Table 6.4.1.4.3-1.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:CSRS on page 258
```

B Hop ← SRS Resource Set Settings

Sets the parameter b_{Hop} that defines the frequency hopping of the SRS.

Set "B Hop" > "B SRS" to disable frequency hopping ("Rep. Factor" = 1).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:BHOP on page 258
```

Rep. Factor ← SRS Resource Set Settings

If **B Hop** < **B SRS**, defines how many times the SRS symbols are repeated.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:REFactor on page 258
```

Trans. Comb ← SRS Resource Set Settings

Transmission comb (k_{TC}) is a method for interleaving SRS symbols by using alternating subcarriers (only the even or odd subcarriers).

The value of k_{TC} defines the maximum number of cyclic shifts $n_{\text{SRS}}^{\text{CS}}$ (**Cyclic Shift**).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:TRTComb on page 259
```

Comb Offset ← SRS Resource Set Settings

Sets the parameter transmission comb offset.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:COFFset on page 259
```

Sequence Settings > Config ← SRS Resource Set Settings

Access: "User/BWP Settings > UL BWP Config > SRS > Resource Set = 1", "Resource > Config" and select "Sequence > Config".

5G New Radio A (U0/C0/B0/RS0/R0) SRS Sequence Settings	
Cyclic Shift	1
Sequence ID	0
Group or Sequence Hopping	Neither

Cyclic Shift ← Sequence Settings > Config ← SRS Resource Set Settings

Sets the number of cyclic shifts $n_{\text{SRS}}^{\text{CS}}$, required for the SRS sequence generation according to TS 38.211.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:SEQ:CYCShift` on page 259

Sequence ID ← Sequence Settings > Config ← SRS Resource Set Settings

Sets the higher-layer parameter `sequenceId` ($n_{\text{ID}}^{\text{SRS}}$), required for the SRS sequence generation according to TS 38.211.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:SEQ:ID` on page 259

Group or Sequence Hopping ← Sequence Settings > Config ← SRS Resource Set Settings

Sets the higher-layer parameter `groupOrSequenceHopping` that defines the sequence group, required for the SRS sequence generation according to TS 38.211.

In this firmware version, group and sequence hopping are not supported.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:SEQ:HOPping` on page 260

No. Ports ← SRS Resource Set Settings

If `TxConfig` = "Codebook", sets how many antenna ports (AP) are used for every SRS transmission. The same number of antenna ports is used for all SRS resources.

If "Non-Codebook" transmission is used, the value of the parameter "No. Ports" is set automatically.

Remote command:

`[ :SOURCE<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:NAPort` on page 260

PTRS Port Idx ← SRS Resource Set Settings

Sets the PTRS antenna port index. The maximum number of configured PTRS ports is set with the parameter [Max Number of Ports](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:PTRS on page 261
```

Power ← SRS Resource Set Settings

Sets the SRS power.

The value is set relative to the PUSCH power.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:POWER on page 260
```

Antenna Ports > Config ← SRS Resource Set Settings

Access: "User/BWP Settings > UL BWP Config > SRS > Resource Set = 1", "Resource > Config" and select "Antenna Ports > Config".

	AP 1000	
	Real	Imag
BB A	1.000+j0.000	

For details on the antenna ports mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Mapping Coordinates ← Antenna Ports > Config ← SRS Resource Set Settings

Switches representation between the "Cartesian" (Real/Imag) and "Cylindrical" (Magn./Phase) coordinates.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:APMap:MAPCoord on page 261
```

Mapping table ← Antenna Ports > Config ← SRS Resource Set Settings

Defines the mapping of the antenna ports (AP) to the physical antennas.

Set the number of antenna ports (AP) with the parameter [No. Ports](#).

For details on the antenna ports mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Remote command:

"Mapping Coordinates = Cartesian":

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:REAL on page 261
```

```
[ :SOURce<hw> ] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:IMAGinary on page 262
```

"Mapping Coordinates = Cylindrical":

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:MAGNitude

on page 263

[:SOURCE<hw>] :BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:PHASe

on page 263

5.4 Scheduling Settings

Access:

- Select "Baseband > 5G NR > Scheduling".

5G New Radio A: Scheduling Settings													
Cell		Subframe											
0		0											
	Content	No. Alloc	SC Spacing / CP	Slot	Map Type	No. Sym.	Sym. Offset	No. RBs	RB Offset	Settings	Power /dB	State	Repetition
▼	Common												
	SS/PBCH		30 kHz NCP	0		4	4	20	-	Config...	0.00	On	
	SS/PBCH		30 kHz NCP	0		4	8	20	-	Config...	0.00	On	
▼	User 0, BWP 0	2	Config...										
	CORESET		30 kHz NCP	0		1	0	270	0	Config...	0.00	On	Slot
	PDSCH		30 kHz NCP	0	A	13	1	273	0	Config...	0.00	On	Slot
	CORESET		30 kHz NCP	1		1	0	270	0	Info...	0.00		of SF 0...
	PDSCH		30 kHz NCP	1		13	1	273	0	Info...	0.00		of SF 0...

This dialog displays an overview of distribution of the configured allocations in the time domain.

Because the slot is the basis time unit in 5G NR and the basis granularity in the numerologies, the **5G NR scheduling is performed on a per slot basis**. In comparison, the LTE scheduling is subframe-based. However, the scheduling dialog still shows the configuration of a complete subframe, as the subframe (besides the frame) is the common time boundary between the different numerologies.

Scheduling table: Understanding the displayed information

The allocation table lists all BWPs of all users configured for the corresponding link direction. Some of the allocations, in the BWPs or in the common group, are automatically configured. You can add additional allocations in the user-specific groups.

Allocations belonging to the same user and BWP are grouped in "**User x BWP x**" groups. Select the group name to unfold the PDSCH/CORESET or PUSCH/PUCCH allocations. You find the **non-user-specific** allocations in the dedicated "**Common**" group.

The allocations can be divided into two types: configurable allocations and read-only allocations.

- *Configurable allocations*
You can configure these allocations in the first 100 subframes. The configurable allocations are always the first allocations in each BWP per user group.
To append or remove allocations of this type, set the number of allocations ([No. of Allocations](#)) for the corresponding group.
Once configured, these allocations are repeated as defined in the [Repetition/Repetition Type](#) field.
- *Read-only allocations*
These allocations are configured automatically, based on: scheduling mode, DCI, repetition of an allocation configured in previous/same subframe, ongoing allocations, aggregated PDSCH, or because they are configured elsewhere, like for example SS/PBCH, SRS, CSI-RS, dummy RE.
You can observe the allocations in all available subframes, where the number of subframes depends on the ARB sequence length.

Allocating channels in the time and frequency domains

The following settings define the allocation position:

- [SC Spacing/CP](#) (defines the available resources)
- Time domain: [Slot](#), [No. Sym.](#) and [Sym. Offset](#)
- Frequency domain: [No. RB](#) and [Offs. RB](#)
- [Repetition/Repetition Type](#)

The allocation table shows information on the selected channel or signal. Additionally, for each "User x BWP x" summary information is also displayed, if this information affects the channel allocation within the BWP, e.g. [SC Spacing/CP](#) or [No. RB](#).

If allocations are overlapping, a conflict is indicated in the most left column of the allocation table.

Settings:

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Expand/Collapse	104
Conflict	104
Content	104
No. of Allocations	105
SC Spacing/CP	105
Slot	105
Mapping Type	106
Format	106
No. Sym.	106
Sym. Offset	106
No. RB	107
Offs. RB	107
Transform Precoding	107

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Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at CELL<ch>

Subframe

Sets the subframe to be configured.

The maximum number of available subframes N_{SF} is calculated as follows:

$$N_{SF} = 10 * \text{"ARB Sequence Length"} - 1$$

Remote command:

via suffix at SUBF<st0>

Next/Prev

Navigates through the subframes.

Remote command:

n.a.

Expand/Collapse

Shows/hides the group of settings.

Remote command:

n.a.

Conflict

A conflict is indicated if two allocations overlap.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CONFLICT? on page 270

Content

Sets or indicates the allocated channel or signal.

- Configurable allocations
 - DL: CORESET, PDSCH, CSI-RS or LTE-CRS
 If **PDSCH Scheduling** = "Auto/DCI", the "Content" = "CORESET".
 (PDSCH allocations are configured automatically)
 If "Users/BWP > DL BWP Config > ZP/NZP CSI-RS > State = On", CSI-RS
 allocations are configured automatically.

If "Node > LTE-CRS Coexistence" > **State** = "On", LTE-CRS allocations are configured automatically.

- UL: PUSCH, PUCCH, PRACH
- Read-only allocations
Same as for the configurable allocations and additionally: Dummy REs, SS/PBCH, SRS

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CONTent on page 269

No. of Allocations

In the "User x BWP x" groups, sets the number of configurable allocations per "User x BWP x" group.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:NALLoc on page 270

SC Spacing/CP

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP), where the available values depend on the "Deployment".

See [Table 3-2](#).

The provided values depend on the channel type.

The SCS of the selected BWP is also displayed, because the channel allocation in the frequency domain depends on this value.

Remote command:

PRACH: [:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:
USER<dir0>:BWPart<gr0>:ALLoc<user0>:PRACH:SCSPacing on page 317

Slot

Sets the start slot for the selected allocation, where the available slots depend on the channel-specific [SC Spacing/CP](#) as follows:

Slot	SCS / CP (as selected for the specific channel)
0	15 kHz NCP
0 to 1	30 kHz NCP
0 to 3	60 kHz NCP 60 kHz ECP
0 to 7	120 kHz NCP
0 to 15	240 kHz NCP

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:SLOT on page 270

Mapping Type

Defines to which symbols of the PDSCH/PUSCH allocation the demodulation reference signals (DMRS) are mapped.

"A" Location relative to the start of the slot.

"B" Location relative to the start of the PDSCH resources.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:MAPType on page 271

Format

In uplink, selects the PUCCH format.

3GPP defines several PUCCH formats. The PUCCH format affects several PUCCH characteristics, like the mapping to physical resource blocks.

You can select a different PUCCH format for each PUCCH allocation. Depending on the format, you can access different enhanced settings for the PUCCH.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:FMT on page 271

No. Sym.

Indicates the time duration of the allocation inside the slot.

The available values depend on the CP and the [Mapping Type](#), see [Table 5-2](#).

Table 5-2: No. Sym. (L)

Mapping Type	Normal cyclic prefix	Extended cyclic prefix
Mapping type A	3, 14	3, 12
Mapping type B	2, 4, 7	2, 4, 6

The parameters "No. Sym." and "Sym. Offset" are interdependent; their sum must not exceed the maximum of 14 or 12 symbols, for normal and extended CP respectively.

For SRS allocations, the value is set with "Users/BWP > UL BWP Config > SRS > SRS Resource Table > Resource > Config" > [No. Sym.](#).

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:SYMNumber on page 271

Sym. Offset

Sets the allocation's start (OFDM symbol) within the slot.

The available values depend on the channel-specific [SC Spacing/CP](#) and the [Mapping Type](#), see [Table 5-3](#)

Table 5-3: Sym. Offset

Mapping Type	Normal cyclic prefix	Extended cyclic prefix
Mapping type A	0, 1, 2, 3	0, 1, 2, 3
Mapping type B	0, 12	0, 10

For SRS allocations, the value is set with "Users/BWP > UL BWP Config > SRS > SRS Resource Table > Resource > Config" > [Offset](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:SYMoffset on page 272
```

No. RB

Sets the bandwidth (frequency size) of selected allocation in terms of resource blocks (RB).

The number of resource blocks generally depends on the SCS of the corresponding BWP, set with the parameter "Users/BWP > DL/UL BWPs" > [SC Spacing/CP](#). Observe the number of *total* available RBs in the selected "User x, BWP x".

The number of *RB per channel* also depends on the channel type as follows:

- CORESET allocations span always a multiple of 6 resource blocks. Thus, only values that are multiple of 6 are allowed for these allocations.
- The number of resource blocks that are available for PUSCH depends on whether the transform precoding is enabled or not (that is if DFT-s-OFDM is applied), see ["Transform Precoding"](#) on page 107.
- The number of resource blocks of the PRACH allocation are set automatically, depending on the selected [SC Spacing/CP](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:RBNumber on page 272
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:RBNumber? on page 273
```

Offs. RB

Adds a frequency offset and sets the start resource block of the selected allocation.

The value range depends on the selected channel type and the parameters [SC Spacing/CP](#) and [No. RB](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:RBOffset on page 273
```

Transform Precoding

In UL, you can define which access method is used, the optional DFT-s-OFDM or the CP-OFDM.

See also [Chapter 3.3, "Multiple Access Schemes"](#), on page 13.

"On" Enables the precoding operation and thus enables the DFT-s-OFDM.

"Off" Disables the precoding operation and thus enables the CP-OFDM.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:TPState on page 271
```

Settings

Opens dialogs with additional setting or information.

Config ← Settings

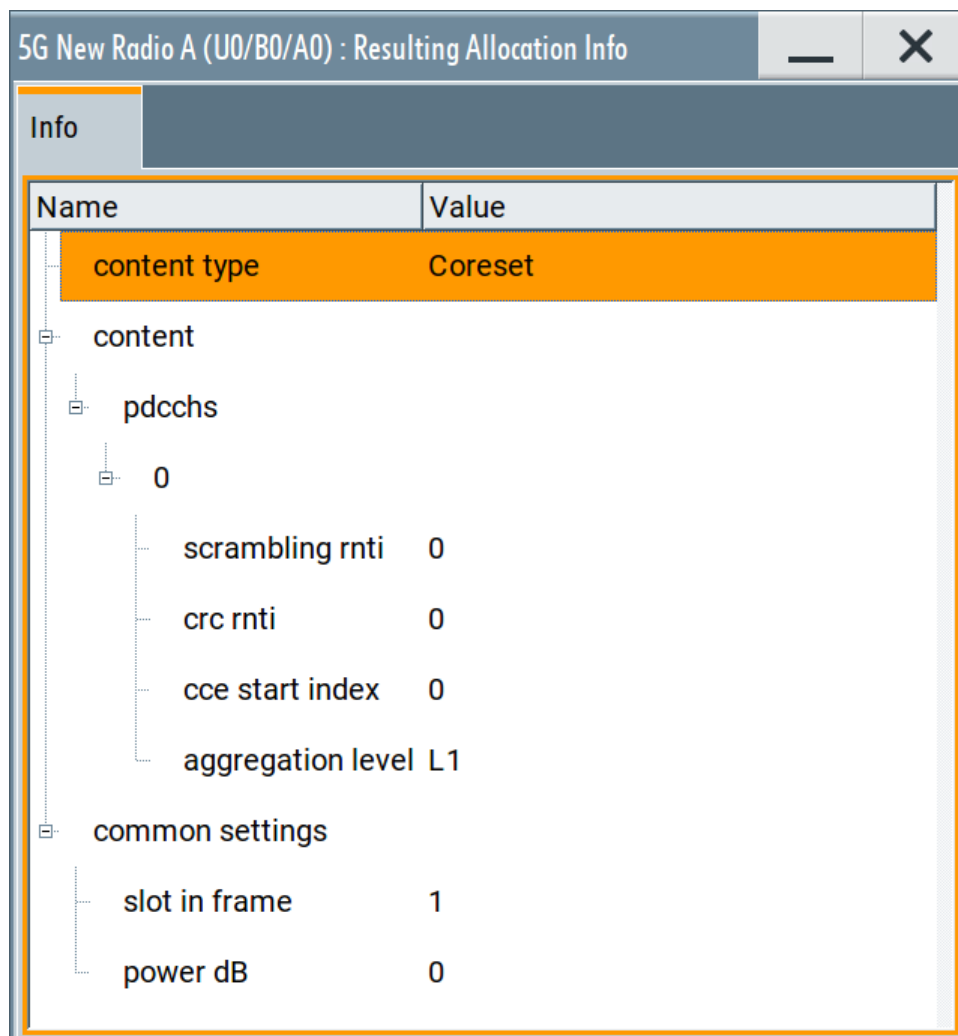
Opens the dialogs for configuration of the corresponding settings.

See:

- [Chapter 5.2.5, "SS/PBCH Settings"](#), on page 37
- [Chapter 5.2.6, "Dummy REs Settings"](#), on page 43
- [Chapter 5.3.3, "DL/UL BWPs Settings"](#), on page 53
- [Chapter 5.5, "PDSCH/PUSCH Settings"](#), on page 111
- [Chapter 5.6, "CORESET Settings"](#), on page 128
- [Chapter 5.7, "PUCCH Settings"](#), on page 147

Info ← Settings

Opens the "Resulting Allocation Info" dialog with details on the automatically allocated and repeated resources, like, for example, PDSCH created from DCIs or repeated PDSCH or CORESET.



Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:REsulting:ALLoc<user0>:INFO? on page 275
```

Power

Sets the power for the selected allocation relative to the power of the other allocations.

Allocations with "Power = 0 dB" use the same power level.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:POWer on page 273
```

State

Sets the allocation to active or inactive state.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:STATe on page 273
```

Repetition/Repetition Type

- Configurable allocations
Displays if and how often the allocation is repeated.
Select the field to open the [Repetition Settings](#) dialog where you can reconfigure the repetition.
- Read-only allocations
Indicates the read-only reason, e.g. ongoing allocation, from SF#, etc.
Select this indication to access the setting that causes the configuration.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:REPetitions on page 274
```

Repetition Settings

Access: "Baseband > 5G NR > Scheduling > Repetition".

Repetition	
Repetition Type	Custom
Period	10 Slots
Duration	1 Slots

Repetition Type ← Repetition Settings

Sets if and how the allocation is repeated, e.g. per slot, per subframe, on a per frame basis or at a user-defined period.

"Off" The allocation is transmitted once.

"Slot", "Subframe", "Frame"

Available for "Content = PDSCH/PUSCH/PUCCH/CORESET".
The allocation is transmitted every slot, subframe, or frame.

"Custom" User-defined repetition pattern, set as [Duration](#) and [Period](#).
Use the "Time Plan" to observe the current scheduling.

Example:

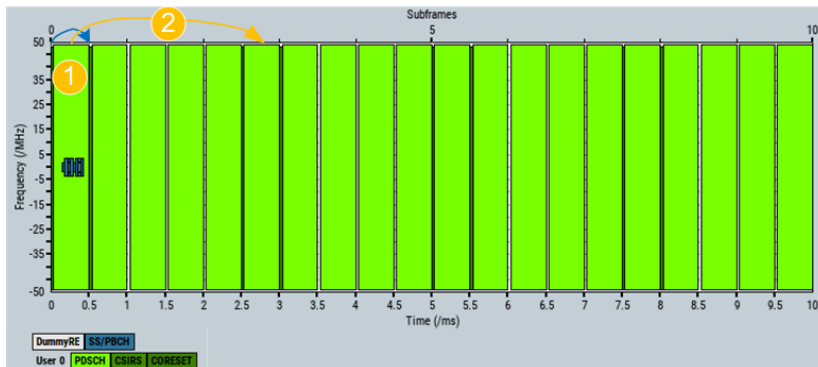


Figure 5-3: Repetition Type = Custom: Representation on the Time Plan

1 = [Duration](#) = "2 Slots"

2 = [Period](#) = "5 Slots"

Period ← Repetition Settings

Sets after how many slots the group of configurable allocation and its repetitions is repeated, see [User-defined repetition](#).

The maximum repartition period is calculated as follows:

$\text{RepPeriod}_{\max} = \text{\#SlotsPerFrame} * \text{"SequenceLength"}$, where

- **#SlotsPerFrame**: is the number of available slots within a frame and depends on the used subcarrier spacing
- **"SequenceLength"**: is the ARB sequence length as set with the parameter "Output/Power > Output" > [Sequence Length](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PERiod on page 274
```

Duration ← Repetition Settings

Sets in how many consecutive slots within a frame the allocation is repeated.

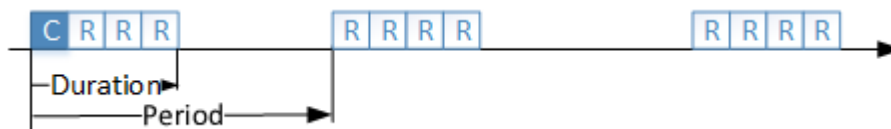


Figure 5-4: User-defined repetition

C = Configurable allocation

R = Repetition

Duration = Number of consecutive slots in that the allocation is transmitted

Period = Sets after how many slots the group of configurable allocation and its repetitions is repeated

Remote command:

[:SOURCE<hw>] :BB:NR5G:SchEd:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:DURation on page 274

5.5 PDSCH/PUSCH Settings

Access:

1. PDSCH settings:

- Select "5G New Radio > General > Link Direction > Downlink".
- Select "5G New Radio > Scheduling".
- Select "User x BWP x > Content > PDSCH".
- Select "PDSCH > Settings > Config".

5G New Radio A (U0/B0/A1) PDSCH Settings	
General	TxScheme DMRS PTRS Manual Antenna Ports Info
PDSCH Type	DCI Format 1_1
Number of Codewords	2
Scheduled by CORESET 0	☒
Rate Match Pattern Group	None
Transport Block 1	
Modulation	16QAM
Number of Physical Bits	272 160
Transport Block 2	
Modulation	64QAM
Number of Physical Bits	612 360

2. PUSCH settings:

- Select "5G New Radio > General > Link Direction > Uplink".
- Select "5G New Radio > Scheduling".
- Select "User x BWP x > Content > PUSCH".
- Select "PUSCH > Settings > Config".

5G New Radio A (U0/B0/A0) PUSCH Settings	
General	DMRS TxScheme UCI
PUSCH Type	DCI Format 0_1
Modulation	QPSK
Number of Physical Bits	44 226

This dialog comprises the settings of the physical downlink shared channel (PDSCH) or the physical uplink shared channel (PUSCH).

The PDSCH carries the general user data and is therefore the most prominent channel in a radio frame; it occupies the most resources.

Settings:

• General Settings	112
• TxScheme Settings	114
• DMRS Settings	116
• Channel Coding Settings	117
• UCI Settings	119
• PTRS Settings	121
• PTRS Settings with Enabled Transform Precoding	123
• Antenna Ports	125
• Info	127

5.5.1 General Settings

Access:

1. Select "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config".
2. Select "General".

5G New Radio A (U0/B0/A1) PDSCH Settings	
General	TxScheme DMRS PTRS Manual Antenna Ports Info
PDSCH Type	DCI Format 1_1
Number of Codewords	2
Scheduled by CORESET 0	☒
Rate Match Pattern Group	None
Transport Block 1	
Modulation	16QAM
Number of Physical Bits	272 160
Transport Block 2	
Modulation	64QAM
Number of Physical Bits	612 360

Settings:

PDSCH Type/PUSCH Type	113
Number of Codewords	113
Scheduled by CORESET 0	113
Rate Match Pattern Group	113
Modulation	113
Phys. Bits	114

PDSCH Type/PUSCH Type

Defines the PDSCH/PUSCH type by selecting the DCI format by that the PDSCH/PUSCH content is defined.

The available settings depend on the current configuration. To select, for example, "PDSCH Type = DCI Format 1_0", select "Users/BWP > DL BWP Config > PDSCH > Additional Position Index = 2" and "Scheduling > PDSCH > Map Type = A".

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:TYPE on page 279
```

Number of Codewords

For PDSCH, sets if one or two codewords are used.

The combination of number of codewords and [Number of Layers](#) determines the layer mapping for the selected precoding scheme.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh:NCW on page 279
```

Scheduled by CORESET 0

For PDSCH allocations, defines if the PDSCH settings are scheduled by CORESET 0 or not.

You can enable several PDSCH allocations, also a combination of PDSCH scheduled by CORESET 0 and such that are not. To enable the CORESET 0 itself, set the parameter [CORESET ID](#) = "0".

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh:SBCZero:STAtE on page 279
```

Rate Match Pattern Group

For PDSCH allocations, selects one of the rate match pattern groups, configured in the "Users/BWP Settings > DL BWP Config > Rate Match" dialog.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh:PATGrp on page 279
```

Modulation

Selects the modulation scheme.

- PDSCH: QPSK, 16QAM, 64QAM, 256QAM
- PUSCH: QPSK, 16QAM, 64QAM, 256QAM, and if [Transform Precoding](#) > "Off" also pi/2-BPSK.

See also [Table 3-1](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:MOD on page 280
```

Phys. Bits

Indicates the size of the selected allocation in bits.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>] :PHYSbits? on page 280

5.5.2 TxScheme Settings

Access:

1. Select "5G New Radio > General > Link Direction" > **"Downlink"**.
2. Select "Scheduling > User x BWP x > Content > PDSCH > Settings > Config".
3. Select "TxScheme".

5G New Radio A (U0/B0/A1) PDSCH Settings	
General	TxScheme
DMRS	PTRS Manual
Antenna Ports	
CDM Groups w/o Data	2
Number of Layers	4
Interleaved VRB-to-PRB	☐

4. Select "5G New Radio > General > Link Direction" > **"Uplink"**.
5. Select "Scheduling > User x BWP x > Content > PUSCH > Settings > Config".
6. Select "General > PUSCH Type = DCI Format 0_1".
7. Select "TxScheme".

5G New Radio A (U0/B0/A0) PUSCH Settings	
General	TxScheme
DMRS	PTRS Auto
Channel Coding	Antenna Ports
CDM Groups w/o Data	2
Number of Layers	4
SRS Resource Indicator	0

Settings:

CDM Groups w/o Data.....	115
Number of Layers.....	115
TPMI.....	115
SRS Resource Indicator.....	115
Interleaved VRB-to-PRB.....	115

CDM Groups w/o Data

Selects the number of CDM groups that are reserved and contain no data. They are therefore not used by the PDSCH/PUSCH for data transmission. In the resource grid, the resource elements for CDM (between PDSCH/PUSCH DMRS resource elements) remain empty.

The value is required to select the number of DMRS ports and hence the number of antenna ports. For details, see [TS 38.212](#).

If "User/BWPs > UL BWP Config > PUSCH" > [Transform Precoding](#) = "On", this parameter is set automatically.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSch|PUSCh:TXSCHEME:CDMData on page 282
```

Number of Layers

Sets the number of precoding layers.

The combination of number of codewords and number of layers determines the layer mapping. Each layer is transmitted on a separate [antenna port](#). The minimum number of layers corresponds to the selected [Number of Codewords](#).

If "User/BWPs > UL BWP Config > PUSCH" > [TxConfig](#) = "Codebook", this parameter is set automatically.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSch|PUSCh:TXSCHEME:NLAYers on page 282
```

TPMI

Access: "User/BWPs > UL BWP Config > PUSCH" > [TxConfig](#) = "Codebook".

Sets the TPMI index, required to select the precoding matrix, as defined in [TS 38.211](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:TXSCHEME:TPMidx on page 283
```

SRS Resource Indicator

Access: "User/BWPs > UL BWP Config > SRS > Resource > Config" > [Resources](#) ≥ 1.

Selects the SRS resource to be used. SRS resources are defined in the SRS resource table and the Resource indicator refers to a table row number.

See "[SRS Resource Set Settings](#)" on page 97.

If only one SRS resource is defined, the "SRS Resource Indicator = 0".

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:TXSCHEME:TPMidx on page 283
```

Interleaved VRB-to-PRB

If "User/BWP Settings > DL BWP Config > PDSCH" > [VRB-to-PRB Interleaver](#) = "2 or 4", the VRB-to-PRB interleaving is enabled for all PDSCH allocation.

With this parameter you can disable the VRB-to-PRB interleaving for the particular PDSCH.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSch:TXSCHEME:INTervp on page 284

5.5.3 DMRS Settings

Access:

1. Select "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config".
2. Select "DMRS".

5G New Radio A (U0/B0/A1) PDSCH Settings				
General	TxScheme	DMRS	PTRS Manual	Antenna Ports
Sequence Generation		N_ID ^{Cell}	N_SCID	
			0	
Length		2	Slot Symbols	
Power		0.00 dB		

Settings:

Sequence Generation	116
N_SCID	117
Length	117
Slot Symbols	117
Power	117

Sequence Generation

The initialization sequence of the pseudo-random sequence generator is generated based among others depending on a parameter called scrambling ID ($N_{ID}^{n_SCID}$).

This parameter defines how the scrambling ID is derived:

"N_ID^{Cell}" Scrambling ID (i.e. $N_{ID}^{n_SCID}$) = cell ID (i.e. N_{ID}^{Cell})

"N_ID^{DMRS}" Scrambling ID (i.e. $N_{ID}^{n_SCID}$) = DMRS scrambling ID (i.e. N_{ID}^{DMRS})

See also "About PDSCH and DMRS sequence generation" on page 60.

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSch|PUSCh:DMRS:SEQGen on page 280

N_SCID

Sets which variable, N_{ID}^0 or N_{ID}^1 , is used by the generation of the initialization sequence of the pseudo-random sequence generator. The value of the parameters N_{ID}^0 or N_{ID}^1 is set per DMRS type with the parameters "User/BWP > DL BWP > PDSCH" > 63.

See also ["About PDSCH and DMRS sequence generation"](#) on page 60.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:DMRS:NSID on page 281
```

Length

Defines whether single- or double-symbol DMRS is used.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:DMRS:LENGth on page 281
```

Slot Symbols

Indicates the number of DMRS slot symbols. The value is set automatically, depending on the selected "Length".

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:DMRS:SLTSymbols? on page 281
```

Power

Sets the power of the PDSCH/PUSCH, relative to the power of the other channels.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:DMRS:POWer on page 282
```

5.5.4 Channel Coding Settings

Access:

1. Select "5G New Radio > General > Users/BWPs > Properties > DSCH/USCH Channel Coding > On".
2. Select "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config".
3. Select "Channel Coding".

The dialog comprises the settings for the LDPC channel coding for PDSCH and PUSCH.

5G New Radio A (U0/B0/A1) PDSCH Settings	
General	TxScheme
Channel Coding	
Antenna Ports	
Transport Block 1	
I_MCS	0
Target Code Rate	0.117
Number of Physical Bits	36 960
Transport Block Size	21 528
Redundancy Version Index	0
Transport Block 2	
I_MCS	10
Target Code Rate	0.332
Number of Physical Bits	110 880
Transport Block Size	24
Redundancy Version Index	0

Channel coding parameters determine the code rate of the PDSCH/PUSCH, which is the ratio between transmitted bits and maximum possible bits in a subframe (or 1 ms transport block). Because the number of bits in a subframe is variable, the target code rate has to be derived from the modulation order in combination with an index I_{MCS} .

You can select the modulation order for the PDSCH/PUSCH (one of several tables, one for each modulation type) by setting the corresponding index value "I_MCS". The target code rates for modulation order and index are defined in [TS 38.214](#).

In addition to the modulation order and I_{MCS} , the number of transmitted bits depends on the "Redundancy Version Index". Depending on the redundancy version index, the PDSCH/PUSCH contains a different amount of parity bits for error detection: Index 0 adds 1 sequence of parity bits, index 1 adds two sequences of parity bits etc.

Settings:

I_MCS	118
Target Code Rate	119
Phys. Bits	119
Transport Block Size	119
Redundancy Version Index	119

I_MCS

Sets the modulation and coding scheme index.

The available values depend on the selected [Modulation](#).

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:IMCS
on page 288
```

Target Code Rate

Indicates the target code rate, set automatically depending on the selected modulation and coding scheme ("I_MSC").

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:TCRate?
on page 289
```

Phys. Bits

Indicates the size of the selected allocation in bits.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PHYSbits? on page 280
```

Transport Block Size

Indicates the resulting transport block size (TBS), calculated for the selected modulation and coding scheme ("I_MSC").

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:TBSize?
on page 289
```

Redundancy Version Index

Sets the redundancy version index.

Depending on the redundancy version index, the PDSCH/PUSCH contains a different amount of parity bits for error detection: Index 0 adds 1 sequence of parity bits, index 1 adds two sequences of parity bits etc.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:RVIndex
on page 289
```

5.5.5 UCI Settings

Access:

1. Select "Users/BWPs > Properties > USCH Channel Coding > On".
2. Select "Users/BWPs > UL BWP Config > PUSCH UCI > State > On".
3. Select "Scheduling > User x BWP x > Content > PUSCH > Settings > Config".

4. Select "UCI".

5G New Radio A (U0/B0/A0)PUSCH Settings		—	×		
General	TxScheme	DMRS	Channel Coding	UCI	Antenna Ports
Number of ACK Bits		1		ACK Pattern	
Number of CSI 1 Bits		1		CSI 1 Pattern	
Number of CSI 2 Bits		1		CSI 2 Pattern	
				0...	

Settings:

Number of ACK Bits.....	120
ACK Pattern.....	120
Number of CSI x Bits.....	120
CSI x Pattern.....	121

Number of ACK Bits

Sets the number of ACK bits and defines the length of the "ACK Pattern" field.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:ACK:BITS on page 290
```

ACK Pattern

Sets the ACK bits in pattern form, where the pattern length is set with the parameter "Number of ACK Bits".

A "1" indicates an ACK, a "0" - a NACK.

The pattern is read out once.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:ACK:PATtern on page 290
```

Number of CSI x Bits

Sets the number of CSI 1/CSI 2 bits and defines the length of the "CSI Pattern" fields.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI1:BITS on page 290
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI2:BITS on page 290
```


CSI x Pattern

Sets the CSI bits in pattern form, where the pattern length is set with the parameter "Number of CSI x Bits".

A "1" indicates an ACK, a "0" - a NACK.

The pattern is read out once.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI1:PATtern on page 290
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI2:PATtern on page 290
```

5.5.6 PTRS Settings

Access:

1. Select "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config".
2. Select "PTRS".

The screenshot shows the 'PTRS Manual' tab in the '5G New Radio A (U0/B0/A1) PDSCH Settings' dialog. The 'Mode' is set to 'Manual'. The 'State' is set to 'On' (indicated by a blue bar). The 'Time Density (L_PTRS)' is set to 'Every 4th OFDM Symbol'. The 'RE-offset' is set to '10'. The 'Frequency Density (K_PTRS)' is set to 'Every 4th RB'. The 'EPRE Ratio' is set to '1'.

The dialog comprises the settings of the phase-tracking reference signal (PTRS). The phase tracking reference signal (PTRS) is a UE-specific reference signal that is used to compensate for the phase noise of the oscillator. Phase-tracking reference signals can be transmitted only in the resource blocks used for the P_xSCH and if the transmission is enabled.

If the PTRS "State" is on, you can define its "Power" relative to the P_xSCH and its location in the resource grid.

Settings:

Mode.....	122
PTRS-DMRS Association	122
State.....	122
Time Density (L_PTRS).....	122
RE-offset.....	122
Frequency Density (K_PTRS).....	123

EPRE Ratio.....	123
Power.....	123
Max Number of Ports.....	123

Mode

Sets how the PTRS configuration is defined:

- "Auto" If **DSCH/USCH Channel Coding** > "On" and "PTRS" > **State** > "On", the PTRS configuration is derived from the current PDSCH DMRS settings.
- "Manual" You configure PTRS manually, with the settings provided in the "PxSCH Settings > PTRS" dialog.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:MODE on page 284
```

PTRS-DMRS Association

If "Scheduling > User x BWP x > Content = PUSCH", indicates the PTRS-DMRS association.

The association between UL PTRS ports and DMRS ports is signalled by PTRS-DMRS association field in DCI format 0_1.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:PTDMrs on page 285
```

State

Enables the transmission of phase-tracking reference signals.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:STATe on page 285
```

Time Density (L_PTRS)

Sets the higher-layer parameter `timeDensity`.

The "L_PTRS" defines distance between the PTRS in terms of OFDM symbols (transmission every 1, 2 or 4 OFDM symbols). If the subcarrier used by the PTRS also contains a DMRS, the distance can be larger.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:TMDen on page 285
```

RE-offset

Sets the parameter `resourceElementOffset` $k_{\text{ref}}^{\text{RE}}$ required to define the precoding and mapping to the physical resources.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:REOF on page 285
```

Frequency Density (K_PTRS)

Sets the higher-layer parameter `frequencyDensity`.

The "K_PTRS" and "RE-offset" define the location of the PTRS in the frequency domain. K_{PTRS} defines the distances between the PTRS in terms of subcarrier. You can also define an additional frequency offset for the PTRS relative to the first subcarrier.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSch:PTRS:FRQDen on page 286
```

EPRE Ratio

If "Scheduling > User x BWP x > Content = PDSCH", sets the higher layer parameter `epre-Ratio`. It is needed to define the ratio of PTRS EPRE to PDSCH EPRE per layer per RE for PTRS port.

The abbreviation EPRE stands for energy per resource element.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSch:PTRS:EPRE on page 286
```

Power

If "Scheduling > User x BWP x > Content = PUSCH", sets the parameter `ptrs-Power` and thus defines the PUSCH to PTRS power ratio per layer per resource element.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSch:PTRS:POWer on page 286
```

Max Number of Ports

If "Scheduling > User x BWP x > Content = PUSCH", sets the maximum number of configured PTRS ports, given by the higher level parameter `maxNrofPorts`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSch:PTRS:PORT on page 287
```

5.5.7 PTRS Settings with Enabled Transform Precoding

Access:

1. Select "Users/BWPs > Properties > USCH Channel Coding > On".
2. Select "General > Users/BWPs > UL BWP Config > PUSCH > Transform Precoding > On"
3. Select "Scheduling > User x BWP x > Content > PUSCH > Settings > Config".

4. Select "PTRS".

5G New Radio A (U0/B0/A0) PUSCH Settings

General TxScheme DMRS **PTRS Manual** Antenna Ports Info

Mode Manual

State ☒

PTRS Transform Precoding Settings

Time Density (L_PTRS) 1

Number of PTRS Groups 2 Samples Per PTRS Group 2

The dialog comprises the settings of the phase-tracking reference signal (PTRS) if transform precoding is enabled.

Settings:

Mode	124
State	124
Time Density (L_PTRS)	124
Number of PTRS Groups	125
Samples Per PTRS Group	125

Mode

Sets how the PTRS configuration is defined:

"Auto"	If DSCH/USCH Channel Coding > "On" and "PTRS (Transform Precoding)" > State > "On", the PTRS configuration is derived from the current PUSCH DMRS settings.
"Manual"	You configure PTRS manually, with the settings provided in the "PUSCH Settings > PTRS" dialog.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:MODE on page 287
```

State

Enables the transmission of phase-tracking reference signals.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:STAtE on page 287
```

Time Density (L_PTRS)

Sets the higher-layer parameter `timeDensity`.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:TMDensity on page 287
```

Number of PTRS Groups

Sets the number of PTRS groups according to TS 38.214, required to define the PTRS group pattern.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:NGRPs on page 288
```

Samples Per PTRS Group

Sets the number of samples per PTRS group according to TS 38.214, required to define the PTRS group pattern.

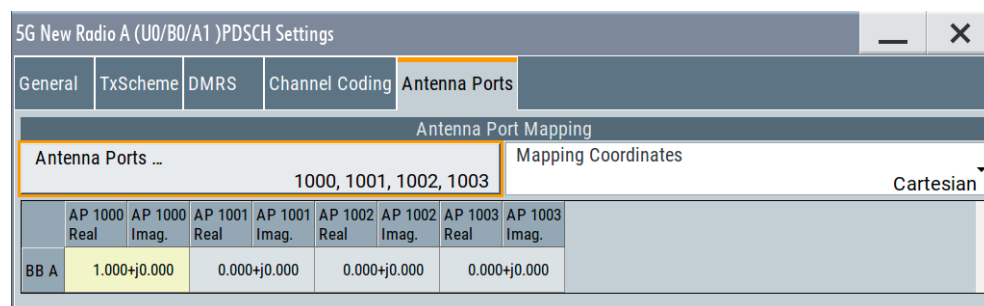
Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:SPPG on page 288
```

5.5.8 Antenna Ports

Access:

1. Select "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config".
2. Select "Antenna Ports".



With the provided settings you can map the antenna ports arbitrarily to the base-band outputs. The available antenna ports depend on the current configuration (see "TxScheme" > [Antenna Ports](#)).

For details on the antenna port mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Settings:

Antenna Ports	125
Mapping Coordinates	127
Mapping Table	127

Antenna Ports

Sets the antenna ports.

The values are calculated automatically, based on several parameters, like for example:

- Channel, see [Table 5-4](#).
- Selected [CDM Groups w/o Data](#)
- "DMRS" > [Length](#)
- [Number of Layers](#)
- DCI format.

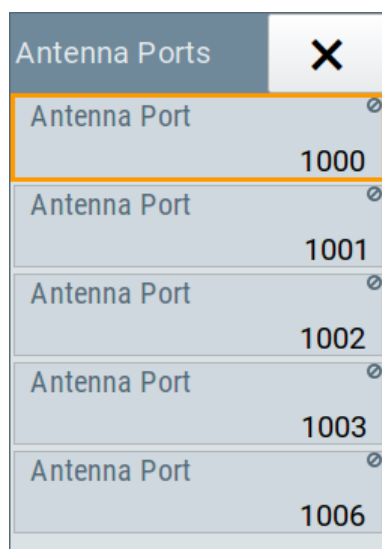
For details, see [TS 38.212](#):

- PUSCH: Tables 7.3.1.1.2-6 to 7.3.1.1.2-23
- PDSCH: Tables 7.3.1.2.2-1 to 7.3.1.2.2-4

Table 5-4: Available APs per channel/signal

Direction	Channel/signal	Antenna ports
Downlink	PDSCH+DMRS	1000 to 1011
	PDCCH	2000
	CSI-RS	3000 to 3031
	SS/PBCH	4000
Uplink	PUSCH+DMRS	0 to 11
	PUCCH	2000
	PRACH	4000
	SRS	1000 to 1004

Each layer of a PDSCH allocation is mapped to a certain antenna port. Select the button "Antenna Ports" to select the antenna ports that are used for the transmission of the PDSCH allocation.



For details on the antenna port mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:APSelect<s2us0> on page 283
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh:APSelect<s2us0> on page 283
```

Mapping Coordinates

Switches representation between the "Cartesian" (Real/Imag) and "Cylindrical" (Magn./Phase) coordinates.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:APMap:MAPCoordinates on page 291
```

Mapping Table

Defines the mapping of the antenna ports (AP) to the physical antennas.

The available antenna ports depend on the current configuration (see "TxScheme" > [Antenna Ports](#)).

For details on the antenna port mapping concept, see [Chapter 3.9, "Antenna Port Mapping"](#), on page 20.

Remote command:

"Mapping Coordinates = Cartesian":

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:REAL
on page 291
```

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:IMAGinary
on page 291
```

"Mapping Coordinates = Cylindrical":

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:MAGNitude
on page 292
```

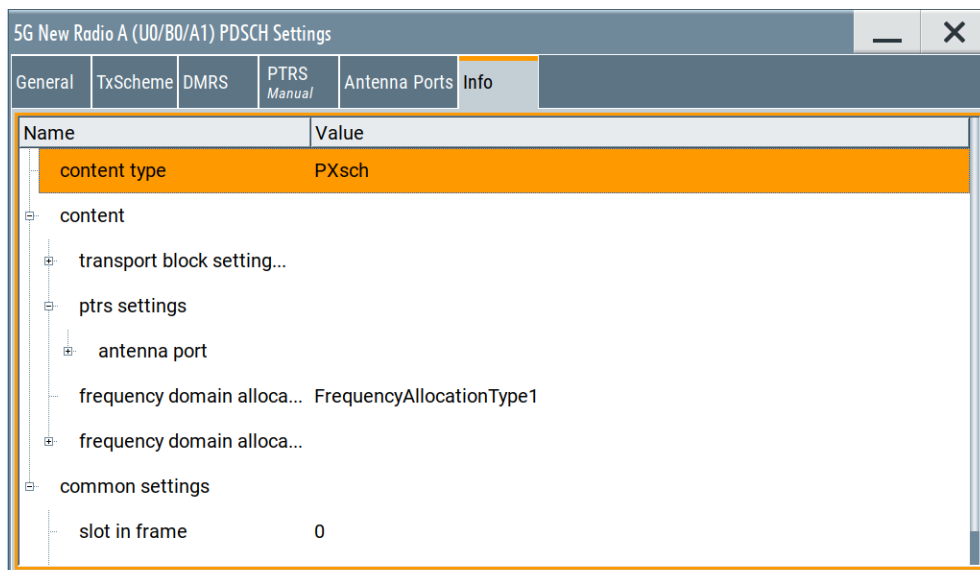
```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:PHASe
on page 292
```

5.5.9 Info

Access:

1. Select "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config".

2. Select "Info".



The dialog displays details on the PDSCH/PUSCH configuration.

Info

The dialog displays details on the configuration.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:INFO? on page 276
```

5.6 CORESET Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".
2. Select "5G New Radio > Scheduling".
3. Select "User x BWP x > Content > CORESET".
4. Select "CORESET > Settings > Config".

CORESET

The CORESET settings are defined per user and per BWP. The CORESET size in the frequency domain is given by number of resource blocks $N_{RB}^{CORESET}$ spans; in the time - it can occupy 1, 2 or 3 symbols.

A CCE consists of 6 resource-element groups (REGs). The mapping between the CEE and the REGs is unique per CORESET. It can be interleaved or non-interleaved and is described by the so called REG bundles with different sizes (L).

Settings:

• General Settings	129
• Payload Settings	133
• Info	147

5.6.1 General Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".
2. Select "5G New Radio > Scheduling".
3. Select "User x BWP x > Content > CORESET".
4. Select "CORESET > Settings > Config" > "General".

5G New Radio A (U0/B0/A0) CORESET Settings		
General Payload Info		
Precoder Granularity	REG Bundle	CORESET ID 1
Use DMRS scrambling ID	☐	ID 0
Interleaving		
State	0	Bundle Size 6
Shift Index	0	Size 2
Search Space		
Aggregation Level	Max. Candidates	
0	1	1
1	2	4
2	4	6
3	8	1
4	16	1

This dialog comprises the settings of the Control Resource Set (CORESET). Each CORESET is assigned to a specific user and comprises of a set of one or more PDCCH channels.

The PDCCH channel carries the downlink control information.

Precoder Granularity	130
CORESET ID	130
Use DMRS Scrambling ID	130
ID	130
Interleaving State	131
Bundle Size	131
Shift Index	131

Size.....	131
Search Space.....	131
L Aggregation Level.....	132
L Max Candidates.....	132

Precoder Granularity

Sets the value of the higher-layer parameter `precoderGranularity`, as defined in TS 38.211.

"REG Bundle" The same precoding is used within a REG bundle; corresponds to `precoderGranularity = sameAsREG-bundle`.

"All Contiguous RBs" The same precoding is used across all REG within the set of contiguous resource blocks in the CORESET; corresponds to `precoderGranularity = allContiguousRBs`.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:PREGran on page 295
```

CORESET ID

Sets the CORESET ID.

Set "CORESET ID = 0" to simulate CORESET 0 (`controlResourceSetZero`). If CORESET 0 is used, the reference point changes and is according to TS 38.211 the subcarrier 0 of the lowest-numbered resource block in the CORESET.

For "CORESET ID ≥ 1 ", the reference point is subcarrier 0 in common resource block 0.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:ID on page 295
```

Use DMRS Scrambling ID

Defines the parameter used to calculate the PUCCH scrambling sequence:

- If disabled, the [Cell ID](#) is used.
- If enabled, the demodulation reference signal DMRS ID is used.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DMRS:SCRam:STATE on page 296
```

ID

For [Use DMRS Scrambling ID](#) > "On", sets the value of the higher-level parameter `PDCCH-DMRS-Scrambling-ID`, according to TS 38.211.

This DMRS scrambling ID is used to calculate the PDCCH scrambling sequence instead of [Cell ID](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DMRS:SCRam:ID on page 296
```

Interleaving State

If enabled, the CCE-to-REG mapping is interleaved.

This parameter corresponds to the higher-level parameter
CORESET-CCE-REG-mapping-type (TS 38.211).

See "CORESET" on page 128.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:STAtE on page 296
```

Bundle Size

For **Interleaving State** > "On", sets the higher-level parameter bundle size L
(CORESET-REG-bundle-size).

The value is used to calculate the CCE-to-REG mapping, according to TS 38.211.

See also "CORESET" on page 128.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:BUNSize on page 296
```

Shift Index

For **Interleaving State** > "On", sets the higher-level parameter shift index n_{shift}
(CORESET-shift-index).

The value is used to calculate the CCE-to-REG mapping, according to TS 38.211.

See also "CORESET" on page 128.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:SHIDx on page 297
```

Size

For **Interleaving State** > "On", sets higher-level parameter interleaver size R
(CORESET-interleaver-size).

The value is used to calculate the CCE-to-REG mapping, according to TS 38.211.

See also "CORESET" on page 128.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:SIZE on page 297
```

Search Space

The term search space describes a set of CCEs that a UE monitors. The UE can decode only the control information on a PDCCH/CORESET that is transmitted over CCEs within the search space this UE monitors.

A physical downlink control channel PDCCH consists of one or more control-channel elements (CCEs), where their number is defined by the "Aggregation level".

The following parameters influence the way the search space is defined:

- If **Restrict to Search Space** > "On", you define the search space by selecting the **aggregation level** and the **number of a particular candidate**. The available can-

didates are defined as a combination aggregation level and maximum number of candidates per aggregation level.

See:

- "Scheduling > CORESET > Payload" > [Aggregation Level](#)
- "Scheduling > CORESET > Payload" > [Candidate](#)
- "Scheduling > CORESET > General" > [Aggregation Level](#) and [Max Candidates](#)

The **CCE start index** [CCE Index](#) is selected automatically so that the CCE is within the search space.

- If [Restrict to Search Space](#) > "Off", you define the search space manually, by setting the CCE start index, see [CCE Index](#).

Example:

Select:

- [Restrict to Search Space](#) > "On".
- For "Scheduling > CORESET > General" > [Aggregation Level](#) = "1", select [Max Candidates](#) = "2".
- Select "Scheduling > CORESET > Payload" > [Aggregation Level](#) = "1" and [Candidate](#) = "1"

Observe the CCE Index ("Scheduling > CORESET > Payload" > [CCE Index](#)). The value is set automatically.

Aggregation Level ← Search Space

Displays the possible aggregation levels, as defined in [TS 38.213](#).

For each aggregation level, you can set the maximum number of candidates that are monitored, see [Max Candidates](#).

"Aggregation level"	Number of CCEs
1	1
2	2
4	4
8	8
16	16

For more information, see ["Search Space"](#) on page 131.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DMRS:SPACe<s2us0>:AGGLLevel?
on page 297
```

Max Candidates ← Search Space

Sets the maximum number of candidates allowed for the selected aggregation level, as specified in [TS 38.213](#).

For more information, see ["Search Space"](#) on page 131.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:CS:DMRS:SPACe<s2us0>:MAXCandidate
on page 298
```

5.6.2 Payload Settings

Access:

1. Select "5G New Radio > General > Link Direction > Downlink".
2. Select "5G New Radio > Scheduling".
3. Select "User x BWP x > Content > CORESET".
4. Select "CORESET > Settings > Config" > "Payload".

5G New Radio A (U0/B0/A0) CORESET Settings

General **Payload** Info

Unused CCEs: [Dropdown] Dummy CCEs: [Dropdown] Dummy CCE Data Source: PN9

Active UL BWP: 0

TCI Present in DCI: ☒

Number of DCIs: 3

	Usage	RNTI	DCI Format	Search Space	Aggregation Level	CCE Index	Content	Create PDSCH
0	C-RNTI	0	0_0	USS	1	0	Config...	
1	SI-RNTI	0	1_0	Type3 CSS	1	0	Config...	
2	C-RNTI	0	Custom	USS	1	0	Config...	

Unused CCEs.....	134
Dummy CCEs Data Source.....	134
Active UL BWP.....	135
TCI Present In DCI.....	135
Number of DCIs.....	135
DCI Table.....	135
L Usage.....	135
L RNTI.....	135
L DCI Format.....	136
L Search Space.....	136
L Aggregation Level.....	137
L Candidate.....	137
L CCE Index.....	137
L Content.....	137
L Create PDSCH.....	138
DCI Content Configuration.....	138
L Bit Data > Pattern.....	138
L DCI Format 0_0.....	138

L DCI Format 0_1.....	139
L DCI Format 1_0.....	141
L DCI Format 1_1.....	143
L Custom.....	145
L Channel Coding.....	146
L Data Source.....	146
L Pattern Length.....	147

Unused CCEs

Defines if unused CORESET resources (CCEs) are used for data transmission or not.

"No Data"	No data is transmitted in the unused CCEs.
"Dummy CCEs"	Unused CCEs are filled in with dummy data.
"Allow PDSCH"	PDSCH transmission is allowed in the unused CCEs.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCCes:STATe on page 298
```

Dummy CCEs Data Source

Selects the data source for the dummy CCEs.

The following standard data sources are available:

- "All 0, All 1"
An internally generated sequence containing 0 data or 1 data.
- "PNxx"
An internally generated pseudo-random noise sequence.
- "Pattern"
An internally generated sequence according to a bit pattern.
Use the "Pattern" box to define the bit pattern.
- "Data List/Select DList"
A binary data from a data list, internally or externally generated.
Select "Select DList" to access the standard "Select List" dialog.
 - Select the "Select Data List > navigate to the list file *.dm_iqd > Select" to select an existing data list.
 - Use the "New" and "Edit" functions to create internally new data list or to edit an existing one.
 - Use the standard "File Manager" function to transfer external data lists to the instrument.

See also:

- Section "Modulation Data" in the R&S SMBV100B user manual.
- Section "File and Data Management" in the R&S SMBV100B user manual.
- Section "Data List Editor" in the R&S SMBV100B user manual

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCCes:DATA on page 298
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCCes:DLISt on page 299
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCCes:PATtern on page 299
```

Active UL BWP

If more than one UL BWPs are configured ("Users/BWP Settings > UL BWPs" > [Number of UL BWPs](#) ≥ 2), sets the index of the UL BWP to be used by the calculation of the bit size of the DCI formats.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:AULBwp on page 299
```

TCI Present In DCI

Sets the higher-layer parameter `tcj-PresentInDCI` according to [TS 38.212](#).

- "On" [Transmission configuration indication](#) (TCI) field in DCI format 1_1 is 3 bits long.
- "Off" Transmission configuration indication field in DCI format 1_1 is not transmitted; field is disabled in the DCI format 1_1.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:TCI on page 299
```

Number of DCIs

Sets the number of DCIs and thus defines the number of rows in the DCI table.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:NDCI on page 300
```

DCI Table

Comprises the DCI settings. There is one row per DCI, where the number of rows depends on the selected "Number of DCIs".

Usage ← DCI Table

Sets the RNTI with that the CRC is scrambled, see [Table 5-6](#).

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:USAGe on page 300
```

RNTI ← DCI Table

Indicates the C-RNTI of the associated user.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:RNTI on page 300
```

DCI Format ← DCI Table

Sets the DCI format.

The downlink control information (DCI) is a message used to control the physical layer resource allocations. It carries scheduling information and power control commands.

Available are the DCI formats "0_0", "1_0", "0_1", "1_1" and the "Custom" format with user-definable content. The DCI formats available in particular situation depend on the selected [Usage](#) and [Search Space](#).

To configure the parameters per DCI format, select "Content > Config".

Table 5-5: DCI formats overview [TS 38.212]

DCI format	Usage
0_0	Scheduling of PUSCH in one UL cell See "DCI Format 0_0" on page 138.
0_1	See "DCI Format 0_1" on page 139.
1_0	Scheduling of PDSCH in one DL cell (allocation type 1) See "DCI Format 1_0" on page 141.
1_1	Scheduling of PDSCH in one DL cell See "DCI Format 1_1" on page 143.
2_0	Notifying a group of UEs of the slot format
2_1	Notifying a group of UEs of the PRBs and OFDM symbols where UE can assume that no transmission is intended for the UE
2_2	Transmission of TPC commands for PUCCH and PUSCH
2_3	Transmission of a group of TPC commands for SRS transmissions by one or more UEs

Remote command:

[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:FMT on page 301

Search Space ← DCI Table

Defines the search space for the selected DCI.

The search space defines the set of PDCCH candidates that a UE has to monitor. Search spaces can be common search space (CSS) and UE-specific search space (USS), where the non-common DCIs are mapped to the USS. The UE can decode only the control information that is transmitted within one or more search spaces this UE monitors.

[TS 38.213](#) defines the search spaces listed in ["Search Space"](#) on page 136.

Table 5-6: Search space sets

Search space	Designation	Parameter	CRC scrambled with
CSS (Common search space)	Type0-PDCCH	"Type0 CSS"	SI-RNTI
	Type0A-PDCCH		
	Type1-PDCCH	"Type1 CSS"	RA-RNTI or TC-RNTI

Search space	Designation	Parameter	CRC scrambled with
	Type2-PDCCH	"Type2 CSS"	P-RNTI
	Type3-PDCCH	"Type3 CSS"	INT-RNTI, SFI-RNTI, TPC-PUSCH/PUCCH/ SRS-RNTI, C-RNTI, CS-RNTI
UE-specific search space	USS	"USS"	C-RNTI, CS-RNTI

The available search spaces depend on the selected [Usage](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:  
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:SSP on page 301
```

Aggregation Level ← DCI Table

Sets the aggregation level for the selected DCI.

A physical downlink control channel PDCCH consists of one or more control-channel elements (CCEs), where their number is defined by the "Aggregation level".

This and several other parameters influence the way the search space is defined. For details, see ["Search Space"](#) on page 131.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:  
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:AGGLevel on page 301
```

Candidate ← DCI Table

Selects one of the candidates configured for the selected [Aggregation Level](#), where the number of candidates per aggregation level is set by the parameter "CORESET > General" > [Max Candidates](#).

This and several other parameters influence the way the search space is defined. For details, see ["Search Space"](#) on page 131.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:  
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CANDidate on page 301
```

CCE Index ← DCI Table

Sets or displays the CCE start index and thus the position of the first CCE of the PDCCH.

This and several other parameters influence the way the search space is defined. For details, see ["Search Space"](#) on page 131.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:  
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:INDEX on page 302
```

Content ← DCI Table

Access a dialog for configuring the parameters per DCI format, see ["DCI Content Configuration"](#) on page 138

Create PDSCH ← DCI Table

If enabled, the content and scheduling of the PDSCH is created automatically, as configured with the DCI format 1_0 and 1_1 parameters.

If **PDSCH Scheduling** = "Auto/DCI", the "Create PDSCH" = "On".

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CPDSch on page 302
```

DCI Content Configuration

Comprises the parameters of the selected DCI format.

The fields defined in the DCI format are mapped to the information bits according to **TS 38.212**. The resulting **Bit Data > Pattern** is displayed.

The following parameters have an impact on the DCI bit data length and the DCI fields:

- Number of RB **No. RBs** of the first UL BWP
- Number of RB of the DL BWP, in that the DCI is configured
- **SUL** indicator of the selected cell
- DCI **Search Space**

Bit Data > Pattern ← DCI Content Configuration

Displays the resulting bit data as selected with the DCI format parameters.

Mapping of the information bits is according to **TS 38.212**.

Padding bits are added in certain cases, that is zeros are append to the DCI format until the payload size is equal to a specified size.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:HBTS? on page 302
```

DCI Format 0_0 ← DCI Content Configuration

The DCI format 0_0 is used for scheduling of PUSCH in one UL cell.

5G New Radio A: DCI Settings (DCI 0)

Bit Data

Pattern

0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

DCI Fields

Frequency Domain Resource Assignment

0

Time Domain Resource Assignment

0

Frequency Hopping Flag

☐

Modulation and Coding Scheme (TB1)

0

New Data Indicator (TB1)

☐

Redundancy Version (TB1)

0

HARQ Process Number

0

TPC Command for Scheduled PUSCH

0

DCI format 0_0 is scrambled by C-RNTI or CS-RNTI and carries the following information:

DCI field	Remote command:	Remark
Frequency Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:FRDRes on page 303	
Time Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TIDRes on page 303	
Frequency Hopping Flag	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:FRHFlag on page 303	
Modulation and Coding Scheme (TB1/TB2)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:MCS on page 304 [:SOURce<hw>] :BB:NR5G:SCHeD: CELL<ch0>:SUBF<st0>:USER<dir0>: BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>: TB2:MCS on page 304	
New data Indicator (TB1/TB2)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:NDI on page 304 [:SOURce<hw>] :BB:NR5G:SCHeD: CELL<ch0>:SUBF<st0>:USER<dir0>: BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>: TB2:NDI on page 304	
Redundancy Version (TB1/TB2)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:RV on page 304 [:SOURce<hw>] :BB:NR5G:SCHeD: CELL<ch0>:SUBF<st0>:USER<dir0>: BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>: TB2:RV on page 304	
HARQ Process Number	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:HAPRoc on page 305	
TPC Command for Scheduling PUSCH	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TPUSch on page 305	
UL/SUL Indicator	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:USINd on page 307	If "Node > Carrier > SUL = On"

DCI Format 0_1 ← DCI Content Configuration

The DCI format 0_1 is used for scheduling of PUSCH in one UL cell.

5G New Radio A: DCI Settings (DCI 0)	
Bit Data	
Pattern	0 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
DCI Fields	
Frequency Domain Resource Assignment	Time Domain Resource Assignment
0	0
Modulation and Coding Scheme (TB1)	New Data Indicator (TB1)
0	☐
Redundancy Version (TB1)	HARQ Process Number
0	0
(1st) Downlink Assignment Index	2nd Downlink Assignment Index
0	0
TPC Command for Scheduled PUSCH	Antenna Ports
0	0
SRS Request	PTRS-DMRS Association
0	0
DMRS Sequence Initialization	☐

DCI format 0_1 is scrambled by C-RNTI, CS-RNTI or SP-CSI-RNTI and carries the following information:

DCI field	Remote command:	Remark
UL/SUL Indicator	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:USINd on page 307	If "Node > Carrier > SUL = On"
Frequency Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:FRDRes on page 303	
Time Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TIDRes on page 303	
Modulation and Coding Scheme (TB1)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TB1:MCS on page 304	
New data Indicator (TB1)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TB1:NDI on page 304	
Redundancy Version (TB1)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TB1:RV on page 304	
HARQ Process Number	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:HAPRoc on page 305	

DCI field	Remote command:	Remark
(1st) Downlink Assignment Index	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DAI1 on page 306	
(2nd) Downlink Assignment Index	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DAI2 on page 306	
TPC Command for Scheduling PUSCH	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TPUSch on page 305	
Antenna Ports	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:ANTPorts on page 306	
SRS Request	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:SRSRequest on page 306	
PTRS-DMRS Association	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PTDMrs on page 307	If "User/ BWPs > UL BWP Config > PUSCH > Max Rank ≥2"
DMRS Sequence Initialization	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DMSQinit on page 307	If "User/ BWPs > UL BWP Config > PUSCH > Transform Precoding = Off"

DCI Format 1_0 ← DCI Content Configuration

The DCI format 1_0 is used for scheduling of PDSCH in one DL cell.

5G New Radio A: DCI Settings (DCI 2)	
Bit Data	
Pattern	1000 0000 0000 0000 0000 0000 0000 0000 0000 0000
DCI Fields	
Frequency Domain Resource Assignment	Time Domain Resource Assignment
0	0
VRB-to-PRB Mapping	Modulation and Coding Scheme (TB1)
☐	0
New Data Indicator (TB1)	Redundancy Version (TB1)
☐	0
HARQ Process Number	(1st) Downlink Assignment Index
0	0
TPC command for Scheduled PUCCH	PUCCH Resource Indicator
0	0
PDSCH-to-HARQ Feedback Timing Indicator	
0	

DCI format 1_0 is scrambled by C-RNTI, CS-RNTI or SI-RNTI and carries the following information:

DCI field	Remote command:	Remark
Frequency Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:FRDRes on page 303	
Time Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TIDRes on page 303	
VRB-to-PRB Mapping	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:VTPRb on page 305	
Modulation and Coding Scheme (TB1)	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TB1:MCS on page 304	
New Data Indicator (TB1)	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TB1:NDI on page 304	If "Usage = C-RNTI"
Redundancy Version (TB1)	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TB1:RV on page 304	
HARQ Process Number	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:HAPRoc on page 305	If "Usage = C-RNTI"
(1st) Downlink Assignment Index	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:DAI1 on page 306	If "Usage = C-RNTI"
TPC Command for Scheduling PUCCH	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:TPUCch on page 307	If "Usage = C-RNTI"
PUCCH Resource Indicator	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:PUCResind on page 308	If "Usage = C-RNTI"
PDSCH-to-HARQ Feedback Timing Indicator	[:SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:PDSHarq on page 308	If "Usage = C-RNTI"

DCI field	Remote command:	Remark
System Information Indicator	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:SIINd on page 308	If "Usage = SI-RNTI"
Padding Bits, Reserved Bits	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:RESVed on page 309	If "Usage = SI-RNTI"

DCI Format 1_1 ← DCI Content Configuration

The DCI format 1_1 is used for scheduling of PDSCH in one DL cell.

5G New Radio A: DCI Settings (DCI 3)		—	×
DCI Fields			
BWP Indicator	0	Frequency Domain Resource Assignment	0
Time Domain Resource Assignment	0	VRB-to-PRB Mapping	☐
PRB Bundling Size Indicator	☐	Rate Matching Indicator	0
ZP CSI-RS Trigger	0	Modulation and Coding Scheme (TB1)	0
New Data Indicator (TB1)	☐	Redundancy Version (TB1)	0
Modulation and Coding Scheme (TB2)	0	New Data Indicator (TB2)	☐
Redundancy Version (TB2)	0	HARQ Process Number	0
(1st) Downlink Assignment Index	0	TPC command for Scheduled PUCCH	0
PUCCH Resource Indicator	0	PDSCH-to-HARQ Feedback Timing Indicator	0
Antenna Ports	0	Transmission Configuration Indication	0
SRS Request	0	CBG Transmission Information (CBGTI)	0
CBG Flushing out Information (CBGFI)	☐	DMRS Sequence Initialization	☐

DCI format 1_1 is scrambled by C-RNTI or CS-RNTI and carries the following information:

DCI field	Remote command:	Remark
Bandwidth Part Indicator	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:BWIND on page 309	
Frequency Domain Resource Assignment	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0> : SUBF<st0>:USER<dir0>:BWPart<gr0> : ALLoc<user0>:CS:DCI<s2us0>:FRDRes on page 303	

DCI field	Remote command:	Remark
Time Domain Resource Assignment	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TIDRes on page 303	
VRB-to-PRB Mapping	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:VTPRb on page 305	
PRB Bundling Size Indicator	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PRBBundling on page 309	
Rate Matching Indicator	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:RMIND on page 309	
ZP CSI-RS Trigger	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:ZCRTrigg on page 310	
Modulation and Coding Scheme (TB1/TB2)	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:MCS on page 304 [:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB2:MCS on page 304	
New data Indicator (TB1/TB2)	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:NDI on page 304 [:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB2:NDI on page 304	
Redundancy Version (TB1/TB2)	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:RV on page 304 [:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB2:RV on page 304	
HARQ Process Number	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:HAPRoc on page 305	
(1st) Downlink Assignment Index	[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DAI1 on page 306	

DCI field	Remote command:	Remark
TPC Command for Scheduling PUCCH	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TPUCCh on page 307	
PUCCH Resource Indicator	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PUCResind on page 308	
PDSCH-to-HARQ Feedback Timing Indicator	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PDSHarq on page 308	
Antenna Ports	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:ANTPorts on page 306	
Transmission Configuration Indication	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TCI on page 310	Enabled if TCI Present In DCIIdPDbNr 5gUserAl- locCSTCI- PresentInD- CIState> "On"
SRS Request	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:SRSRequest on page 306	
CBG Transmission Information (CBGTI)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:CBGTi on page 310	
CBG Flushing Out Information (CBGFI)	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:CBGFi on page 310	
DMRS Sequence Initialization	[:SOURce<hw>] :BB:NR5G:SCHeD:CELL<ch0>: SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DMSQinit on page 307	

Custom ← DCI Content Configuration

This is a DCI without predefined DCI fields. The DCI content is user-definable.

5G New Radio A: DCI Settings (DCI 0)	
Channel Coding	☒
Bit Data	
Data Source	Pattern
Pattern Length	84
Pattern	0000 0000...

Channel Coding ← Custom ← DCI Content Configuration

Enables channel coding.

If **PDSCH Scheduling** = "Auto/DCI", the "Channel Coding" = "On".

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:STAtE on page 311
```

Data Source ← Custom ← DCI Content Configuration

Selects the data source for the custom DCI.

The following standard data sources are available:

- "All 0, All 1"
An internally generated sequence containing 0 data or 1 data.
- "PNxx"
An internally generated pseudo-random noise sequence.
- "Pattern"
An internally generated sequence according to a bit pattern.
Use the "Pattern" box to define the bit pattern.
- "Data List/Select DList"
A binary data from a data list, internally or externally generated.
Select "Select DList" to access the standard "Select List" dialog.
 - Select the "Select Data List > navigate to the list file *.dm_iqd > Select" to select an existing data list.
 - Use the "New" and "Edit" functions to create internally new data list or to edit an existing one.
 - Use the standard "File Manager" function to transfer external data lists to the instrument.

See also:

- Section "Modulation Data" in the R&S SMBV100B user manual.
- Section "File and Data Management" in the R&S SMBV100B user manual.
- Section "Data List Editor" in the R&S SMBV100B user manual

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DATA on page 311
```

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DLISt on page 311
```

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:PATtern on page 312
```

Pattern Length ← Custom ← DCI Content Configuration

If **Channel Coding** > "On", sets the number of DCI bits, i.e. the number of bits in the "Pattern" field.

If **Channel Coding** > "Off", the number of DCI bits is calculated based on the selected **Aggregation Level** for the corresponding DCI.

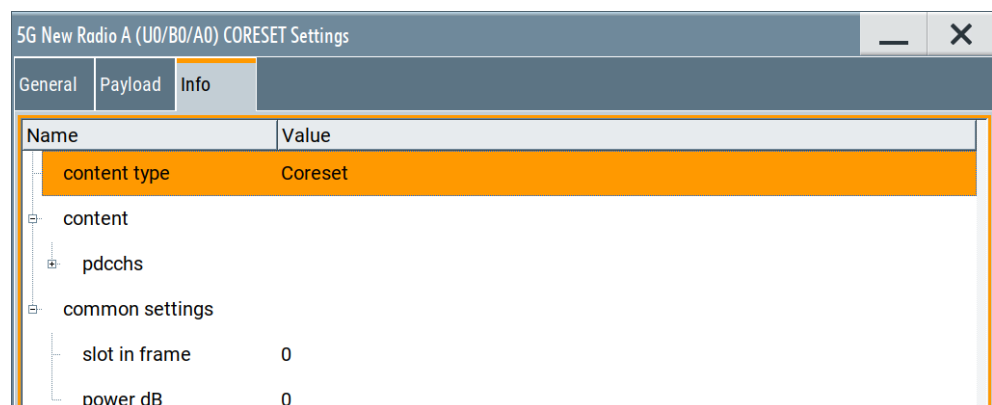
Remote command:

[:SOURCE<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:BITLength on page 312

5.6.3 Info

Access:

1. Select "Scheduling > User x BWP x > Content > CORESET > Settings > Config".
2. Select "Info".



The dialog displays details on the CORESET configuration.

Info

The dialog displays details on the configuration.

Remote command:

[:SOURCE<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:INFO? on page 276

5.7 PUCCH Settings

Access:

1. Select "5G New Radio > General > Link Direction > Uplink".
2. Select "5G New Radio > Scheduling".
3. Select "User x BWP x > Content > PUCCH".

4. For the PUCCH allocation, select "User x BWP x > Format = e.g. F0"
5. Select "PUCCH > Settings > Config".

This dialog comprises the settings of the physical uplink common channel (PUCCH).

PUCCH and PUCCH formats

The PUCCH supports 5 different formats (F0 to F4), each of them spanning different number of OFDM symbols. PUCCH can be transmitted on a constant frequency or optionally use frequency hopping, where the latter is signaled by a higher-level parameter. If frequency hopping is enabled, the PUCCH in formats F1, F3 and F4 spans the half of the OFDM symbols available in the constant frequency case.

Settings:

- [General Settings](#)..... 148
- [Payload Settings](#)..... 150
- [Info](#)..... 152

5.7.1 General Settings

Access:

1. Select "Scheduling > User x BWP x > Content > PUCCH > Settings > Config".
2. Select "General".

The screenshot shows a dialog box titled "5G New Radio A (U0/B0/A0) PUCCH Settings". It has three tabs: "General", "Payload", and "Info". The "General" tab is active. The settings are as follows:

Setting	Value
Group Hopping	Neither
Hopping ID	0
Intra Slot Frequency Hopping	☐
Second Hop PRB	0
Format Specific	
Initial Cyclic Shift	0

Settings:

Group Hopping.....	149
Hopping ID.....	149
Intra Slot Frequency Hopping.....	149
Second Hop PRB.....	149
Initial Cyclic Shift.....	149
Time Domain OCC Index.....	150
OCC Length.....	150
OCC Index.....	150

Group Hopping

Available for PUCCH formats 0 and 1.

Sets the higher-layer parameter `pucch-GroupHopping`, as defined in TS 38.211.

You turn group and sequence hopping for the PUCCH demodulation reference signal on and off as follows:

"Neither"	Disables group and sequence hopping
"Enable"	Enables group hopping
"Disable"	Enables sequence hopping

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:GRPHopping on page 313
```

Hopping ID

Sets the parameter hopping ID (n_{ID}) used to select the frequency hopping sequence number and to initialize the pseudo-random sequence generator. In the practice, the value is transmitted by the higher-layer parameter `hoppingId` [TS 38.211].

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:HOPId on page 313
```

Intra Slot Frequency Hopping

Enables PUCCH demodulation reference signal hopping within a slot.

If intra-slot hopping is enabled, you can define the physical resource block offset for the second hop ("Second Hop PRB").

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:ISFHopping on page 314
```

Second Hop PRB

Sets the physical resource block (PRB) of the second hop.

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:SHOPping on page 314
```

Initial Cyclic Shift

For PUCCH formats F0 and F1, sets the initial cyclic shift.

As defined in TS 38.211, the value is required to calculate the cyclic shift for the hopping sequence and thus define the frequency positions of the different hops.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:FS:CYCShift on page 314
```

Time Domain OCC Index

For PUCCH format F1, sets the orthogonal sequence index $\mathfrak{l}$ ($\mathfrak{l} = 0$ to 6). As defined in TS 38.211, the value is required to select the orthogonal sequence, used for the block-wise spreading.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:FS:TDOCC on page 314
```

OCC Length

For PUCCH format F4, sets the value of the parameter $N_{SF}^{PUCCH,4}$ and thus defines which of the orthogonal sequences $w_n(m)$ defined in TS 38.211 is used.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:FS:OCCLength on page 315
```

OCC Index

For PUCCH format F4, sets the orthogonal sequence index n ($n = 0$ to 4). As defined in TS 38.211, the value is required to select the orthogonal sequence, used for the block-wise spreading.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:FS:OCCindex on page 315
```

5.7.2 Payload Settings

Access:

1. Select "Scheduling > User x BWP x > Content > PUCCH > Settings > Config".
2. Select "Payload".

5G New Radio A (U0/B0/A0) PUCCH Settings	
General	Payload
Positive SR Count: 0	
Number of ACK Bits: 1	ACK Pattern: 0...
Number of UCI Bits: 0	UCI Pattern: 0...

Settings:

Positive SR Count.....	151
Number of ACK Bits.....	151
ACK Pattern.....	151
Number of UCI Bits.....	151
UCI Pattern.....	152

Positive SR Count

Sets the number of positive scheduling requests (SR). The value is used to define the mapping of the HARQ-ACK and CSI bits. For details, see [TS 38.213](#).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:PL:SRCount on page 315
```

Number of ACK Bits

Sets the number of ACK bits and defines the length of the "ACK Pattern" field.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:PL:ACK:BITS on page 316
```

ACK Pattern

Sets the ACK bits in pattern form, where the pattern length is set with the parameter "Number of ACK Bits".

A "1" indicates an ACK, a "0" - a NACK.

The pattern is read out once.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:PL:ACK:PATtern on page 316
```

Number of UCI Bits

Sets the number of UCI bits and defines the length of the "UCI Pattern" field.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:PL:UCI:BITS on page 316
```

UCI Pattern

Sets the UCI bit sequence according to TS 38.212. The sequence is set in pattern form, where the pattern length is set with the parameter "Number of UCI Bits".

A "1" indicates an ACK, a "0" - a NACK.

The pattern is read out once.

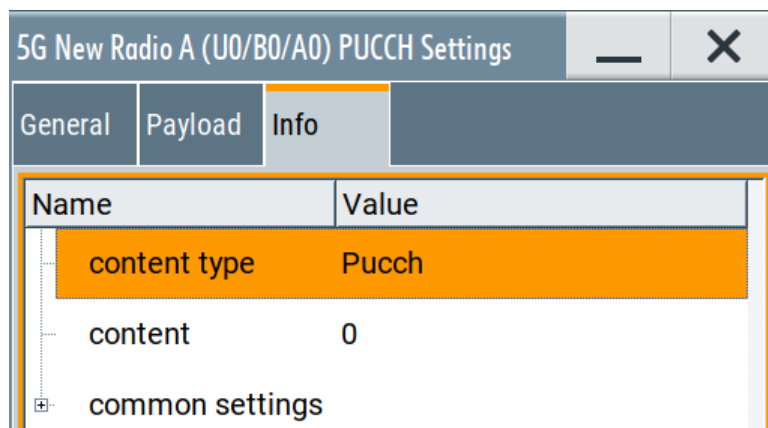
Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:PL:UCI:PATtern on page 316
```

5.7.3 Info

Access:

1. Select "Scheduling > User x BWP x > Content > PUCCH > Settings > Config".
2. Select "Info".



The dialog displays details on the PUCCH configuration.

Info

The dialog displays details on the configuration.

Remote command:

```
[ :SOURce<hw>] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:INFO? on page 276
```


5.8 PRACH Settings

Access:

1. Select "5G New Radio > General > Link Direction > Uplink".
2. Select "5G New Radio > Scheduling".
3. Select "User x BWP x > Content > PRACH".
4. Select "PRACH > Settings > Config".

5G New Radio A (U0/B0/A0) PRACH Settings	
General	
Format	Restricted Set
A1	Unrestricted
Logical Root Sequence	Zero Correlation Zone
0	0
Preamble Index	Time Offset
0	0.00 µs

This dialog comprises the settings of the physical random access channel (PRACH).

According to [TS 38.211](#), the 64 preambles ([Preamble Index](#)) defined in each time-frequency PRACH occasion are numbered starting from the first cyclic shift of a logical root sequence and then following the logical root sequence index, starting with the index set by the parameter [Logical Root Sequence](#).

Settings:

Format	153
Restricted Set	154
Logical Root Sequence	154
Zero Correlation Zone	154
Preamble Index	154
Time Offset	154
Info	154

Format

Selects the PRACH format.

The available formats depend on the [SC Spacing/CP](#) as follows:

- SCS = {1.25 kHz; 5 KHz}: "Format" = 0, 1, 2, 3
- SCS = {15 kHz; 30 kHz; 60 kHz; 120 kHz}: "Format" = A1, A2, A3, B1, B2, B3, B4, C0, C2

Remote command:

```
[ :SOURce<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:FORMat on page 317
```

Restricted Set

Sets the higher-layer parameter `restrictedSetConfig` that defines the type of restricted sets (unrestricted, restricted type A, restricted type B).

Supported restricted sets depend on the PRACH [Format](#) as follows:

- "Format" = 0, 1, 2, 3: "Restricted Type A, Restricted Type B"
- "Format" = A1, A2, A3, B1, B2, B3, B4, C0, C2: "Unrestricted"

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:RSET on page 318
```

Logical Root Sequence

Sets the higher-layer parameter `prach-RootSequenceIndex` that defines the starting logical root sequence index and thus the 64 PRACH preambles.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:RSEquence on page 318
```

Zero Correlation Zone

Sets the parameter `zeroCorrelationZoneConfig` required to select the N_{cs} value that in turn is used for the calculation of the cyclic shift ([TS 38.211](#)).

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:ZCZone on page 318
```

Preamble Index

Sets the PRACH preamble index and thus defines which one of the 64 PRACH preambles is used.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:PRIndex on page 318
```

Time Offset

Sets the timing offset base value for PRACH, required for the PRACH performance requirement tests according to [TS 38.141-1](#).

The supported value range allows you to test PRACH detection also with in case of delay or misalignment greater than the values defined in the specification.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:TOFFset on page 319
```

Info

The dialog displays details on the configuration.

Remote command:

```
[ :SOURCE<hw> ] :BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:INFO? on page 276
```

6 Signal Control and Signal Characteristics

This section lists settings provided for improving the signal and spectrum characteristics of the generated signal, defining the signal power and the signal generation start.

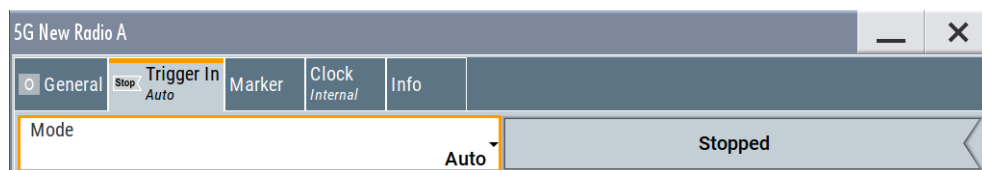
It covers the following topics:

- [Trigger Settings](#)..... 155
- [Marker Settings](#)..... 160
- [Clock Settings](#)..... 162
- [Global Connector Settings](#)..... 163
- [Output Settings](#)..... 163
- [Time Domain Windowing Settings](#)..... 166
- [Adjusting the Signal Power](#)..... 168

6.1 Trigger Settings

Access:

- ▶ Select "5G New Radio > Trigger In".



This tab provides access to the settings necessary to select and configure the trigger, like trigger source and mode, and to arm or trigger an internal trigger manually. The current signal generation status is displayed in the header of the tab together with information on the enabled trigger mode. As in the "Marker" and "Clock" tabs, this tab provides also access to the settings of the related connectors.



This section focuses on the available settings.

For information on how these settings affect the signal, refer to section "Basics on ..." in the R&S SMBV100B user manual.



Routing and enabling a trigger

The provided trigger signals are not dedicated to a particular connector. Trigger signals can be mapped to one or more User x connectors.

The provided trigger signals are not dedicated to a particular connector. Trigger signals can be mapped to one or two User x connectors.

Use the [Global Connector Settings](#) to configure the signal mapping, the polarity, the trigger threshold and the input impedance of the input connectors.

To route and enable a trigger signal, perform the following *general steps*:

- Define the signal source and the effect of a trigger event.
Select the "Trigger In > Mode" and "Trigger In > Source".
- Define the connector where the selected signal is provided.
Use the "Global Connectors" settings.

Settings:

Trigger Mode.....	156
Signal Duration Unit.....	157
Trigger Signal Duration.....	157
Running/Stopped.....	157
Arm.....	157
Execute Trigger.....	157
Trigger Source.....	157
Sync. Output to External Trigger/Sync. Output to Trigger.....	158
External / Trigger Inhibit.....	159
(External) Delay Unit.....	159
(Specified) External Delay/(Specified) Trigger Delay.....	159
Actual Trigger Delay/Actual External Delay.....	159

Trigger Mode

Selects trigger mode, i.e. determines the effect of a trigger event on the signal generation.

- "Auto"
The signal is generated continuously.
- "Retrigger"
The signal is generated continuously. A trigger event (internal or external) causes a restart.
- "Armed Auto"
The signal is generated only when a trigger event occurs. Then the signal is generated continuously.
An "Arm" stops the signal generation. A subsequent trigger event (internal or external) causes a restart.
- "Armed Retrigger"
The signal is generated only when a trigger event occurs. Then the signal is generated continuously. Every subsequent trigger event causes a restart.
An "Arm" stops signal generation. A subsequent trigger event (internal or external) causes a restart.
- "Single"

The signal is generated only when a trigger event occurs. Then the signal is generated once to the length specified at "Signal Duration".

Every subsequent trigger event (internal or external) causes a restart.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G [ :TRIGger ] :SEQuence` on page 325

Signal Duration Unit

Defines the unit for describing the length of the signal sequence to be output in the "Single" trigger mode.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:SLUNit` on page 329

Trigger Signal Duration

Enters the length of the signal sequence to be output in the "Single" trigger mode.

Use this parameter to output part of the signal deliberately, an exact sequence of the signal, or a defined number of repetitions of the signal.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:SLENgth` on page 328

Running/Stopped

With enabled modulation, displays the status of signal generation for all trigger modes.

- "Running"
The signal is generated; a trigger was (internally or externally) initiated in triggered mode.
- "Stopped"
The signal is not generated and the instrument waits for a trigger event.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:RMODE?` on page 328

Arm

Stops the signal generation until subsequent trigger event occurs.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:ARM:EXECute` on page 328

Execute Trigger

For internal trigger source, executes trigger manually.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:EXECute` on page 328

Trigger Source

The following sources of the trigger signal are available:

- "Internal"
The trigger event is executed manually by the "Execute Trigger".
- "External Global Trigger"
The trigger event is the active edge of an external trigger signal provided and configured at the User x connectors.
- "Baseband Sync In"

In master-slave mode, slave instruments are triggered by the active edge of the synchronization signal.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:SOURce` on page 325

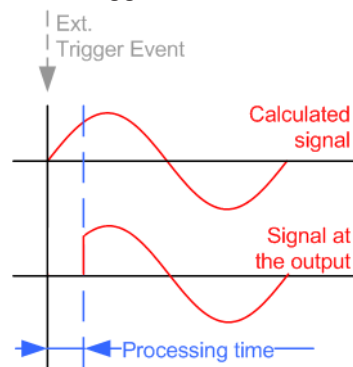
Sync. Output to External Trigger/Sync. Output to Trigger

Enables signal output synchronous to the trigger event.

- "On"

Corresponds to the default state of this parameter.

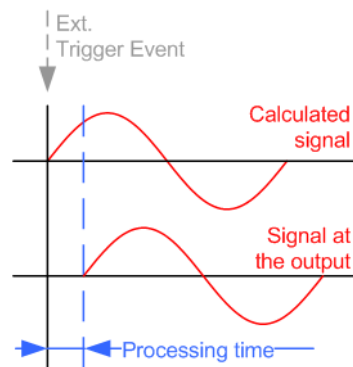
The signal calculation starts simultaneously with the trigger event. Because of the processing time of the instrument, the first samples are cut off and no signal is output. After elapsing of the internal processing time, the output signal is synchronous to the trigger event.



- "Off"

The signal output begins after elapsing of the processing time. Signal output starts with sample 0. The complete signal is output.

This mode is recommended for triggering of short signal sequences. Short sequences are sequences with signal duration comparable with the processing time of the instrument.



In master-slave mode, this setting ensures that once achieved, synchronization is not lost if the baseband signal sampling rate changes.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:EXTeRnal:SYNChronize:OUTPut`
on page 326

External / Trigger Inhibit

Applies for external trigger signal.

Sets the duration with that any following trigger event is suppressed. In "Retrigger" mode, for example, a new trigger event does not cause a restart of the signal generation until the specified inhibit duration does not expire.

For more information, see chapter "Basics" in the R&S SMBV100B user manual.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger[:EXternal]:INHibit` on page 327

(External) Delay Unit

Determine whatever the trigger delay is expressed in samples or directly defined as a time period (seconds).

To specify the delay, use the parameter [\(Specified\) External Delay/\(Specified\) Trigger Delay](#).

The parameter [Actual Trigger Delay/Actual External Delay](#) displays the delay converted in time.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:DElay:UNIT` on page 326

(Specified) External Delay/(Specified) Trigger Delay

The name of the parameter and the units the delay is expressed in, changes depending on the parameter [\(External\) Delay Unit](#).

Delays the trigger event of the signal from:

- The external trigger source

Use this setting to:

- Synchronize the instrument with the device under test (DUT) or other external devices
- Compensate delays and align the signal generation start in multi-instrument setup

For more information, see chapter "Basics on ..." in the R&S SMBV100B user manual.

For "(External) Delay Unit = Samples", the delay is relative to the sample rate of the first output of the respective baseband block.

The parameter displays the delay converted in time.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger[:EXternal]:DElay` on page 326

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:EXternal:TDElay` on page 327

Actual Trigger Delay/Actual External Delay

Indicates the resulting trigger delay in "Time" unit.

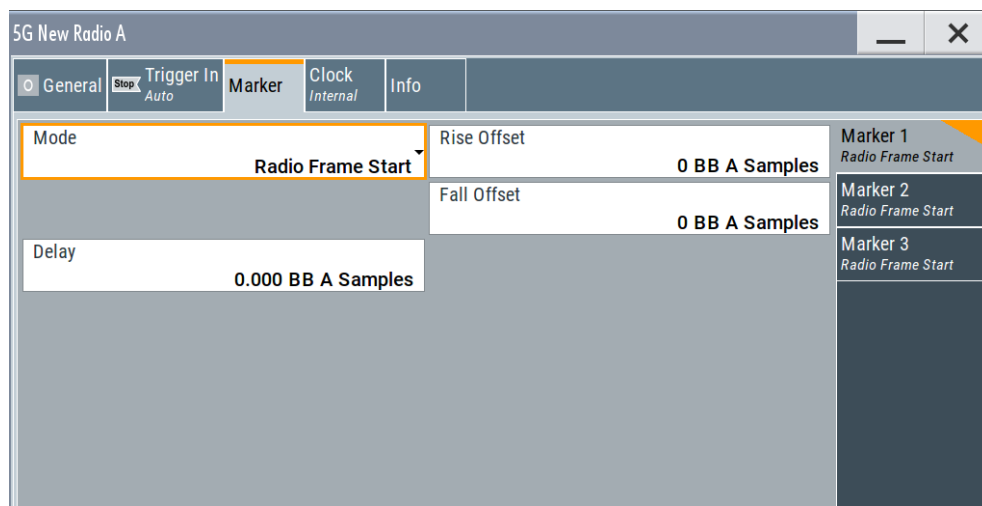
Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:EXternal:RDElay?` on page 327

6.2 Marker Settings

Access:

- Select "5G New Radio > Marker".



This tab provides access to the settings necessary to select and configure the marker output signal, like the marker mode or marker delay settings.



This section focuses on the available settings.

For information on how these settings affect the signal, refer to section "Basics on ..." in the R&S SMBV100B user manual.



Routing and enabling a marker

The provided marker signals are not dedicated to a particular connector. They can be mapped to one or more User x connectors.

To route and enable a marker signal, perform the following *general steps*:

- Define the shape of the generated marker, i.e. select the "Marker > Mode".
- Define the connector where the selected signal is provided.
Use the [Global Connector Settings](#).

Settings:

Marker Mode	160
Rise/Fall Offset	161
Marker x Delay	162

Marker Mode

Marker configuration for up to 3 markers. The settings are used to select the marker mode defining the shape and periodicity of the markers. The contents of the dialog change with the selected marker mode; the settings are self-explanatory.

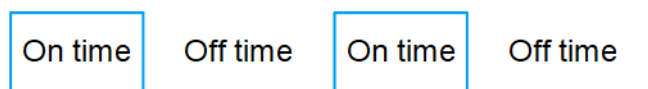
- "Subframe" A marker signal is generated at the start of each subframe.
- "Radio Frame Start" A marker signal is generated at the start of each radio frame.
- "Restart (ARB)" A marker signal is generated at the start of each ARB sequence.
- "User Period" A marker signal is generated at the beginning of every user-defined period, as set with the parameter "Period."

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:OUTPut<ch>:PERiod` on page 330

"On/Off Period"

A regular marker signal that is defined by an on/off ratio. A period lasts one on and off cycle.



Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:OUTPut<ch>:ONTime` on page 330

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:OUTPut<ch>:OFFTime` on page 330

"System Frame Number (SFN) Restart"

A marker signal is generated at the start of every SFN period.

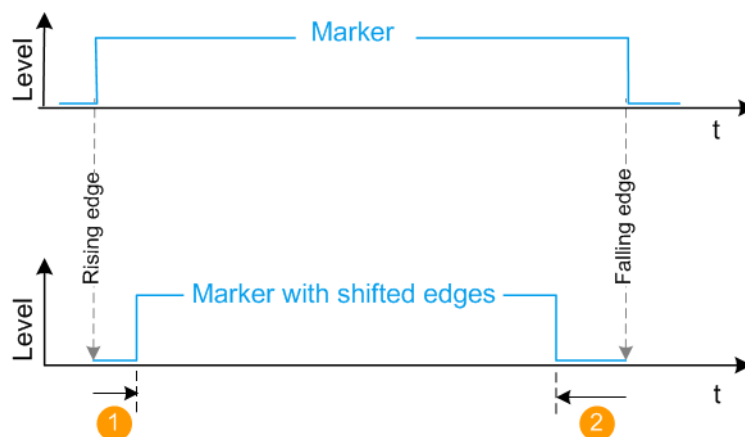
Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:OUTPut<ch>:MODE` on page 329

Rise/Fall Offset

Sets the value for the rise/fall offset.

The ramps of the marker signal are shifted by the specified number of a samples. Positive values delay the rising ramp; negative values - shift it back.



1 = Positive rise offset

2 = Positive fall offset

Remote command:

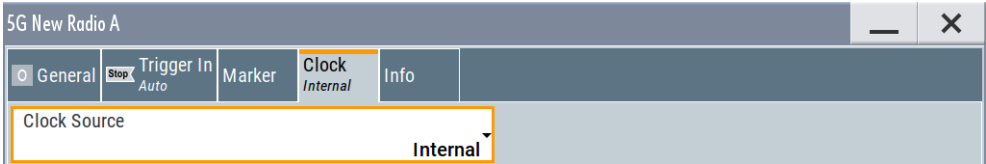
`[ :SOURce<hw> ] :BB:NR5G:TRIGger:OUTPut<ch>:ROFFset` on page 330

`[ :SOURce<hw> ] :BB:NR5G:TRIGger:OUTPut<ch>:FOFFset` on page 330

Marker x Delay
 Delays the marker signal at the marker output relative to the signal generation start.
 Variation of the parameter "Marker x Delay" causes signal recalculation.
 Remote command:
 [:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:DELaY on page 329

6.3 Clock Settings

Access:
 ▶ Select "5G New Radio > Clock".



This tab provides access to the settings necessary to select and configure the clock signal, like the clock source and clock mode.

This section focuses on the available settings.
 For information on how these settings affect the signal, refer to section "Basics on ..." in the R&S SMBV100B user manual.

Defining the clock
 The provided clock signals are not dedicated to a particular connector. They can be mapped to one or more User x connectors.
 Use the [Global Connector Settings](#) to configure the signal mapping, the polarity, the trigger threshold, and the input impedance of the input connectors.
 To route and enable a trigger signal, perform the following *general steps*:

- Define the signal source, that is select the "Clock > Source".
- Define the connector where the selected signal is provided.
 Use the [Global Connector Settings](#).

Settings:

Clock Source.....

162

Clock Source

Selects the clock source.

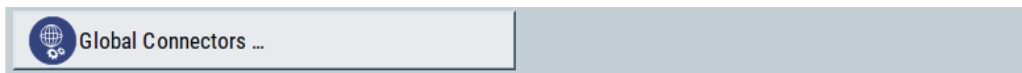
- "Internal"
- The instrument uses its internal clock reference.

Remote command:

[:SOURce<hw>] :BB:NR5G:CLOCK:SOURce on page 331

6.4 Global Connector Settings

Each of the "Trigger In", "Marker" and "Clock" dialogs as well as the "Trigger Marker Clock" dialog provides a quick access to the related connector settings.

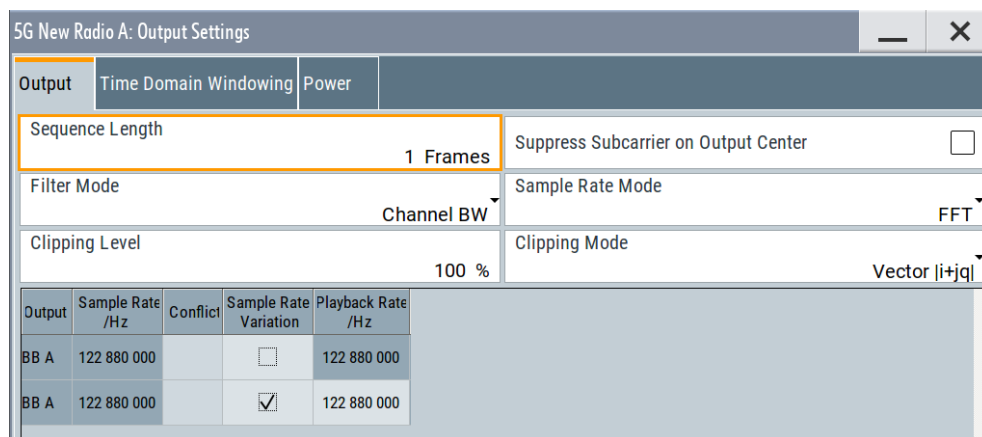


See also chapter "Global Connector Settings" in the user manual.

6.5 Output Settings

Access:

- Select "5G New Radio" > "Output/Power" > "Output".



This dialog comprises the settings required for configuring the output power.

Settings:

Sequence Length.....	164
Suppress Subcarrier on Output Center.....	164
Filter Mode.....	164
Load User Filter.....	164
Clipping Level.....	165
Clipping Mode.....	165
Sample Rate Mode.....	165
Output.....	165
Sample Rate.....	166

Conflict.....	166
Sample Rate Variation.....	166
Playback Rate.....	166

Sequence Length

Sets the sequence length of the signal in number of frames. The signal is calculated in advance and output in the arbitrary waveform generator.

Remote command:

[:SOURce<hw>] :BB:NR5G:OUTPut:SEQLen on page 320

Suppress Subcarrier on Output Center

If enabled, the subcarriers that use the same frequency as the center frequency of the baseband output are not transmitted.

The suppressed subcarrier is not necessarily the center of a carrier (channel bandwidth), as it is if "Node > Carrier Mapping" > [Delta f to Output](#) ≠ 0 MHz is used to shift the center carrier relative to the output center. Not considered are center carrier shifts caused by baseband offsets applied by parameters like offsets in the "I/Q Stream Mapper" or "Baseband Offsets".

Remote command:

[:SOURce<hw>] :BB:NR5G:OUTPut:SSOC:STATe on page 320

Filter Mode

Enables filtering and sets how the filter is applied:

- | | |
|--------------|---|
| "Channel BW" | On the whole channel bandwidth. |
| "Per BWP" | By filtering the individual BWPs separately. |
| "Off" | Disables filtering. |
| "Fast" | Applies fast filtering with reduced number of filter taps.
Supported in the following configuration: <ul style="list-style-type: none"> • "Node > Carriers > Number of Carriers = 1" and "Node > Carrier Mapping > Δf to Output = 0". |
| "User" | You can load you custom filter, see "Load User Filter" on page 164. |

Each of these filters is designed for different application field and optimized for a particular performance.

Depending on the filter implementation, these filters require different calculation time:

- "Channel BW" and "Per BWP": high-quality filters, providing good EVM and ACP but requiring long calculation time
- "Fast": short calculation time combined with real-time processing, good EVM performance while ignoring the effects on ACP

Remote command:

[:SOURce<hw>] :BB:NR5G:OUTPut:FMODE on page 320

Load User Filter

For [Filter Mode](#) > "User", it opens the standard dialog "Select List File User Filter" for loading a user-defined filter file.

User filters are used as offline filters. The following types are supported:

- Files with predefined file format and extensions *.vaf

For information, refer to the description "Introduction to "filtwiz" Filter Editor" on the Rohde & Schwarz web page.

- ASCII files with simple format and file extension *.dat
These files describe filters as a sequence of normalized filter coefficients. Each coefficient is defined as a pair of I and Q samples. The I and Q components alternate at each file line. The I and Q values vary between - 1 and + 1.
A user filter can contain up to 2560 coefficients.
The user filter must be real-valued. For both I and Q components of the coefficients, only real coefficients different than 0 are allowed.

Remote command:

[\[:SOURce<hw>\]:BB:NR5G:FMODE:USRFile](#) on page 321

Clipping Level

Sets the limit for clipping.

This value indicates at what point the signal is clipped. It is specified as a percentage, relative to the highest level. 100% indicates that clipping does not take place.

Remote command:

[\[:SOURce<hw>\]:BB:NR5G:OUTPut:CLeVel](#) on page 322

Clipping Mode

Selects the clipping method. The dialog displays a graphical illustration on how this two methods work.

- "Vector $|i + jq|$ "
The limit is related to the amplitude $|i + q|$. The I and Q components are mapped together, the angle is retained.
- "Scalar $|i|, |q|$ "
The limit is related to the absolute maximum of all the I and Q values $|i| + |q|$.
The I and Q components are mapped separately, the angle changes.

Remote command:

[\[:SOURce<hw>\]:BB:NR5G:OUTPut:CMODE](#) on page 323

Sample Rate Mode

Sets how the sample rate is defined.

- | | |
|-----------|---|
| "FFT" | Sample rate is derived from the used FFT. |
| "Minimum" | Available if "Filter Mode" ≠ "Fast".
Sample rate is set to the minimum possible value.
This mode is set automatically in the following cases: <ul style="list-style-type: none"> • If more than two carriers are enabled |

Remote command:

[\[:SOURce<hw>\]:BB:NR5G:OUTPut:SAMRmode](#) on page 321

Output

Indicates the baseband (BB) output.

Sample Rate

Indicates the sample rate value, calculated according to the selected [Sample Rate Mode](#), configured signal contents and several other settings, like the "Channel Bandwidth" or "Deployment".

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:BBConf:ROW<ch0>:SRATe?` on page 321

Conflict

A conflict is indicated if "Sample Rate Variation" is disabled but the "Playback Rate" value is restricted to a value different from "Sample Rate" because of limitations of the instrument's capabilities (installed options).

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:BBConf:ROW<ch0>:CONFLict` on page 322

Sample Rate Variation

Activates sample rate variation and allows you to change the [Playback Rate](#) manually.

Enabling "Sample Rate Variation" is useful for chipset testing, where the chipset runs with a modified clock rate for test reasons.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:BBConf:ROW<ch0>:VARiation` on page 322

Playback Rate

Indicates the playback **speed**.

Per default, the "Playback Rate = Sample Rate" but the "Playback Rate" value is also limited by the instrument's capabilities (installed options).

If the "Sample Rate" and the "Playback Rate" are different, a conflict is indicated.

Changing the "Playback Rate" does not do resampling. If you need a resampled signal, generate a waveform and resample it, for example, with software like ARB Toolbox.

Remote command:

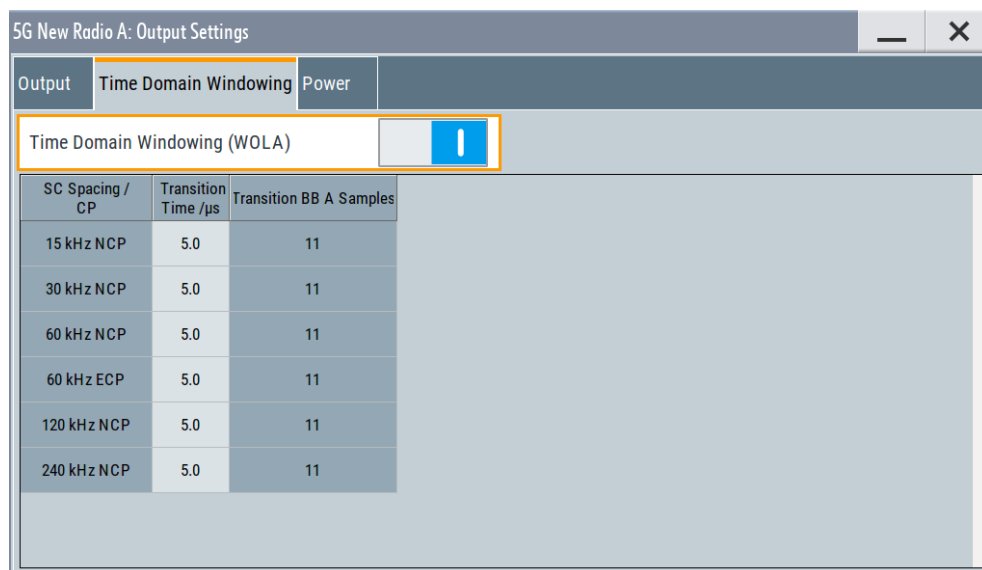
`[ :SOURce<hw> ] :BB:NR5G:OUTPut:BBConf:ROW<ch0>:SRATe?` on page 321

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:BBConf:ROW<ch0>:PBRate` on page 322

6.6 Time Domain Windowing Settings

Access:

- Select "5G New Radio" > "Output/Power" > "Time Domain Windowing".



This dialog comprises the settings required for configuring the time domain windowing. You can set different transition time for each of the specified numerologies.

Settings:

Time Domain Windowing (WOLA).....	167
SC Spacing/CP.....	167
Transition Time.....	167
Transition BB A Samples.....	168

Time Domain Windowing (WOLA)

Activates/deactivates the time domain windowing.

Time domain windowing is a method that influences the spectral characteristics of the signal. The method removes the spikes caused by the OFDM; it does not replace oversampling and subsequent signal filtering.

The abbreviation WOLA stands for weighted-overlap-and-add.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:TDWind:STATe` on page 323

SC Spacing/CP

Indicates all allowed combinations of the subcarrier spacing (SSC) and the cyclic prefix (CP), see [Table 3-2](#).

Transition Time

Sets the transition time when time domain windowing is active.

The transition time defines the overlap range of two OFDM symbols. You can define different transition times for each of the numerologies ([SC Spacing/CP](#)).

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:TDWind:S240K:TRTime` on page 323

etc.

Transition BB A Samples

Indicates the number of samples during the [Transition Time](#).

Remote command:

[\[:SOURce<hw>\]:BB:NR5G:OUTPut:TDWind:S240K:TRTSamples?](#) on page 324 etc.

6.7 Adjusting the Signal Power

In the R&S SMBV100B, you can adjust the power level of the generated signal in different ways.

This section explains the different power calculation modes and outlines the signal settings influencing the relations between the power levels of the channels and signals in the signal itself.

Overview of the power-related settings

The following settings influence the power level of the generated signal:

- Output level (P_{out}) of the instrument
To adjust the value, select "Status bar > Level".
- Selected power calculation method
To define it, select "General > Output/Power > Power > Power Mode"
- Selected relative signal and channel power values
 - "5G New Radio > General > Node > SS/PBCH" > ["PSS/SSS Power"](#) on page 42
 - "5G New Radio > General > Node > SS/PBCH" > ["PBCH Power"](#) on page 42
 - "5G New Radio > General > Node > Dummy REs" > ["Power"](#) on page 43
 - "Scheduling > User x BWP x > Content > PDSCH/PUSCH > Settings > Config" > ["Power"](#) on page 117

General principles

By selecting the power calculation method ("Power Mode"), you define the **power level at the first instrument's output**. The signal power level of the other outputs is calculated automatically according to the current power relations of all channels and signals.

Relative vs absolute power

All the relative channel and signal power values are set **relative** to each other.

Example:

If "PSS Power = 2 dB" and "SSS Power = -3 dB", then the power level of the SSS subcarriers is with 5 dB higher than the power level of the PSS subcarriers.

The **absolute power level** of one resource element depends on the selected "Power Mode" and on the configuration during the remaining subframes.

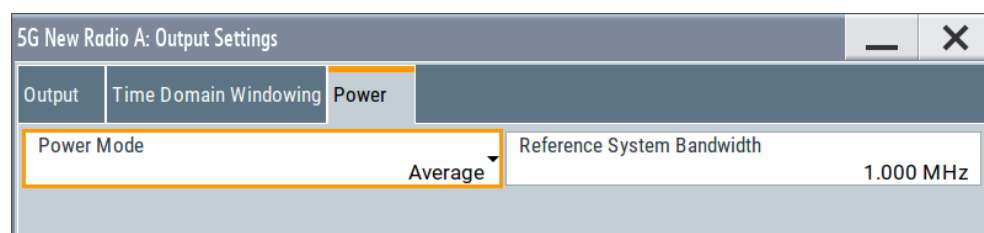
Power mode and the different power calculation methods

The value displayed in the "Status bar > Level" is the current RMS level at the output. The RMS and PEP values are calculated different, depending on the selected "5G New Radio" > "Output/Power" > "Power" > **Power Mode**.

6.7.1 Power Settings

Access:

- Select "5G New Radio" > "Output/Power" > "Power".



This dialog comprises the settings required for configuring the output power.

Settings:

Power Mode.....	169
Reference System Bandwidth.....	173

Power Mode

Defines which method is used to calculate the signal power at the first instrument output. The signal power at the other outputs is calculated according to the current power relations of all channels and signals.

The different power modes are illustrated for the two signals, shown on [Figure 6-1](#).

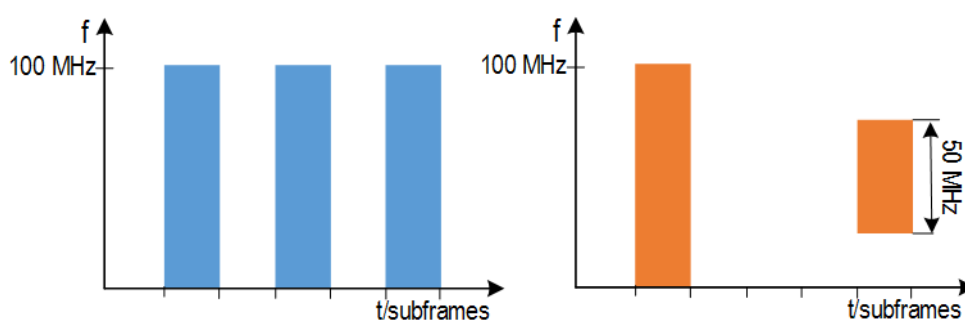


Figure 6-1: Example: time plans of 2 signals with different allocated subframes and channel bandwidth per allocation

left = Every second subframe is allocated; for all subframes, channel bandwidth BW = 100 MHz
 right = 2 allocated subframes, channel bandwidth of the first allocation $BW_1 = 100$ MHz and of the second - $BW_2 = 50$ MHz

"Average"

The power level set with the parameter "Status Bar > Level" is the average **RMS power of the complete signal**.

The absolute power of a subframe depends on the configuration of the remaining signal. In fully allocated frames, the average RMS value applies for all subframes. If the frames are partly allocated, the absolute RMS power within an allocated subframe is greater than the displayed RMS value ("Status Bar > Level").

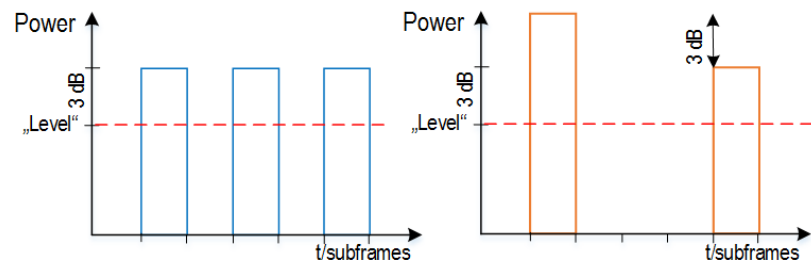


Figure 6-2: Signal power level for Power Mode = Average

Level = "Status Bar > Level"

Example:

If "Status Bar > Level" = -10 dBm, the RMS power measured with a power sensor in both cases is:

- $\text{RMS}_1 = \text{RMS}_2 = -10 \text{ dBm}$

"Average Active Subframes"

The power level set with the parameter "Status Bar > Level" is the **average power in all allocated subframes**.

Thus, the absolute power of the allocated subframe depends only on the configuration of the remaining allocated signal.

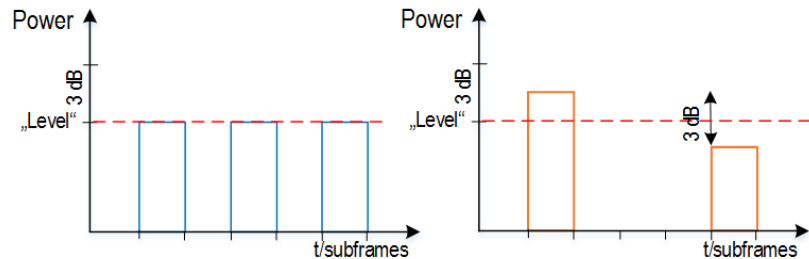


Figure 6-3: Signal power level for Power Mode = Average Active Subframes

Level = "Status Bar > Level"

Example:

If "Status Bar > Level" = -10 dBm, the RMS power measured with a power sensor in both cases is:

- $\text{RMS}_1 = -13 \text{ dBm}$
- $\text{RMS}_2 = -14.5 \text{ dBm}$

This mode is usefully if a specific SNR is required for a particular channel but this channel is not transmitted in every subframe.

"Constant PSD"

The power level set with the parameter "Status Bar > Level" is the **power of a fully allocated, fictitious signal with a specific bandwidth (Reference System Bandwidth)**. The power spectral density (PSD) of this fictitious signal is equal to the PSD of the current signal. In this mode, you configure a specific power spectral density per resource element rather than the RMS of the complete signal. The absolute power of a particular allocation is then calculated by multiplying the configured PSD with the bandwidth of the allocation.

Thus, for a given "Reference System Bandwidth", the PSD and the signal power for a particular bandwidth (SignalPower_{BW}) are calculated as follows:

- $\text{PSD} = \text{"Status Bar > Level"} / \text{"Reference System Bandwidth"}$
- $\text{SignalPower}_{\text{BW}} = \text{PSD} * \text{BW}$, where BW is the channel bandwidth.

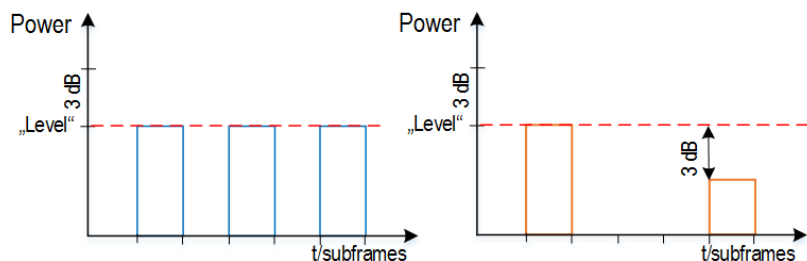
Example:

Figure 6-4: Signal power level for Power Mode = Constant PSD

Level = "Status Bar > Level"

For "Reference System Bandwidth" = 100 MHz, the signal power of a signal with 100 MHz bandwidth (BW = 100 MHz) is:

$$\text{SignalPower}_{100 \text{ MHz}} = (\text{"Status Bar > Level"} / 100 \text{ MHz}) * 100 \text{ MHz} = \text{"Status Bar > Level"}$$

The signal power of a signal with 50 MHz bandwidth (BW = 50 MHz) is then:

$$\text{SignalPower}_{50 \text{ MHz}} = (\text{"Status Bar > Level"} / 100 \text{ MHz}) * 50 \text{ MHz} = \text{"Status Bar > Level"} / 2 \text{ or}$$

$$\text{SignalPower}_{50 \text{ MHz}} = \text{"Status Bar > Level"} - 3 \text{ dB}$$

If "Status Bar > Level" = -10 dBm, the RMS power measured with a power sensor in both cases is:

- $\text{RMS}_1 = -13 \text{ dBm}$
- $\text{RMS}_2 = -16 \text{ dBm}$

This mode is useful if you need to configure an absolute SNR. To achieve an absolute SNR, set the "AWGN > System Bandwidth" value to be equal to the "5G New Radio > Output/Power > Power > Reference System Bandwidth".

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:POWer:MODE` on page 324

Reference System Bandwidth

In "Power Mode > Constant PSD", sets the system bandwidth used as reference for the calculation of the output power.

Remote command:

`[ :SOURce<hw> ] :BB:NR5G:OUTPut:POWer:RSBW` on page 325

7 Observing Current Allocations On the Time Plan

You can observe the current allocations on the time plan.

The time plan shows active channels and signals, the allocations of the active and selected UEs and indicates the cell it applies for. The time plan shows also an overview of the configured BWPs. It indicates conflicting allocations, too.

Per default, the time plan shows the allocation per used channel bandwidth and one frame but you can change the displayed time region to up to 20 subframes. You can also scroll over all available subframes.

Access:

- In the "General" dialog, select "Time Plan".

Allocations are calculated as configured in the "Scheduling" dialog and depend on the "Link Direction" and configured BWPs.

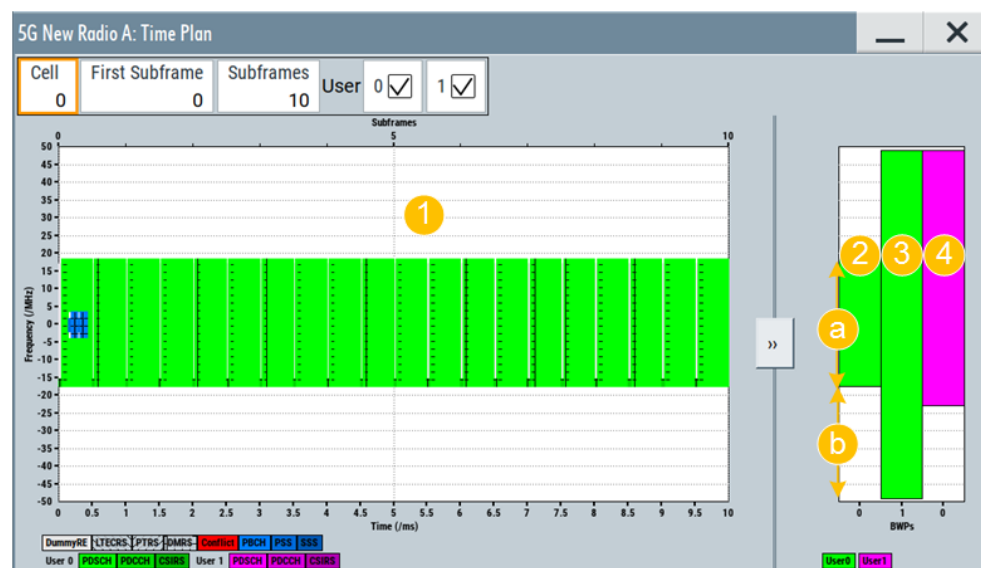


Figure 7-1: Time Plan: understanding the displayed information

- 1 = User 1, BWP 1 (displayed are the SS/PBCH blocks, the fully allocated carrier with dummy REs and the PDSCH/CORESET)
- 2 = User 1, BWP 1 ("No. RBs = 100", "RB Offset in TxBW = 88")
- 3 = User 1, BWP 2 ("No. RBs = 273", "RB Offset in TxBW = 0")
- 4 = User 2, BWP 1 ("No. RBs = 200", "RB Offset in TxBW = 73")
- a = "No. RBs"
- b = "RB Offset in TxBW" - illustrated as principle (because the time plan shows the entire channel bandwidth and the different Tx BWs use different SCSs, the RB offset in TxBW is not given as an offset relative to the lower edge of the channel bandwidth)

Settings:

Cell.....	175
Show.....	175
First Subframe.....	175
Subframe.....	175
User.....	175

Cell

Selects/indicates to which carrier/cell the settings apply.

Remote command:

via suffix at `CELL<ch>`

Show

Defines which portion of the carrier bandwidth is displayed:

"Carrier" Displayed is the entire carrier bandwidth.

Remote command:

n.a.

First Subframe

Selects the first subframe to be displayed.

Remote command:

n.a.

Subframe

Selects the number of subframes to be displayed.

Change the value to zoom in and out in the time plan. A maximum of 20 subframes can be displayed, where more than 10 Subframe are allowed if the "Output > Sequence Length ≥ 2 ".

Remote command:

n.a.

User

Enable a User to display its allocations on the time plan.

Remote command:

n.a.

8 Remote-Control Commands

The following commands are required to perform signal generation with the option R&S SMBV100B-K144 in a remote environment. We assume that the R&S SMBV100B has already been set up for remote operation in a network as described in the R&S SMBV100B documentation. A knowledge about the remote control operation and the SCPI command syntax are assumed.



Conventions used in SCPI command descriptions

For a description of the conventions used in the remote command descriptions, see section "Remote Control Commands" in the R&S SMBV100B user manual.

Common Suffixes

The following common suffixes are used in the remote commands:

Suffix	Value range	Description
ENTity<ch>	1	Optional keyword, provided for compatibility with R&S®SMW200A ENTity1:SOURce1 = SOURce1
SOURce<hw>	1	Available baseband signals
OUTPut<ch>	1 to 3	Available markers
USER<ch>	0 to 9	Available users
ALLoc<ch0>	0 to 15	Number of allocations per user/BWP section
SUBF<st0>	0 to 100	Configurable subframes
CW<s2us0>	0 to 1	Codeword
SET<gr0>	0 to 15	SRS resource sets
RES<user0>	0 to 63	SRS resources within a resource set
TD<gr>	0 to 64	PDSCH time domain allocations

The following commands specific to the R&S SMBV100B-K144 option are described here:

• Programming Examples	177
• General Commands	195
• Carrier Commands	199
• TxBW Commands	202
• LTE-CRS Coexistence Commands	204
• SS/PBCH Commands	206
• Dummy Resource Elements Commands	214
• Carrier Mapping Commands	216
• User Commands	218
• DL/UL BWP Commands	220
• BWP Configuration Commands	222

• Scheduling Commands.....	268
• PDSCH/PUSCH Commands.....	277
• CORESET Commands.....	292
• PUCCH Commands.....	312
• PRACH Commands.....	317
• Output Commands.....	319
• Trigger Commands.....	325
• Marker Commands.....	329
• Clock Commands.....	331

8.1 Programming Examples

This description provides simple programming examples. The purpose of the examples is to present **all** commands for a given task. In real applications, one would rather reduce the examples to an appropriate subset of commands.

The programming examples have been tested with a software tool which provides an environment for the development and execution of remote tests. To keep the example as simple as possible, only the "clean" SCPI syntax elements are reported. Non-executable command lines (e.g. comments) start with two // characters.

At the beginning of the most remote control program, an instrument (p)reset is recommended to set the instrument to a definite state. The commands ***RST** and **SYSTem:PRESet** are equivalent for this purpose. ***CLS** also resets the status registers and clears the output buffer.

Example: Storing current configuration

```
SOURcel:BB:NR5G:SETTing:STORe "/var/user/my_nr"
*RST
SOURcel:BB:NR5G:SETTing:CATalog?
// my_nr, 5gnr
SOURcel:BB:NR5G:SETTing:LOAD "/var/user/5gnr"
SOURcel:BB:NR5G:STATe 1
SOURcel:BB:NR5G:NODE:NCARrier?
// 1
SOURcel:BB:NR5G:SETTing:DEL "my_nr"

SOURcel:BB:NR5G:WAVEform:CREate "/var/user/my_nr_wv"
```

Example: Loading predefined test signals

```
SOURcel:BB:NR5G:LINK DOWN
// SOURcel:BB:NR5G:SETting:TMOdel:FILTer:TMOdel TM1_1
// SOURcel:BB:NR5G:SETting:TMOdel:FILTer:DUPLexing FDD
// SOURcel:BB:NR5G:SETting:TMOdel:FILTer:FREQ FR1
// SOURcel:BB:NR5G:SETting:TMOdel:FILTer:BW F20
// SOURcel:BB:NR5G:SETting:TMOdel:FILTer:SCS 30

SOURcel:BB:NR5G:SETting:TMOdel:CATalog?
// Example 100MHz Full 30kHz QPSK Sub6GHz,
// Example 100MHz TwoBWP 60kHz 120kHz QPSK mmWave,
// Example 400MHz Full 120kHz 256QAM mmWave,
// ...
SOURcel:BB:NR5G:SETting:TMOdel:DL "Example 100MHz Full 30kHz QPSK Sub6GHz"

SOURcel:BB:NR5G:STATe 1
SOURcel:BB:NR5G:PRESet
```

Example: Configuring the carrier

```
SOURcel:BB:NR5G:NODE:NCARrier 1
SOURcel:BB:NR5G:NODE:CELL0:CINDicator 1
SOURcel:BB:NR5G:NODE:CELL0:CELLId 1
SOURcel:BB:NR5G:NODE:CELL0:N1ID?
// 0
SOURcel:BB:NR5G:NODE:CELL0:N2ID?
// 1
SOURcel:BB:NR5G:NODE:CELL0:CARDepl LT3
SOURcel:BB:NR5G:NODE:CELL0:CBW BW50
SOURcel:BB:NR5G:NODE:CELL0:TAPos 2
SOURcel:BB:NR5G:NODE:CELL0:SUL 0
```

Example: Configuring the TxBWs

```
// enable 15 kHz SCS
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S15K:USE 1
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S15K:NRB?
// 270
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S15K:OFFSet 0
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S15K:KOMU?
// -1700

// enable 30 kHz SCS
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S30K:USE 1
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S30K:NRB?
// 133
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S30K:OFFSet?
// 0
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S30K:KOMU?
// -840
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:POINTa?
// -49140000
// both k_0_u values are not allowed values

SOURCE1:BB:NR5G:NODE:CELL0:TXBW:CONFLICT?
// 1
// execute resolve conflicts to obtain valid values
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:RESolve
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S30K:OFFSet?
// 1
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S30K:KOMU?
// 0
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:POINTa?
// -24300000
```

Example: LTE-CRS Coexistence

```
SOURCE1:BB:NR5G:NODE:CELL0:CBW BW50
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S15K:USE 1
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:RESolve

SOURCE1:BB:NR5G:NODE:CELL0:LTE:STATe 1
SOURCE1:BB:NR5G:NODE:CELL0:LTE:POINTa 100
SOURCE1:BB:NR5G:NODE:CELL0:LTE:CBW N25
SOURCE1:BB:NR5G:NODE:CELL0:LTE:VSHift 2
SOURCE1:BB:NR5G:NODE:CELL0:LTE:NAP AP2
```

Example: Configuring the SS/PBCH

```

SOURCE1:BB:NR5G:NODE:CELL0:NSSPBch 1
SOURCE1:BB:NR5G:NODE:CELL0:OFFSet TXBW
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:STAtE 1

SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:SCSPacing N15
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:RBOffset 125
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:SCOffset 0
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:DFReq?
// -540000
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:CASe A
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:L L4
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:POStion #H5,4
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:BSPeriodicty BS10
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:PSSPow 0
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:SSSPow 0

// PBCH configuration
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:POWeR 0
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:STAtE 1
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:CCODing:STAtE 1
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:CCODing:DATA PN9

SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:STAtE 0
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:SCSC N15_60
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:ASOF:STAtE 1
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:SCOffset?
// 0
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:SFOffset 0
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:CSZero 1
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:SSZero 1
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:CBARred NBAR
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:IFRResel ALWD
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:SPARe:STAtE 1
SOURCE1:BB:NR5G:NODE:CELL0:TAPos?
// 2
SOURCE1:BB:NR5G:NODE:CELL0:SSPBch0:MIB:SCOffset?
// 0

```

Example: Filling in the frame with dummy REs

```

SOURCE1:BB:NR5G:NODE:CELL0:DUMRes:STAtE 1
SOURCE1:BB:NR5G:NODE:CELL0:DUMRes:POWeR 0
SOURCE1:BB:NR5G:NODE:CELL0:DUMRes:MODulation QPSK
SOURCE1:BB:NR5G:NODE:CELL0:DUMRes:SCSPacing N15

// load a data list from the default directory; the file must exist
SOURCE1:BB:NR5G:NODE:CELL0:DUMRes:DATA DLIS
SOURCE1:BB:NR5G:NODE:CELL0:DUMRes:DSElect "/var/user/dummyRE"

```

Example: Mapping the carriers to the outputs

```
SOURcel:BB:NR5G:NODE:NCARrier 1
SOURcel:BB:NR5G:NODE:CARMapping:CELL:ROW0 0
SOURcel:BB:NR5G:NODE:CARMapping:MAPPed:ROW0 1
SOURcel:BB:NR5G:NODE:CARMapping:AP4000:ROW0 1
SOURcel:BB:NR5G:NODE:CARMapping:DFReq:ROW0 0
```

Example: Configuring the users and the BWPs

```
// enable 2 users
SOURcel:BB:NR5G:UBWP:NUSer 2

SOURcel:BB:NR5G:UBWP:USER0:UEID 10
SOURcel:BB:NR5G:UBWP:USER0:DSCH:CCODing:STAtE 0
SOURcel:BB:NR5G:UBWP:USER0:DSCH:DATA PN9
// SOURcel:BB:NR5G:UBWP:USER0:DSCH:DATA PATT
// SOURcel:BB:NR5G:UBWP:USER0:DSCH:PATtern #H1E15,14
SOURcel:BB:NR5G:UBWP:USER1:UEID 20
SOURcel:BB:NR5G:UBWP:USER1:DSCH:DATA DLIS
SOURcel:BB:NR5G:UBWP:USER1:DSCH:DLISt "/var/user/user"

// configure the BWPs per user
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:NBWParts 1
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:INDicator 1
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:SCSPacing N30
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RBNumber 100
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RBOffset 10
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PRBOffset?
// 10
SOURcel:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:DFReq?
// -2340000
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:NBWParts 1
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:INDicator 0
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SCSPacing N30
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:RBNumber 11
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:RBOffset 11
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:DFReq?
// -18000000

SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:NBWParts 2
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP0:INDicator 1
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP0:RBNumber 110
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP0:RBOffset 20
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP0:DFReq?
// 3060000
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP1:INDicator 2
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP1:RBNumber 120
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP1:RBOffset 10
SOURcel:BB:NR5G:UBWP:USER1:CELL0:DL:BWP1:DFReq?
// 1260000
```

Example: Fixed reference channels

```

SOURCE1:BB:NR5G:LINK UP
SOURCE1:BB:NR5G:UBWP:USER0:USCH:CCODing:StAtE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:StAtE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:TYpE FR1A15
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:SCS?
// N30
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:ALRB?
// 51
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:MODulation?
// QPSK
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:PASize?
// 4352
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:RBOffset 10

SOURCE1:BB:NR5G:NODE:CELL0:CARDeplY GT6
SOURCE1:BB:NR5G:UBWP:USER0:USCH:CCODing:StAtE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:StAtE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:TYpE FR2A58
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:FRC:PTRS:StAtE?
// 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:MAPType?
// B

```

Example: Configuring the configurable allocations in the users and BWPs groups

```

SOURCE1:BB:NR5G:LINK DOWN
SOURCE1:BB:NR5G:SCHeDuling:MODE MANual

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:NALLoc 1
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:CONTEnt PDSCh
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:SLOT 1

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:SYMNumber 14
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:SYMOffset 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:RBNumber 20
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:RBOffset 100
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:POWEr 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:StAtE 1
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart0:ALLoc0:REPetitions SUBF

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:NALLoc 1
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CONTEnt COR
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:RBNumber 30
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:RBOffset 100
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:REPetitions FRAM

```

Example: Configuring the PDSCH incl. PTRS settings

```

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:CTyPe T1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:APIndex 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:MLENgtH 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:SID0 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:SID1 20
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTB:CTyPe T2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTB:APIndex 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTB:MLENgtH 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTB:SID0 100
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTB:SID1 200

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:SCRambling:STATe 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DSID 100
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:MCWDci 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:MCSTable QAM256
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:RESalloc?
// T1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:VPINter?
// VPN
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:RESalloc T0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:RBGSize C1

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:STATe 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:REOF RE01
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:EPRE RAT1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:MCS1 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:MCS2 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:MCS3 3
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:RB0 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTA:PTRS:RB1 100

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:DMTB:PTRS:STATe 0

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:CONTent PDSC
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:SLOT 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:MAPType A
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:SYMNumber 7
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:SYMoffset 1
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:RBNumber 273
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:REPetitions SLOT

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PDSch:TYPE?
// F11
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PDSch:NCW?
// 1
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PDSch:PATGrp N
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:CW0:MOD QAM16
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:CW0:PHYsbits?
// 61488

```

```

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:DMRS:SEQGen CELL
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:DMRS:NSID 0

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:DMRS:LENGth 2
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:DMRS:SLTSymbols?
// "2,3"
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:DMRS:POWer 0
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:TXScheme:NLAYer?
// 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:TXScheme:CDMData 2
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:TXScheme:INTERvp?
// 0

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:APSelect0?
// AP1000
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:APMap:MAPCoordinates CART
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:APMap:AP0:ROW0:REAL 0.7
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:APMap:AP0:ROW0:IMAGinary 0.7

SOURCE1:BB:NR5G:UBWP:USER0:USCH:CCODing:STATE 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:CW0:PDSch:CCODing:IMCS 5
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:CW0:PDSch:CCODing:TCRate?
// 0.369140625
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:CW0:PDSch:CCODing:TBSIZE?
// 22536
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:CW0:PDSch:CCODing:RVIndex 1

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:PTRS:MODE MAN
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:PTRS:STATE 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:PTRS:TMDen TD1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:PTRS:REOF RE01
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:PTRS:FRQDen FD2
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLOcl:PDSch:PTRS:EPRE RAT1

```

Example: Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list

```

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TDANum 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD0:KNULl 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD0:MAPPing A
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD0:START 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD0:LENGth 3
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD0:SLIV?
// 28
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD1:KNULl 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD1:MAPPing B
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD1:START 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD1:LENGth 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSch:TD1:SLIV?
// 14

```


Example: Configuring the CSI-RS settings

```
// periodic ZP CSI-RS
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:STATE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:NRESources 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:PER 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:OFFSet 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:RBNumber 273
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:SRBNumber 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:ROW 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:BITMap #HAF,12
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:IO 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:ZP:SET0:RES0:I1 2

// periodic NZP CSI-RS
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:STATE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:NRESources 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:PER 4
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:OFFSet 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:RBNumber 100
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:SRBNumber 50
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:ROW 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:BITMap #HF,4
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:IO 0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:I1 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:SCID 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:PWR 3

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:APMap:MAPCoord CARTesian
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:APMap:AP3000:ROW0:REAL 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:APMap:AP3000:ROW0:IMAGinary 0.7
// SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:APMap:MAPCoord CYLindrical
// SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:APMap:AP3000:ROW0:PHASe 270
// SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:CSIRs:NZP:SET0:RES0:APMap:AP3000:ROW0:MAGNitude 0
```

Example: Antenna port settings

```
SOURCE1:BB:NR5G:LINK DOWN
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPPart0:ALLoc1:APMap:MAPCoordinates CART
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPPart0:ALLoc1:APMap:COL0:ROW0:REAL 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPPart0:ALLoc1:APMap:COL0:ROW0:IMAGinary?
// 0

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPPart0:ALLoc1:APMap:MAPCoordinates CYL
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPPart0:ALLoc1:APMap:COL0:ROW0:MAGNitude 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPPart0:ALLoc1:APMap:COL0:ROW0:PHASe 360
```

Example: Configuring one SRS resource set

```

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:NRSets 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RSType PER
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:TXConfig NCB
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:USAGe?
// NCB

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:NRESources 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:PER SL10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:OFFSet 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:SPOS 3
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:SYMNumber SYM2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:FPOS 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:FQShift 20
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:BSRS 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:CSRS 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:BHOP 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:REFactor?
// REP1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:TRTComb TC4
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:COFFset 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:SEQ:CYCShift 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:SEQ:ID 10
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:POWEr 3
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:NAPort AP1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:PTRS P0
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:APMap:MAPCoord CYL
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:APMap:AP1000:ROW0:MAGNitude 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:APMap:AP1000:ROW0:PHASe 0
// SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:APMap:MAPCoord CART
// SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:APMap:AP1000:ROW0:REAL?
// SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:RES0:APMap:AP1000:ROW0:IMAGinary?

SOURCE1:BB:NR5G:LINK UP
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPpart0:ALLOc0:STATE 0
// open the timeplan and observe the SRS allocations

```

Example: Configuring the DL BWP rate match settings

```

SOURCE1:BB:NR5G:LINK DOWN

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:STATe 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:GRPNumber 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:NRESources 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS0:GRID G1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS0:SLTPatt #H1001,14
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS0:PER 2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS0:PERPatt #H2,2
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS0:RBDList "/var/user/dl"
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS0:RBPatt?

```

```

SOURce1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS1:SLOT 2
SOURce1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS1:PER 4
SOURce1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS1:PERPatt #H5,4
SOURce1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:RATM:RS1:SLTPatt #H8888888,28

SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc1:PDSCH:PATGrp G1

```

Example: Search space configuration

For example on how the CORESET is configured, see [Example "Configuring the CORESET settings"](#) on page 187.

```

SOURce1:BB:NR5G:SCHeDuling:MODE MAN
SOURce1:BB:NR5G:SCHeDuling:RSSpace 1

SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe0:AGGLeveL?
// AL1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe0:MAXCandidate 2
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe1:AGGLeveL?
// AL2
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe1:MAXCandidate 2
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe2:AGGLeveL?
// AL4
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe2:MAXCandidate 4
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe3:AGGLeveL?
// AL8
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe3:MAXCandidate 6
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe4:AGGLeveL?
// AL16
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SPACe4:MAXCandidate 8

SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DCI0:AGGLeveL AL1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DCI0:CANDidate 1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWP1:ALLOc0:CS:DCI0:INDex?
// 22

```

Example: Configuring the CORESET settings

For example on how the search space is configured, see [Example "Search space configuration"](#) on page 187.

```

SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CONTent COR

SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:PREGran REG
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:ID 0
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SCRam:STATe 1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWP0:ALLOc0:CS:DMRS:SCRam:ID 0

SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWP1:ALLOc0:CS:IL:STATe 1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWP1:ALLOc0:CS:IL:BUNSize 6

```

```

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:IL:SHIDx 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:IL:SIZE IS6

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCCes:StAtE 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCCes:DATA DLIS
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCCes:DLISt "/var/user/dl"
// SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCCes:DATA PATtern
// SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCCes:PATtern #H9,4

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:DMRS:SCRam:StAtE 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:DMRS:SCRam:ID 10
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:AULBwp?

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:NDCI 5

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:USAGe?
// C
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:RNTI?
// 0
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:FMT F00
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:SSP USS
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:AGGLeveL AL1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:CANDidate 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:DCI0:INDeX?
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:FRDRes 0
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:FRHFlag 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:TB1:MCS 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:TB1:NDI 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:TB1:RV 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:HAPRoc 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:TPUSch 2
SOURcel:BB:NR5G:NODE:CELL0:SUL:StAtE?
// 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:USINd 0
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:HBTS?
// #H00000434600,44

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:USAGe?
// C
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:RNTI?
// 0
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:DCI1:AGGLeveL AL2
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPart1:ALLoc0:CS:DCI1:INDeX 0
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:FMT F10
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:SSP CSS3
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:FRDRes 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:TIDRes 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:VTPRb?
// 0
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:TB1:MCS 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI1:TB1:NDI 1

```

```

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:TB1:RV 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:HAPRoc 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:DAI1 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:TPUCch 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:PUCResind 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:PDSHarq 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:HBTS?
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:CPDSch 1

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI2:USAGe?
// C
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI2:FMT F01
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI2:ANTPorts 15
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI2:SRSRequest 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:TPStAte 0
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI2:DMSQinit 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:MRANK 2
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI2:PTDMrs 1
...

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:USAGe SI
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:RNTI?
// 0
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:FMT F10
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:SSP CSS0
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:AGGLeveL AL1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:CANDidate 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER1:BWP0:ALLoc0:CS:DCI3:INDeX?
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:FRDRes 0
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:TIDRes 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI1:VTPRb 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:TB1:MCS 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:TB1:RV 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:SIINd 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:RESVed?
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:HBTS?
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI3:CPDSch 1

SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:FMT CUSTom
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:STATe 1
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:DATA PATTeRn
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:BITLength 12
SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:PATTeRn #HCA3,12
// SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:DATA DLISt
// SOURCE1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWP0:ALLoc0:CS:DCI5:DLISt "var/user/nr_dl"

```

Example: Configuring the PUSCH settings

```

SOURCE1:BB:NR5G:LINK UP

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:CTYPe T1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:APIIndex 1

```

```

SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:MLENgtH 2
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:SID0 0
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:SID1 0
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTB:CTYPe T2
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTB:APINdex 2
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTB:MLENgtH 1
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTB:SID0 10
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTB:SID1 10
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:TPStAtE 0
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:MRANk 2
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:TXConfig?
// CB
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:MCSTable QAM256
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:MTTPrecoding QAM256
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:CBSubset FPMC
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:SCRambling:STATe 1
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DSID 100

SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:STATe 1
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:MODE UCI
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:MODE UCLS
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:ALPHA A0_8
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:ALPHA A1_0
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:HARQ:OFF1 1
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:HARQ:OFF2 5
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:CSI:OF10 0
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:CSI:OF11 5
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:CSI:OF20 1
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:UCI:CSI:OF21 10

SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:SLOT 0
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:MAPType A
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:SYMNumber 14
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:RBNumber 273
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:POWer 0
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:REPetitions SLOT

SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUSCh:TYPE?
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:MOD QAM64
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:CW0:PHYsbits?
// 196560
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:DMRS:LENGth 2
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:DMRS:SLTSymbols?
// "2,3,10,11"
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:DMRS:POWer -3
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUSCh:TXScheme:NLAYer?
// 1
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:TXScheme:CDMData 2

SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:TXScheme:TPMidx 1

```

```
// enable at least one SRS resource set with at least one SRS resource
// SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:NRSets 1
// SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:SRS:RS:SET0:NRESources 1
// ... configure the SRS
// SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:TXSCHeM:SRI 1

SOURcel:BB:NR5G:UBWP:USER0:USCH:CCODing:STATe 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:CCODing:IMCS 11
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CW0:PUSCh:CCODing:TCRate?
// 0.455078125
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CW0:PUSCh:CCODing:TBSizE?
// 90176
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:CCODing:RVINdex 1

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:ACK:BITS 4
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:ACK:PATtern #HC,4
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:CSI1:BITS 5
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:CSI1:PATtern #H15,5
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:CSI2:PATtern #H0,1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:CSI2:BITS 3
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:UCI:CSI2:PATtern #H2,3
```

Example: Configuring the PUSCH PTRS with enabled transform precoding

```
SOURcel:BB:NR5G:LINK UP

SOURcel:BB:NR5G:UBWP:USER0:USCH:CCODing:STATe 1
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:TPSTate 1

SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:STATe 1
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:TMDensity TD2
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:RB0 1
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:RB1 270
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:RB2 275
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:RB3 276
SOURcel:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:DMTA:PTRS:TP:RB4?
// 276

SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CONTent PUSC
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:PTRS:TP:MODE MAN
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:PTRS:TP:STATe 1
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:PTRS:TP:TMDensity TD2
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:PTRS:TP:NGRPs G4
SOURcel:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:PUSCh:PTRS:TP:SPPG S4
```

Example: Configuring the PUCCH settings

```

SOURCE1:BB:NR5G:LINK UP

SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:ADMRs:STATe 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:BPSK:STATe 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:HACK:STATe?
// 0

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:NALLoc 2
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:CONTent PUCc
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:SLOT 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:FMT F4
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:SYMNumber 4
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:SYMoffset 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:RBNumber 10
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:RBOffset 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:POWer 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:REPetitions SUBF

SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:GRPHopping ENA
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:HOPid 100
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:ISFHopping 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:SHOPping 20
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:FS:OCCLength L4
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:FS:OCCindex 1
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCChPL:SRCcount?
// 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:PL:UCI:BITS 3
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:PL:UCI:PATtern #H5,3
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:PL:ACK:BITS?
// 0
SOURCE1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:PL:ACK:PATtern?
// #H0,0

```


Example: Configuring the PRACH settings

```

SOURCE1:BB:NR5G:LINK UP
SOURCE1:BB:NR5G:NODE:CELL0:CARDePLY GT6
SOURCE1:BB:NR5G:NODE:CELL0:TXBW:S120K:USE 1
SOURCE1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:SCSPACING N120

SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:NALLOC 2
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:CONTENT PRACH
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:SCSPACING N120
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:FORMAT FA1
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:RSET URES
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:RSEQUENCE 1
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:ZCZONE 1
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:PRINDEX 1
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:TOFFSET 0
SOURCE1:BB:NR5G:SCHED:CELL0:SUBF0:USER0:BWPART0:ALLOCL:PRACH:RBNUMBER?
// 12

```

Example: Time domain windowing

```

SOURCE1:BB:NR5G:TDWIND:STATE?
// 0
SOURCE1:BB:NR5G:OUTPut:TDWIND:TRTime:S15K:TRTime 5
...
SOURCE1:BB:NR5G:OUTPut:TDWIND:TRTime:S240K:TRTime 5
SOURCE1:BB:NR5G:OUTPut:TDWIND:TRTime:S240K:TRTSamples?
// 11

```

Example: Output settings

```

SOURCE1:BB:NR5G:OUTPut:SEQLen 10
SOURCE1:BB:NR5G:OUTPut:SSOC:STATE 0
SOURCE1:BB:NR5G:OUTPut:OUTPut:FMODE CBW
SOURCE1:BB:NR5G:OUTPut:SAMRmode FFT
SOURCE1:BB:NR5G:OUTPut:BBConf:ROW0:SRATE?
// 122880000
SOURCE1:BB:NR5G:OUTPut:BBConf:ROW0:CONFLICT?
// 0
SOURCE1:BB:NR5G:OUTPut:BBConf:ROW0:PBRate?
// 122880000
SOURCE1:BB:NR5G:OUTPut:BBConf:ROW0:VARIATION 0

SOURCE1:BB:NR5G:OUTPut:CLEVEL 80
SOURCE1:BB:NR5G:OUTPut:CMODE VECT

```

Example: Configure and enable triggering

```

SOURCE:BB:NR5G:TRIGger:SEQuence SINGLE
SOURCE:BB:NR5G:TRIGger:SLENgth 200

// the first 200 samples of the current waveform will be output after
// the next trigger event
// SOURCE:BB:NR5G:TRIGger:SEQuence ARETrigger
// SOURCE:BB:NR5G:TRIGger:SOURce EGT1
// external trigger signal must be provided at the User connector
// SOURCE:BB:NR5G:TRIGger:EXternal:SYNChronize:OUTPut ON
// SOURCE:BB:NR5G:TRIGger:EXternal:DELay 200
// SOURCE:BB:NR5G:TRIGger:EXternal:INHibit 100

SOURCE:BB:NR5G:TRIGger:SEQuence AAUTO
SOURCE:BB:NR5G:TRIGger:SOURce INTERNAL
SOURCE:BB:NR5G:STAT ON
SOURCE:BB:NR5G:TRIGger:EXEC

```

Example: Specifying delay and inhibit values in time units

```

SOURCE1:BB:NR5G:CLOCK 1000000
SOURCE1:BB:NR5G:TRIGger:SEQuence AAUT
SOURCE1:BB:NR5G:TRIGger:SOURce EGT1
SOURCE1:BB:NR5G:TRIGger:DELay:UNIT SAMP
SOURCE1:BB:NR5G:TRIGger:EXternal:DELay 100
SOURCE1:BB:NR5G:TRIGger:EXternal:RDELay?
// Response: 100

SOURCE1:BB:NR5G:TRIGger:DELay:UNIT TIME
SOURCE1:BB:NR5G:TRIGger:EXternal:TDELay 0.00001
SOURCE1:BB:NR5G:TRIGger:EXternal:RDELay?
// Response: 0.00001

SOURCE1:BB:NR5G:TRIGger:DELay:UNIT SAMP
SOURCE1:BB:NR5G:TRIGger:EXternal:DELay 10

```

Example: Configure and enable standard marker signals

```

SOURCE:BB:NR5G:TRIGger:OUTPut1:MODE RAT
SOURCE:BB:NR5G:TRIGger:OUTPut1:ONTime 10
SOURCE:BB:NR5G:TRIGger:OUTPut1:OFFTime 20

SOURCE:BB:NR5G:TRIGger:OUTPut2:MODE?
// REStart
SOURCE:BB:NR5G:TRIGger:OUTPut2:FOFFset 10
SOURCE:BB:NR5G:TRIGger:OUTPut2:ROFFset 20

SOURCE:BB:NR5G:TRIGger:OUTPut2:DELay 16

SOURCE:BB:NR5G:TRIGger:OUTPut3:MODE PER
SOURCE:BB:NR5G:TRIGger:OUTPut3:PERiod 30

```

Example: Clock settings

SOURce:BB:NR5G:CLOCK:SOURce INT

8.2 General Commands

[SOURce<hw>]:BB:NR5G:PRESet.....	195
[SOURce<hw>]:BB:NR5G:LINK.....	195
[SOURce<hw>]:BB:NR5G:SETting:CATalog?	196
[SOURce<hw>]:BB:NR5G:SETting:STORe.....	196
[SOURce<hw>]:BB:NR5G:SETting:LOAD.....	196
[SOURce<hw>]:BB:NR5G:SETting:DEL.....	196
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:UL.....	197
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:DL.....	197
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:FiLTeR:CATalog.....	197
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:FiLTeR:TMODeL.....	197
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:FiLTeR:FREQ.....	197
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:FiLTeR:DUPLexing.....	198
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:FiLTeR:BW.....	198
[SOURce<hw>]:BB:NR5G:SETting:TMODeL:FiLTeR:SCS.....	198
[SOURce<hw>]:BB:NR5G:STATe.....	198
[SOURce<hw>]:BB:NR5G:WAVEform:CREate.....	198
[SOURce]:BB:NR5G:VERsion?.....	199

[SOURce<hw>]:BB:NR5G:PRESet

Sets the parameters of the digital standard to their default values (*RST values specified for the commands).

Not affected is the state set with the command SOURce<hw>:BB:NR5G:STATe.

Example: See [Example "Storing current configuration"](#) on page 177.

Usage: Event

Manual operation: See ["Set to Default"](#) on page 24

[SOURce<hw>]:BB:NR5G:LINK <LinkDir>

Selects the transmission direction.

Parameters:

<LinkDir> DOWN | UP
*RST: DOWN

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["Link Direction"](#) on page 25

[SOURce<hw>]:BB:NR5G:SETTing:CATalog?

Queries the files with settings in the default directory. Listed are files with the file extension *.nr5g.

Return values:

<FileNames> <filename1>,<filename2>,...
Returns a string of filenames separated by commas.

Example: See [Example "Storing current configuration"](#) on page 177.

Usage: Query only

Manual operation: See ["Save/Recall"](#) on page 24

[SOURce<hw>]:BB:NR5G:SETTing:STORe <Filename>

Saves the current settings into the selected file; the file extension (*.nr5g) is assigned automatically.

Parameters:

<Filename> "<filename>"
Filename or complete file path

Example: See [Example "Storing current configuration"](#) on page 177.

Manual operation: See ["Save/Recall"](#) on page 24

[SOURce<hw>]:BB:NR5G:SETTing:LOAD <Filename>

Loads the selected file from the default or the specified directory. Loaded are files with extension *.nr5g.

Parameters:

<Filename> "<filename>"
Filename or complete file path; file extension can be omitted

Example: See [Example "Storing current configuration"](#) on page 177.

Manual operation: See ["Save/Recall"](#) on page 24

[SOURce<hw>]:BB:NR5G:SETTing:DEL <Filename>

Deletes the selected file from the default or the specified directory. Deleted are files with extension *.nr5g.

Parameters:

<Filename> "<filename>"
Filename or complete file path; file extension can be omitted

Example: See [Example "Storing current configuration"](#) on page 177.

Manual operation: See ["Save/Recall"](#) on page 24

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeI:UL <TmodUp>

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeI:DL <TmodDn>

Loads a files with predefined settings.

Parameters:

<TmodDn> "<filename>"

Filename as queried with the command [\[:SOURce<hw>\]:BB:NR5G:SETTing:TMODeI:FiLTeR:CATalog](#).

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["User Files"](#) on page 25

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeI:FiLTeR:CATalog

Queries the filenames of predefined files with test signals in the default directory.

Return values:

<FileNames> <filename1>,<filename2>,...

Returns a string of filenames separated by commas.

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["User Files"](#) on page 25

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeI:FiLTeR:TMODeI <FilterTestModel>

Applies the selected filter to narrow down the list of files.

Parameters:

<FilterTestModel> ALL | TM1_1 | TM1_2 | TM2 | TM2a | TM3_1 | TM3_1A |
TM3_2 | TM3_3

*RST: ALL

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["Filter Test Modes"](#) on page 27

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeI:FiLTeR:FREQ <FilterFreqRange>

Applies the selected filter to narrow down the list of files.

Parameters:

<FilterFreqRange> ALL | FR1 | FR2

*RST: ALL

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["Filter Test Modes"](#) on page 27

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeL:FiLTeR:DUPLexing <FilterDuplexing>

Applies the selected filter to narrow down the list of files.

Parameters:

<FilterDuplexing> ALL | FDD | TDD
 *RST: ALL

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["Filter Test Modes"](#) on page 27

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeL:FiLTeR:BW <FilterBandwidth>

Applies the selected filter to narrow down the list of files.

Parameters:

<FilterBandwidth> ALL | F5 | F10 | F15 | F20 | F25 | F30 | F40 | F50 | F60 | F70 |
 F80 | F90 | F100 | F200 | F400
 *RST: ALL

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["Filter Test Modes"](#) on page 27

[:SOURce<hw>]:BB:NR5G:SETTing:TMODeL:FiLTeR:SCS <FilterScs>

Applies the selected filter to narrow down the list of files.

Parameters:

<FilterScs> ALL | F15 | F30 | F60 | F120
 *RST: ALL

Example: See [Example "Loading predefined test signals"](#) on page 178.

Manual operation: See ["Filter Test Modes"](#) on page 27

[:SOURce<hw>]:BB:NR5G:STATe <State>

Activates the standard and deactivates all the other digital standards and digital modulation modes in the same path.

Parameters:

<State> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Storing current configuration"](#) on page 177.

Manual operation: See ["State"](#) on page 24

[:SOURce<hw>]:BB:NR5G:WAVEform:CREate <Filename>

Stores the current settings as an ARB signal in a waveform file (* .wv).

Parameters:

<Filename> string
 Filename or complete file path; file extension is assigned automatically

Example: See [Example "Storing current configuration"](#) on page 177.

Manual operation: See ["Generate Waveform File"](#) on page 25

[:SOURce]:BB:NR5G:VERSion?

Queries the version of the 3GPP standard underlying the definitions.

Return values:

<Version> string

Example: SOURce1:BB:NR5G:VERSion?

Usage: Query only

8.3 Carrier Commands

[:SOURce<hw>]:BB:NR5G:NODE:NCARrier.....	199
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CELLid.....	199
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CINDicator.....	200
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:N1ID?.....	200
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:N2ID?.....	200
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CARDepl.....	200
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CBW.....	200
[:SOURce<hw>]:BB:NR5G:NODE:RFPHase:STATe.....	201
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:PCFReq.....	201
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TAPos.....	201
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SUL:STATe.....	202

[:SOURce<hw>]:BB:NR5G:NODE:NCARrier <NumCarrier>

Sets the number of simulated carriers.

Parameters:

<NumCarrier> integer
 Range: 1 to 16
 *RST: 1

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["Number of Carriers"](#) on page 30

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CELLid <CellId>

Sets the cell identity of the selected cell.

Parameters:

<CellId> integer
 Range: 0 to 1007
 *RST: 0

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["Cell ID"](#) on page 31

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CINDicator <CellIndicator>

Sets the cell indicator.

Parameters:

<CellIndicator> integer
 Range: 0 to 15

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["Cell Indicator"](#) on page 31

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:N1ID?

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:N2ID?

Queries the physical cell indicator group ($N_{ID}^{(1)}$) and the physical layer identity ($N_{ID}^{(2)}$).

Return values:

<CarrierN2Id> integer
 Range: 0 to 2
 *RST: 0

Example: See [Example "Configuring the carrier"](#) on page 178.

Usage: Query only

Manual operation: See ["N1 ID/N2 ID"](#) on page 31

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CARDepl <CarrierDepl>

Selects one of the frequency ranges, specified for 5G NR transmission.

Parameters:

<CarrierDepl> LT3 | BT36 | GT6
 *RST: LT3

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["Deployment"](#) on page 31

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:CBW <ChanBandWidth>

Selects the bandwidth of the node carrier.

Parameters:

<ChanBandWidth> BW5 | BW10 | BW15 | BW20 | BW25 | BW40 | BW50 | BW60 |
 BW100 | BW80 | BW400 | BW200 | BW30 | BW70 | BW90
 *RST: BW100

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["Channel Bandwidth"](#) on page 32

[:SOURce<hw>]:BB:NR5G:NODE:RFPHase:STATe <RFPhaseComp>

If enabled, applies a frequency related phase compensation after each symbol, as specified in [TS 38.211](#).

Parameters:

<RFPhaseComp> 0 | 1 | OFF | ON
 *RST: !

Example: SOURce1:BB:NR5G:NODE:CELL0:PCFReq 1E9
 SOURce1:BB:NR5G:NODE:RFPHase:STATe 1

Manual operation: See ["RF Phase Compensation"](#) on page 31

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:PCFReq <CarrierFreq>

Sets the carrier frequency of the selected carrier at which the frequency phase compensation is applied.

Parameters:

<CarrierFreq> float
 Range: 0.0 to 44E9
 Increment: 1E3
 *RST: 1E9

Example: See [\[:SOURce<hw>\]:BB:NR5G:NODE:RFPHase:STATe](#) on page 201.

Manual operation: See ["Frequency in GHz"](#) on page 32

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TAPos <DmrsTypeAPos>

Sets the position of the first DMRS symbol within the slot, if mapping type A is used.

Parameters:

<DmrsTypeAPos> integer
 Range: 2 to 3
 *RST: 2

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["DMRS Type A Position"](#) on page 32

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SUL:STATe <CarrierSULState>

Defines if the carrier supports supplementary uplink (SUL) or not.

Parameters:

<CarrierSULState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the carrier"](#) on page 178.

Manual operation: See ["SUL"](#) on page 32

8.4 TxBW Commands

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:POINta.....	202
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:CONFlit?.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:RESolve.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:USE.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:USE.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:USE.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:USE.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:NRB?.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:NRB?.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:NRB?.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:NRB?.....	203
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:OFFSet.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:OFFSet.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:OFFSet.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:OFFSet.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:KOMU?.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:KOMU?.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:KOMU?.....	204
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:KOMU?.....	204

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:POINta <CarrierPointA>

Sets the frequency offset between the reference Point A and the center carrier frequency.

Parameters:

<CarrierPointA> float
 Range: -50e6 to 0
 Increment: 0.015
 *RST: 49140000

Example: See [Example "Configuring the TxBW"](#) on page 179.

Manual operation: See ["Point A to Carrier Center"](#) on page 33

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:CONFLICT?

Queries whether there is a conflict.

Return values:

<Conflict> 0 | 1 | OFF | ON

*RST: n.a. (no preset. default: 0)

Example: See [Example "Configuring the TxBW"](#) on page 179.

Usage: Query only

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:RESolve

Recalculates the frequency-dependent settings and thus redefines the frequency position of the TxBW.

Example: See [Example "Configuring the TxBW"](#) on page 179.

Usage: Event

Manual operation: See ["Resolve Conflicts"](#) on page 34

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:USE <ScsUse>**[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:USE <ScsUse>****[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:USE <ScsUse>****[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:USE <ScsUse>**

Enables a SCS in the particular cell.

Parameters:

<ScsUse> 0 | 1 | OFF | ON

Example: See [Example "Configuring the TxBW"](#) on page 179.

Manual operation: See ["Use"](#) on page 34

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:NRB?**[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:NRB?****[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:NRB?****[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:NRB?**

Queries the number of available resource blocks (N_{RB}).

Return values:

<NRB> integer

Range: 0 to 273

*RST: 0

Example: See [Example "Configuring the TxBW"](#) on page 179.

Usage: Query only

Manual operation: See ["N_{RB}"](#) on page 34

```
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:OFFSet <ScsOffset>
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:OFFSet <ScsOffset>
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:OFFSet <ScsOffset>
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:OFFSet <ScsOffset>
```

Sets the offset between the usable RB and the common RBs.

Parameters:

<ScsOffset> float
 Range: 0 to 9
 Increment: 1
 *RST: 0

Example: See [Example "Configuring the TxBWs"](#) on page 179.

Manual operation: See ["TxBW Offset"](#) on page 35

```
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S120K:KOMU?
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S15K:KOMU?
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S60K:KOMU?
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:TXBW:S30K:KOMU?
```

Queries the value of the parameter k_0^{μ} .

Return values:

<KoMu> integer
 Range: -6; 0; 6 further values indicate conflict
 *RST: 0

Example: See [Example "Configuring the TxBWs"](#) on page 179.

Usage: Query only

Manual operation: See ["k0u"](#) on page 35

8.5 LTE-CRS Coexistence Commands

```
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:STATe.....204
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:POINta..... 205
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:VSHift.....205
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:CBW..... 205
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:NAP..... 205
```

```
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:STATe <LteCrsState>
```

Enable the LTE-CRS.

Parameters:

<LteCrsState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "LTE-CRS Coexistence"](#) on page 179.

Manual operation: See ["State"](#) on page 36

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:POINta <LteOffsToPointA>

Sets the LTE carrier center subcarrier location.

Parameters:

<LteOffsToPointA> integer
 Range: 0 to 30300
 *RST: 0

Example: See [Example "LTE-CRS Coexistence"](#) on page 179.

Manual operation: See ["Offset to Point A / 15 kHz SCS"](#) on page 37

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:VSHift <LteVShift>

Sets the parameter *v-Shift*.

Parameters:

<LteVShift> integer
 Range: 0 to 5
 *RST: 0

Example: See [Example "LTE-CRS Coexistence"](#) on page 179.

Manual operation: See ["vShift"](#) on page 37

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:CBW <LteCarrierBW>

Sets the LTE bandwidth.

Parameters:

<LteCarrierBW> N6 | N15 | N25 | N50 | N75 | N100
 *RST: N6

Example: See [Example "LTE-CRS Coexistence"](#) on page 179.

Manual operation: See ["LTE Bandwidth / RBs"](#) on page 37

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:LTE:NAP <LteAntennaPorts>

Sets the number of antenna ports used for LTE-CRS.

Parameters:

<LteAntennaPorts> AP1 | AP2 | AP4
 *RST: AP1

Example: See [Example "LTE-CRS Coexistence"](#) on page 179.

Manual operation: See ["LTE Antenna Ports"](#) on page 37

8.6 SS/PBCH Commands

[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:OFFSet.....	206
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:NSSPbch.....	207
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:STATe.....	207
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SCSPacing.....	207
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:RBOFset.....	207
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SCOFset.....	208
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:DFReq?.....	208
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CASE.....	208
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:L.....	209
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:POSition.....	209
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:BSPeriodicty.....	209
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:POWer.....	210
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:PSSPow.....	210
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SSSPow.....	210
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:STATe.....	210
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:DATA.....	210
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:DLISt.....	210
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:PATtern.....	211
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:STATe.....	211
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SCSC.....	211
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:ASOF:STATe.....	212
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SCOFset.....	212
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SFOFset.....	212
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:CSZero.....	212
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SSZero.....	213
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:CBARred.....	213
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:IFRResel.....	213
[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SPARe:STATe.....	213

[SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:OFFSet <OffsetRelativeT>

Defines the reference point, relative to which the SS/PBCH is allocated in frequency domain.

Parameters:

<OffsetRelativeT> TXBW | POINta

TXBW

The frequency position of the SS/PBCH is set relative to the usable RBs that apply for the current numerology, i.e. to the start of the TxBWs.

POINta

The frequency position of the SS/PBCH is set relative to the position of point A.

*RST: TXBW

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Offset Relative to"](#) on page 39

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:NSSPBch <NuOfPbchPattern>

Sets the number of SS/PBCH patterns to be configured.

Parameters:

<NuOfPbchPattern> integer
 Range: 0 to 4
 *RST: 1

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Number of SS/PBCH Patterns"](#) on page 39

**[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:STATe
 <sspbchState>**

Enables the transmission of SS/PBCH.

Parameters:

<sspbchState> 0 | 1 | OFF | ON
 *RST: 1

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["State"](#) on page 42

**[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SCSPacing
 <SCSPacing>**

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP), where the available values depend on the "Deployment".

See [Table 3-2](#).

Parameters:

<SCSPacing> N15 | N30 | N60 | E60 | N120 | N240
 *RST: N30

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["SC Spacing/CP"](#) on page 39

**[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:RBOffset
 <RBOffset>**

Sets the start resource block of the selected allocation as offset to the start of usable RBs that apply for the current numerology.

Parameters:

<RBOffset> integer
 Range: 0 to 126
 *RST: 10

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["RB Offset/RB Offset \(15 kHz SCS\)/RB Offset \(60 kHz SCS\)"](#) on page 40

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SCOffset<SCOffset>

Sets the start subcarrier of the selected allocation within the resource block.

Parameters:

<SCOffset> integer
 The value range depends on the selected RB offset ([:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:RBOffset).
 Range: 0 to 11
 *RST: 6

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["SC Offset/SC Offset \(15 kHz SCS\)/SC Offset \(60 kHz SCS\)"](#) on page 40

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:DFReq?

Queries the frequency offset between the center of the SS/PBCH block to the center of the carrier.

Return values:

<PbchDeltaF> float
 Range: -40E6 to 40E6
 Increment: 1.0
 *RST: 0.0

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Usage: Query only

Manual operation: See ["Delta f to Carrier \(Centers\)"](#) on page 41

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CASE <PbschCase>

Selects one of the SS/PBCH cases, as specified in [TS 38.213](#).

Parameters:

<PbschCase> A | B | C | D | E
 *RST: B

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Case"](#) on page 41


```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:L <PositionBitLen>
```

Sets the number of SS/PBCH blocks, transmitted per half-frame.

Parameters:

<PositionBitLen>	L4 L8 L64
------------------	---------------

*RST: L4

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See "L" on page 41

```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:POSition <Pattern>,  
    <BitCount>
```

Sets which of the SS/PBCH blocks are transmitted in each burst set.

Parameters:

<Pattern>	64 bits
-----------	---------

Each bit applies to one SS/PBCH block

1: SS/PBCH block is present

0: SS/PBCH transmission is suppressed.

*RST: #HC

<BitCount>	integer
------------	---------

Range: 4 to 64

*RST: 4

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See "Positions" on page 41

**[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:BSPeriodicty
<BurstSetPer>**

Sets the burst set periodicity.

Parameters:

<BurstSetPer>	BS5	BS10	BS20	BS40	BS80	BS160
---------------	-----	------	------	------	------	-------

*RST: BS10

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See "Burst Set Periodicity" on page 42

```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:POWer <PbchPower>
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:PSSPow
    <PssPower>
```

```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:SSSPow
    <SssPower>
```

Sets the power of the SSS/PSS/PBCH allocations relative to the power of the other resource elements.

Parameters:

```
<SssPower>          float
                    Range:    -80.0 to 10.0
                    Increment: 0.01
                    *RST:     0.0
```

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["PSS/SSS Power"](#) on page 42

```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:STATe
    <PbchChanCoding>
```

Enables PBCH channel coding.

Parameters:

```
<PbchChanCoding>  0 | 1 | OFF | ON
                    *RST:    1
```

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Channel Coding"](#) on page 47

```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:DATA
    <PbchDataSource>
```

Selects the PBCH data source.

Parameters:

```
<PbchDataSource>  PN9 | PN11 | PN15 | PN16 | PN20 | PN21 | PN23 | PATtern |
                    DLISt | ZERO | ONE
                    *RST:    PN9
```

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Data Source"](#) on page 47

```
[ :SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:DLISt
    <DataList>
```

Selects an existing data list file from the default directory or from the specific directory.

Parameters:

<DataList> string
Filename incl. file extension or complete file path

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Data Source"](#) on page 47

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:CCODing:PATtern
 <Pattern>, <BitCount>

Sets a bit pattern as a data source.

Parameters:

<Pattern> 64 bits
 *RST: #H0
 <BitCount> integer
 Range: 1 to 64
 *RST: 1

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Data Source"](#) on page 47

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:STATe
 <SSPbchMIBState>

Defines how the MIB is defined.

Parameters:

<SSPbchMIBState> 0 | 1 | OFF | ON
1|ON
 This is a faster way to define the MIB.
 You can define if channel coding is used or not and select an arbitrary data source; further settings are not required.
0|OFF
 Allows you to configure the MIB content according to [TS 38.331](#).
 *RST: 1

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Dummy Content for MIB"](#) on page 47

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SCSC
 <SSPbchSCSCCommon>

Selects the common SCS (subcarrier spacing).

Parameters:

<SSPbchSCSCCommon> N15_60 | N30_120
 *RST: N15_60

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["SCS Common"](#) on page 49

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:ASOF:STATe
<SSPbchASCOFStat>

If enabled, the SSPBCH subcarrier offset is set automatically, depending on the selected common subcarrier spacing.

Parameters:

<SSPbchASCOFStat> 0 | 1 | OFF | ON

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Auto Subcarrier Offset"](#) on page 49

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SCOFset
<SSPbchCustomSsb>

Queries the SS/PBCH subcarrier offset.

Parameters:

<SSPbchCustomSsb>float

Range: 0 to 31

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["SSPBCH Subcarrier Offset"](#) on page 50

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SFOffset
<SSPbchSFStartOf>

By default, the counting of the SFN (system frame number) starts with 0. Use this parameter to set a different start SFN value.

Parameters:

<SSPbchSFStartOf> float

Range: 0 to 1023

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["System Frame Number Start Offset"](#) on page 50

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:CSZero
<SSPbchCSZero>

Sets the common control resource set (CORESET) of the initial downlink BWP.

Parameters:

<SSPbchCSZero> integer

Range: 0 to 15

*RST: 0

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["CORESET Zero"](#) on page 50

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SSZero
<SSPbchSSZero>

Sets the common search space of the initial downlink BWP.

Parameters:

<SSPbchSSZero> integer
 Range: 0 to 15
 *RST: 0

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Search Space Zero"](#) on page 50

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:CBARred
<SSPbchCellBarre>

Cell barring is system information that indicates if UEs can camp on the particular cell NBAR or not BARR.

Parameters:

<SSPbchCellBarre> BARR | NBAR
 *RST: BARR

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Cell Barred"](#) on page 50

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:IFRResel
<SSPbchInFreqSel>

Sets the value of the system information parameter intraFreqReselection.

Parameters:

<SSPbchInFreqSel> ALWD | NALW
 *RST: ALWD

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Intra Freq Reselection"](#) on page 50

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SSPBch<st0>:MIB:SPARe:STATe
<PbchSpareState>

Sets the system information bit spare, as defined in [TS 38.331](#).

Parameters:

<PbchSpareState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the SS/PBCH"](#) on page 180.

Manual operation: See ["Spare"](#) on page 51

8.7 Dummy Resource Elements Commands

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:STATe	214
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:POWer	214
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:SCSPacing	215
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:MODulation	215
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:TPSTate	215
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:DATA	215
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:DLISt	216
[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:PATtern	216

[\[:SOURce<hw>\]:BB:NR5G:NODE:CELL<ch0>:DUMRes:STATe](#) <DummyResState>

Enables filling in unallocated resource elements (RE) with dummy data.

Parameters:

<DummyResState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Filling in the frame with dummy REs"](#) on page 180.

Manual operation: See ["State"](#) on page 43

[\[:SOURce<hw>\]:BB:NR5G:NODE:CELL<ch0>:DUMRes:POWer](#) <Power>

Sets the power of the resource elements filled in with dummy data.

Parameters:

<Power> float
 Range: -80 to 10
 Increment: 0.001
 *RST: 0

Example: See [Example "Filling in the frame with dummy REs"](#) on page 180.

Manual operation: See ["Power"](#) on page 43

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:SCSPacing <SCSPacing>

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP), where the available values depend on the "Deployment".

See [Table 3-2](#).

Parameters:

<SCSPacing> N15 | N30 | N60 | E60 | N120 | N240
*RST: N30

Example: See [Example "Filling in the frame with dummy REs"](#) on page 180.

Manual operation: See ["SC Spacing/CP"](#) on page 44

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:MODulation <Modulation>

Sets the modulation scheme for the dummy REs.

Parameters:

<Modulation> BPSK | BPSK2 | QPSK | QAM16 | QAM64 | QAM256
*RST: QPSK

Example: See [Example "Filling in the frame with dummy REs"](#) on page 180.

Manual operation: See ["Modulation"](#) on page 44

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:TPSTate <TrPrecState>

In uplink, enables using the optional DFT-S scheme.

Parameters:

<TrPrecState> 0 | 1 | OFF | ON
*RST: 0

Example: SOURce1:BB:NR5G:LINK UP
SOURce1:BB:NR5G:NODE:CELL0:DUMRes:TPSTate 1

Manual operation: See ["Transform Precoding"](#) on page 44

[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:DATA <DataSource>

Sets the data source for the dummy REs.

Parameters:

<DataSource> PN9 | PN11 | PN15 | PN16 | PN20 | PN21 | PN23 | PATtern |
DLISt | ZERO | ONE
*RST: PN9

Example: See [Example "Filling in the frame with dummy REs"](#) on page 180.

Manual operation: See ["Data Source"](#) on page 44

[[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:DLISt <DSelect>

Selects an existing data list file from the default directory or from the specific directory.

Parameters:

<DSelect> string
 Filename incl. file extension or complete file path

Example: See [Example "Filling in the frame with dummy REs"](#)
 on page 180.

Manual operation: See ["Data Source"](#) on page 44

[[:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:DUMRes:PATtern <Pattern>, <BitCount>

Sets a bit pattern as a data source.

Parameters:

<Pattern> 64 bits
 *RST: #H0

<BitCount> integer
 Range: 1 to 64
 *RST: 1

Example: See [Example "Filling in the frame with dummy REs"](#)
 on page 180.

Manual operation: See ["Data Source"](#) on page 44

8.8 Carrier Mapping Commands

[\[:SOURce<hw>\]:BB:NR5G:NODE:CARMapping:CELL\[:ROW<ch0>\]](#).....216
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[[:SOURce<hw>]:BB:NR5G:NODE:CARMapping:CELL[:ROW<ch0>] <MapCellNum>

Queries which of the carriers is output at which baseband.

Parameters:

<MapCellNum> integer
 Indicates the carrier.
 Carriers are numbered automatically, where the maximum number of carriers is set with the command `[ :SOURce<hw> ] :BB:NR5G:NODE:NCARrier`.
 Range: 0 to 15
 *RST: 0

Example: See [Example "Mapping the carriers to the outputs"](#) on page 181.

Manual operation: See ["Carrier"](#) on page 45

`[:SOURce<hw>] :BB:NR5G:NODE:CARMapping:MAPPed[:ROW<ch0>]
 <CellMapped>`

If enabled, the signal of the selected cell is mapped to the output.

Parameters:

<CellMapped> 0 | 1 | OFF | ON
 *RST: 1

Example: See [Example "Mapping the carriers to the outputs"](#) on page 181.

Manual operation: See ["Cell Mapped"](#) on page 46

`[ :SOURce<hw> ] :BB:NR5G:NODE:CARMapping:AP4000[:ROW<ch0>] <AP4000>`

Sets if the SS/PBCHs are output at the selected output.

Parameters:

<AP4000> 0 | 1 | OFF | ON
 *RST: 1

Example: See [Example "Mapping the carriers to the outputs"](#) on page 181.

Manual operation: See ["Antenna Port"](#) on page 38
 See ["SS/PBCH AP 4000"](#) on page 45

`[:SOURce<hw>] :BB:NR5G:NODE:CARMapping:DFReq[:ROW<ch0>]
 <DeltaFToDc>`

Shifts the signal relative to the center output frequency.

Parameters:

<DeltaFToDc> float
 Range: -40E6 to 40E6
 Increment: 1.0
 *RST: 0.0

Example: See [Example "Mapping the carriers to the outputs"](#) on page 181.

Manual operation: See ["Delta f to Output"](#) on page 45

8.9 User Commands

<code>[SOURce<hw>]:BB:NR5G:UBWP:NUSer</code>	218
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<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:CCODing:STATe</code>	218
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:CCODing:STATe?</code>	218
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:DATA</code>	219
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:DATA</code>	219
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:DLISt</code>	219
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:DLISt</code>	219
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:PATTern</code>	219
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:PATTern</code>	219

`[SOURce<hw>]:BB:NR5G:UBWP:NUSer <NumUsers>`

Sets the number of simulated users.

Parameters:

`<NumUsers>` integer
 Range: 1 to 10
 *RST: 1

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["Number of Users"](#) on page 51

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:UEID <UserId>`

Sets the RNTI of the user.

Parameters:

`<UserId>` integer
 Range: 0 to 65535
 *RST: 0

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["UE ID \(RNTI\)"](#) on page 52

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:CCODing:STATe`
 `<UschChaCod>`

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:CCODing:STATe?`
 `<DschChaCod>`

Enables DSCH/USCH channel coding.

Parameters:

`<DschChaCod>` 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Usage: Query only

Manual operation: See ["DSCH/USCH Channel Coding"](#) on page 52

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:DATA <UschDataSource>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:DATA <DschDataSource>

Sets the DSCH/USCH data source.

Parameters:

<DschDataSource> PN9 | PN11 | PN15 | PN16 | PN20 | PN21 | PN23 | PATtern |
 DList | ZERO | ONE
 *RST: PN9

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["\(P\)DSCH/\(P\)USCH Data Source"](#) on page 53

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:DList <UschDataList>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:DList <DschDataList>

Selects an existing data list file from the default directory or from the specific directory.

Parameters:

<DschDataList> string
 Filename incl. file extension or complete file path

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["\(P\)DSCH/\(P\)USCH Data Source"](#) on page 53

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:USCH:PATtern <Pattern>,
 <BitCount>**

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:DSCH:PATtern <Pattern>,
 <BitCount>**

Sets a bit pattern as a data source.

Parameters:

<Pattern> 64 bits
 *RST: #H0
 <BitCount> integer
 Range: 1 to 64
 *RST: 1

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See "(P)DSCH/(P)USCH Data Source" on page 53

8.10 DL/UL BWP Commands

<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:NBWParts</code>	220
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:NBWParts</code>	220
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<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:RBNumber</code>	221
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<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:RBOffset</code>	221
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<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PRBOffset?</code>	222
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PRBOffset?</code>	222
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:DFReq?</code>	222
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:DFReq?</code>	222

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:NBWParts`

`<UserUIBwp>`

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:NBWParts`

`<UserDIBwp>`

Sets the number of downlink bandwidth parts (BWP).

Parameters:

`<UserDIBwp>` integer
 Range: 1 to 4
 *RST: 1

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["Number of DL/UL BWPs"](#) on page 56

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:INDicator` `<BWPIndicator>`

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:INDicator` `<BwpIndicator>`

Sets the indicator for the selected BWP.

Parameters:

`<BwpIndicator>` integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["BWP Indicator"](#) on page 56

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
SCSPacing <SCSPacing>
```

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
SCSPacing <SCSPacing>
```

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP).

Parameters:

<SCSPacing> N15 | N30 | N60 | E60 | N120 | N240
 *RST: N30

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["SC Spacing/CP"](#) on page 56

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
RBNumber <UIBwpRBNumber>
```

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RBNumber <DIBwpRBNum>
```

Sets the number of resource blocks (RB) the bandwidth part (BWP) occupies.

Parameters:

<DIBwpRBNum> integer
 The number of available RBs depends on the SCS and channel bandwidth.
 CORESET allocations span always a multiple of 6 resource blocks. Thus, for CORESET allocations the minimum value is 6 and the allowed values are multiple of 6.
 Range: 1 to 273
 *RST: 11

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["No. RBs"](#) on page 56

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
RBOffset <RBOffset>
```

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
RBOffset <RBOffset>
```

Sets the offset between the first resource block in the BWP relative to the first usable resource blocks (URB).

Parameters:

<RBOffset> integer
 Range: 0 to 255
 *RST: 0

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Manual operation: See ["RB Offset in TxBW"](#) on page 57

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PRBOffset?

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PRBOffset?

Queries the frequency starting position of the BWP relative to the point A.

Return values:

<PointAOffset> integer
 Range: 0 to 500
 *RST: 0

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Usage: Query only

Manual operation: See ["RB Offset to Point A"](#) on page 57

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:DFReq?

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:DFReq?

Queries the frequency offset between the BWP center and the center frequency of the carrier.

Return values:

<DeltaFCarrier> float
 Range: -40E6 to 40E6
 Increment: 1.0
 *RST: 0.0

Example: See [Example "Configuring the users and the BWPs"](#) on page 181.

Usage: Query only

Manual operation: See ["Δf to Carrier \(Centers\)"](#) on page 57

8.11 BWP Configuration Commands

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8.11.1 PDSCH Commands

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[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:SID0.....	225
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<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: TD<grp0>:SLIV?</code>	229

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:CTYPe <TypeBConfigType>`

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:CTYPe <TypeAConfigType>`

Sets the configuration type (type 1 or type 2) and defines the mapping of the DMRS to the physical resource elements.

Parameters:

`<TypeAConfigType>` T1 | T2
*RST: T1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Config Type"](#) on page 63

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:APINdex <TypeBAddPosIdx>`

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:APINdex <TypeAAddPosIdx>`

Sets the parameter additional position index, required to select the positions of the DMRS symbols.

Parameters:

`<TypeAAddPosIdx>` integer
Range: 0 to 3
*RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Additional Position Index"](#) on page 63

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:MLENgtH <TypeBMaxLength>`

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:MLENgtH <DmrsMaxLength>`

Sets if single- or double-symbol DMRS is used.

Parameters:

<DmrsMaxLength> integer
 Range: 1 to 2
 *RST: 2

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Max Length"](#) on page 63

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:SID0 <ScramID0>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:SID1 <ScramID1>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTB:SID0 <ScramID0>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTB:SID1 <ScramID1>
```

Sets the scrambling ID.

Parameters:

<ScramID1> integer
 Range: 0 to 65535
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Scrambling ID 0/Scrambling ID 1"](#) on page 63

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:SCRambling:STATe <PdschScrambling>
```

Sets if a data scrambling ID is used for the initialization of the generator of the PDSCH scrambling sequence.

Parameters:

<PdschScrambling> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Use PDSCH Scrambling ID"](#) on page 61

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DSID <DataScrambleId>
```

If `[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh:SCRambling:STATe 1`, sets the data scrambling ID.

Parameters:

<DataScrambleId> integer
 Range: 0 to 1023
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Data Scrambling ID"](#) on page 61

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
 PDSCh:MCWDci <MaxCodeWordsPer>**

Sets the number of codewords.

Parameters:

<MaxCodeWordsPer>integer
 Range: 1 to 2
 *RST: 1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Max. Number of Codewords per DCI"](#) on page 61

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
 PDSCh:MCSTable <PdschMcsTable>**

Sets the MCS table and defines the used modulation scheme.

Parameters:

<PdschMcsTable> QAM64 | QAM256 | QAM64LSE
 *RST: QAM64

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["MCS Table"](#) on page 62

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
 PDSCh:VPINter <VrbToPrblInterle>**

Queries the mapping method used for the mapping of the virtual resource blocks (VRB) to the physical resource blocks (PRB).

Parameters:

<VrbToPrblInterle> VPN | VP2 | VP4
VPN
 Non-interleaved
VP2|VP4
 Interleaving is enabled. The value defines the interleaving unit size: $VP2 = 2$ and $VP4 = 4$.

*RST: VPN

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["VRB-to-PRB Interleaver"](#) on page 61

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:RESalloc <ResourceAlloc>**

Sets the used resource allocation type.

Parameters:

<ResourceAlloc> T0 | T1 | DS

*RST: T1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Resource Allocation"](#) on page 62

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:RBGSize <UrDIBpPdschRBGS>**

Sets the size of the resource block group.

Parameters:

<UrDIBpPdschRBGS> C1 | C2

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Resource Block Group Size"](#) on page 62

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TDANum <NumTDAllocs>**

Sets the number of PDSCH time domain allocations.

Parameters:

<NumTDAllocs> integer

Range: 0 to 16

*RST: 0

Example: See [Example "Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list"](#) on page 184.

Manual operation: See ["Time Domain Allocations"](#) on page 66

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:TD<grp0>:KNULI <K0>**

Sets the slot offset K_0 .

Parameters:

<K0> integer
 Range: 0 to 32
 *RST: 0

Example: See [Example "Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list"](#) on page 184.

Manual operation: See "[K0](#)" on page 66

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
 PDSCh:TD<grp0>:MAPPING <MappingType>**

Sets the DMRS-mapping type A and B.

Parameters:

<MappingType> A | B
 *RST: A

Example: See [Example "Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list"](#) on page 184.

Manual operation: See "[Mapping Type](#)" on page 66

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
 PDSCh:TD<grp0>:START <Start>**

Sets the start ODFM symbol of the PDSCH allocation.

Parameters:

<Start> integer
 Range: 0 to 12
 *RST: 0

Example: See [Example "Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list"](#) on page 184.

Manual operation: See "[Start](#)" on page 66

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
 PDSCh:TD<grp0>:LENGTH <Length>**

Sets the PDSCH allocation length.

Parameters:

<Length> 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14
 *RST: 2

Example: See [Example "Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list"](#) on page 184.

Manual operation: See "[Length](#)" on page 67

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh:TD<grp0>:SLIV?

Queries the resulting start and length indicator SLIV.

Return values:

<SLIV> integer
Range: 0 to 32
*RST: 0

Example: See [Example "Configuring PDSCH time domain scheduling by means of PDSCH time domain allocation list"](#) on page 184.

Usage: Query only

Manual operation: See ["SLIV"](#) on page 67

8.11.2 ZP and NZP CSI-RS Commands

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:STATE	230
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:STATE	230
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:NRESources	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:NRESources	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:RES<user0>:PER	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:RES<user0>:PER	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:RES<user0>:OFFSet	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:RES<user0>:OFFSet	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:RES<user0>:RBNumber	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:RES<user0>:RBNumber	231
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:RES<user0>:SRBNumber	232
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:RES<user0>:SRBNumber	232
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:RES<user0>:ROW	232
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:RES<user0>:ROW	232
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP:SET<grp0>:RES<user0>:PORTs?	232
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP:SET<grp0>:RES<user0>:PORTs?	232

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP: SET<gr0>:RES<user0>:DENSity.....	233
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:DENSity.....	233
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP: SET<gr0>:RES<user0>:CDMType?.....	233
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:CDMType?.....	233
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP: SET<gr0>:RES<user0>:BITMap.....	233
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:BITMap.....	233
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP: SET<gr0>:RES<user0>:I0.....	234
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:ZP: SET<gr0>:RES<user0>:I1.....	234
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:I0.....	234
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:I1.....	234
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:SCID.....	234
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:PWR.....	234
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:APMap:MAPCoord.....	235
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:REAL.....	235
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:IMAGinary.....	235
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:MAGNitude.....	236
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:NZP: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:PHASe.....	236

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
ZP:STATe <State>

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
NZP:STATe <State>

Enables the NZP/ZP CSI-RS transmission.

Parameters:

<State> 0 | 1 | OFF | ON
*RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["State"](#) on page 71

```
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:NRESources <Resources>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:NRESources <Resources>
```

Sets the number of CSI-RS resources.

Parameters:

<Resources> integer
 Range: 0 to 64
 *RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Resources"](#) on page 71

```
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:PER <Periodicity>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:PER <Periodicity>
```

Sets the periodicity $T_{\text{CSI-RS}}$ (in slots).

Parameters:

<Periodicity> 4 | 5 | 8 | 10 | 16 | 20 | 32 | 40 | 64 | 80 | 160 | 320 | 640
 *RST: 4

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Periodicity"](#) on page 71

```
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:OFFSet <Offset>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:OFFSet <Offset>
```

Sets the slot offset T_{offset} .

Parameters:

<Offset> integer
 Range: 0 to 639
 *RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Offset"](#) on page 72

```
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:RBNumber <CsiRsZpNumRB>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:RBNumber <CsiRsNZpNumRB>
```

Sets the number of resource blocks the CSI-RS spans.

Parameters:

<CsiRsNZpNumRB> integer
 Range: 24 to 273
 *RST: = number of RB in the BWP

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["No. RBs"](#) on page 72

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:SRBNumber <CsiRsZpStartRb>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:SRBNumber <CsiRsNZpStartRB>
```

Sets the first resource block the CSI-RS occupies.

Parameters:

<CsiRsNZpStartRB> integer
 Range: 0 to 273
 *RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Start RB"](#) on page 72

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:ROW <Row>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:ROW <Row>
```

Sets the parameter row.

Parameters:

<Row> integer
 Range: 1 to 18
 *RST: 1

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Row"](#) on page 72

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:PORTs?
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:PORTs?
```

Sets the number of ports x.

Return values:

<Ports> 1 | 2 | 4 | 8 | 12 | 16 | 24 | 32
 *RST: 1

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Usage: Query only
Manual operation: See ["Ports"](#) on page 73

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:DENSity <Density>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:DENSity <Density>
```

Sets the parameter density.

Parameters:

<Density> EVE5 | ODD5 | DEN1 | DEN3
 *RST: DEN3

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Density"](#) on page 72

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:CDMType?
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:CDMType?
```

Queries the higher-level parameter `cdm-Type`.

The value is automatically, depending on the value set with the command [\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs: NZP:SET<gr0>:RES<user0>:ROW](#) on page 232

Return values:

<CDMType> NOCDm | CDM2 | CDM4 | CDM8
 *RST: NOCDm

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Usage: Query only

Manual operation: See ["CDM-Type"](#) on page 73

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:BITMap <Pattern>, <BitCount>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:BITMap <Pattern>, <BitCount>
```

Sets the location of the ZP CSI-RS in the frequency domain.

Parameters:

<Pattern> 12 bits
 *RST: #H0

<BitCount> integer
 Range: 1 to 12
 *RST: 1

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Bitmap"](#) on page 73

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:I0 <I0>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  ZP:SET<gr0>:RES<user0>:I1 <I1>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:I0 <I0>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:I1 <I1>
```

Sets the parameters I_0 and I_1 and define the ZP CSI-RS location relative to the start of as slot.

Parameters:

<I1> integer
 Range: 2 to 12
 *RST: 11

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["I0/I1"](#) on page 74

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:SCID <ScramblingId>
```

Sets the parameter n_{ID} (scrambling ID).

Parameters:

<ScramblingId> integer
 Range: 0 to 1023
 *RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Scrambling ID"](#) on page 74

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:PWR <NZpPower>
```

Sets the power of the NZP resource relative to the PDSCH power.

Parameters:

<NZpPower> float
 Range: -80 to 10
 Increment: 0.01
 *RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Power"](#) on page 73

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:APMap:MAPCoord <APMapMode>
```

Switches between the cartesian and cylindrical coordinates representation.

Parameters:

<APMapMode> CARTesian | CYLindrical
 *RST: CARTesian

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Mapping Coordinates"](#) on page 75

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:REAL
  <APMapDataReal>
```

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:IMAGinary
  <APMapDataImag>
```

Sets the mapping of the antenna ports (AP).

Suffix:

<s2us0> AP3000
 Antenna port
 <s3us0> 0

Parameters:

<APMapDataImag> float
 The REAL (Magnitude) and IMAGinary (Phase) values are interdependent. Their value ranges change depending on each other and so that the resulting complex value is as follows:
 $|\text{REAL} + j \cdot \text{IMAGinary}| \leq 1$
 Otherwise, the values are normalized to Magnitude = 1.
 Range: -1 to 1
 Increment: 0.001
 *RST: 0

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Mapping table"](#) on page 75

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:MAGNitude
  <APMapDataMagn>
```

Sets the mapping of the antenna ports (AP).

Parameters:

```
<APMapDataMagn> float
                    Range:    0 to 1
                    Increment: 0.001
                    *RST:     0
```

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Mapping table"](#) on page 75

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:CSIRs:
  NZP:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:PHASe
  <APMapDataPhase>
```

Sets the mapping of the antenna ports (AP).

Parameters:

```
<APMapDataPhase> float
                    Range:    0 to 360
                    Increment: 0.1
                    *RST:     0
```

Example: See [Example "Configuring the CSI-RS settings"](#) on page 185.

Manual operation: See ["Mapping table"](#) on page 75

8.11.3 Rate Match Commands

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:STATe.....237
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
  NRESources..... 237
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
  GRPNumber.....237
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
  RS<gr0>:RBDList..... 238
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
  RS<gr0>:RBPatt..... 238
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
  RS<gr0>:SLOT.....238
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
  RS<gr0>:SLTPatt.....239
```

<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:PER</code>	239
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:PERPatt</code>	239
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:GRID</code>	239

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:STATE <RateMatchState>`

Enables the configuration of rate match groups.

Parameters:

`<RateMatchState>` 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["State"](#) on page 76

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:NRESources <RateMatResource>`

Sets the number of rate match resources.

Parameters:

`<RateMatResource>` integer
 Range: 0 to 8
 *RST: 0

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Resources"](#) on page 77

`[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:GRPNumber <RateMatNoOfGrps>`

Sets the number of rate match pattern groups

Parameters:

`<RateMatNoOfGrps>` integer
 Range: 0 to 2
 *RST: 0

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Number of Groups"](#) on page 76

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:RBDList <RateRBPattDList>

Selects an existing data list file from the default directory or from the specific directory.

Parameters:

<RateRBPattDList> string
 Filename incl. file extension or complete file path

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["RB Pattern"](#) on page 77

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:RBPatt <Pattern>, <Bitcount>

Sets the resource block level bitmap pattern directly, alternatively to loadign a data list with the command [\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:RBDList](#).

Parameters:

<Pattern> 275 bits
 *RST: #H0

<Bitcount> integer
 Range: 275 to 275
 *RST: 0

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["RB Pattern"](#) on page 77

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:RS<gr0>:SLOT <RateMatchSlot>

Sets the number of slots.

Parameters:

<RateMatchSlot> integer
 Range: 1 to 2
 *RST: 1

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Slot"](#) on page 77

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
RS<gr0>:SLTPatt <RateMatSlotPatt>, <Bitcount>**

Set the slots to be used as a pattern.

Parameters:

<RateMatSlotPatt> 28 bits
*RST: #H0

<Bitcount> integer
Range: 14 to 28
*RST: 14

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Slot Pattern"](#) on page 78

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
RS<gr0>:PER <RateMatchPeriod>**

Sets the time-domain pattern.

Parameters:

<RateMatchPeriod> 1 | 2 | 5 | 4 | 8 | 10 | 20 | 40
*RST: 2

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Periodicity"](#) on page 78

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
RS<gr0>:PERPatt <RateMatPerdPatt>, <BitCount>**

Sets the periodicity in a pattern form.

Parameters:

<RateMatPerdPatt> 40 bits

<BitCount> integer
Range: 1 to 40
*RST: 2

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Periodicity Pattern"](#) on page 78

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:RATM:
RS<gr0>:GRID <RateMatchGrpID>**

Sets the group identifier for the selected rate match pattern.

Parameters:

<RateMatchGrpID> N | G1 | G2
 *RST: N

Example: See [Example "Configuring the DL BWP rate match settings"](#) on page 186.

Manual operation: See ["Group ID"](#) on page 78

8.11.4 PUCCH Commands

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:ADMRs:STATe	240
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:BPSK:STATe	240
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:HACK:STATe	241
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:PDSHarq:NTMentry	241
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:PDSHarq:TMIValue<gr0>	241

[\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:ADMRs:STATe](#) <AdditionalDMRS>

Defines if additional DMRS is used for PUCCH format 3 and 4.

Parameters:

<AdditionalDMRS> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Additional DMRS"](#) on page 80

[\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:BPSK:STATe](#) <PiBpsk>

If enabled, the $\pi/2$ - BPSK modulation order is used for PUCCH, instead of the default QPSK.

Parameters:

<PiBpsk> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["π/2 - BPSK"](#) on page 80

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:HACK:STATe <PucchHarAckCSI>**

Reserved for future use.

Parameters:

<PucchHarAckCSI> 0
*RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Simultaneous HARQ-ACK-CSI"](#) on page 80

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:PDSHarq:NTMentry <NumUIBpPucchACK>**

Sets the number of entries in the sequence of PDSCH to HARQ timing values.

The value set here defines the available suffixes in the command [\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:PDSHarq:TMIValue<gr0>](#).

Parameters:

<NumUIBpPucchACK> float
Range: 0 to 8

Example: See [\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:PDSHarq:TMIValue<gr0>](#) on page 241.

Manual operation: See ["Number of Entries"](#) on page 81

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUCCh:PDSHarq:TMIValue<gr0> <UserUIBwpPucchD>**

Sets the individual timing values.

Suffix:

<gr0> 0 to 7
Indicator number
Value range depends on the value set with the command [\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUCCh:PDSHarq:NTMentry](#).

Parameters:

<UserUIBwpPucchD> float
Range: 0 to 15

Example:

```
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:PDSHarq:NTMentry 3
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:PDSHarq:TMIValue0 1
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:PDSHarq:TMIValue1 0
SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUCCh:PDSHarq:TMIValue2 15
```

Manual operation: See "Value of Indicator x" on page 81

8.11.5 FRC Commands

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:STATe.....	242
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:TYPE.....	242
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:SCS?.....	243
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:ALRB?.....	244
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC: MODulation?.....	244
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:PASize?....	244
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:RBOffset..	244
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:MAPType..	245
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:PTRS: STATe.....	245

**[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
STATe <FrcState>**

Enables the fixed reference measurement channels (FRC), as defined in TS 38.141.

Parameters:

<FrcState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Fixed reference channels"](#) on page 182.

Manual operation: See "[FRC State](#)" on page 82

**[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
TYPE <FrcType>**

Selects the fixed reference measurement channel (FRC).

The currently available FRCs depend on the selected BWP subcarrier spacing SCS) and selected number of resource blocks.

Parameters:

<FrcType> FR1A11 | FR1A12 | FR1A13 | FR1A14 | FR1A15 | FR1A16 |
 FR1A17 | FR1A18 | FR1A19 | FR2A11 | FR2A12 | FR2A13 |
 FR2A14 | FR2A15 | FR1A21 | FR1A22 | FR1A23 | FR1A24 |
 FR1A25 | FR1A26 | NA | FR1A31 | FR1A32 | FR1A33 |
 FR1A34 | FR1A35 | FR1A36 | FR1A37 | FR1A38 | FR1A39 |
 FR1A310 | FR1A311 | FR1A312 | FR1A313 | FR1A314 |
 FR1A315 | FR1A316 | FR1A317 | FR1A318 | FR1A319 |
 FR1A320 | FR1A321 | FR1A322 | FR1A323 | FR1A324 |
 FR1A325 | FR1A326 | FR1A327 | FR1A328 | FR1A41 |
 FR1A42 | FR1A43 | FR1A44 | FR1A45 | FR1A46 | FR1A47 |
 FR1A48 | FR1A49 | FR1A410 | FR1A411 | FR1A412 |
 FR1A413 | FR1A414 | FR1A415 | FR1A416 | FR1A417 |
 FR1A418 | FR1A419 | FR1A420 | FR1A421 | FR1A422 |
 FR1A423 | FR1A424 | FR1A425 | FR1A426 | FR1A427 |
 FR1A428 | FR1A51 | FR1A52 | FR1A53 | FR1A54 | FR1A55 |
 FR1A56 | FR1A57 | FR1A58 | FR1A59 | FR1A510 | FR1A511 |
 FR1A512 | FR1A513 | FR1A514 | FR1A329 | FR1A330 |
 FR1A331 | FR1A332 | FR2A31 | FR2A32 | FR2A33 | FR2A34 |
 FR2A35 | FR2A36 | FR2A37 | FR2A38 | FR2A39 | FR2A310 |
 FR2A311 | FR2A312 | FR2A41 | FR2A42 | FR2A43 | FR2A44 |
 FR2A45 | FR2A46 | FR2A47 | FR2A48 | FR2A49 | FR2A410 |
 FR2A51 | FR2A52 | FR2A53 | FR2A54 | FR2A55 | FR2A313 |
 FR2A314 | FR2A315 | FR2A316 | FR2A317 | FR2A318 |
 FR2A319 | FR2A320 | FR2A321 | FR2A322 | FR2A323 |
 FR2A324 | FR2A411 | FR2A412 | FR2A413 | FR2A414 |
 FR2A415 | FR2A416 | FR2A417 | FR2A418 | FR2A419 |
 FR2A420 | FR2A56 | FR2A57 | FR2A58 | FR2A59 | FR2A510
 *RST: 1A12

Example: See [Example "Fixed reference channels"](#) on page 182.

Manual operation: See ["FRC"](#) on page 82

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
SCS?**

Queries the SCS of the selected FRC.

Return values:

<SubcarrierSpac> N15 | N30 | N60 | E60 | N120 | N240
 *RST: N30

Example: See [Example "Fixed reference channels"](#) on page 182.

Usage: Query only

Manual operation: See ["Subcarrier Spacing"](#) on page 83

**[[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
ALRB?**

Queries the number of used RBs of the selected FRC.

Return values:

<AllocRB> integer
 Range: 6 to 273
 *RST: 11

Example: See [Example "Fixed reference channels"](#) on page 182.

Usage: Query only

Manual operation: See ["Allocated Resource Blocks"](#) on page 83

**[[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
MODulation?**

Queries the modulation scheme used by the selected FRC.

Return values:

<Modulation> BPSK | BPSK2 | QPSK | QAM16 | QAM64 | QAM256 |
 NSQAM1024 | NSQAM2048 | NSQAM4096
 *RST: QPSK

Example: See [Example "Fixed reference channels"](#) on page 182.

Usage: Query only

Manual operation: See ["Modulation"](#) on page 84

**[[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
PASize?**

Queries the payload size of the selected FRC.

Return values:

<PayloadSize> integer
 Range: 528 to 217128
 *RST: 984

Example: See [Example "Fixed reference channels"](#) on page 182.

Usage: Query only

Manual operation: See ["Payload Size"](#) on page 84

**[[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:
RBOffset <RbOffset>**

Sets the RB offset for the selected FRC.

Parameters:

<RbOffset> integer
 Range: 0 to dynamic
 *RST: 0

Example: See [Example "Fixed reference channels"](#) on page 182.

Manual operation: See ["RB Offset"](#) on page 84

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:MAPType <MappingType>

Sets the PUSCH mapping type for the selected FRC.

Parameters:

<MappingType> A | B
 *RST: A

Example: See [Example "Fixed reference channels"](#) on page 182.

Manual operation: See ["Mapping Type"](#) on page 84

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:FRC:PTRS:STATe <FrcPtrsState>

Enables PTRS for the selected FRC.

Parameters:

<FrcPtrsState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Fixed reference channels"](#) on page 182.

Manual operation: See ["PTRS"](#) on page 84

8.11.6 PUSCH Commands

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTB:CTYPE.....	247
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTA:CTYPE.....	247
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTB:MLENgt.....	247
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTA:MLENgt.....	247
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTB:APIIndex.....	247
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTA:APIIndex.....	247
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DMTA:SID0.....	247

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:SID1.....	247
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:SID0.....	248
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:SID1.....	248
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: TPSTate.....	248
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:MRANK.	248
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: TXConfig.....	248
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: CBSubset.....	249
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: MCSTable.....	249
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: MTTPrecoding.....	249
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: SCRambling:STATe.....	250
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh:DSID....	250
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:STATe.....	250
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:STATe.....	250
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:TMDensity.....	250
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:TMDensity.....	250
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:RB0.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:RB1.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:RB2.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:RB3.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:TP:RB4.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:RB0.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:RB1.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:RB2.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:RB3.....	251
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:TP:RB4.....	251

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:CTYPe <TypeBConfigType>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:CTYPe <TypeAConfigType>
```

Sets the configuration type (type 1 or type 2) and defines the mapping of the DMRS to the physical resource elements

Parameters:

<TypeAConfigType> T1 | T2
*RST: T1

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Config Type"](#) on page 86

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:MLENgtH <TypeBMaxLength>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:MLENgtH <TypeAMaxLength>
```

Sets if single- or double-symbol DMRS is used.

Parameters:

<TypeAMaxLength> integer
Range: 1 to 2
*RST: 2

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Max Length"](#) on page 86

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:APINdEx <TypeBAddPosIdx>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:APINdEx <TypeAAddPosIdx>
```

Sets the additional position index.

Parameters:

<TypeAAddPosIdx> integer
Range: 0 to 3
*RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Additional Position Index"](#) on page 86

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:SID0 <ScramID0>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:SID1 <ScramID1>
```

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:SID0 <ScramID0>**

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:SID1 <ScramID1>**

Sets the scrambling ID.

Parameters:

<ScramID1> integer
Range: 0 to 65535
*RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Scrambling ID 0/Scrambling ID 1"](#) on page 86

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:TPSTate <TransformPrec>**

Enables the precoding operation and thus enables the DFT-s-OFDM.

Parameters:

<TransformPrec> 0 | 1 | OFF | ON
0|OFF
Enables the CP-OFDM.
1|ON
Enables the DFT-s-OFDM.
*RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Transform Precoding"](#) on page 90

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:MRANK <MaximumRank>**

Sets the parameter max rank.

Parameters:

<MaximumRank> integer
Range: 1 to 4
*RST: 4

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Max Rank"](#) on page 91

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:TXConfig <TxConfig>**

Sets if codebook-based (CB) or non-codebook-based (NCB) transmission is used.

Parameters:

<TxConfig> NCB | CB
 *RST: NCB

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["TxConfig"](#) on page 91

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
 PUSCh:CBSubset <CodebookSubset>**

Sets the higher layer parameter `codebookSubset`.

Parameters:

<CodebookSubset> FPNC | PNC | NC
FPNC
 Fully-and-partial-and-non-coherent
PNC
 Partial-non-coherent
NC
 Non-coherent
 *RST: FPNC

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Codebook Subset"](#) on page 91

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
 PUSCh:MCSTable <PuschMcsTable>**

Sets the MSC table and defines the used modulation scheme.

Parameters:

<PuschMcsTable> QAM64 | QAM256 | QAM64LSE
 *RST: QAM64

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["MCS Table"](#) on page 92

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
 PUSCh:MTTPrecoding <PuschMcsTableTP>**

Sets the MCS table.

Parameters:

<PuschMcsTableTP> QAM64 | QAM256 | QAM64LSE
 *RST: QAM64

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["MCS Table Transform Precoding"](#) on page 92

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:SCRambling:STATe <UseScramblingId>**

Sets if a data scrambling ID is used for the initialization of the generator of the PUSCH scrambling sequence.

Parameters:

<UseScramblingId> 0 | 1 | OFF | ON
*RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Use PUSCH Scrambling ID"](#) on page 91

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DSID <DataScrambleId>**

If [\[:SOURce<hw>\]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:SCRambling:STATe](#) 1, sets the data scrambling ID.

Parameters:

<DataScrambleId> integer
Range: 0 to 1023
*RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Data Scrambling ID"](#) on page 92

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:STATe <PtrsTpState>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:STATe <PtrsTPState>**

Simulates the transmission of the higher-layer parameter
transformPrecoderEnabled.

Parameters:

<PtrsTPState> 0 | 1 | OFF | ON
*RST: 0

Example: See [Example "Configuring the PUSCH PTRS with enabled
transform preconding"](#) on page 191.

Manual operation: See ["State"](#) on page 90

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:TMDensity <PtrsTpTimeDens>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:TMDensity <PtrsTpTimeDens>**

Sets the PTRS time density.

Parameters:

<PtrsTpTimeDens> TD1 | TD2
 *RST: TD1

Example:

See [Example "Configuring the PUSCH PTRS with enabled transform precoding"](#) on page 191.

Manual operation:

See ["Time Density"](#) on page 90

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB0 <PtrsTpNumRB0>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB1 <PtrsTpNumRB1>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB2 <PtrsTpNumRB2>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB3 <PtrsTpNumRB3>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:TP:RB4 <PtrsTpNumRB4>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB0 <PtrsTpNumRB0>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB1 <PtrsTpNumRB0>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB2 <PtrsTpNumRB2>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB3 <PtrsTpNumRB3>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:TP:RB4 <PtrsTpNumRB4>
```

Sets the corresponding parameter `sampleDensity`.

Parameters:

<PtrsTpNumRB4> integer
 Range: 1 to 276
 *RST: $RB0 = 1/RB4 = 276$

Example:

See [Example "Configuring the PUSCH PTRS with enabled transform precoding"](#) on page 191.

Manual operation:

See ["N_RB_x"](#) on page 90

8.11.7 PUSCH UCI Commands

```
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:STATe..... 252
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:MODE..... 252
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:ALPHa..... 252
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:HARQ:
OFF0..... 253
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:HARQ:
OFF1..... 253
```

```

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:HARQ:
  OFF2.....253
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF10...253
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF11...253
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF20...253
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:CSI:OF21...253

```

```

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
  STATE <UCIState>

```

Enables the transmission and the configuration of the uplink control information (UCI).

Parameters:

<UCIState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["State"](#) on page 93

```

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
  MODE <PuschUciMode>

```

Defines the information transmitted on the PUSCH.

Parameters:

<PuschUciMode> UCLSch | UCInly
 UCLSch
 Control information and data are multiplexed into the PUSCH.
 UCInly
 Only uplink control information is transmitted on PUSCH.
 *RST: UCLSch

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Mode"](#) on page 93

```

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
  ALPHA <PuschUciAlpha>

```

Sets the scaling parameter alpha.

Parameters:

<PuschUciAlpha> A0_5 | A0_65 | A0_8 | A1_0
 *RST: A05

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Scaling Alpha"](#) on page 93

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    HARQ:OFF0 <PuschUciHarqOff>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    HARQ:OFF1 <PuschUciHarqOf1>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    HARQ:OFF2 <PuschUciHarq2Of>
```

Sets the HARQ information offsets `OFF0` | `OFF1` | `OFF2`, each used for the multiplexing of defined number of HARQ-ACK information bits.

Parameters:

<PuschUciHarq2Of> integer
 Range: 0 to 15
 *RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See "[I_HARQ_Offset,x](#)" on page 94

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    CSI:OF10 <PuschUciCsiOf10>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    CSI:OF11 <PuschUciCsiOf11>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    CSI:OF20 <PuschUciCsiOf20>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:UCI:
    CSI:OF21 <PuschUciCsiOf21>
```

Sets the CSI part 1 (`OF10` | `OF11`) and CSI part 2 (`OF20` | `OF21`) information offsets, where the `OF10` | `OF20` are used if up to 11 CSI bits are multiplexed and `OF11` | `OF21` otherwise.

Parameters:

<PuschUciCsiOf21> integer
 Range: 0 to 18
 *RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See "[I_CSI1_Offset,x](#), [I_CSI2_Offset,x](#)" on page 93

8.11.8 SRS Commands

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:STATe..... 255
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:
    NRSetS..... 255
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:
    SET<gr0>:RSType..... 255
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:
    SET<gr0>:USAGe..... 255
```

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:NRESources.....	256
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:PER.....	256
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:OFFSet.....	256
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:SPOS.....	256
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:SYMNumber.....	257
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:FPOS.....	257
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:FQSHift.....	257
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:BSRS.....	258
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:CSRS.....	258
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:BHOP.....	258
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:REFactor.....	258
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:TRTComb.....	259
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:COFFset.....	259
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:SEQ:CYCShift.....	259
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:SEQ:ID.....	259
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:SEQ:HOPping.....	260
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:NAPort.....	260
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:POWer.....	260
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:PTRS.....	261
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:APMap:MAPCoord.....	261
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:REAL.....	261
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:IMAGinary.....	262
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:MAGNitude.....	263
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS: SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:PHASe.....	263

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:STATE <SrsState>

Enables the SRS transmission.

Parameters:

<SrsState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["State"](#) on page 96

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:NRSets <SrsResourceSets>

Sets the number of SRS sets.

Parameters:

<SrsResourceSets> integers
 Range: 0 to 15
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Resource Sets"](#) on page 97

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:RSType <SrsRsSetRsType>

Sets the SRS transmission type.

Parameters:

<SrsRsSetRsType> PER | APER | SP
 Supported is only PER (periodic) SRS transmission.
 *RST: PER

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Resource Type"](#) on page 97

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:USAGe <SrsRsSetUsage>

Queries the type of the SRS transmission.

Parameters:

<SrsRsSetUsage> NCB | CB | BM | ASW
 NCB = non-codebook based
 CB = codebook
 BM = beam management (reserved for future use)
 ASW = antenna switching (reserved for future use)

*RST: NCB

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Usage"](#) on page 97

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:NRESources <SrsResPerSet>**

Sets the number of SRS resources in the SRS resource set.

Parameters:

<SrsResPerSet> integer
Range: 0 to 4
*RST: 1

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Resources"](#) on page 97

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:PER <SrsRsPeriodicit>**

Sets the SRS repetition factor.

Parameters:

<SrsRsPeriodicit> SL1 | SL2 | SL4 | SL5 | SL8 | SL10 | SL16 | SL20 | SL32 | SL40 |
SL64 | SL80 | SL160 | SL320 | SL640 | SL1280 | SL2560
*RST: SL1

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Periodicity"](#) on page 98

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:OFFSet <SrsRsOffset>**

Sets time position of first SRS allocation within an SRS periodicity.

Parameters:

<SrsRsOffset> integer
Range: 0 to 1279
*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Offset"](#) on page 98

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:SPOS <SrsRsSPosition>**

Sets the starting position (the OFDM symbol number) of the SRS allocation within a slot.

Parameters:

<SrsRsSPostion> integer
 Range: 0 to 5
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Start Pos."](#) on page 98

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:SYMNumber <SrsRsNumSymbols>**

Sets the number of consecutive OFDM symbols the SRS resource spans.

Parameters:

<SrsRsNumSymbols> SYM1 | SYM2 | SYM4
 *RST: SYM1

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["No. Sym."](#) on page 98

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:FPOS <SrsRsFPosition>**

Sets the starting position of the SRS allocation in the frequency domain.

Parameters:

<SrsRsFPosition> integer
 Range: 0 to 67
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Freq. Pos."](#) on page 98

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:FQSHift <SrsRsFreqShift>**

Sets the frequency shift.

Parameters:

<SrsRsFreqShift> integert
 Range: 0 to 268
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Freq. Shift"](#) on page 98

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
  RS:SET<gr0>:RES<user0>:BSRS <SrsRsBSrs>
```

Sets the parameter b-SRS sed to define the length of the SRS sequence.

Parameters:

<SrsRsBSrs> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["B SRS"](#) on page 99

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
  RS:SET<gr0>:RES<user0>:CSRS <SrsRsCSrs>
```

Sets the parameter c-SRS sed to define the length of the SRS sequence.

Parameters:

<SrsRsCSrs> integer
 Range: 0 to 63
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["C SRS"](#) on page 99

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
  RS:SET<gr0>:RES<user0>:BHOP <SrsRsBHop>
```

Sets the parameter b_{Hop} that defines the frequency hopping of the SRS.

Parameters:

<SrsRsBHop> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["B Hop"](#) on page 99

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
  RS:SET<gr0>:RES<user0>:REFactor <SrsRsRepFactor>
```

Sets how many times the SRS symbols are repeated.

Parameters:

<SrsRsRepFactor> REP1 | REP2 | REP4
 *RST: REP1

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Rep. Factor"](#) on page 99

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:TRTComb <SrsRsTransCombo>**

Sets the transmission comb (k_{TC}).

Parameters:

<SrsRsTransCombo> TC2 | TC4

*RST: TC2

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Trans. Comb"](#) on page 99

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:COFFset <SrsRsCombOffset>**

Sets the parameter transmission comb offset.

Parameters:

<SrsRsCombOffset> integer

Range: 0 to 3

*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Comb Offset"](#) on page 99

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:SEQ:CYCShift <SrsRsCyclicShif>**

Sets the number of cyclic shifts, required for the SRS sequence generation.

Parameters:

<SrsRsCyclicShif> float

Range: 0 to 11

*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Cyclic Shift"](#) on page 100

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:SEQ:ID <SrsRsSequenceld>**

Sets the higher-layer parameter sequence ID required for the SRS sequence generation.

Parameters:

<SrsRsSequenceId> float
 Range: 0 to 1023
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Sequence ID"](#) on page 100

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:SEQ:HOPPING <SrsRsHopping>**

Sets the higher-layer parameter `groupOrSequenceHopping` that defines the sequence group, required for the SRS sequence generation.

Parameters:

<SrsRsHopping> N | GRP | SEQ
N
 Neither
GRP
 Group hopping (reserved for future use)
SEQ
 Sequence hopping (reserved for future use)
 *RST: N

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Group or Sequence Hopping"](#) on page 100

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:NAPort <SrsNumAp>**

Sets how many antenna ports (AP) are used for every SRS transmission.

Parameters:

<SrsNumAp> AP1 | AP2 | AP4
 *RST: AP1

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["No. Ports"](#) on page 100

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:POWER <SrsRsPower>**

Sets the SRS power.

Parameters:

<SrsRsPower> float
 Range: -80 to 10
 Increment: 0.01
 *RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Power"](#) on page 101

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:PTRS <SrsPtrsPortIdx>**

Sets the PTRS antenna port index.

Parameters:

<SrsPtrsPortIdx> P1 | P0
 *RST: P0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["PTRS Port Idx"](#) on page 100

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:APMap:MAPCoord <SrsAPMapMode>**

Switches between the cartesian and cylindrical coordinates representation.

Parameters:

<SrsAPMapMode> CARTesian | CYLindrical
 *RST: CARTesian

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Mapping Coordinates"](#) on page 101

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
 RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:REAL
 <SrAPMapDataReal>**

Define the mapping of the antenna ports to the physical antennas.

Suffix:

<s2us0> 1000 to 1003
 Defines the antenna port
 Value range depends on the selected number of antenna ports
 as follows
 $\text{<s2us0>} = 1000 + \{0 \text{ to } \text{NAPort} - 1\}$
 <s3us0> 0
 Row number in the antenna port mapping table

Parameters:**<SrAPMapDataReal>** float

The **REAL** (Magnitude) and **IMAGinary** (Phase) values are interdependent. Their value ranges change depending on each other and so that the resulting complex value is as follows:

$$|\text{REAL} + j \cdot \text{IMAGinary}| \leq 1$$

Otherwise, the values are normalized to Magnitude = 1.

Range: -1 to 1

Increment: 0.001

*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Mapping table"](#) on page 101

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:IMAGinary
<SrsAPMapDataIma>**

Define the mapping of the antenna ports to the physical antennas.

Suffix:**<s2us0>**

1000 to 1003

Defines the antenna port

Value range depends on the selected number of antenna ports as follows

$$\text{<s2us0>} = 1000 + \{0 \text{ to } \text{NAPort} - 1\}$$

<s3us0>

0

Row number in the antenna port mapping table

Parameters:**<SrsAPMapDataIma>** float

The **REAL** (Magnitude) and **IMAGinary** (Phase) values are interdependent. Their value ranges change depending on each other and so that the resulting complex value is as follows:

$$|\text{REAL} + j \cdot \text{IMAGinary}| \leq 1$$

Otherwise, the values are normalized to Magnitude = 1.

Range: -1 to 1

Increment: 0.001

*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Mapping table"](#) on page 101

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
  RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:MAGNitude
  <SrsAPMapDataMag>
```

Defines the mapping of the antenna ports to the physical antennas, cylindrical mapping coordinates are used.

Parameters:

<SrsAPMapDataMag>float

Range: 0 to 1

Increment: 0.001

*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Mapping table"](#) on page 101

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:
  RS:SET<gr0>:RES<user0>:APMap:AP<s2us0>:ROW<s3us0>:PHASe
  <SrsAPMapDataPha>
```

Defines the mapping of the antenna ports to the physical antennas, if cylindrical mapping coordinates are used.

Parameters:

<SrsAPMapDataPha>float

Range: 0 to 360

Increment: 0.1

*RST: 0

Example: See [Example "Configuring one SRS resource set"](#) on page 186.

Manual operation: See ["Mapping table"](#) on page 101

8.11.9 PTRS Commands

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:STaTe.....	265
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:STaTe.....	265
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:STaTe.....	265
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:STaTe.....	265
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:REOF.....	265
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:REOF.....	265
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:REOF.....	265

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:REOF.....	265
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:EPRE.....	266
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:EPRE.....	266
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:POWer.....	266
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:POWer.....	266
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:PORT.....	266
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:PORT.....	266
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:MCS1.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:MCS2.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:MCS3.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:MCS1.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:MCS2.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:MCS3.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:MCS1.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:MCS2.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:MCS3.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:MCS1.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:MCS2.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:MCS3.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:RB0.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTA:PTRS:RB1.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:RB0.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:PUSCh: DMTB:PTRS:RB1.....	267
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:RB0.....	267

<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTA:PTRS:RB1.....</code>	267
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:RB0.....</code>	268
<code>[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh: DMTB:PTRS:RB1.....</code>	268

```

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:STATe <TypeAPtrsState>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:STATe <PtrsState>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:PTRS:STATe <TypeAPtrsState>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:PTRS:STATe <PtrsState>

```

Enables the transmission of phase-tracking reference signals.

Parameters:

<PtrsState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["State"](#) on page 64

```

[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTA:PTRS:REOF <ResElemOffset>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:DMTB:PTRS:REOF <ResElemOffset>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTA:PTRS:REOF <ResElemOffset>
[SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:PTRS:REOF <ResElemOffset>

```

Sets the parameter resource element offset.

Parameters:

<ResElemOffset> RE00 | RE01 | RE10 | RE11
 *RST: RE00

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["RE-offset"](#) on page 65

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:PTRS:EPRE <EpreRatio>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTB:PTRS:EPRE <PtrsEpreRatio>
```

Sets the parameter EPRE ratio.

Parameters:

<PtrsEpreRatio> RAT0 | RAT1
 *RST: RAT0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["EPRE Ratio"](#) on page 65

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:POWer <PtrsPower>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:POWer <PtrsPower>
```

Sets the parameter `ptrs-Power` and thus defines the PUSCH to PTRS power ratio per layer per resource element.

Parameters:

<PtrsPower> P00 | P01
 *RST: P00

Example: `SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:`
 `DMTB:PTRS:POWer P00`

Manual operation: See ["Power"](#) on page 88

```
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:PORT <PtrsMaxNrofPort>
[ :SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:PORT <PtrsMaxNrofPort>
```

Sets the maximum number of configured PTRS ports.

Parameters:

<PtrsMaxNrofPort> P1 | P2
 *RST: P1

Example: `SOURce1:BB:NR5G:UBWP:USER0:CELL0:UL:BWP0:PUSCh:`
 `DMTB:PTRS:PORT P1`

Manual operation: See ["Max Number of Ports"](#) on page 88

```

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:MCS1 <Ptrslmcs1>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:MCS2 <Ptrslmcs2>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:MCS3 <Ptrslmcs3>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:MCS1 <Ptrslmcs1>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:MCS2 <Ptrslmcs2>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:MCS3 <Ptrslmcs3>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:PTRS:MCS1 <Ptrslmcs1>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:PTRS:MCS2 <Ptrslmcs2>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:PTRS:MCS3 <Ptrslmcs3>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTB:PTRS:MCS1 <Ptrslmcs1>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTB:PTRS:MCS2 <Ptrslmcs2>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTB:PTRS:MCS3 <Ptrslmcs3>

```

Sets the threshold values ptrs-MCS_l with $l = \{1, 2, 3\}$.

Parameters:

<Ptrslmcs3> integer
 Range: 0 to 29
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["I_MCS_1/I_MCS_2/I_MCS_3"](#) on page 65

```

[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:RB0 <PtrsNrb0>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTA:PTRS:RB1 <PtrsNrb1>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:RB0 <PtrsNrb0>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
  PUSCh:DMTB:PTRS:RB1 <PtrsNrb1>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:PTRS:RB0 <PtrsNrb0>
[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
  PDSCh:DMTA:PTRS:RB1 <PtrsNrb1>

```

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:PTRS:RB0 <PtrsNrb0>**

**[:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:
PDSCh:DMTB:PTRS:RB1 <PtrsNrb1>**

Sets the threshold values NRB_l with $l = \{0, 1\}$.

Parameters:

<PtrsNrb1> integer
Range: 1 to 276
*RST: 276

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["N_RB_1/N_RB_2"](#) on page 65

8.12 Scheduling Commands

[:SOURce<hw>]:BB:NR5G:SCHeduling:MODE	269
[:SOURce<hw>]:BB:NR5G:SCHeduling:RSSPace	269
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CONTent	269
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CONFlct?	270
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: NALLoc	270
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:SLOT	270
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:MAPType	271
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:FMT	271
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:TPSTate	271
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:SYMNumber	271
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:SYMoffset	272
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:RBNumber	272
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PRACH:RBNumber?	273
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:RBOffset	273
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:POWer	273
[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:STATe	273

<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:REPetitions</code>	274
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:DURation</code>	274
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PERiod</code>	274
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:RESulting:ALLOc<user0>:INFO?</code>	275
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:INFO?</code>	276

`[SOURce<hw>]:BB:NR5G:SCHeduling:MODE <SchedulingMode>`

Defines how the scheduling and the content of the different PDSCH allocations is defined and performed.

Parameters:

`<SchedulingMode>` MANual | AUTO

MANual

No cross-reference between the settings made for the CORE-SET DCIs and the PDSCHs settings.
Configure the PDSCH allocations manually.

AUTO

Content and scheduling of the PDSCH according to the configuration of the CORESET DCIs.

*RST: MANual

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["PDSCH Scheduling"](#) on page 29

`[SOURce<hw>]:BB:NR5G:SCHeduling:RSSPace <RestrToSSpace>`

If enabled, the CCE start index is selected automatically to be within the current search space.

Parameters:

`<RestrToSSpace>` 0 | 1 | OFF | ON

*RST: 0

Example: See [Example "Search space configuration"](#) on page 187.

Manual operation: See ["Restrict to Search Space"](#) on page 29

`[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:CONTent <Content>`

Sets the allocated channel or signal.

Parameters:

<Content> PDSCch | COReset | PUSCh | PUCCh | DUMRe | SPBCh |
 PRACH | CSIRs | SRS | LTERCs
 *RST: PSS

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["Content"](#) on page 104

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CONFLICT?**

Queries if allocations overlap.

Return values:

<Conflict> 0 | 1 | OFF | ON
 *RST: 0

Example: SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CONFLICT?
 // 0
 // there is no conflict, i.e. allocations are not overlapping

Usage: Query only

Manual operation: See ["Conflict"](#) on page 104

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:NALLoc <NumUserAlloc>**

Sets the number of configurable allocations in the selected
 USER<dir0>:BWPart<gr0> group.

Parameters:

<NumUserAlloc> integer
 Range: 0 to 64
 *RST: 0

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["No. of Allocations"](#) on page 105

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:SLOT <Slot>**

Sets the start slot for the selected allocation.

Parameters:

<Slot> integer
 Range: 0 to 31
 *RST: 0

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["Slot"](#) on page 105

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPpart<gr0>:ALLoc<user0>:MAPType <MappingType>**

Defines to which symbols of the PDSCH/PUSCH allocation the demodulation reference signals (DMRS) are mapped.

Parameters:

<MappingType> A | B
*RST: A

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Mapping Type"](#) on page 106

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPpart<gr0>:ALLoc<user0>:FMT <AllocPucchForma>**

In uplink, selects the PUCCH format.

Parameters:

<AllocPucchForma> F0 | F1 | F2 | F3 | F4
*RST: F0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Format"](#) on page 106

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPpart<gr0>:ALLoc<user0>:TPStAte <TrPrecState>**

In UL, enables the optional DFT-s-OFDM instead of CP-OFDM per PUSCH.

Parameters:

<TrPrecState> 0 | 1 | OFF | ON
*RST: 0

Example:
SOURce1:BB:NR5G:LINK UP
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER1:BWPpart0:ALLoc0:TPStAte 1

Manual operation: See ["Transform Precoding"](#) on page 107

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPpart<gr0>:ALLoc<user0>:SYMNumber <UserAllocSymNum>**

Sets the number of symbols the allocation spans.

Parameters:

<UserAllocSymNum> integer

Range: 1 to 14

*RST: 1

Example:See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.**Manual operation:** See ["No. Sym."](#) on page 106

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:SYMoffset <UserAllocSymOff>**

Sets the allocation's start (OFDM symbol) within the slot.

Parameters:

<UserAllocSymOff> integer

Range: 0 to 13

*RST: 0

Example:See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.**Manual operation:** See ["Sym. Offset"](#) on page 106

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:RBNumber <UserAllocRBNum>**

Sets the number of resource blocks (RB) the allocation spans.

Parameters:

<UserAllocRBNum> integer

CORESET allocations span always a multiple of 6 resource blocks. Thus, only values that are multiple of 6 are allowed for these allocations.

The number of resource blocks that are available for PUSCH depends on whether the transform precoding is enabled or not (that is if DFT-s-OFDM is applied), see [\[:SOURce<hw>\]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:TPState](#).

Query the number of resource blocks used by PRACH with the command [\[:SOURce<hw>\]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PRACH:RBNumber?](#).

Range: 20 to 275

*RST: 20

Example:See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.**Manual operation:** See ["No. RB"](#) on page 107

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:RBNumber?**

Queries the number of resource blocks the PRACH allocation spans.

Return values:

<UserAllocRBNum> integer
Range: 3 to 24
*RST: 12

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Usage: Query only

Manual operation: See ["No. RB"](#) on page 107

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:RBOffset <RbOff>**

Adds a frequency offset and sets the start resource block of the selected allocation.

Parameters:

<RbOff> integer
Range: 0 to 255
*RST: 0

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["Offs. RB"](#) on page 107

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:POWer <Power>**

Sets the power for the selected allocation relative to the power of the other allocations.

Parameters:

<Power> float
Range: -80 to 10.0
Increment: 0.01
*RST: 0.0

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["Power"](#) on page 109

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:STATe <State>**

Activates the allocation.

Parameters:

<State> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Manual operation: See ["State"](#) on page 109

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:REPetitions <Repetition>**

Defines if and how often the allocation is repeated.

Parameters:

<Repetition> OFF | SUBFrame | FRAME | SLOT | CUSTom
 *RST: depends on allocation number

Example: See [Example "Configuring the configurable allocations in the users and BWPs groups"](#) on page 182.

Example:

```
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER1:BWPart0:ALLoc0:REPetitions CUSTom
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER1:BWPart0:ALLoc0:PERiod 2
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER1:BWPart0:ALLoc0:DURation 5
```

Manual operation: See ["Repetition/Repetition Type"](#) on page 109

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:DURation <RepDuration>**

Sets in how many consecutive slots within a frame the allocation is repeated.

Parameters:

<RepDuration> integer
 Range: 1 to 10
 *RST: 1

Example: See [\[:SOURce<hw>\]:BB:NR5G:SCHed:CELL<ch0>:
 SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:
 REPetitions.](#)

Manual operation: See ["Duration"](#) on page 110

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PERiod <RepPeriod>**

Sets the repetition periodicity.

Parameters:

<RepPeriod>

integer

Range: 1 to max*

*RST: 10

*) Max repetition period is calculated as follows:

RepPeriod_{max} = #SlotsPerFrame * "SequencLength", where:

#SlotsPerFrame: is the number of available slots and depends on the used subcarrier spacing.

"SequencLength": is the ARB sequence length as set with the command [:SOURce<hw>]:BB:NR5G:OUTPut:SEQLen.

Example:

See [:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:
SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:
REPetitions.

Manual operation: See "Period" on page 110

[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:RESulting:ALLoc<user0>:INFO?

Queries scheduling information on the selected automatically created allocation.

Return values:

<ResultingInfo>

string

Syntax follows the [JSON schema](#) format.

Retrieved information depends on the channel type.

Example:

```
SOURce1:BB:NR5G:SCHed:CELL0:SUBF1:USER0:
BWPart0:RESulting:ALLoc0:INFO?
{
  "commonSets": {
    "powerDb": 0.0,
    "slotInFrame": 2
  },
  "content_type": "Coreset",
  "content": {
    "pdcchs": [
      {
        "scramblingRnti": 0,
        "crcRnti": 65535,
        "aggregationLevel": "L1",
        "cceStartIndex": 0
      }
    ]
  }
}
```

Usage:

Query only

Manual operation: See "Info" on page 108

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:INFO?**

Queries information on the current configuration.

Return values:

<UserAllocInfo> string

Example:

SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:INFO?

Response:

```
{
  "common_settings": {
    "power_dB": 0.0,
    "slot_in_frame": 0
  },
  "content_type": "PXsch",
  "content": {
    "frequency_domain_allocation_type": "FrequencyAllocationType1",
    "frequency_domain_allocation": {
      "vrb_offset": 0,
      "vrb_num": 273
    },
    "transport_block_settings_no_coding": [
      {
        "modulation": "QAM16"
      },
      {
        "modulation": "QAM64"
      }
    ],
    "ptrs_settings": {
      "antenna_port": [
        1002
      ]
    }
  }
}
```

Usage: Query only

Manual operation: See ["Info"](#) on page 128

8.13 PDSCH/PUSCH Commands

[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:TYPE.....	279
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh:NCW.....	279
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh:SBCZero:STATe.....	279
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh:PATGrp.....	279
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>[:CW<s2us0>]:MOD.....	280
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>[:CW<s2us0>]:PHYSBits?.....	280
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:DMRS:SEQGen.....	280
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:DMRS:NSID.....	281
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:DMRS:LENGth.....	281
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:DMRS:SLTSymbols?.....	281
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:DMRS:POWEr.....	282
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:TXSCHEME:NLAYers.....	282
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:TXSCHEME:CDMData.....	282
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh:APSelect<s2us0>.....	283
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:APSelect<s2us0>.....	283
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:TXSCHEME:TPMidx.....	283
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh:TXSCHEME:INTervp.....	284
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:TXSCHEME:SRI.....	284
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:PTRS:MODE.....	284
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:PTDMrs.....	285
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:PTRS:STATe.....	285
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:PTRS:TMDen.....	285
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCCh PUSCh:PTRS:REOF.....	285

[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCh:PTRS:FRQDen.....	286
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PDSCh:PTRS:EPRE.....	286
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:POWer.....	286
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:PORT.....	287
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:TP:MODE.....	287
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:TP:STATe.....	287
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:TP:TMDensity.....	287
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:TP:NGRPs.....	288
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:PTRS:TP:SPPG.....	288
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>[:CW<s2us0>]:PDSCh PUSCh:CCODing:IMCS.....	288
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>[:CW<s2us0>]:PDSCh PUSCh:CCODing:TCRate?.....	289
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>[:CW<s2us0>]:PDSCh PUSCh:CCODing:TBSIZE?.....	289
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>[:CW<s2us0>]:PDSCh PUSCh:CCODing:RVIndex.....	289
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:UCI:CSI1:BITS.....	290
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:UCI:CSI2:BITS.....	290
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:UCI:ACK:BITS.....	290
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:UCI:CSI1:PATtern.....	290
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:UCI:CSI2:PATtern.....	290
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:PUSCh:UCI:ACK:PATtern.....	290
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:APMap:MAPCoordinates.....	291
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:IMAGinary.....	291
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:REAL.....	291
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:MAGNitude.....	292
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:PHASe.....	292

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:TYPE <PXschType>
```

Defines the PDSCH/PUSCH type by selecting the DCI format by that the PDSCH/PUSCH content is defined.

Parameters:

<PXschType> F00 | F01 | F10 | F11
 *RST: F00

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["PDSCH Type/PUSCH Type"](#) on page 113

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh:NCW <NumUserAllocCW>
```

For PDSCH, sets if one or two codewords are used.

Parameters:

<NumUserAllocCW> integer
 Range: 1 to 2
 *RST: 1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Number of Codewords"](#) on page 113

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh:SBCZero:STATe <UserAllocPdschS>
```

Sets if the PDSCH settings are scheduled by CORESET 0 or not.

Parameters:

<UserAllocPdschS> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Scheduled by CORESET 0"](#) on page 113

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh:PATGrp <RateMatPattGrp>
```

For PDSCH allocations, selects one of the configured rate match patten groups.

Parameters:

<RateMatPattGrp> N | G1 | G2
 N = none
 *RST: N

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Rate Match Pattern Group"](#) on page 113

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:MOD <Modulation>**

Sets the modulation scheme.

Parameters:

<Modulation> BPSK | BPSK2 | QPSK | QAM16 | QAM64 | QAM256
 Supported modulations depend on the channel.
 *RST: BPSK

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Modulation"](#) on page 113

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PHYSbits?**

Queries the size of the selected allocation in bits.

Return values:

<AllocPhysBits> integer
 Range: 0 to 65535
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Usage: Query only

Manual operation: See ["Phys. Bits"](#) on page 114

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:DMRS:SEQGen <SeqGen>**

Sets how the scrambling ID for DMRS is derived.

Parameters:

<SeqGen> CELLid | DMRSid
DMRSid
 N_{ID}^{DMRS} : Scrambling ID (i.e. n_{ID}) = DMRS scrambling ID (i.e. N_{ID}^{DMRS})

CELLid

N_{ID}^{Cell} : Scrambling ID (i.e. n_{ID}) = cell ID (i.e. N_{ID}^{Cell})

*RST: CELLid

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Sequence Generation"](#) on page 116

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:PDSCh|PUSCh:DMRS:NSID <NScid>**

Sets the scrambling ID (N_{ID}).

Parameters:

<NScid> integer
Range: 0 to 1
*RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["N_SCID"](#) on page 117

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:PDSCh|PUSCh:DMRS:LENGth <DmrsLen>**

Defines whether single- or double-symbol DMRS is used.

Parameters:

<DmrsLen> integer
Range: 1 to 2
*RST: 1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Length"](#) on page 117

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:PDSCh|PUSCh:DMRS:SLTSymbols?**

Queries the slot number of DMRS symbols.

Return values:

<PxschDmrsSlotSy> string
Range: 0 char to 256 char

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Usage: Query only

Manual operation: See ["Slot Symbols"](#) on page 117

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:DMRS:POWer <DmrsPower>**

Sets the power of the PDSCH/PUSCH, relative to the power of the other channels.

Parameters:

<DmrsPower> float
Range: -80.00 to 10.00
Increment: 0.001
*RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Power"](#) on page 117

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:TXSCHEME:NLAYers
<TxNumLayers>**

For DCI format 0_1, sets the number of precoding layers.

Parameters:

<TxNumLayers> integer
Range: 1 to 8
*RST: 1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Number of Layers"](#) on page 115

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:TXSCHEME:CDMData
<CDMGroupData>**

For DCI format 0_1, sets the number of DMRS CDM groups without data.

Parameters:

<CDMGroupData> integer
Range: 0 to 2
*RST: 1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["CDM Groups w/o Data"](#) on page 115

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh:APSelect<s2us0> <PdschAPSel>
```

Queries the antenna ports.

Suffix:

<s2us0> 0 to (Number of layers - 1)
 Antenna port

Parameters:

<PdschAPSel> AP1000 | AP1001 | AP1002 | AP1003 | AP1004 | AP1005 |
 AP1006 | AP1007
*RST: AP1000

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#)
 on page 183.

Manual operation: See ["Antenna Ports"](#) on page 125

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:APSelect<s2us0> <PuschAPSel>
```

Queries the antenna ports.

Parameters:

<PuschAPSel> AP0 | AP1 | AP2 | AP3 | AP4 | AP5 | AP6 | AP7
*RST: AP0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Antenna Ports"](#) on page 125

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:TXSCHEME:TPMidx <TxTPMIIndex>
```

For PUSCH DCI format 0_1, sets the TPMI index, required to select the precoding matrix.

Parameters:

<TxTPMIIndex> integer
 Range: 0 to 27
 *RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["TPMI"](#) on page 115
 See ["SRS Resource Indicator"](#) on page 115

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh:TXSCHEME:INTervp
  <UserAllocPdschV>
```

If [:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:DL:BWP<dir0>:PDSCh:VPINter VP2 | VP4, enables interleaved VRB-to-PRB mapping.

Parameters:

<UserAllocPdschV> 0 | 1 | OFF | ON

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Example: SOURce1:BB:NR5G:UBWP:USER0:CELL0:DL:BWP0:PDSCh:VPINter VP2
 SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PDSCh:TXSCHEME:INTervp 1

Manual operation: See ["Interleaved VRB-to-PRB"](#) on page 115

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:TXSCHEME:SRI <AllocPuschSri>
```

Selects the SRS resource to be used.

Parameters:

<AllocPuschSri> integer

The max value depends on the number of SRS resources, as set with the command [:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:SRS:RS:SET<gr0>:NRESources.

Range: 0 to 15

*RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:MODE <PtrsMode>
```

Sets if the PTRS configuration is defined, manually (with the . . . PDSCH:PTRS: . . . commands) or automatically (from the current PDSCH DMRS settings).

Parameters:

<PtrsMode> MANual | AUTO

*RST: MANual

Manual operation: See ["Mode"](#) on page 122

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:PTDMrs <PtrsDmrsAssocia>
```

Sets the PTRS-DMRS association.

Parameters:

<PtrsDmrsAssocia> integer
 Range: 0 to 3
 *RST: 0

Example: SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:
 BWPart0:ALLoc0:PUSCh:PTRS:PTDMrs 0

Manual operation: See ["PTRS-DMRS Association "](#) on page 122

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:STATe
  <PXschPtrsState>
```

Enables the transmission of phase-tracking reference signals.

Parameters:

<PXschPtrsState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#)
 on page 183.

Manual operation: See ["State"](#) on page 122

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:TMDen
  <PtrsTimeDensity>
```

Sets the parameter time density.

Parameters:

<PtrsTimeDensity> TD1 | TD2 | TD4
 *RST: TD1

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#)
 on page 183.

Manual operation: See ["Time Density \(L_PTRS\)"](#) on page 122

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PDSCh|PUSCh:PTRS:REOF <PtrsREOffset>
```

Sets the parameter resource element offset.

Parameters:

<PtrsREOffset> RE00 | RE01 | RE10 | RE11
 *RST: RE00

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["RE-offset"](#) on page 122

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PDSCCh:PTRS:FRQDen <PtrsFreqDensity>**

Sets the parameter frequency density.

Parameters:

<PtrsFreqDensity> FD2 | FD4
 *RST: FD2

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Frequency Density \(K_PTRS\)"](#) on page 123

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PDSCCh:PTRS:EPRE <EpreRatio>**

Sets the parameter EPRE ratio.

Parameters:

<EpreRatio> RAT0 | RAT1
 *RST: RAT0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["EPRE Ratio"](#) on page 123

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:POWer <PtrsPower>**

Sets the parameter `ptrs-Power` and thus defines the PUSCH to PTRS power ratio per layer per resource element.

Parameters:

<PtrsPower> P00 | P01
 *RST: P00

Example: SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:
 BWPart0:ALLoc0:PUSCh:PTRS:POWer P00

Manual operation: See ["Power"](#) on page 123

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:PORT <MaxNrofPorts>
```

Sets the maximum number of configured PTRS ports.

Parameters:

<MaxNrofPorts> P1 | P2
 *RST: P1

Example: SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:
 BWPart0:ALLoc0:PUSCh:PTRS:PORT P1

Manual operation: See ["Max Number of Ports"](#) on page 123

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:MODE <PtrsTpMode>
```

Sets if the PTRS configuration is defined, manually (with the ... PUSCH:PTRS:TP: ... commands) or automatically (from the current PUSCH DMRS settings).

Parameters:

<PtrsTpMode> MANual | AUTO
 *RST: MANual

Example: See [Example "Configuring the PUSCH PTRS with enabled transform preconding"](#) on page 191.

Manual operation: See ["Mode"](#) on page 124

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:STATE <PtrsTPState>
```

Enables the transmission of phase-tracking reference signals.

Parameters:

<PtrsTPState> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PUSCH PTRS with enabled transform preconding"](#) on page 191.

Manual operation: See ["State"](#) on page 124

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:TMDensity
  <PtrsTpTimeDens>
```

Sets the higher-layer parameter `timeDensity`.

Parameters:

<PtrsTpTimeDens> TD1 | TD2

TD1 $L_{PT-RS} = 1$ **TD2** $L_{PT-RS} = 2$

*RST: TD1

Example: See [Example "Configuring the PUSCH PTRS with enabled transform precoding"](#) on page 191.

Manual operation: See ["Time Density \(\$L_{PTRS}\$ \)"](#) on page 124

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:NGRPs <PtrsTpNumGroups>**

Sets the number of PTRS groups.

Parameters:

<PtrsTpNumGroups> G2 | G4 | G8

*RST: G2

Example: See [Example "Configuring the PUSCH PTRS with enabled transform precoding"](#) on page 191.

Manual operation: See ["Number of PTRS Groups"](#) on page 125

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUSCh:PTRS:TP:SPPG <PtrsTpSamplesPG>**

Sets the number of samples per PTRS group.

Parameters:

<PtrsTpSamplesPG> S4 | S2

*RST: S2

Example: See [Example "Configuring the PUSCH PTRS with enabled transform precoding"](#) on page 191.

Manual operation: See ["Samples Per PTRS Group"](#) on page 125

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:IMCS
<ChanCodingIMCS>**

Sets the modulation and coding scheme index.

Parameters:

<ChanCodingIMCS> integer

Range: 0 to 28

*RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See "[I_MCS](#)" on page 118

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:
TCRate?**

Queries the target code rate for the selected modulation and coding scheme.

Return values:

<TargetCodeRate> float
 Range: 0.0 to 1.0
 Increment: 0.001
 *RST: 0.0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Usage: Query only

Manual operation: See "[Target Code Rate](#)" on page 119

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:
TBSize?**

Queries the resulting transport block size (TBS).

Return values:

<TransBlockSize> integer
 Range: 0 to 3824
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Usage: Query only

Manual operation: See "[Transport Block Size](#)" on page 119

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>[:CW<s2us0>]:PDSCh|PUSCh:CCODing:
RVIndex <RVIndex>**

Sets the redundancy version index.

Parameters:

<RVIndex> integer
 Range: 1 to 2
 *RST: 0

Example: See [Example "Configuring the PDSCH incl. PTRS settings"](#) on page 183.

Manual operation: See ["Redundancy Version Index"](#) on page 119

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI1:BITS <PuschUciCSI1Bit>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI2:BITS <PuschUCICSI2Bit>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:ACK:BITS <PuschUciAckBit>
```

Sets the number of ACK/CSI 1/CSI 2 bits.

Parameters:

<PuschUciAckBit> float
 Range: 0 to 128
 Increment: 1
 *RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["Number of ACK Bits"](#) on page 120

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI1:PATtern <CSI1Pattern>,
  <CSI1Bitcount>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:CSI2:PATtern <CSI2Pattern>,
  <CSI2Bitcount>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUSCh:UCI:ACK:PATtern <AckPattern>,
  <AckBitcount>
```

Sets the ACK/CSI 1/CSI 2 bits in pattern form.

Parameters:

<AckPattern> 128 bits
 Bit pattern
 *RST: #H0

<AckBitcount> integer
 Pattern length, should be the same as the length set with the
 command `[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:`
 `SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:`
 `PUSCh:UCI:ACK:BITS`.
 Range: 0 to 128
 *RST: 0

Example: See [Example "Configuring the PUSCH settings"](#) on page 189.

Manual operation: See ["ACK Pattern"](#) on page 120

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:APMap:MAPCoordinates <APCoordMapMode>
```

Switches between the cartesian and cylindrical coordinates representation.

Parameters:

<APCoordMapMode> CARTesian | CYLindrical

*RST: CARTesian

Example: See [Example "Antenna port settings"](#) on page 185.

Manual operation: See ["Mapping Coordinates"](#) on page 127

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:
  IMAGinary <APMapDataImag>
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:APMap:COL<s2us0>:ROW<s3us0>:REAL
  <APMapDataReal>
```

Define the mapping of the antenna ports to the physical antennas.

Suffix:

<s2us0> 0 to 7
Defines the antenna port (i.e. the column in the antenna port mapping table)
Value range depends on the selected number of layers
See also [Table 5-4](#).

<s3us0> 0
Row number in the antenna port mapping table

Parameters:

<APMapDataReal> float
The REAL (Magnitude) and IMAGinary (Phase) values are interdependent. Their value ranges change depending on each other and so that the resulting complex value is as follows:
 $|\text{REAL} + j \cdot \text{IMAGinary}| \leq 1$
Otherwise, the values are normalized to Magnitude = 1.
Range: -1 to 1
Increment: 0.001
*RST: 0

Example: See [Example "Antenna port settings"](#) on page 185.

Manual operation: See ["Mapping Table"](#) on page 127

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLOc<user0>:APMap:COL<s2us0>:ROW<s3us0>:
  MAGNitude <APMapDataMagn>
```

Defines the mapping of the antenna ports to the physical antennas, cylindrical mapping coordinates are used.

Parameters:

<APMapDataMagn> float

Range: 0 to 1
Increment: 0.001
*RST: 0

Example: See [:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:
SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:
APMap:COL<s2us0>:ROW<s3us0>:PHASe on page 292.

Manual operation: See "Mapping Table" on page 127

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLOc<user0>:APMap:COL<s2us0>:ROW<s3us0>:PHASe
  <APMapDataPhase>
```

Defines the mapping of the antenna ports to the physical antennas, if cylindrical mapping coordinates are used.

Parameters:

<APMapDataPhase> float

Range: 0 to 360
Increment: 0.1
*RST: 0

Example: SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:
BWPart0:ALLOc1:APMap:MAPCoordinates CYL
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:
BWPart0:ALLOc1:APMap:COL0:ROW0:PHASe 0
SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:
BWPart0:ALLOc1:APMap:COL0:ROW0:MAGNitude 1

Manual operation: See "Mapping Table" on page 127

8.14 CORESET Commands

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
  ALLOc<user0>:CS:PREGran.....295
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
  ALLOc<user0>:CS:ID.....295
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
  ALLOc<user0>:CS:DMRS:SCRam:STATE.....296
```

[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DMRS:SCRam:ID.....	296
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:IL:STATe.....	296
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:IL:BUNSize.....	296
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:IL:SHIDx.....	297
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:IL:SIZE.....	297
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DMRS:SPACe<s2us0>:AGGLeve!.....	297
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DMRS:SPACe<s2us0>:MAXCandidate.....	298
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCCes:STATe.....	298
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCCes:DATA.....	298
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCCes:DLISt.....	299
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCCes:PATTern.....	299
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:TCI.....	299
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:AULBwp.....	299
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:NDCl.....	300
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:USAGe.....	300
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:RNTI.....	300
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:FMT.....	301
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:SSP.....	301
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:AGGLeve!.....	301
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:CANDidate.....	301
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:INDex.....	302
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:CPDSch.....	302
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:HBTS?.....	302
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:FRDRes.....	303
[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TIDRes.....	303

[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:FRHFlag.....	303
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:MCS.....	304
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB2:MCS.....	304
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:NDI.....	304
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB2:NDI.....	304
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB1:RV.....	304
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TB2:RV.....	304
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:HAPRoc.....	305
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TPUSch.....	305
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:VTPRb.....	305
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DAI2.....	306
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DAI1.....	306
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:ANTPorts.....	306
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:SRSRequest.....	306
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PTDMrs.....	307
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:DMSQinit.....	307
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:USINd.....	307
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:TPUCch.....	307
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PUCResind.....	308
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PDSHarq.....	308
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:SIINd.....	308
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:RESVed.....	309
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:BWINd.....	309
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:PRBBundling.....	309
[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>: ALLoc<user0>:CS:DCI<s2us0>:RMINd.....	309

<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:ZCRTrigg</code>	310
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TCI</code>	310
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CBGTi</code>	310
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CBGFi</code>	310
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:STaTe</code>	311
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DAta</code>	311
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DLISt</code>	311
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:BITLength</code>	312
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:PATtern</code>	312

`[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:PREGran <PrecoderGranula>`

Sets the value of the higher-layer parameter `precoderGranularity`, as defined in TS 38.211.

Parameters:

`<PrecoderGranula>` REG | ACRB

REG
REG bundle (`sameAsREG-bundle`)

ACRB
All contiguous RBs (`allContiguousRBs`)

*RST: REG

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Precoder Granularity"](#) on page 130

`[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:ID <ID>`

Sets the CORESET ID.

Parameters:

`<ID>` integer

Range: 0 to 11

*RST: 1

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["CORESET ID"](#) on page 130

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DMRS:SCRam:STATe <CSDmrsScramStat>**

If enabled, the DMRS ID is used to calculate the PUCCH scrambling sequence.

Parameters:

<CSDmrsScramStat> 0 | 1 | OFF | ON

*RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Use DMRS Scrambling ID"](#) on page 130

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DMRS:SCRam:ID <DmrsScramId>**

Sets the DMRS scrambling ID.

Parameters:

<DmrsScramId> integer

Range: 0 to 65535

*RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["ID"](#) on page 130

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:STATe <CSIntlState>**

If enabled, the CCE-to-REG mapping is interleaved.

Parameters:

<CSIntlState> 0 | 1 | OFF | ON

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Interleaving State"](#) on page 131

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:BUNSize <BundleSize>**

Sets the bundle size L.

Parameters:

<BundleSize> BS2 | BS3 | BS6

*RST: BS6

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Bundle Size"](#) on page 131

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:SHIDx <CSIntlShiftIdx>**

Sets the shift index n_{shift} .

Parameters:

<CSIntlShiftIdx> integer
Range: 0 to 274
*RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Shift Index"](#) on page 131

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:IL:SIZE <CSIntlSize>**

Sets the interleaver size R.

Parameters:

<CSIntlSize> IS2 | IS3 | IS6
*RST: IS2

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Size"](#) on page 131

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DMRS:SPACe<s2us0>:AGGLLevel?**

Queries the possible aggregation levels.

Return values:

<AggrLevel> AL1 | AL2 | AL4 | AL8 | AL16
*RST: AL1

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Usage: Query only

Manual operation: See ["Aggregation Level"](#) on page 132

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLOc<user0>:CS:DMRS:SPACe<s2us0>:MAXCandidate
  <MaxCandidate>
```

Sets the maximum number of candidates allowed for the selected aggregation level.

Parameters:

<MaxCandidate> 1 | 2 | 3 | 4 | 5 | 6 | 8
 *RST: 1

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["Max Candidates"](#) on page 132

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLOc<user0>:CS:DCCes:STATe <DummyCceState>
```

Defines if unused CORESET resources (CCEs) are used for data transmission or not.

Parameters:

<DummyCceState> 0 | 1 | ALLowpdsch
 0
 Disables data transmission in the unused CCEs.
 1
 Fills unused CCEs with dummy data, as set with the command
 [\[:SOURce<hw>\]:BB:NR5G:SCHeD:CELL<ch0>:](#)
 [SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:](#)
 [CS:DCCes:DATA.](#)
 ALLowpdsch
 Allows PDSCH transmission in the unused CCEs.
 *RST: 1

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["Unused CCEs"](#) on page 134

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLOc<user0>:CS:DCCes:DATA <DummyCceDataSrc>
```

Sets the data source for the dummy CCEs.

Parameters:

<DummyCceDataSrc> PN9 | PN11 | PN15 | PN16 | PN20 | PN21 | PN23 | PATtern |
 DLISt | ZERO | ONE
 *RST: PN9

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["Dummy CCEs Data Source"](#) on page 134

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCCes:DLISt <DummyCceDataLst>
```

Selects an existing data list file from the default directory or from the specific directory.

Parameters:

<DummyCceDataLst> string

Filename incl. file extension or complete file path

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Dummy CCEs Data Source"](#) on page 134

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCCes:PATtern <Pattern>, <BitCount>
```

Sets a bit pattern as a data source.

Parameters:

<Pattern> 64 bits

*RST: #H0

<BitCount> integer

Range: 1 to 64

*RST: 1

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Dummy CCEs Data Source"](#) on page 134

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:TCI <Nr5gCSTCInDCI>
```

Enables support of quasi co-location TCI field in DCI format 1_1.

Parameters:

<Nr5gCSTCInDCI> 0 | 1 | OFF | ON

Example:

```
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:TC 1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:BWPart0:ALLoc0:CS:DCI0:TCI 1
```

Manual operation: See ["TCI Present In DCI"](#) on page 135

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:AULBwp <ActiveULBwp>
```

Sets the index of the UL BWP used for the calculation of the DCI bit size.

Parameters:

<ActiveULBwp> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Active UL BWP"](#) on page 135

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:NDCI <NumDci>**

Sets the number of DCIs.

Parameters:

<NumDci> integer
 Range: 1 to 4
 *RST: 1

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Number of DCIs"](#) on page 135

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:USAGe <Usage>**

Sets the RNTI with that the CRC is scrambled.

Parameters:

<Usage> C | CS | P | SI | RA | TC | SPCS | SFI | INT | TPUS | TPUC |
 TSRS
 *RST: C

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Usage"](#) on page 135

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:RNTI <RNTI>**

Queries the C-RNTI of the associated user.

Parameters:

<RNTI> integer
 Range: 0 to 65535
 *RST: 0

Manual operation: See ["RNTI"](#) on page 135

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:FMT <AllocDciFmt>
```

Sets the DCI format.

Parameters:

<AllocDciFmt> F00 | F01 | F10 | F11 | F20 | F21 | F22 | F23 | CUSTom
 F20 | F21 | F22 | F23 are reserved for future use.
 *RST: F00

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format"](#) on page 136

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:SSP <SearchSpace>
```

Sets the search space for the selected DCI.

Parameters:

<SearchSpace> USS | CSS0 | CSS1 | CSS2 | CSS3
 *RST: USS

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Search Space"](#) on page 136

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:AGGLLevel <DciAggLvl>
```

Sets the aggregation level for the selected DCI.

Parameters:

<DciAggLvl> AL1 | AL2 | AL4 | AL8 | AL16
 *RST: 4

Example: See [Example "Search space configuration"](#) on page 187.

Manual operation: See ["Aggregation Level"](#) on page 137

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CANDIdate <Candidate>
```

Selects the candidate number for the selected aggregation level, where the maximum number of candidates per aggregation level are set with the command [:

[SOURce<hw>\]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DMRS:SPACe<s2us0>:MAXCandidate.](#)

Parameters:

<Candidate> integer
 Range: 0 to 8
 *RST: 0

Example: See [Example "Search space configuration"](#) on page 187.

Manual operation: See ["Candidate"](#) on page 137

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:INDex <DciCCEIdx>**

If [\[:SOURce<hw>\]:BB:NR5G:SCHeduling:RSSpace 0](#), sets the index and thus the position of the first CCE of the PDCCH.

Parameters:

<DciCCEIdx> integer
 Range: 0 to 10
 *RST: 0

Example: See [Example "Search space configuration"](#) on page 187.

Manual operation: See ["CCE Index"](#) on page 137

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CPDSch <CreatePDSCH>**

Use this command to create the content and scheduling of the PDSCH automatically, as configured with the DCI format 1_0 parameters.

Parameters:

<CreatePDSCH> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Create PDSCH"](#) on page 138

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:HBTS?**

Queries the resulting bit data as selected with the DCI format parameters.

Return values:

<PatternHigh> 64 bits

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Usage: Query only

Manual operation: See ["Bit Data > Pattern"](#) on page 138

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:FRDRes <FreqDomResAssig>
```

Sets the DCI field frequency domain resource assignment.

Parameters:

<FreqDomResAssig> integer

Range: 0 to 262143

*RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 0_1"](#) on page 139
 See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TIDRes <TimeDomResAssig>
```

Sets the DCI field time domain resource assignment.

Parameters:

<TimeDomResAssig> integer

Range: 0 to 15

*RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 0_1"](#) on page 139
 See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:FRHFlag <FreqHopFlag>
```

Sets the DCI field frequency hopping flag.

Parameters:

<FreqHopFlag> 0 | 1 | OFF | ON

*RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TB1:MCS <MCSTB1>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TB2:MCS <MCSTB2>
```

Sets the DCI field modulation and coding scheme per transport block (TB).

Parameters:

<MCSTB2> integer
 Range: 0 to 31
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TB1:NDI <NDITB1>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TB2:NDI <NDITB2>
```

Sets the DCI field new data indicator per transport block (TB).

Parameters:

<NDITB2> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TB1:RV <RVTB1>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TB2:RV <RVTB2>
```

Sets the DCI field redundancy version per transport block (TB).

Parameters:

<RVTB2> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:HAPRoc <HarqProc>
```

Sets the DCI field HARQ process number.

Parameters:

<HarqProc> integer
 Range: 0 to 15
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 0_1"](#) on page 139
 See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TPUSch <TpcPusch>
```

Sets the DCI field TPC command for scheduling PUSCH.

Parameters:

<TpcPusch> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
 See ["DCI Format 0_1"](#) on page 139

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:VTPRb <VrbToPrb>
```

Sets the DCI field VRB-to-PRB mapping.

Parameters:

<VrbToPrb> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DAI2 <DAI2>
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DAI1 <DAI1>
```

Sets the DCI filed DL assignment index.

Parameters:

<DAI1> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_1"](#) on page 139
 See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:ANTPorts <AntPorts>
```

Sets the DCI field antenna ports.

Parameters:

<AntPorts> integer
 Range: 0 to 31
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_1"](#) on page 139
 See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:SRSRequest <SRSRequest>
```

Sets the DCI field SRS request.

Parameters:

<SRSRequest> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
 on page 187.

Manual operation: See ["DCI Format 0_1"](#) on page 139
 See ["DCI Format 1_1"](#) on page 143

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:CS:DCI<s2us0>:PTDMrs <PtrsDmrs>**

If [:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:MRANk#1, sets the DCI field PTRS-DMRS association.

Parameters:

<PtrsDmrs> integer
Range: 0 to 3
*RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
on page 187.

Manual operation: See ["DCI Format 0_1"](#) on page 139

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:CS:DCI<s2us0>:DMSQinit <DmrsSeqInit>**

If [:SOURce<hw>]:BB:NR5G:UBWP:USER<ch0>:CELL<st0>:UL:BWP<dir0>:
PUSCh:TPStAtE 0, sets the DCI field DMRS sequence initialization.

Parameters:

<DmrsSeqInit> 0 | 1 | OFF | ON
*RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
on page 187.

Manual operation: See ["DCI Format 0_1"](#) on page 139
See ["DCI Format 1_1"](#) on page 143

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:CS:DCI<s2us0>:USINd <UISullInd>**

If [:SOURce<hw>]:BB:NR5G:NODE:CELL<ch0>:SUL:STATe 1, sets the DCI field
UL/SUL indicator.

Parameters:

<UISullInd> 0 | 1 | OFF | ON
*RST: 0

Example: See [Example "Configuring the CORESET settings"](#)
on page 187.

Manual operation: See ["DCI Format 0_0"](#) on page 138
See ["DCI Format 0_1"](#) on page 139

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLOc<user0>:CS:DCI<s2us0>:TPUCch <TpcPucch>**

Sets the DCI field TPC command for scheduling PUCCH.

Parameters:

<TpcPucch> integer
 Range: 0 to 3
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:PUCResind <PucchResInd>**

Sets the DCI field PUCCH resource indicator.

Parameters:

<PucchResInd> integer
 Range: 0 to 7
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:PDSHarq <PdschToHarq>**

Sets the DCI field PDSCH-to-HARQ feedback timing Indicator.

Parameters:

<PdschToHarq> integer
 Range: 0 to 7
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 1_0"](#) on page 141
 See ["DCI Format 1_1"](#) on page 143

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:SIInd <Silnd>**

Sets the DCI field system information indicator.

Parameters:

<Silnd> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 1_0"](#) on page 141

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:RESVed <Pattern>,
  <BitCount>
```

Sets the DCI padding and reserved bits.

Parameters:

<Pattern>	32 bits
*RST:	#H0
<BitCount>	integer
Range:	1 to 32
*RST:	1

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["DCI Format 1_0"](#) on page 141

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:BWIND <BWPInd>
```

Sets the DCI field BWP indicator.

Parameters:

<BWPInd>	integer
Range:	0 to 3
*RST:	0

Manual operation: See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:PRBBundling <PRBBundling>
```

Sets the DCI field PRB bundling size indicator.

Parameters:

<PRBBundling>	0 1 OFF ON
*RST:	0

Manual operation: See ["DCI Format 1_1"](#) on page 143

```
[ :SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:RMIND <RMInd>
```

Sets the DCI field rate matching indicator.

Parameters:

<RMInd> integer
 Range: 0 to 3
 *RST: 0

Manual operation: See "DCI Format 1_1" on page 143

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:ZCRTrigg <ZpCsiRsTrigg>**

Sets the DCI field ZP CSI-RS trigger.

Parameters:

<ZpCsiRsTrigg> integer
 Range: 0 to 3
 *RST: 0

Manual operation: See "DCI Format 1_1" on page 143

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:TCI <TCI>**

Sets the DCI field transmission configuration indication.

Parameters:

<TCI> integer
 Values 1 to 7 are available if [:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:TCI 1.
 Range: 0 to 7
 *RST: 0

Manual operation: See "DCI Format 1_1" on page 143

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CBGTi <CBGTi>**

Sets the DCI field CBG transmission information (CBGTI).

Parameters:

<CBGTi> integer
 Range: 0 to 255
 *RST: 0

Manual operation: See "DCI Format 1_1" on page 143

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:CBGFi <CBGFi>**

Sets the DCI field CBG flushing out information (CBGFI).

Parameters:

<CBGFI> 0 | 1 | OFF | ON
 *RST: 0

Manual operation: See ["DCI Format 1_1"](#) on page 143

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:STATe <ChanCodingState>**

Enables channel coding for the DCI type custom.

Parameters:

<ChanCodingState> 0 | 1 | OFF | ON
 *RST: 1

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Channel Coding"](#) on page 146

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DATA <DataSource>**

Sets the data source for the custom DCI.

Parameters:

<DataSource> PN9 | PN11 | PN15 | PN16 | PN20 | PN21 | PN23 | PATtern |
 DLIst | ZERO | ONE
 *RST: PN9

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Data Source"](#) on page 146

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:DLIst <DataList>**

Selects an existing data list file from the default directory or from the specific directory.

Parameters:

<DataList> string
 Filename incl. file extension or complete file path

Example: See [Example "Configuring the CORESET settings"](#) on page 187.

Manual operation: See ["Data Source"](#) on page 146

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:BITLength <PatternLength>**

If channel coding is enabled ([:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:
SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:
STATe 1), sets the number of DCI bits in the custom DCI.

Parameters:

<PatternLength> integer
Range: 12 to 128
*RST: 84

Example: See [Example "Configuring the CORESET settings"](#)
on page 187.

Manual operation: See ["Pattern Length"](#) on page 147

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:CS:DCI<s2us0>:PATtern <Pattern>,
<BitCount>**

Sets the content of the custom DCI as a bit pattern.

Parameters:

<Pattern> 128 bits
Corresponds to the value set with the command [:
SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:
USER<dir0>:BWPart<gr0>:ALLoc<user0>:CS:
DCI<s2us0>:BITLength.
*RST: #H0

<BitCount> integer
Range: 1 to 128
*RST: 1

Example: See [Example "Configuring the CORESET settings"](#)
on page 187.

Manual operation: See ["Data Source"](#) on page 146

8.15 PUCCH Commands

[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
ALLoc<user0>:PUCCh:GRPHopping..... 313

[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
ALLoc<user0>:PUCCh:HOPid..... 313

[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
ALLoc<user0>:PUCCh:ISFHopping..... 314

[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:
ALLoc<user0>:PUCCh:SHOPping..... 314

<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:FS:CYCShift.....</code>	314
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:FS:TDOCC.....</code>	314
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:FS:OCCIndex.....</code>	315
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:FS:OCCLength.....</code>	315
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:PL:SRCount.....</code>	315
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:PL:UCI:BITS.....</code>	316
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:PL:ACK:BITS.....</code>	316
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:PL:UCI:PATtern.....</code>	316
<code>[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:PL:ACK:PATtern.....</code>	316

`[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:GRPHopping <PucchGrpHopping>`

Sets the higher-layer parameter `pucch-GroupHopping`.

Parameters:

`<PucchGrpHopping>` N | ENA | DIS

N = neither

ENA = enable

DIS = disable

*RST: N

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Group Hopping"](#) on page 149

`[SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:PUCCh:HOPId <PucchHoppingId>`

Sets the parameter hopping ID.

Parameters:

`<PucchHoppingId>` integer

Range: 0 to 1024

Increment: 0

*RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Hopping ID"](#) on page 149

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:ISFHopping <PucchISFreqHop>
```

Enables intra-slot frequency hopping.

Parameters:

<PucchISFreqHop> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Intra Slot Frequency Hopping"](#) on page 149

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:SHOPping <SecongHoppingPR>
```

Sets the physical resource block (PRB) of the second hop.

Parameters:

<SecongHoppingPR> float
 Range: 0 to 273
 Increment: 1
 *RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Second Hop PRB"](#) on page 149

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:FS:CYCShift <InitialCyclicSh>
```

For PUCCH format F0 and F1, sets the initial cyclic shift.

Parameters:

<InitialCyclicSh> integer
 Range: 0 to 11
 *RST: 0

Example:

```
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:
BWPart0:ALLoc1:FMT F1
SOURce1:BB:NR5G:SCHeD:CELL0:SUBF0:USER0:
BWPart0:ALLoc1:PUCCh:FS:CYCShift 1
```

Manual operation: See ["Initial Cyclic Shift"](#) on page 149

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:FS:TDOCC <TimeDomainOcclD>
```

For PUCCH format F1, sets the time domain OCC index.

Parameters:

<TimeDomainOcclD> integer

Range: 0 to 6

*RST: 0

Example:

SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:FMT F1

SOURce1:BB:NR5G:SCHed:CELL0:SUBF0:USER0:BWPart0:ALLoc1:PUCCh:FS:TDOCC 1

Manual operation: See ["Time Domain OCC Index"](#) on page 150

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:FS:OCCindex <OCCindex>**

For PUCCH format F4, sets the OCC index.

Parameters:

<OCCindex> integer

Range: 0 to 6

*RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.**Manual operation:** See ["OCC Index"](#) on page 150

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:FS:OCCLength <PucchOCCLength>**

For PUCCH format F4, sets the OCC length.

Parameters:

<PucchOCCLength> L2 | L4

*RST: L2

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.**Manual operation:** See ["OCC Length"](#) on page 150

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PUCCh:PL:SRCount <SrCount>**

Sets the number of positive scheduling requests (SR).

Parameters:

<SrCount> integer

Range: 0 to 10

*RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.**Manual operation:** See ["Positive SR Count"](#) on page 151

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:PL:UCI:BITS <PucchUciBits>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:PL:ACK:BITS <ACKBitSize>
```

Sets the number of ACK/UCI bits.

Parameters:

<ACKBitSize> float
 Range: 0 to 128
 Increment: 1
 *RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["Number of ACK Bits"](#) on page 151

```
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:PL:UCI:PATtern <PucchUciPattern>,
  <BitCount>
[ :SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
  BWPart<gr0>:ALLoc<user0>:PUCCh:PL:ACK:PATtern <AckPattern>,
  <Bitcount>
```

Sets the ACK/UCI bits in pattern form.

Parameters:

<AckPattern> 128 bits
 Bit pattern
 *RST: #H0

<Bitcount> integer
 Pattern length, should be the same as the length set with the
 command [\[:SOURce<hw>\]:BB:NR5G:SCHeD:CELL<ch0>:](#)
 [SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLoc<user0>:](#)
 [PUCCh:PL:ACK:BITS](#).
 Range: 1 to 128
 *RST: 0

Example: See [Example "Configuring the PUCCH settings"](#) on page 192.

Manual operation: See ["ACK Pattern"](#) on page 151

8.16 PRACH Commands

<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:SCSPacing</code>	317
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:FORMat</code>	317
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:RSET</code>	318
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:RSEquence</code>	318
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:ZCZone</code>	318
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:PRINdex</code>	318
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:RBNumber?</code>	319
<code>[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:TOFFset</code>	319

`[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:SCSPacing <PrachNumerology>`

Selects a combination of the subcarrier spacing (SCS) and the cyclic prefix (CP) for PRACH.

Parameters:

<PrachNumerology> N1_25 | N5 | N15 | N30 | N60 | N120
 *RST: N30

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["SC Spacing/CP"](#) on page 105

`[SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:BWPart<gr0>:ALLOc<user0>:PRACH:FORMat <PrachFormat>`

Selects the PRACH format.

Parameters:

<PrachFormat> F0 | F1 | F2 | F3 | FA1 | FA2 | FA3 | FB1 | FB2 | FB3 | FB4 | FC0 | FC2
 *RST: FA1

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["Format"](#) on page 153

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:RSET <RestrictedSet>**

Sets the higher-layer parameter `restrictedSetConfig` that defines the type of restricted sets (unrestricted, restricted type A, restricted type B).

Parameters:

<RestrictedSet> URES | ARES | BRES
 URES = Unrestricted
 ARES = Restricted Type A
 BRES = Restricted Type B
 *RST: URES

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["Restricted Set"](#) on page 154

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:RSEquence <LogicalRootSeq>**

Sets the starting logical root sequence index.

Parameters:

<LogicalRootSeq> integer
 Range: 0 to 837
 *RST: 0

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["Logical Root Sequence"](#) on page 154

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:ZCZone <ZeroCorrZone>**

Sets the parameter `zeroCorrelationZoneConfig` according to [TS 38.211](#).

Parameters:

<ZeroCorrZone> integer
 Range: 0 to 15
 *RST: 0

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["Zero Correlation Zone"](#) on page 154

**[:SOURce<hw>]:BB:NR5G:SCHed:CELL<ch0>:SUBF<st0>:USER<dir0>:
BWPart<gr0>:ALLoc<user0>:PRACH:PRIndex <PreambleIndex>**

Sets the PRACH preamble index and thus defines which one of the 64 PRACH preambles is used.

Parameters:

<PreambleIndex> integer
 Range: 0 to 63
 *RST: 0

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["Preamble Index"](#) on page 154

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PRACH:RBNumber?**

Queries the number of resource blocks the PRACH allocation spans.

Return values:

<UserAllocRBNum> integer
 Range: 3 to 24
 *RST: 12

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Usage: Query only

Manual operation: See ["No. RB"](#) on page 107

**[:SOURce<hw>]:BB:NR5G:SCHeD:CELL<ch0>:SUBF<st0>:USER<dir0>:
 BWPart<gr0>:ALLoc<user0>:PRACH:TOFFset <PrachTimeOffset>**

Sets the timing offset base value for PRACH.

Parameters:

<PrachTimeOffset> float
 Range: -3E-4 to 3E-4
 Increment: 1E-8
 *RST: 0

Example: See [Example "Configuring the PRACH settings"](#) on page 193.

Manual operation: See ["Time Offset"](#) on page 154

8.17 Output Commands

[:SOURce<hw>]:BB:NR5G:OUTPut:SEQLen	320
[:SOURce<hw>]:BB:NR5G:OUTPut:SSOC:STATe	320
[:SOURce<hw>]:BB:NR5G:OUTPut:FMODe	320
[:SOURce<hw>]:BB:NR5G:FMODe:USRFile	321
[:SOURce<hw>]:BB:NR5G:OUTPut:SAMRmode	321
[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:SRATe?	321
[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:CONFLict	322
[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:PBRate	322
[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:VARiation	322

[:SOURce<hw>]:BB:NR5G:OUTPut:CLEVel.....	322
[:SOURce<hw>]:BB:NR5G:OUTPut:CMODE.....	323
[:SOURce<hw>]:BB:NR5G:TDWind:STATe.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:SE60K:TRTime.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S15K:TRTime.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S30K:TRTime.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S60K:TRTime.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S120K:TRTime.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S240K:TRTime.....	323
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:SE60K:TRTSamples?.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S15K:TRTSamples?.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S30K:TRTSamples?.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S120K:TRTSamples?.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S60K:TRTSamples?.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S240K:TRTSamples?.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:POWEr:MODE.....	324
[:SOURce<hw>]:BB:NR5G:OUTPut:POWEr:RSBW.....	325

`[:SOURce<hw>]:BB:NR5G:OUTPut:SEQLen <SeqLen>`

Sets the sequence length of the signal in number of frames.

Parameters:

`<SeqLen>` integer
 Range: 1 to depends on settings
 *RST: 1

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Sequence Length"](#) on page 164

`[:SOURce<hw>]:BB:NR5G:OUTPut:SSOC:STATe <SupSCOnOPCtr>`

If enabled, the subcarriers that use the same frequency as the center frequency of the baseband output are not transmitted.

Parameters:

`<SupSCOnOPCtr>` 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Suppress Subcarrier on Output Center"](#) on page 164

`[:SOURce<hw>]:BB:NR5G:OUTPut:FMODE <FilterBWP>`

Defines if and how the filter is applied, on the whole channel bandwidth or on the individual BWPs separately.

Parameters:

`<FilterBWP>` CBW | BWP | OFF | FAST | 2 | 1 | 0 | USER
 OFF = "Off"


```
CBW | 0 = "Channel BW"
BWP | 1 = "Per BWP"
FAST | 2 = "Fast"
USER = "User"
*RST:      0
```

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Filter Mode"](#) on page 164

[[:SOURce<hw>]:BB:NR5G:FMODE:USRFile <FilterFileName>

Loads the file from the default or the specified directory.

Loaded are files with extension *.vaf or *.dat.

Parameters:

<FilterFileName> string
 Complete file path incl. filename and extension

Example:

```
SOURce:BB:NR5G:OUTPut:FMODE USER
SOURce:BB:NR5G:FMODE:USRFile "/var/user/my_filter.dat"
```

Manual operation: See ["Load User Filter"](#) on page 164

[[:SOURce<hw>]:BB:NR5G:OUTPut:SAMRmode <SampRateMode>

Sets the sample rate mode.

Parameters:

<SampRateMode> MIN | FFT
 *RST: FFT

Manual operation: See ["Sample Rate Mode"](#) on page 165

[[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:SRATe?

Queries the resulting sample rate.

Suffix:

<ch0> 0 to 7

Return values:

<SampleRate> integer
 Among others, value range depends on the selected deployment scenario and channel bandwidth.
 Range: 4E6 to 5E8
 *RST: 4E6

Example: See [Example "Output settings"](#) on page 193.

Usage: Query only

Manual operation: See ["Sample Rate"](#) on page 166
See ["Playback Rate"](#) on page 166

[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:CONFLICT <Conflict>

Queries if there are conflicting settings.

Parameters:

<Conflict> 0 | 1 | OFF | ON

*RST: 0

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Conflict"](#) on page 166

[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:PBRate <PlaybackRate>

For [\[:SOURce<hw>\]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:VARIation 1](#), sets the playback speed.

Parameters:

<PlaybackRate> integer

Per default, the playback rate is the same as the calculated sample rate but the value range also depends on the installed options.

Range: 0 to 24e8

*RST: same as the calculated sample rate

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Playback Rate"](#) on page 166

**[:SOURce<hw>]:BB:NR5G:OUTPut:BBConf:ROW<ch0>:VARIation
<SRateVariation>**

Activates sample rate variation.

Parameters:

<SRateVariation> 0 | 1 | OFF | ON

*RST: 0

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Sample Rate Variation"](#) on page 166

[:SOURce<hw>]:BB:NR5G:OUTPut:CLeVel <ClippingLevel>

Sets the limit for level clipping.

Parameters:

<ClippingLevel> integer
 Range: 1 to 100
 *RST: 100

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Clipping Level"](#) on page 165

[[:SOURce<hw>]:BB:NR5G:OUTPut:CMODE <ClippingMode>

Sets the method for level clipping.

Parameters:

<ClippingMode> VECTor | SCALar
 *RST: VECTor

Example: See [Example "Output settings"](#) on page 193.

Manual operation: See ["Clipping Mode"](#) on page 165

[[:SOURce<hw>]:BB:NR5G:TDWind:STATe <TDWindow>

Enables time domain windowing.

Parameters:

<TDWindow> 0 | 1 | OFF | ON
 *RST: 0

Example: See [Example "Time domain windowing"](#) on page 193.

Manual operation: See ["Time Domain Windowing \(WOLA\)"](#) on page 167

[[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:SE60K:TRTime <TransitionTime>
[[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S15K:TRTime <TransitionTime>
[[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S30K:TRTime <TransitionTime>
[[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S60K:TRTime <TransitionTime>
[[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S120K:TRTime <TransitionTime>
[[:SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S240K:TRTime <TransitionTime>

Sets the transition time when time domain windowing is active.

The next to last block in the command syntax indicates the used SCS and CP combination.

- DL: SE<SCS>K, where E indicates the extended CP or for normal CP, the designation is omitted

Parameters:

<TransitionTime> float
 Range: 0 to 1E-5
 Increment: 1E-7
 *RST: 5E-6

Example: See [Example "Time domain windowing"](#) on page 193.

Manual operation: See ["Transition Time"](#) on page 167

```
[ :SOURce<hw>]:BB:NR5G:OUTPut:TDWind:SE60K:TRTSamples?
[ :SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S15K:TRTSamples?
[ :SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S30K:TRTSamples?
[ :SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S120K:TRTSamples?
[ :SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S60K:TRTSamples?
[ :SOURce<hw>]:BB:NR5G:OUTPut:TDWind:S240K:TRTSamples?
```

Queries the number of transition samples.

The next to last block in the command syntax indicates the used SCS and CP combination.

- DL: SE<SCS>K, where E indicates the extended CP or for normal CP, the designation is omitted

Return values:

<TransitionSampl> integer
 Range: 0 to 1000
 *RST: 0

Example: See [Example "Time domain windowing"](#) on page 193.

Usage: Query only

Manual operation: See ["Transition BB A Samples"](#) on page 168

```
[ :SOURce<hw>]:BB:NR5G:OUTPut:POWer:MODE <PowerMode>
```

Sets how the first output is leveled.

Parameters:

<PowerMode> AVG | ACTvsf | PSDConst

AVG

Average RMS power

ACTvsf

Average power in the active subframes

PSDConst

The absolute power of a particular allocation is set by multiplying the configured power spectral density (PSD) with the bandwidth of the particular allocation.

*RST: AVG

Example: SOURce1:BB:NR5G:OUTPut:POWer:MODE PSDConst

 SOURce1:BB:NR5G:OUTPut:POWer:RSBW 1000000

Manual operation: See ["Power Mode"](#) on page 169

[:SOURce<hw>]:BB:NR5G:OUTPut:POWer:RSBW <RefSystemBW>

Sets the reference bandwidth used by the leveling of the output signal at the first output.

Parameters:

<RefSystemBW> float
 Range: 15E3 to 400E6
 Increment: 1
 *RST: 1E6

Example: See [:SOURce<hw>]:BB:NR5G:OUTPut:POWer:MODE on page 324.

Manual operation: See "Reference System Bandwidth" on page 173

8.18 Trigger Commands

[:SOURce<hw>]:BB:NR5G:TRIGger:SEquence.....	325
[:SOURce<hw>]:BB:NR5G:TRIGger:SOURce.....	325
[:SOURce<hw>]:BB:NR5G:TRIGger:EXternal:SYNChronize:OUTPut.....	326
[:SOURce<hw>]:BB:NR5G:TRIGger:DElay:UNIT.....	326
[:SOURce<hw>]:BB:NR5G:TRIGger[:EXternal]:DElay.....	326
[:SOURce<hw>]:BB:NR5G:TRIGger[:EXternal]:INHibit.....	327
[:SOURce<hw>]:BB:NR5G:TRIGger:EXternal:RDElay?.....	327
[:SOURce<hw>]:BB:NR5G:TRIGger:EXternal:TDElay.....	327
[:SOURce<hw>]:BB:NR5G:TRIGger:ARM:EXECute.....	328
[:SOURce<hw>]:BB:NR5G:TRIGger:EXECute.....	328
[:SOURce<hw>]:BB:NR5G:TRIGger:RMODE?.....	328
[:SOURce<hw>]:BB:NR5G:TRIGger:SEnGth.....	328
[:SOURce<hw>]:BB:NR5G:TRIGger:SLUNit.....	329

[:SOURce<hw>]:BB:NR5G:TRIGger:SEquence <TrigMode>

Sets the trigger mode.

Parameters:

<TrigMode> AUTO | RETRigger | AAUTo | ARETrigger | SINGLE
 *RST: AUTO

Example: See Example "Configure and enable triggering" on page 194.

Manual operation: See "Trigger Mode" on page 156

[:SOURce<hw>]:BB:NR5G:TRIGger:SOURce <TrigSource>

Selects the trigger signal source and determines the way the triggering is executed. Provided are:

- Internal triggering by a command (INTernal)

- External trigger signal via one of the User x connectors
 - EGT1: External global trigger
 - EGC1: External global clock
- In master-slave mode, the external baseband synchronization signal (BBSY)
- EXternal: Setting only
 Provided only for backward compatibility with other Rohde & Schwarz signal generators.
 The R&S SMBV100B accepts this value and maps it automatically as follows:
 EXternal = EGT1

Parameters:

<TrigSource> INTernal|EGT1|EGC1|EXternal|BBSY
 *RST: INTernal

Example: See [Example "Configure and enable triggering"](#) on page 194.

Manual operation: See ["Trigger Source"](#) on page 157

**[[:SOURce<hw>]:BB:NR5G:TRIGger:EXternal:SYNChronize:OUTPut
 <TrigSyncOutpSta>**

Enables signal output synchronous to the trigger event.

Parameters:

<TrigSyncOutpSta> 0 | 1 | OFF | ON
 *RST: 1

Example: See [Example "Configure and enable triggering"](#) on page 194.

Manual operation: See ["Sync. Output to External Trigger/Sync. Output to Trigger"](#) on page 158

[[:SOURce<hw>]:BB:NR5G:TRIGger:DELay:UNIT <TrigDelUnit>

Sets the units the trigger delay is expressed in.

Parameters:

<TrigDelUnit> SAMPlE | TIME
 *RST: SAMPlE

Example: See [Example "Specifying delay and inhibit values in time units"](#) on page 194.

Manual operation: See ["\(External\) Delay Unit"](#) on page 159

[[:SOURce<hw>]:BB:NR5G:TRIGger[:EXternal]:DELay <TrigExtDelay>

Sets the trigger delay.

Parameters:

<TrigExtDelay> float
 Range: 0 to 2147483647
 Increment: 0.01
 *RST: 0

Example: See [Example "Configure and enable triggering"](#) on page 194.

Manual operation: See ["\(Specified\) External Delay/\(Specified\) Trigger Delay"](#) on page 159

[:SOURce<hw>]:BB:NR5G:TRIGger[:EXternal]:INHibit <TrigExtInhibit>

Specifies the duration by which a restart is inhibited.

Parameters:

<TrigExtInhibit> integer
 Range: 0 to dynamic
 *RST: 0

Example: See [Example "Configure and enable triggering"](#) on page 194.

Manual operation: See ["External / Trigger Inhibit"](#) on page 159

[:SOURce<hw>]:BB:NR5G:TRIGger:EXternal:RDELay?

Queries the time (in seconds) an external trigger event is delayed for.

Return values:

<ResExtDelaySec> float
 Range: 0 to 688
 Increment: 250E-12
 *RST: 0

Example: See [Example "Specifying delay and inhibit values in time units"](#) on page 194.

Usage: Query only

Manual operation: See ["Actual Trigger Delay/Actual External Delay"](#) on page 159

[:SOURce<hw>]:BB:NR5G:TRIGger:EXternal:TDELay <TrigExtTimeDel>

Specifies the trigger delay for external triggering. The value affects all external trigger signals.

Parameters:

<TrigExtTimeDel> float
 Range: 0 to 688
 Increment: 250E-12
 *RST: 0

Example: See [Example "Specifying delay and inhibit values in time units"](#) on page 194.

Manual operation: See ["\(Specified\) External Delay/\(Specified\) Trigger Delay"](#) on page 159

[:SOURce<hw>]:BB:NR5G:TRIGger:ARM:EXECute

Stops signal generation; a subsequent trigger event restarts signal generation.

Example:

```
SOURce1:BB:NR5G:TRIGger:SOURce INT
SOURce1:BB:NR5G:TRIGger:SEquence ARETrigger
SOURce1:BB:NR5G:TRIGger:EXECute
// executes a trigger, signal generation starts
SOURce1:BB:NR5G:TRIGger:ARM:EXECute
// signal generation stops
SOURce1:BB:NR5G:TRIGger:EXECute
// executes a trigger, signal generation starts again
```

Usage: Event

Manual operation: See ["Arm"](#) on page 157

[:SOURce<hw>]:BB:NR5G:TRIGger:EXECute

Executes a trigger.

Example: See [Example "Configure and enable triggering"](#) on page 194.

Usage: Event

Manual operation: See ["Execute Trigger"](#) on page 157

[:SOURce<hw>]:BB:NR5G:TRIGger:RMODE?

Queries the signal generation status.

Return values:

```
<TrigRunMode>    STOP | RUN
*RST:             STOP
```

Example:

```
SOURce1:BB:NR5G:TRIGger:RMODE?
```

Usage: Query only

Manual operation: See ["Running/Stopped"](#) on page 157

[:SOURce<hw>]:BB:NR5G:TRIGger:SLENgth <TrigSeqLen>

Defines the length of the signal sequence that is output in the `SINGLE` trigger mode.

Parameters:

<TrigSeqLen> integer
 Range: 1 to dynamic
 *RST: 1

Example: See [Example "Configure and enable triggering"](#) on page 194.

Manual operation: See ["Trigger Signal Duration"](#) on page 157

[:SOURce<hw>]:BB:NR5G:TRIGger:SLUNit <TrigSeqLenUnit>

Defines the unit for the entry of the signal sequence length.

Parameters:

<TrigSeqLenUnit> SEQUENCE | SAMPLE
 *RST: SEQUENCE

Example:

SOURce1:BB:NR5G:TRIGger:SEQUence SING
 SOURce1:BB:NR5G:TRIGger:SLUNit SEQ
 SOURce1:BB:NR5G:TRIGger:SLENgth 2

Manual operation: See ["Signal Duration Unit"](#) on page 157

8.19 Marker Commands

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:MODE	329
[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:DElay	329
[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:PERiod	330
[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:ROFFset	330
[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:FOFFset	330
[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:ONTIME	330
[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:OFFTime	330

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:MODE <MarkMode>

Defines the signal for the selected marker output.

Parameters:

<MarkMode> SUBFram | FRAM | REStart | PERiod | RATio | SFNRestart
 *RST: FRAM

Example: See [Example "Configure and enable standard marker signals"](#) on page 194.

Manual operation: See ["Marker Mode"](#) on page 160

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:DElay <MarkDelay>

Defines the delay between the signal on the marker outputs and the start of the signals.

Parameters:

<MarkDelay> float
 Range: 0 to 16777215
 Increment: 1E-3
 *RST: 0

Example: See [Example "Configure and enable standard marker signals"](#) on page 194.

Manual operation: See ["Marker x Delay"](#) on page 162

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:PERiod <MarkUsrPer>

Sets the period of the user-defined period marker.

Parameters:

<MarkUsrPer> integer
 Range: 1 to 4294967295
 *RST: 2

Example: See [Example "Configure and enable standard marker signals"](#) on page 194.

Manual operation: See ["Marker Mode"](#) on page 160

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:ROFFset <MarkRiseOffs>

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:FOFFset <MarkFallOffs>

Sets the rise/fall offset.

Parameters:

<MarkFallOffs> integer
 Range: -640000 to 640000
 *RST: 0

Example: See [Example "Configure and enable standard marker signals"](#) on page 194.

Manual operation: See ["Rise/Fall Offset"](#) on page 161

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:ONTime <MarkTimeOn>

[:SOURce<hw>]:BB:NR5G:TRIGger:OUTPut<ch>:OFFTime <MarkTimeOff>

Sets the number of samples during which the marker output is on or off.

Parameters:

<MarkTimeOff> integer
 Range: 1 to 16777215
 *RST: 1

Example: See [Example "Configure and enable standard marker signals"](#) on page 194.

Manual operation: See ["Marker Mode"](#) on page 160

8.20 Clock Commands

[\[:SOURce<hw>\]:BB:NR5G:CLOCK:SOURce](#).....331

[:SOURce<hw>]:BB:NR5G:CLOCK:SOURce <ClockSour>

Selects the clock source:

- **INTernal:** Internal clock reference

Parameters:

<ClockSour> INTernal
 *RST: INTernal

Example: See [Example "Clock settings"](#) on page 195.

Manual operation: See ["Clock Source"](#) on page 162

Glossary: Abbreviations

Symbols

5G NR: 5G new radio

B

BW: Bandwidth

BWP: Carrier bandwidth part

C

CP: Cyclic prefix

CP-OFDM: Cyclic prefix OFDM, as known from LTE

D

DFT-s-OFDM: Discrete Fourier transform spread OFDM

DMRS: Demodulation reference signal

E

ECP: Extended cyclic prefix

F

FR: Frequency range

G

gNB: 5G NR base station

N

NCP: Normal cyclic prefix

P

PSS: Primary synchronisation signal

PTRS: Phase tracking reference signal

R

RB: Resource block

RE: Resource element

RS: Reference signal

S

SCS: Subcarrier spacing

SRS: Sounding reference signal

SS: Synchronisation signal

SSS: Secondary synchronisation signal

Glossary: 5G NR 3GPP Specifications, Documents with Further Information

Symbols

[1]: Rohde & Schwarz
Poster "Be ahead in 5G. Demystifying 5G NR"

T

TS 38.101: "User Equipment (UE) radio transmission and reception"

TS 38.104: "Base Station (BS) radio transmission and reception"

TS 38.133: "Requirements for support of radio resource management"

TS 38.141: "Base Station (BS) conformance testing"

TS 38.141-1: "Base Station (BS) conformance testing Part 1: Conducted conformance testing"

TS 38.141-2: "Base Station (BS) conformance testing Part 2: Radiated conformance testing"

TS 38.201: "Physical layer; General description"

TS 38.211: "Physical channels and modulation"

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